ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2525 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

CONTENTS

Set 1	**HSN-CN.A.1**	Write complex numbers in the form a + bi	7
Set 2	**HSN-CN.A.2**	Add and subtract complex numbers	44
Set 3	**HSN-CN.A.2**	Multiply complex numbers and simplify powers of i	81
Set 4	**HSN-CN.C.7**	Solve quadratic equations with complex solutions	118
Set 5	**HSN-CN.C.8**	Factor polynomials over the complex numbers	155
Set 6	**HSN-CN.C.9**	Use the Fundamental Theorem of Algebra	192
Set 7	**HSA-SSE.A.1.a**	Interpret the parts of polynomial, rational, and exponential expressions in context	229
Set 8	**HSA-SSE.A.1.b**	Interpret complicated expressions by viewing parts as single entities	266
Set 9	**HSA-SSE.A.2**	Factor higher-degree expressions by recognizing differences of squares and sums	303
Set 10	**HSA-SSE.A.2**	Rewrite expressions in quadratic form and factor	340
Set 11	**HSA-SSE.B.4**	Derive the formula for the sum of a finite geometric series	377
Set 12	**HSA-SSE.B.4**	Use the finite geometric series formula to solve problems	414
Set 13	**HSA-APR.A.1**	Add and subtract polynomials of higher degree	451
Set 14	**HSA-APR.A.1**	Multiply polynomials of higher degree	488
Set 15	**HSA-APR.B.2**	Find remainders with the Remainder Theorem	525
Set 16	**HSA-APR.B.2**	Use p(a) = 0 to find factors	562
Set 17	**HSA-APR.B.3**	Identify zeros and multiplicities from a factorization	599

CONTENTS (continued)

Set 18	**HSA-APR.B.3**	Construct a rough graph of a polynomial from its zeros	636
Set 19	**HSA-APR.C.4**	Prove polynomial identities	679
Set 20	**HSA-APR.C.4**	Use polynomial identities to describe numerical relationships	716
Set 21	**HSA-APR.C.5**	Expand with the Binomial Theorem	753
Set 22	**HSA-APR.D.6**	Rewrite rational expressions by polynomial division	790
Set 23	**HSA-APR.D.6**	Rewrite simple rational expressions by inspection	827
Set 24	**HSA-APR.D.7**	Multiply and divide rational expressions	864
Set 25	**HSA-APR.D.7**	Add and subtract rational expressions	901
Set 26	**HSA-CED.A.1**	Create and solve rational, radical, and exponential equations from contexts	938
Set 27	**HSA-CED.A.1**	Create and solve nonlinear inequalities from contexts	975
Set 28	**HSA-CED.A.2**	Create and graph nonlinear two-variable models	1012
Set 29	**HSA-CED.A.3**	Represent nonlinear constraints and judge viable options	1049
Set 30	**HSA-CED.A.4**	Rearrange rational, radical, and exponential formulas	1086
Set 31	**HSA-REI.A.2**	Solve rational equations and identify extraneous solutions	1123
Set 32	**HSA-REI.A.2**	Solve radical equations and identify extraneous solutions	1160
Set 33	**HSA-REI.D.11**	Explain why intersections of nonlinear graphs solve f(x) = g(x)	1197
Set 34	**HSA-REI.D.11**	Approximate solutions of f(x) = g(x)	1234

CONTENTS (continued)

Set 35	**HSF-TF.A.1**	Use radian measure as arc length on the unit circle	1271
Set 36	**HSF-TF.A.2**	Find exact trigonometric values beyond the first quadrant	1308
Set 37	**HSF-TF.A.2**	Explain the extension of sine and cosine to all real numbers	1345
Set 38	**HSF-TF.B.5**	Write sinusoidal models with given amplitude, period, and midline	1382
Set 39	**HSF-TF.B.5**	Interpret the parameters of sinusoidal models	1419
Set 40	**HSF-TF.C.8**	Prove the Pythagorean identity	1456
Set 41	**HSF-TF.C.8**	Use the Pythagorean identity to find trigonometric values	1493
Set 42	**HSF-IF.B.4**	Interpret key features of polynomial and rational models	1530
Set 43	**HSF-IF.B.4**	Interpret periodic models from graphs and tables	1567
Set 44	**HSF-IF.B.5**	State domains of radical, rational, logarithmic, and polynomial models	1604
Set 45	**HSF-IF.B.6**	Calculate and interpret average rates of change of nonlinear models	1641
Set 46	**HSF-IF.C.7.b**	Graph square root and cube root functions with transformations	1678
Set 47	**HSF-IF.C.7.b**	Graph piecewise-defined and absolute value functions	1715
Set 48	**HSF-IF.C.7.c**	Graph polynomial functions showing zeros and end behavior	1752
Set 49	**HSF-IF.C.7.e**	Graph exponential functions with intercepts, asymptote, and end behavior	1789
Set 50	**HSF-IF.C.7.e**	Graph logarithmic functions with intercept, asymptote, and end behavior	1826
Set 51	**HSF-IF.C.7.e**	Graph trigonometric functions showing period, midline, and amplitude	1863

CONTENTS (continued)

Set 52	**HSF-IF.C.8.b**	Rewrite exponential functions to reveal rates over other time units	1900
Set 53	**HSF-IF.C.9**	Compare properties of Algebra II functions in different representations	1937
Set 54	**HSF-BF.A.1.b**	Build models by combining Algebra II function types	1974
Set 55	**HSF-BF.B.3**	Identify vertical shifts, vertical stretches, and horizontal shifts of Algebra II functio	2011
Set 56	**HSF-BF.B.3**	Horizontal stretches, compressions, and reflections f(kx)	2048
Set 57	**HSF-BF.B.3**	Recognize even and odd functions	2085
Set 58	**HSF-BF.B.4.a**	Write inverses of radical, rational, cubic, exponential, and logarithmic functions	2122
Set 59	**HSF-LE.A.4**	Write exact logarithmic solutions of exponential equations	2159
Set 60	**HSF-LE.A.4**	Evaluate logarithmic solutions of exponential models with technology	2196
Set 61	**HSS-ID.A.4**	Estimate population percentages with a normal distribution	2233
Set 62	**HSS-ID.A.4**	Recognize data sets for which a normal model is not appropriate	2270
Set 63	**HSS-IC.A.1**	Distinguish population, sample, parameter, and statistic	2307
Set 64	**HSS-IC.A.2**	Decide whether a model is consistent with observed results using a simulation	2344
Set 65	**HSS-IC.B.3**	Identify sample surveys, experiments, and observational studies	2381
Set 66	**HSS-IC.B.3**	Explain random sampling versus random assignment	2418
Set 67	**HSS-IC.B.4**	Estimate a population proportion with a margin of error from a simulation	2455
Set 68	**HSS-IC.B.4**	Estimate a population mean with a margin of error from a simulation	2492

CONTENTS (continued)

Set 69	**HSS-IC.B.5**	Compare two treatments with a re-randomization simulation	2529
Set 70	**HSS-IC.B.6**	Evaluate claims in reports based on data	2566
Set 71	**HSS-MD.B.6**	Design and judge fair decision procedures with probabilities	2603
Set 72	**HSS-MD.B.7**	Analyze decisions and strategies with probabilities	2640
Set 73	**HSN-CN.A.3**	Use conjugates to find moduli and quotients of complex numbers	2677
Set 74	**HSF-IF.C.7.d**	Graph rational functions with zeros and asymptotes	2714
Set 75	**HSF-IF.C.7.d**	Distinguish holes from vertical asymptotes	2751
Set 76	**HSF-BF.A.1.c**	Compose functions and interpret compositions with domains	2788
Set 77	**HSF-BF.B.4.b**	Verify inverses by composition on the stated domains	2825
Set 78	**HSF-BF.B.4.c**	Read values of an inverse function from graphs and tables	2862
Set 79	**HSF-BF.B.4.d**	Restrict a domain to make a function invertible and write the inverse	2899
Set 80	**HSF-BF.B.5**	Use the inverse relationship of exponents and logarithms to solve equations	2936
Set 81	**HSF-TF.A.3**	Derive values from special triangles and relate π - x, π + x, 2π - x to x	2973
Set 82	**HSF-TF.A.4**	Explain the symmetry and periodicity of trigonometric functions with the unit circ	3010
Set 83	**HSF-TF.B.6**	Choose and justify domain restrictions that make trigonometric functions invertib	3047
Set 84	**HSF-TF.B.7**	Solve trigonometric equations in models with inverse functions	3084

Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write √(−49) in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. √(−49) = √49 · √(−1) = 7i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write √(−18) in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

3i√2 (that is, 3√2 i)

GRADING NOTE Form required: simplest radical form. √(−18) = √9 · √2 · i = 3√2 i. Not accepted: √18 i (not simplified).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form $a + bi$. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

$6 - 2\sqrt{5}\,i$; real part 6, imaginary part $-2\sqrt{5}$

GRADING NOTE Form required: $a + bi$ with both parts named. $\sqrt{(-20)} = 2\sqrt{5}\,i$. The imaginary part is the real number $-2\sqrt{5}$, not $-2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write (−4 + √(−75)) / 2 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + (5√3/2)i; real part −2, imaginary part 5√3/2

GRADING NOTE Form required: a + bi with both parts named. √(−75) = 5√3 i; divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write (10 + √(−64)) / 4 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

5/2 + 2i; real part 5/2, imaginary part 2

GRADING NOTE Form required: a + bi with both parts named (5/2 or 2.5 accepted). √(−64) = 8i; 10/4 = 5/2 and 8/4 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5√2

GRADING NOTE Form required: simplest radical form. 50 = 25 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify √72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6√2

GRADING NOTE Form required: simplest radical form. 72 = 36 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 4√3

B. 3√2

C. 6√2

D. 2√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2√3

GRADING NOTE $12 = 4 \cdot 3$, so $\sqrt{12} = 2\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. A square root of a sum is not the sum of the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify √28.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2√7

GRADING NOTE Form required: simplest radical form. 28 = 4 · 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q5**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 9 or x = −9

GRADING NOTE Any order. Both 9^2 and $(-9)^2$ equal 81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 = (-3)(-3) = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Short answer

CORRECT ANSWER

No: the square of any real number is 0 or positive.

GRADING NOTE Any equivalent reason (a positive times a positive and a negative times a negative are both positive).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. 18

B. $\sqrt{-36}$

C. -6

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −6

GRADING NOTE $(-6)^2 = 36$. $\sqrt{-36}$ is not a real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

49

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

Write (8 + 6√2) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

27

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 3√2

GRADING NOTE Any equivalent form. Divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (−6 + 4√5) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

28

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3 + 2√5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

29

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both terms must be divided: $(10 + \sqrt{7})/5 = 2 + \sqrt{7}/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

(12 − 9√3) / 3 equals

A. 4 − 3√3

B. 4 − 9√3

C. 4 − 3√1

D. 12 − 3√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

30

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 − 3√3

GRADING NOTE 12/3 = 4 and 9√3/3 = 3√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write (−4 + 2√3) / (−2) as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

31

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2 − √3

GRADING NOTE Any equivalent form. −4/(−2) = 2; 2√3/(−2) = −√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and (x − 3) + 2yi = 5 − 8i.
Find x and y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

33

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 8, y = −4

GRADING NOTE Form required: both values. Equal complex numbers have equal real parts (x − 3 = 5) and equal imaginary parts (2y = −8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and 3a + (b + 1)i = −12 + 4i. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

34

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = −4, b = 3

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

35

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + yi = 7 + 0i, so x = 7 and y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and (p + q) + (p − q)i = 10 + 2i. What is p?

A. 6

B. 5

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

36

QUESTION ID

S1-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE p + q = 10 and p − q = 2 give p = 6 and q = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

37

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

m = −3; the number is 5

GRADING NOTE Form required: both. The imaginary part m + 3 must be 0, so m = −3 and $m^2 - 4 = 9 - 4 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for −3 + 5i.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

39

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 5)

GRADING NOTE Form required: an ordered pair (real part, imaginary part).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point (4, −2) in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

40

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 − 2i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

41

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{(6^2 + 8^2)} = 10$ (the modulus $|6 + 8i|$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

42

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i, the point (a, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. -7

B. $7 + i$

C. $-7i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

43

QUESTION ID

S1-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $-7i$

GRADING NOTE $-7i = 0 - 7i$, the point $(0, -7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

46

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7i

GRADING NOTE Form required: a + bi. 3 − 7 = −4 and 5i + 2i = 7i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify (6 − 4i) − (−2 + 9i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

47

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13i

GRADING NOTE Form required: a + bi. Subtract both parts: 6 − (−2) = 8 and −4i − 9i = −13i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify (−1 + √(−16)) + (5 − √(−9)). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

48

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

4 + i

GRADING NOTE Form required: a + bi. (−1 + 4i) + (5 − 3i) = 4 + i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

49

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 (that is, 4 + 0i)

GRADING NOTE Form required: a + bi or the real number 4. Real: 2.5 + 1.5 = 4; imaginary: −1 − 3 + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the complex number w for which
(4 − 7i) + w = −2 + 3i.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

50

QUESTION ID

S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

−6 + 10i

GRADING NOTE Form required: a + bi. w = (−2 + 3i) − (4 − 7i) = −6 + 10i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (3 + 5x) + (−7 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

52

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

Simplify (6 − 4x) − (−2 + 9x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

53

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13x

GRADING NOTE Any equivalent form. The minus sign applies to both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

(2 − 3y) − (5 − y) equals

A. −3 − 2y

B. 7 − 4y

C. −3 + 2y

D. −3 − 4y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

54

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3 − 2y

GRADING NOTE 2 − 5 = −3 and −3y + y = −2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 + 4t.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

55

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2t - 6t = -4t$, so the sum is $5 - 4t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (0.5 − x) − (1.5 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

56

QUESTION ID

S2-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

58

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

59

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

60

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE $-1 - 3 = -4$ and $-4 + 4 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

61

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2.5 + 1.5 = 4.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

62

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−16) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

64

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt{(-9)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

65

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

√(−25) equals

A. −5

B. 25i

C. −5i

D. 5i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

66

QUESTION ID

S2-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5i

GRADING NOTE √(−25) = √25 · i = 5i (the principal root).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

67

QUESTION ID

S2-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of i as the principal square root of −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

68

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

k is a real number. For what value of k is (3 + ki) + (k − 5i) a real number? Give that real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

70

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

k = 5; the sum is 8

GRADING NOTE Form required: k and the number. The imaginary part k − 5 must be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
(2k − 6) + 4i − (k + 1 − i) a pure imaginary number
(real part 0)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

71

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. Real part: 2k − 6 − k − 1 = k − 7 = 0. The sum is then 5i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its conjugate is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

72

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (a + bi) + (a − bi) = 2a, which is 0 only when a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and (a + 3i) − (2 + ai) is real. What is a?

A. 0

B. 2

C. 3

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

73

QUESTION ID

S2-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Imaginary part 3 − a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

Explain why the sum of a + bi and a − bi is always a real number when a and b are real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

74

QUESTION ID

S2-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The imaginary parts b and −b add to 0, leaving 2a, a real number.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

Plot 2 + i and 1 + 3i as points in the complex plane (real axis horizontal). What point represents their sum?

Sum point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

76

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4), the number 3 + 4i

GRADING NOTE Form required: the point or the number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral. What kind of quadrilateral is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

77

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A parallelogram

GRADING NOTE Any equivalent answer. Adding complex numbers adds their vectors (parallelogram rule).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

78

QUESTION ID

S2-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE z − w is the displacement from w to z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points 5 + 2i and 1 − i in the complex plane?

A. 25

B. √13

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

79

QUESTION ID

S2-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE |(5 + 2i) − (1 − i)| = |4 + 3i| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

80

QUESTION ID

S2-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2 + i$

GRADING NOTE Form required: a + bi. The average $((6 - 2i) + (-2 + 4i))/2 = 2 + i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply (3 + 2i)(4 − 5i). Write the product in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

83

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

22 − 7i

GRADING NOTE Form required: a + bi. $12 - 15i + 8i - 10i^2 = 12 - 7i + 10 = 22 - 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

84

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

3 − 4i

GRADING NOTE Form required: $a + bi$. $4 - 4i + i^2 = 4 - 4i - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

85

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

34 (that is, 34 + 0i)

GRADING NOTE Form required: $a + bi$ or the real number 34. $25 + 15i - 15i - 9i^2 = 25 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(1 + i)^3$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

86

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + 2i

GRADING NOTE Form required: a + bi. $(1 + i)^2 = 2i$, and $2i(1 + i) = 2i + 2i^2 = -2 + 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

87

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 4i

GRADING NOTE Form required: a + bi. 23 = 4 · 5 + 3, so $i^{23} = i^3 = -i$; $-i(4 - i) = -4i + i^2 = -1 - 4i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply (3 + 2x)(4 − 5x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

89

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$12 - 7x - 10x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

90

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$4 - 4x + x^2$

GRADING NOTE Any equivalent form. Not $x^2 + 4$ (missing the middle term).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(5 - 3x)(5 + 3x)$ equals

A. $25 + 9x^2$

B. $10 - 9x^2$

C. $25 - 9x^2$

D. $25 - 30x - 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

91

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $25 - 9x^2$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

92

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(1 + x)^2 = 1 + 2x + x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply x(4 − x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

93

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$4x - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

95

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. Definition of i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

96

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-i$

GRADING NOTE Any equivalent form. $i^3 = i^2 \cdot i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

97

QUESTION ID

S3-B2-Q3

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. $i^4 = (i^2)^2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

98

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-10i^2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^8

B. i^9

C. i^3

D. i^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

99

QUESTION ID

S3-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. i^8

GRADING NOTE Powers of i repeat every 4; 8 is a multiple of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

101

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 23 = 4 · 5 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

102

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 50 = 4 · 12 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

103

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE $36 = 4 \cdot 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

104

QUESTION ID

S3-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $101 = 4 \cdot 25 + 1$, so the remainder is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

105

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

39 = 4 · 9 + 3

GRADING NOTE Form required: q = 9 and r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is (a + 2i)(3 − i) a real number? Give the product.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

107

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

a = 6; the product is 20

GRADING NOTE Form required: both. (a + 2i)(3 − i) = (3a + 2) + (6 − a)i; the imaginary part 6 − a = 0. Product 18 + 2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

108

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Product: (2 − b) + (2 + b)i; real part 2 − b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

109

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example (1 + i)(1 − i) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. $c = 0$

B. $c = 1$

C. $c = -1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

110

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. c = 0

GRADING NOTE $(c + i)^2 = c^2 - 1 + 2ci$; imaginary part $2c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

Show that (a + bi)(a − bi) is a real number for all real a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

111

QUESTION ID

S3-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - abi + abi - b^2i^2 = a^2 + b^2$, which is real.

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

113

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11/5 − (2/5)i

GRADING NOTE Form required: a + bi. Multiply by (1 − 2i)/(1 − 2i): (3 + 4i)(1 − 2i)/5 = (11 − 2i)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write 10 / (3 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

114

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 + i

GRADING NOTE Form required: a + bi. 10(3 + i)/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by 2 − 5i, multiply numerator and denominator by 2 − 5i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

115

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the conjugate 2 + 5i; $(2 - 5i)^2$ is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. −1 + 2i

B. 2 + i

C. 1 − 2i

D. 1 + 2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

116

QUESTION ID

S3-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 − 2i

GRADING NOTE (2 + i)(−i)/(i · −i) = (−2i + 1)/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

Explain why multiplying by the conjugate of the denominator always gives a real denominator.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

117

QUESTION ID

S3-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$(c + di)(c - di) = c^2 + d^2$, a real number (and positive unless $c = d = 0$).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 49 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

120

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7i or x = −7i

GRADING NOTE Form required: both solutions, exact. $x^2 = -49$, $x = \pm 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

121

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2 + 3i or

x = 2 − 3i

GRADING NOTE Form required: both, a + bi. Discriminant 16 − 52 = −36; x = (4 ± 6i)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

122

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = −1/2 + (3/2)i or
x = −1/2 − (3/2)i

GRADING NOTE Form required: both, a + bi. Discriminant 4 − 40 = −36; x = (−2 ± 6i)/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

123

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x = -3 + 2\sqrt{5}\,i$ or
$x = -3 - 2\sqrt{5}\,i$

GRADING NOTE Form required: both, exact. $x + 3 = \pm\sqrt{-20} = \pm 2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

124

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 1 + (2\sqrt{3}/3)i$ or
$x = 1 - (2\sqrt{3}/3)i$

GRADING NOTE Form required: both, exact, a + bi. Discriminant $36 - 84 = -48 = (4\sqrt{3}\, i)^2$; $x = (6 \pm 4\sqrt{3}\, i)/6$. Check: product 7/3 and sum 2 agree with c/a and −b/a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

126

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 16 − 52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B1-Q1**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

127

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 4 − 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

128

QUESTION ID

S4-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A negative discriminant gives two non-real solutions and no real ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

129

QUESTION ID

S4-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3, −6, 7

GRADING NOTE Standard form $ax^2 + bx + c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the quadratic formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

130

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−36) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

132

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

Write √(−48) in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

133

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4√3 i

GRADING NOTE Form required: simplest radical form. 48 = 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

√(−20) equals

A. 2√5 i

B. 4√5 i

C. −2√5

D. 10i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

134

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2√5 i

GRADING NOTE √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

135

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(-49)} = 7i.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write (−2 + √(−36)) / 4 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

136

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2 + (3/2)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B2-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $(x - 2)^2 = 9$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

138

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 or x = −1

GRADING NOTE Form required: both. $x - 2 = \pm 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

139

QUESTION ID

S4-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 + √5 or
x = −1 − √5

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B3-Q2**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

140

QUESTION ID

S4-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Also $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

141

QUESTION ID

S4-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 and −9

GRADING NOTE $x + 4 = \pm 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

Solve $x^2 = 50$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

142

QUESTION ID

S4-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 5\sqrt{2}$ or $x = -5\sqrt{2}$

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

144

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

b = −6, c = 13

GRADING NOTE Form required: both. With real coefficients the other solution is 3 + 2i; sum 6 = −b, product 9 + 4 = 13 = c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Write the equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

145

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x + 17 = 0$

GRADING NOTE Any equivalent equation. Sum −2, product 1 + 16 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

146

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Non-real solutions of real quadratics come in conjugate pairs (± in the formula).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

147

QUESTION ID

S4-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 25 = 0$

GRADING NOTE $(5i)^2 + 25 = -25 + 25 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Explain why $x^2 - (2 + i)x = 0$ has the solution $2 + i$ but not the solution $2 - i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

148

QUESTION ID

S4-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Its coefficient 2 + i is not real, so solutions need not come in conjugate pairs: the solutions are 0 and 2 + i.

GRADING NOTE Any equivalent explanation naming the non-real coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

150

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6 < k < 6$

GRADING NOTE Form required: an inequality. Discriminant $k^2 - 36 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

For which real values of c does $x^2 - 4x + c = 0$ have non-real solutions? Give them for c = 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

151

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

c > 4; for c = 8, x = 2 ± 2i

GRADING NOTE Form required: both parts. Discriminant $16 - 4c < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

152

QUESTION ID

S4-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its discriminant $k^2 + 4$ is always positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when m =

A. 3

B. 36

C. −9

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

153

QUESTION ID

S4-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE 36 − 4m = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For c > 4, write the solutions of $x^2 - 4x + c = 0$ in terms of c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

154

QUESTION ID

S4-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 2 \pm i\sqrt{(c - 4)}$

GRADING NOTE Form required: exact in terms of c. $(4 \pm \sqrt{(16 - 4c)})/2 = 2 \pm \sqrt{(4 - c)} = 2 \pm i\sqrt{(c - 4)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

157

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5i)(x - 5i)$

GRADING NOTE Any order of factors. $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

158

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + i\sqrt{7})(x - i\sqrt{7})$

GRADING NOTE Any order. $x^2 - (i\sqrt{7})^2 = x^2 + 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

159

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(2x + 3i)(2x - 3i)$

GRADING NOTE Any equivalent factorization (for example $4(x + 3i/2)(x - 3i/2)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

160

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)(x - 2i)(x + 2i)$

GRADING NOTE Any order. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

161

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - (3 + 2i))(x - (3 - 2i))$

GRADING NOTE Any equivalent form. Its zeros are $3 \pm 2i$ (discriminant -16). Check: sum 6, product $9 + 4 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

163

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 7)(x − 7)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

164

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3x + 2)(3x − 2)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 - 1)(x^2 + 1)$

D. $(x^2 + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

165

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x^2 - 1)(x^2 + 1)$

GRADING NOTE $(x^2)^2 - 1^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

166

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + \sqrt{5})(x - \sqrt{5})$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

169

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 25$

GRADING NOTE Any equivalent form. $-25i^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

170

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$4x^2 + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

171

QUESTION ID

S5-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3i)^2 = 9i^2 = -9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 - 2x + 2$

C. $x^2 + 2$

D. $x^2 - 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

172

QUESTION ID

S5-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 - 2x + 2$

GRADING NOTE Sum 2, product 1 + 1 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

173

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 25 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

175

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±5i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 6x + 13 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

176

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 ± 2i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

177

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Factor theorem, also for complex r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. 2i and −2i

B. 4i and −4i

C. −4

D. 2 and −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

178

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2i and −2i

GRADING NOTE $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 5 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

179

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x = -1 \pm 2i$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 2)(x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$

GRADING NOTE Any order. $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$, and $x^2 - 2x + 4$ has zeros $1 \pm i\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(x + i)(x − i)(x + 2i)(x − 2i)

GRADING NOTE Any order. $(x^2 + 1)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Discriminant $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - 3i$

D. $x - (-3/2 + (3\sqrt{3}/2)i)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D.

$x - (-3/2 + (3\sqrt{3}/2)i)$

GRADING NOTE $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$; $x^2 + 3x + 9$ has zeros $-3/2 \pm (3\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include 2i and 1 − i.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 4)(x^2 - 2x + 2) = x^4 - 2x^3 + 6x^2 - 8x + 8$

GRADING NOTE Any equivalent form. Conjugates −2i and 1 + i must also be zeros, so degree 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. −4i and 2 − i are also zeros: 1 + 2 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Non-real zeros come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and −i. Its third zero is

A. −5

B. 5i

C. 1

D. i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. i

GRADING NOTE Conjugate of −i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Explain why a polynomial with real coefficients and degree 3 must have at least one real zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It has 3 complex zeros with multiplicity; non-real zeros pair up, so they come two at a time, leaving at least one real zero.

GRADING NOTE Any equivalent explanation using the Fundamental Theorem of Algebra and conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

SET 6 • HSN-CN.C.9 • THE COMPLEX NUMBER SYSTEM

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? Name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

194

QUESTION ID
S6-M-Q1

TYPE
Short answer

CORRECT ANSWER
5, by the Fundamental Theorem of Algebra (degree 5)

GRADING NOTE Form required: the number and the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

List every complex zero of $p(x) = (x - 2)^3(x^2 + 9)$ with its multiplicity, and check that the multiplicities add to the degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

195

QUESTION ID
S6-M-Q2

TYPE
Short answer

CORRECT ANSWER
2 (multiplicity 3), 3i (multiplicity 1), −3i (multiplicity 1); 3 + 1 + 1 = 5 = degree

GRADING NOTE Form required: all zeros with multiplicities and the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Which argument shows that every quadratic $p(x) = ax^2 + bx + c$ (real a, b, c, $a \neq 0$) has exactly two complex zeros, counted with multiplicity?

A. $x^2 - 5x + 6 = (x - 2)(x - 3)$ has two zeros, so every quadratic has two zeros.

B. By the quadratic formula the zeros are $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. If $b^2 - 4ac < 0$ the square root does not exist, so there are no zeros; otherwise there are two, so a quadratic never has more than two zeros.

C. The graph of every quadratic is a parabola, and a parabola that opens up or down always crosses the x-axis in exactly two places, so every quadratic $ax^2 + bx + c$ has exactly two zeros.

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

GRADING NOTE Every case of the discriminant gives two zeros counted with multiplicity (for $D < 0$, $\sqrt{D} = i\sqrt{(-D)}$ is a complex number). The others check one example, use a false graph fact ($x^2 + 1$ never crosses), or ignore complex zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$ and check that the count agrees with the Fundamental Theorem of Algebra.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

3, −3, 3i, −3i; four zeros for degree 4

GRADING NOTE Form required: all four and the check. $(x^2 - 9)(x^2 + 9)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Which statement is correct?

A. The student is right: a polynomial that does not factor over the integers has no zeros.

B. It has exactly 3 real zeros, because the Fundamental Theorem of Algebra says a polynomial of degree 3 has 3 zeros, and a polynomial with real coefficients has only real zeros.

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

D. It has 3 complex zeros counted with multiplicity, and all three could be non-real, because a polynomial that does not factor over the integers need not cross the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

GRADING NOTE Degree 3 gives 3 zeros (Fundamental Theorem of Algebra); conjugate pairs make the number of non-real zeros even, so at least 1 is real. The others confuse "no integer factor" with "no zero", assume all zeros are real, or ignore conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S6-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 3 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 4

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 1 + 1 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

203

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The highest power is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(x - 2)(x + 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 and −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B2-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

Find the zeros of $(x - 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 (repeated three times)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Zero-product property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of x(x + 4) are

A. 0 and 4

B. 0 and −4

C. −4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 and −4

GRADING NOTE x = 0 or x + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(2x - 3)(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/2 and −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 9 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

212

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±3i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

Solve $x^2 + 2x + 2 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

213

QUESTION ID

S6-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x = -1 \pm i$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B3-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

214

QUESTION ID

S6-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ±i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $1 \pm 2i$

D. $2 \pm i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

215

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 ± 2i

GRADING NOTE Discriminant −16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = -81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

216

QUESTION ID

S6-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = ±9i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

218

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 4 zeros in all, 2 real (the crossings), so 2 non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

The graph of $y = x^3 - 4x^2 + 5x - 2$ touches the x-axis at $x = 1$ and crosses it at $x = 2$. List all complex zeros with multiplicity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

219

QUESTION ID
S6-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1 (multiplicity 2) and 2 (multiplicity 1)

GRADING NOTE Form required: zeros with multiplicities. $(x - 1)^2(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

220

QUESTION ID

S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 6 zeros are non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 4

B. 5

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

221

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE 5 − 1 = 4 (two conjugate pairs).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

222

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the fourth zero would be a single non-real zero, but non-real zeros come in conjugate pairs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Which argument proves that every polynomial of odd degree with real coefficients has at least one real zero?

A. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come one at a time, so there is an odd number of them, and an odd number of non-real zeros leaves at least one zero that is real.

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

C. The Fundamental Theorem of Algebra says that every polynomial of degree $n \geq 1$ with real coefficients has at least one real zero; a polynomial of odd degree has degree at least 1, so it has at least one real zero as well.

D. $x^3 - 1$ has the real zero 1, so every polynomial of odd degree has a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

224

QUESTION ID

S6-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

GRADING NOTE Count zeros with the theorem and pair the non-real ones. The others check one example, misstate the theorem (it guarantees complex zeros, and $x^2 + 1$ has no real zero), or claim an odd number of non-real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. What are all of its possible zero lists?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

225

QUESTION ID
S6-F2-Q2

TYPE
Short answer

CORRECT ANSWER
1 + i, 1 − i, 3 and one more real zero r (r may be 3 again, giving 3 with multiplicity 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients must have at least one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

226

QUESTION ID

S6-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^4 + 1$ has no real zeros; only odd degree guarantees a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 1, 3, 5, 7

B. 7 only

C. 0, 2, 4, 6

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

227

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1, 3, 5, 7

GRADING NOTE The number of real zeros is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Explain why the Fundamental Theorem of Algebra says nothing about whether a polynomial's zeros can be found exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

228

QUESTION ID
S6-F2-Q5

TYPE
Short answer

CORRECT ANSWER
It guarantees that the zeros exist and counts them; it gives no method or formula for finding them.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

Interpret the parts of polynomial, rational, and exponential expressions in context

MAIN	Algebra II main question	**HSA-SSE.A.1.a**
BACKWARD 1	Name terms, factors, and coefficients	**HSA-SSE.A.1.a**
BACKWARD 2	Translate between a percent change and a multiplying factor	**7.RP.A.3**
BACKWARD 3	Interpret a quotient as an amount per unit	**6.RP.A.2**
FORWARD 1	Build an expression whose parts have given meanings	**HSA-SSE.A.1.a**
FORWARD 2	Rewrite an exponential expression to reveal a different rate	**HSF-IF.C.8.b**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A town's population is modeled by $P(t) = 12{,}500(0.97)^t$, where t is the number of years after 2020. What do 12,500 and 0.97 represent, and what does $1 - 0.97$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

231

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

12,500 is the population in 2020; 0.97 is the yearly factor (each year the population is 97% of the year before); 1 − 0.97 = 0.03 means the population decreases by 3% per year.

GRADING NOTE Any equivalent interpretation of all three parts, with units (people, per year).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The average cost per item of making x items is C(x) = (4,000 + 6x) / x dollars. Interpret 4,000, 6x, and the division by x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

232

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

4,000 is a fixed cost in dollars; 6x is the variable cost ($6 per item for x items); dividing the total cost by x gives the cost per item.

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

An open box is made by cutting squares of side x cm from the corners of a 20 cm by 20 cm sheet. Its volume is $V(x) = x(20 - 2x)^2$ cm³. Interpret x, $20 - 2x$, and $(20 - 2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

233

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

x is the height of the box; $20 - 2x$ is the side length of the square base (20 cm minus two cuts); $(20 - 2x)^2$ is the base area in cm².

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

The amount of a medicine in the body t hours after a dose is $A(t) = 200(1/2)^{t/6}$ mg. Interpret 200, 1/2, and t/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

234

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

200 mg is the initial dose; the amount is halved repeatedly; t/6 counts 6-hour periods, so the amount halves every 6 hours (half-life 6 hours).

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

Maya keeps $1,500 in an account that earns 4% interest compounded once a year and also puts $200 per year in a jar (no interest). Her total after n years is $T(n) = 1500(1.04)^n + 200n$. Interpret $1500(1.04)^n$, 1.04, and 200n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

235

QUESTION ID
S7-M-Q5

TYPE
Short answer

CORRECT ANSWER

$1500(1.04)^n$ is the account balance after n years; 1.04 is the yearly growth factor (4% interest added each year); 200n is the cash in the jar after n years.

GRADING NOTE Any equivalent interpretation of all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Name terms, factors, and coefficients • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Name the terms of 4,000 + 6x.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

237

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000 and 6x

GRADING NOTE Form required: both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of x in 7x − 3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

238

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

In 5x(x − 2), which is a factor?

A. 5x − 2

B. x − 10

C. x − 2

D. −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

239

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x − 2

GRADING NOTE 5x(x − 2) is the product of 5, x and x − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 3 + 2x, the number 3 is a factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

240

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 is a term (it is added).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

Name the factors of $200(0.5)^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

241

QUESTION ID

S7-B1-Q5

TYPE

Short answer

CORRECT ANSWER

200 and $(0.5)^t$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Translate between a percent change and a multiplying factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A price is multiplied by 0.97. By what percent does it change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

243

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

It decreases by 3%.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What factor multiplies an amount that increases by 4%?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

244

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.04

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

Multiplying by 1.25 is the same as

A. increasing by 2.5%

B. increasing by 25%

C. increasing by 125%

D. decreasing by 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

245

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. increasing by 25%

GRADING NOTE 1 + 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 0.8 decreases an amount by 80%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

246

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 20%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

Find 1 − 0.85 and explain what it means for an amount multiplied by 0.85.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

247

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.15; the amount decreases by 15%.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 3

Interpret a quotient as an amount per unit • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A total cost of $600 is shared by 30 people. What does 600/30 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

249

QUESTION ID

S7-B3-Q1

TYPE

Short answer

CORRECT ANSWER

The cost per person, $20.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

A car travels 150 miles in 3 hours. What does 150/3 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

250

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

The average speed, 50 miles per hour.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

If T is a total cost for n items, then T/n is

A. the profit

B. the number of items

C. the cost per item

D. the total cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

251

QUESTION ID

S7-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the cost per item

GRADING NOTE Unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Dividing a total by the number of items gives an average per item.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

252

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a mean or unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

What does (4,000 + 6x)/x mean when x = 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

253

QUESTION ID

S7-B3-Q5

TYPE

Short answer

CORRECT ANSWER

The average cost per item when 100 items are made: $46.

GRADING NOTE Any equivalent answer. 4,600/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Build an expression whose parts have given meanings • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write an expression for the value of a $24,000 car that loses 15% of its value each year, t years after purchase.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

255

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$24{,}000(0.85)^t$

GRADING NOTE Any equivalent expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $90 joining fee and $35 per month. Write an expression for the average cost per month over m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

256

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(90 + 35m)/m

GRADING NOTE Any equivalent expression (for example 35 + 90/m).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

True or false?

An expression for a quantity that triples every 5 years, starting at 40, is $40(3)^{5t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

257

QUESTION ID

S7-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The exponent counts 5-year periods: $40(3)^{t/5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

A bacteria count starts at 500 and doubles every 3 hours. Which expression gives the count after h hours?

A. $2(500)^{h/3}$

B. $500(2)^{3h}$

C. $500(3)^{h/2}$

D. $500(2)^{h/3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

258

QUESTION ID

S7-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $500(2)^{h/3}$

GRADING NOTE h/3 doubling periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $90 joining fee and $35 per month, so the average cost per month over m months is 35 + 90/m dollars. Explain what the part 90/m means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

259

QUESTION ID

S7-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The joining fee spread over m months: the part of the monthly average that comes from the fee, which shrinks as m grows.

GRADING NOTE Any equivalent explanation (the gym charges $90 to join and $35 per month).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Rewrite an exponential expression to reveal a different rate • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $12{,}500(0.97)^t$ (t in years) to show the monthly factor, rounded to 5 decimal places.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

261

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$12{,}500(0.99746)^{12t}$

GRADING NOTE Rounding to 5 decimal places required for the factor. $0.97^{1/12} \approx 0.997465$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

$A(t) = 200(1/2)^{t/6}$. Write A as $200 \cdot b^t$ and give b to 4 decimal places. What is the hourly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

262

QUESTION ID
S7-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$b = (1/2)^{1/6} \approx 0.8909$; about 10.91% decrease per hour

GRADING NOTE Rounding to 4 decimal places for b; percent to 0.01.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

True or false?

$1.04^{n} = (1.04^{1/12})^{12n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

263

QUESTION ID

S7-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power: (1/12)(12n) = n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

Which expression shows that $500(2)^{t/3}$ grows by about 26% per time unit?

A. $1.26(500)^t$

B. $500(1.26)^{3t}$

C. $500(1.26)^t$

D. $500(2.26)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

264

QUESTION ID

S7-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $500(1.26)^t$

GRADING NOTE $2^{1/3} \approx 1.26$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

Explain why rewriting does not change the values of the function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

265

QUESTION ID

S7-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The new form is equivalent by the properties of exponents, so it gives the same output for every input; it only shows another rate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b • SEEING STRUCTURE IN EXPRESSIONS

Interpret complicated expressions by viewing parts as single entities

MAIN	Algebra II main question	**HSA-SSE.A.1.b**
BACKWARD 1	Squares are never negative	**8.EE.A.2**
BACKWARD 2	Powers of numbers greater than 1 and between 0 and 1	**8.EE.A.1**
BACKWARD 3	A fraction gets smaller as its denominator grows	**3.NF.A.3.d**
FORWARD 1	Find the set of all values of an expression on a domain	**HSA-SSE.A.1.b**
FORWARD 2	Use single-entity reasoning to find horizontal asymptotes	**HSF-IF.C.7.d**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

$P(r) = 5000(1 + r)^{10}$ with $r > 0$ is the value of an investment. Without calculating, explain why $P(r) > 5000$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

268

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

$1 + r > 1$, so $(1 + r)^{10} > 1$, and 5000 times a number greater than 1 is greater than 5000.

GRADING NOTE Any equivalent explanation that treats $(1 + r)^{10}$ as one quantity greater than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

The height of a ball is $h(t) = 80 - 16(t - 2)^2$ feet after t seconds. Without expanding, find the greatest height and when it occurs, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

269

QUESTION ID
S8-M-Q2

TYPE
Short answer

CORRECT ANSWER
80 ft at t = 2: $(t - 2)^2 \geq 0$, so $16(t - 2)^2 \geq 0$ and $h(t) \leq 80$, with equality only when $t = 2$.

GRADING NOTE Form required: the height, the time, and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

$C(x) = 10 + 40/(x + 1)$ is a cost in dollars for $x \geq 0$. Explain why $C(x)$ is always more than 10 and never more than 50.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

270

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x + 1 \geq 1$, so $40/(x + 1)$ is positive and at most 40; adding 10 gives more than 10 and at most 50 (equal to 50 at $x = 0$).

GRADING NOTE Any equivalent explanation treating $40/(x + 1)$ as one positive quantity at most 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

The temperature of a drink is $T(t) = 70 - 30(0.8)^t$ °F after t minutes. Explain why T(t) is always below 70 and gets closer to 70 as t grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

271

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

$30(0.8)^t$ is positive, so T is below 70; $(0.8)^t$ shrinks toward 0 as t grows, so the subtracted part shrinks and T approaches 70.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

$f(x) = 3/(x^2 + 1)$. Find the greatest value of $f(x)$ and explain why no input gives a greater value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

272

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

3, at $x = 0$: $x^2 + 1 \geq 1$, so $3/(x^2 + 1) \leq 3$, with equality only when $x = 0$.

GRADING NOTE Form required: the value and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Squares are never negative • Grade 8 • Expressions & Equations • 8.EE.A.2

True or false?

$(t - 2)^2 \geq 0$ for every real t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

274

QUESTION ID

S8-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A square of a real number is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the smallest value of $(x + 5)^2$, and for which x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

275

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0, when x = −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

Which expression can be negative for some real x?

A. $x^2 + 1$

B. $x - 1$

C. x^2

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

276

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $x - 1$

GRADING NOTE $x - 1 < 0$ for $x < 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

What is the smallest value of $x^2 + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

277

QUESTION ID

S8-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$-(x - 3)^2$ is never positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

278

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the opposite of a nonnegative number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Powers of numbers greater than 1 and between 0 and 1 • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Is 1.05^{10} greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

280

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Greater than 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

Is 0.8^5 greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

281

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Less than 1 (about 0.33)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As n increases, 0.5^n

A. becomes negative

B. stays 0.5

C. gets closer to 0

D. grows without bound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

282

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. gets closer to 0

GRADING NOTE Repeated halving.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

True or false?

A negative number raised to any whole-number power is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

283

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-2)^2 = 4$ is positive; even powers of negatives are positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

Order from least to greatest: 0.9^2, 0.9^5, 0.9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

284

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.9^5, 0.9^2, 0.9

GRADING NOTE Any equivalent listing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 3

A fraction gets smaller as its denominator grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, 40/2 or 40/5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

286

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

40/2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

True or false?

If $x \geq 0$, then $40/(x + 1) \leq 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

287

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The denominator is at least 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As x grows, 6/(x + 3)

A. increases

B. becomes negative

C. decreases toward 0

D. stays 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

288

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. decreases toward 0

GRADING NOTE The denominator grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

What is the greatest value of 12/(x + 4) for x ≥ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

289

QUESTION ID

S8-B3-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. At x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

8/10 is greater than 8/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

290

QUESTION ID

S8-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Same numerator, larger denominator gives a smaller fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Find the set of all values of an expression on a domain • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

Find every value that $h(t) = 80 - 16(t - 2)^2$ takes for $0 \le t \le 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

292

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$16 \le h \le 80$

GRADING NOTE Form required: an inequality or interval. $(t - 2)^2$ goes from 0 to 4, so $16(t - 2)^2$ from 0 to 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every value of $f(x) = 3/(x^2 + 1)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

293

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 < f(x) \leq 3$

GRADING NOTE Form required: an inequality or interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

True or false?

For $x \geq 0$, $C(x) = 10 + 40/(x + 1)$ takes every value in $10 < C \leq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

294

QUESTION ID

S8-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 40/(x + 1) takes every value in (0, 40].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

For $t \geq 0$, which set contains every value of $70 - 30(0.8)^t$?

A. $T \geq 70$

B. $40 \leq T < 70$

C. $40 < T \leq 70$

D. $0 < T < 70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

295

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $40 \leq T < 70$

GRADING NOTE $30(0.8)^t$ goes from 30 (at $t = 0$) toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

Find every value of $g(x) = 5 - 2/(x^2 + 4)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

296

QUESTION ID

S8-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$4.5 \le g(x) < 5$

GRADING NOTE Form required: an inequality or interval. $2/(x^2 + 4)$ is in $(0, 0.5]$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Use single-entity reasoning to find horizontal asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Explain why y = 10 + 40/(x + 1) has the horizontal asymptote y = 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

298

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

As x grows without bound, 40/(x + 1) gets closer to 0, so y gets closer to 10.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

Rewrite (2x + 7)/(x + 3) as 2 + 1/(x + 3) and state its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

299

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2

GRADING NOTE Form required: the equation y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

True or false?

$y = 3/(x^2 + 1)$ has the horizontal asymptote $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

300

QUESTION ID

S8-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3/(x^2 + 1)$ approaches 0 as $|x|$ grows; the asymptote is $y = 0$ (3 is the maximum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

The horizontal asymptote of $y = 5 - 2/(x - 4)$ is

A. $y = -2$

B. $y = 5$

C. $y = 4$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

301

QUESTION ID

S8-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 5

GRADING NOTE $-2/(x - 4)$ approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

Does the graph of y = 10 + 40/(x + 1) ever reach its horizontal asymptote? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

302

QUESTION ID

S8-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No: 40/(x + 1) is never 0, so y is never exactly 10.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

Factor higher-degree expressions by recognizing differences of squares and sums or differences of cubes

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Take out a greatest common factor	**6.EE.A.3**
BACKWARD 2	Recognize perfect squares and perfect cubes	**8.EE.A.2**
BACKWARD 3	Multiply to check a factoring pattern	**HSA-APR.A.1**
FORWARD 1	Factor expressions whose parts are themselves expressions	**HSA-SSE.A.2**
FORWARD 2	Complete the factorization over the complex numbers	**HSN-CN.C.8**

SET 9 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 81$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

305

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 9)(x + 3)(x - 3)$

GRADING NOTE Form required: factored completely over the integers. $(x^2)^2 - 9^2 = (x^2 - 9)(x^2 + 9)$. $x^2 + 9$ does not factor over the integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $8x^3 + 27$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

306

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 3)(4x^2 - 6x + 9)$

GRADING NOTE Form required: factored completely. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 2x$, $b = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

Factor $x^6 - 64$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

307

QUESTION ID

S9-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x^2 + 2x + 4)(x + 2)(x^2 - 2x + 4)$

GRADING NOTE Form required: factored completely, any order. $(x^3 - 8)(x^3 + 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

Factor $16x^4 - y^4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

308

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(4x^2 + y^2)(2x + y)(2x - y)$

GRADING NOTE Form required: factored completely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $2x^5 - 32x$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

309

QUESTION ID

S9-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x^2 + 4)(x + 2)(x - 2)$

GRADING NOTE Form required: factored completely. Common factor 2x first: $2x(x^4 - 16)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 1

Take out a greatest common factor • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor 6x + 18.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

311

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

Factor $2x^5 - 32x$ by taking out the greatest common factor only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

312

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$2x(x^4 - 16)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The greatest common factor of $12x^3$ and $18x$ is

A. $6x^3$

B. $6x$

C. $3x$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

313

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6x

GRADING NOTE 6 and x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$3x^2 + 9x = 3x(x + 9)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

314

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3x(x + 9) = 3x^2 + 27x$; the correct factorization is $3x(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

Factor $5y^2 - 20y$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

315

QUESTION ID

S9-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5y(y − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 2

Recognize perfect squares and perfect cubes • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write $16x^4$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

317

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(4x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $8x^3$ as a cube.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

318

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x)^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a perfect cube?

A. 32

B. 16

C. 36

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

319

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 64

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

True or false?

x^6 is both a perfect square and a perfect cube.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

320

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x^3)^2$ and $(x^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

Find the cube root of 27.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

321

QUESTION ID

S9-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 3

Multiply to check a factoring pattern • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(a + b)(a^2 - ab + b^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

323

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$a^3 + b^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

Multiply $(x - 2)(x^2 + 2x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

True or false?

$(x + 3)(x^2 - 6x + 9) = x^3 + 27.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

325

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle term should be $-3x$: $(x + 3)(x^2 - 3x + 9) = x^3 + 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 - 9)(x^2 + 9)$ equals

A. $x^4 - 18x^2 - 81$

B. $x^4 + 81$

C. $(x - 3)^4$

D. $x^4 - 81$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

326

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x^4 - 81$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(2x + 1)(4x^2 - 2x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

327

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$8x^3 + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 1

Factor expressions whose parts are themselves expressions • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $(x + 1)^2 - 9$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

329

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 4)$

GRADING NOTE Form required: factored completely. $((x + 1) - 3)((x + 1) + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $(2x - 1)^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

330

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(2x + 1)(4x^2 - 8x + 7)$

GRADING NOTE Form required: factored completely. a = 2x − 1, b = 2: $(a + b)(a^2 - 2a + 4) = (2x + 1)((2x - 1)^2 - 2(2x - 1) + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

True or false?

$a^6 - b^6$ can be factored first as a difference of squares or first as a difference of cubes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

331

QUESTION ID

S9-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a^3)^2 - (b^3)^2$ or $(a^2)^3 - (b^2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - y)^2 - (x + y)^2$ equals

A. 0

B. $4xy$

C. $-4xy$

D. $-2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

332

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-4xy$

GRADING NOTE $((x - y) - (x + y))((x - y) + (x + y)) = (-2y)(2x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

Factor $x^2(x + 2) - 4(x + 2)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

333

QUESTION ID

S9-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 2)^2(x - 2)$

GRADING NOTE Form required: factored completely. $(x + 2)(x^2 - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 2

Complete the factorization over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^4 - 81$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

335

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 3)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $16x^4 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

336

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(2x + 1)(2x − 1)(2x + i)(2x − i)

GRADING NOTE Any equivalent factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

True or false?

Over the complex numbers, $x^3 + 8$ has only one linear factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

337

QUESTION ID

S9-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has three: $(x + 2)$ and two from $x^2 - 2x + 4 = (x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

Which polynomial is fully factored over the integers but not over the complex numbers?

A. $x^2 + 9$

B. $x - 9$

C. $x^2 - 9$

D. $(x - 3)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

338

QUESTION ID

S9-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 9$

GRADING NOTE $x^2 + 9 = (x + 3i)(x - 3i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

$2x^5 - 32x = 2x(x^2 + 4)(x + 2)(x - 2)$ over the integers. Complete the factorization over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

339

QUESTION ID

S9-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2x(x + 2i)(x - 2i)(x + 2)(x - 2)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

Rewrite expressions in quadratic form and factor

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Factor quadratic trinomials	**HSA-SSE.A.2**
BACKWARD 2	Write powers as squares of other powers	**8.EE.A.1**
BACKWARD 3	Treat a repeated part as a single variable	**HSA-SSE.A.1.b**
FORWARD 1	Solve equations in quadratic form over the complex numbers	**HSN-CN.C.7**
FORWARD 2	Solve exponential equations in quadratic form	**HSF-LE.A.4**

SET 10 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 5x^2 + 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

342

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. With $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

Factor $x^4 + 3x^2 - 10$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

343

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 5)(x^2 - 2)$

GRADING NOTE Form required: factored completely over the integers ($x^2 - 2$ does not factor over the integers). $u^2 + 3u - 10 = (u + 5)(u - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

Factor $9^x - 4 \cdot 3^x + 3$ by writing it in quadratic form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

344

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(3^x - 1)(3^x - 3)$

GRADING NOTE Form required: a product of two factors. $9^x = (3^x)^2$, so with $u = 3^x$: $u^2 - 4u + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^6 - 9x^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

345

QUESTION ID

S10-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x^2 + x + 1)(x - 2)(x^2 + 2x + 4)$

GRADING NOTE Form required: factored completely. $u = x^3$: $(u - 1)(u - 8)$; then difference of cubes twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

Factor $2x^4 - 7x^2 - 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

346

QUESTION ID

S10-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(2x^2 + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. $2u^2 - 7u - 4 = (2u + 1)(u - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 1

Factor quadratic trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 5u + 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

348

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 4)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

Factor $u^2 + 3u - 10$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

349

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(u + 5)(u − 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-B1-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

$2u^2 - 7u - 4$ equals

A. $(2u + 1)(u - 4)$

B. $(u - 2)(2u + 2)$

C. $(2u + 4)(u - 1)$

D. $(2u - 1)(u + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

350

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2u + 1)(u − 4)

GRADING NOTE Check: $2u^2 - 8u + u - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$u^2 - 9u + 8 = (u + 1)(u - 8)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

351

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(u + 1)(u - 8) = u^2 - 7u - 8$; the correct factorization is $(u - 1)(u - 8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $u^2 - 4u + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

352

QUESTION ID

S10-B1-Q5

TYPE

Short answer

CORRECT ANSWER

(u − 1)(u − 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 2

Write powers as squares of other powers • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write x^4 as a power of x^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

354

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x^2)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 9^x as a power of 3^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

355

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(3^x)^2$

GRADING NOTE Any equivalent form. $9^x = (3^2)^x = (3^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

x^6 equals

A. $(x^4)^2$

B. $(x^3)^2$

C. $x^3 \cdot x^3 \cdot x^3$

D. $(x^2)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

356

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^3)^2$

GRADING NOTE $3 \cdot 2 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$4^x = (2^x)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

357

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = 2^{2x}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 25^x as a power of 5^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

358

QUESTION ID

S10-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5^x)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 3

Treat a repeated part as a single variable • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

In $x^4 - 5x^2 + 4$, what single variable u turns the expression into $u^2 - 5u + 4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

360

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$u = x^2$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2(x - 1)^2 + 3(x - 1) - 5$ using $u = x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

361

QUESTION ID

S10-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$2u^2 + 3u - 5$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

True or false?

In $4^x - 2^x - 6$, the substitution $u = 2^x$ gives $u^2 - u - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

362

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^x = (2^x)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

Which substitution writes $x^6 - 9x^3 + 8$ as a quadratic?

A. $u = x^2$

B. $u = x^3$

C. $u = x^6$

D. $u = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

363

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $u = x^3$

GRADING NOTE $x^6 = (x^3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

After factoring $(u - 1)(u - 8)$ with $u = x^3$, write the result in terms of x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

364

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x^3 - 1)(x^3 - 8)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 1

Solve equations in quadratic form over the complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^4 + 3x^2 - 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

366

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 1, -1, 2i, -2i$

GRADING NOTE Form required: all four. $x^2 = 1$ or $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

Solve $x^4 - 5x^2 + 4 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

367

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = ±1, ±2

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

True or false?

$x^4 + 10x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

368

QUESTION ID

S10-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x^2 + 1)(x^2 + 9) = 0$ has only the solutions $\pm i$ and $\pm 3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

How many complex solutions, with multiplicity, does $x^4 - 13x^2 + 36 = 0$ have?

A. 1

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

369

QUESTION ID

S10-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE ±2 and ±3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^4 + x^2 - 12 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

370

QUESTION ID

S10-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = \pm\sqrt{3}, \pm 2i$

GRADING NOTE Form required: all four, exact. $(x^2 + 4)(x^2 - 3) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 2

Solve exponential equations in quadratic form • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $9^x - 5 \cdot 3^x + 6 = 0$. Give exact solutions.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

372

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \log_3 2$ or $x = 1$

GRADING NOTE Form required: both, exact. $3^x = 2$ or $3^x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

Solve $4^x - 6 \cdot 2^x + 8 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

373

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 1 or x = 2

GRADING NOTE Form required: both. $2^x = 2$ or $2^x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q2**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

True or false?

$4^x + 2^x - 2 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

374

QUESTION ID

S10-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^x = 1$ gives $x = 0$; $2^x = -2$ is impossible since $2^x > 0$. One solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $25^x - 26 \cdot 5^x + 25 = 0$ are

A. 0 and 2

B. 1 and 25

C. 0 and 1

D. 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

375

QUESTION ID

S10-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0 and 2

GRADING NOTE $5^x = 1$ or $5^x = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

Solve $e^{2t} - 7e^{t} + 10 = 0$. Give exact solutions and decimal values to 3 decimal places.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

376

QUESTION ID

S10-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$t = \ln 2 \approx 0.693$ or $t = \ln 5 \approx 1.609$

GRADING NOTE Form required: exact and 3 decimal places. $e^{t} = 2$ or $e^{t} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4 • SEEING STRUCTURE IN EXPRESSIONS

Derive the formula for the sum of a finite geometric series

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Write terms of a geometric sequence	**HSF-LE.A.2**
BACKWARD 2	Distribute a factor over a sum	**6.EE.A.3**
BACKWARD 3	Use the product rule for exponents	**8.EE.A.1**
FORWARD 1	Prove the sum formula from a polynomial identity	**HSA-APR.C.4**
FORWARD 2	Describe how partial sums behave when the ratio is between -1 a...	**HSA-SSE.B.4**

SET 11 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

$S_n = a + ar + ar^2 + \ldots + ar^{n-1}$ with $r \neq 1$. In the derivation of $S_n = a(1 - r^n)/(1 - r)$, which equation results from subtracting rS_n from S_n?

A. $S_n - rS_n = a - ar^n$

B. $S_n - rS_n = ar - ar^n$

C. $S_n - rS_n = a - ar$

D. $S_n - rS_n = a - ar^{n-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

379

QUESTION ID

S11-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $S_n - rS_n = a - ar^n$

GRADING NOTE $rS_n = ar + ar^2 + \ldots + ar^n$, so every middle term cancels and $S_n - rS_n = a - ar^n$. Then $(1 - r)S_n = a(1 - r^n)$, and dividing by $1 - r$ (allowed because $r \neq 1$) gives the formula. The others use the wrong last term of rS_n, cancel too much, or lose the first term a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

$S = 3 + 6 + 12 + \ldots + 3 \cdot 2^9$. With the multiply-and-subtract method, $2S - S$ equals which expression, and what is S?

A. $3 \cdot 2^9 - 3$, so $S = 1533$

B. $3 \cdot 2^{10} + 3$, so $S = 3075$

C. $3 \cdot 2^{10} - 6$, so $S = 3066$

D. $3 \cdot 2^{10} - 3$, so $S = 3069$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

380

QUESTION ID

S11-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $3 \cdot 2^{10} - 3$, so $S = 3069$

GRADING NOTE $2S = 6 + 12 + \ldots + 3 \cdot 2^{10}$; subtracting S leaves the last term of 2S minus the first term of S: $3 \cdot 2^{10} - 3 = 3072 - 3 = 3069$. The others use the wrong last term, subtract the wrong first term, or add it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

Why does the derivation of $S_n = a(1 - r^n)/(1 - r)$ fail when $r = 1$, and what is S_n then?

A. When $r = 1$ the terms a, ar, ar^2, ... grow without bound, so S_n does not exist and no formula for it is possible.

B. Its last step divides by $1 - r$, which is 0 when $r = 1$. Every term then equals a, so $S_n = na$.

C. rS_n cannot be formed when $r = 1$, so $S_n = a$.

D. The formula gives 0/0, which means $S_n = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

381

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Its last step divides by 1 − r, which is 0 when r = 1. Every term then equals a, so $S_n = na$.

GRADING NOTE Division by $1 - r = 0$ is not allowed; with $r = 1$ the series is $a + a + \ldots + a$ (n terms) $= na$. The others treat 0/0 as 0, misread the method, or confuse $r = 1$ with $|r| > 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

$T = 1 + x + x^2 + \ldots + x^n$ ($n + 1$ terms, $x \neq 1$).
Multiplying by x and subtracting T gives which formula?
(Check: $x = 2$, $n = 3$ gives $1 + 2 + 4 + 8 = 15$.)

A. $T = (x^n - 1)/(x - 1)$

B. $T = (x^{n+1} - x)/(x - 1)$

C. $T = (x^{n+1} - 1)/(x - 1)$

D. $T = (x^{n+1} - 1)/(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

382

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C.

$T = (x^{n+1} - 1)/(x - 1)$

GRADING NOTE $xT = x + x^2 + \ldots + x^{n+1}$, so $xT - T = x^{n+1} - 1$ and $T = (x^{n+1} - 1)/(x - 1)$. Check: $(16 - 1)/1 = 15$. The others give 7, 5, and 14 for the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

$T = 100 + 100(1.05) + 100(1.05)^2 + \ldots + 100(1.05)^{19}$ (20 yearly amounts). Using the multiply-and-subtract method with $1.05T - T$, which equation and total are correct? A calculator may be used.

A. $0.05T = 100(1.05^{19} - 1)$, so $T \approx \$3{,}053.90$

B. $1.05T = 100(1.05^{20} - 1)$, so $T \approx \$157.46$

C. $0.05T = 100(1.05^{21} - 1)$, so $T \approx \$3{,}571.93$

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

383

QUESTION ID

S11-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

GRADING NOTE $1.05T = 100(1.05) + \ldots + 100(1.05)^{20}$, so $1.05T - T = 100(1.05)^{20} - 100$ and $T = 100(1.05^{20} - 1)/0.05 \approx 3306.60$. The others use the wrong last power (19 or 21) or forget that $1.05T - T = 0.05T$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 1

Write terms of a geometric sequence • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the first four terms of the geometric sequence with first term 3 and ratio 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

385

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3, 6, 12, 24

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

What is the 10th term of 3, 6, 12, ...?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

386

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$3 \cdot 2^9 = 1536$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B1-Q2**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

The nth term of a geometric sequence with first term a and ratio r is

A. $a + (n - 1)r$

B. $a \cdot n \cdot r$

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

387

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ar^{n-1}

GRADING NOTE The first term is ar^0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

True or false?

100, 105, 110.25, ... is arithmetic with common difference 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

388

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 5 and 5.25; the ratio 1.05 is constant, so it is geometric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

Find the ratio of 81, 27, 9, 3, ...

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

389

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 2

Distribute a factor over a sum • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Distribute: 2(3 + 6 + 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

391

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 + 12 + 24

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distribute: $r(a + ar + ar^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

392

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$ar + ar^2 + ar^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

True or false?

$x(1 + x + x^2) = x + x^2 + x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

393

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

1.05(100 + 105) equals

A. 105 + 105

B. 100 + 110.25

C. 105 + 110.25

D. 205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

394

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 105 + 110.25

GRADING NOTE Each term is multiplied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Subtract: (6 + 12 + 24 + 48) − (3 + 6 + 12 + 24).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

395

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

45 (= 48 − 3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 3

Use the product rule for exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $r \cdot r^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

397

QUESTION ID

S11-B3-Q1

TYPE

Short answer

CORRECT ANSWER

r^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

Simplify $ar^{n-1} \cdot r$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

398

QUESTION ID

S11-B3-Q2

TYPE

Short answer

CORRECT ANSWER

ar^n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2 \cdot 2^9$ equals

A. 2^{10}

B. 4^{10}

C. 4^9

D. 2^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

399

QUESTION ID

S11-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2^{10}

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x \cdot x^n = x^{2n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

400

QUESTION ID

S11-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Add exponents: $x^1 \cdot x^n = x^{n+1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2^{10}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

401

QUESTION ID

S11-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1024

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 1

Prove the sum formula from a polynomial identity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity
$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \ldots + x + 1)$?

A. Divide both sides by $x - 1$ to get $(x^n - 1)/(x - 1) = x^{n-1} + \ldots + x + 1$; this is the geometric sum formula, so the identity is true.

B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

C. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and 1, leaving $x^n + 1$.

D. For $x = 2$ and $n = 3$: $7 = 1 \cdot 7$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

403

QUESTION ID
S11-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

GRADING NOTE Distribute and cancel (telescoping). The others check one example, assume what is to be proved (and divide by $x - 1$, which can be 0), or make a sign error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

Use the identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + 1)$ to write $1 + 3 + 9 + \ldots + 3^5$ as a quotient, and evaluate it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

404

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(3^6 - 1)/(3 - 1)$
$= 728/2 = 364$

GRADING NOTE Form required: the quotient and the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

True or false?

The identity shows that $x - 1$ is a factor of $x^n - 1$ for every whole number $n \geq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

405

QUESTION ID

S11-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It writes $x^n - 1$ as $(x - 1)$ times a polynomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

Which equals $1 + 2 + 4 + \ldots + 2^9$?

A. 2^{10}

B. $2^{10} - 1$

C. $2^9 - 1$

D. $(2^9 - 1)/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

406

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2^{10} - 1$

GRADING NOTE $(2^{10} - 1)/(2 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

Explain how the identity gives $a(1 - r^n)/(1 - r)$ for $a + ar + \ldots + ar^{n-1}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

407

QUESTION ID

S11-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Put x = r and multiply by a: $a(r^n - 1) = a(r - 1)(1 + r + \ldots + r^{n-1})$; divide by r − 1 (r ≠ 1) and rewrite signs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 2

Describe how partial sums behave when the ratio is between −1 and 1 • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

For the series 8 + 4 + 2 + 1 + ..., find S_5 and S_{10} to 4 decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

409

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$S_5 = 15.5000$;

$S_{10} = 15.9844$

GRADING NOTE Rounding to 4 decimal places. $S_n = 8(1 - 0.5^n)/0.5 = 16(1 - 0.5^n)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

Using $S_n = 16(1 - 0.5^n)$, explain why every partial sum of $8 + 4 + 2 + \ldots$ is less than 16 but can be made as close to 16 as you like.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

410

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.5^n is positive, so $S_n < 16$; 0.5^n shrinks toward 0 as n grows, so S_n gets as close to 16 as you like.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

True or false?

For r = 2, the partial sums of $a + ar + ar^2 + \ldots$ ($a > 0$) stay below a fixed number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

411

QUESTION ID

S11-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE r^n grows without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

For 10 − 5 + 2.5 − 1.25 + …, the partial sums get close to

A. 20/3

B. 15

C. 5

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

412

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20/3

GRADING NOTE a/(1 − r) = 10/1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

A ball travels 10 m down, then rebounds and falls in a geometric pattern so the total distance after n bounces is $10 + 2 \cdot 6(1 - 0.6^n)/0.4$. What total can the distance never reach, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

413

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

40 m: $2 \cdot 6/0.4 = 30$ and $10 + 30 = 40$; $(1 - 0.6^n) < 1$ always.

GRADING NOTE Form required: the value and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

Use the finite geometric series formula to solve problems

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Identify the first term, ratio, and number of terms	**HSF-LE.A.2**
BACKWARD 2	Change a percent to a growth factor	**7.RP.A.3**
BACKWARD 3	Round results of calculations correctly	**HSN-Q.A.3**
FORWARD 1	Find the number of terms with logarithms	**HSF-LE.A.4**
FORWARD 2	Compare two payment plans with the series formula	**HSA-SSE.B.4**

SET 12 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 12 terms of 5 + 15 + 45 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

416

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

1,328,600

GRADING NOTE Any equivalent form. $5(3^{12} - 1)/(3 - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

Leah deposits $250 at the end of each month into an account that earns 0.5% interest per month. The balance just after the 36th deposit is $250 + 250(1.005) + \ldots + 250(1.005)^{35}$. Find it to the nearest cent.

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

417

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

$9834.03

GRADING NOTE Rounding to the nearest cent required. $250(1.005^{36} - 1)/0.005 = 9834.03$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

A $180,000 loan is repaid with 360 equal monthly payments P at a monthly interest rate of 0.4%. P satisfies
$180{,}000 = P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$. Use the geometric series formula to find P to the nearest cent.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

418

QUESTION ID
S12-M-Q3

TYPE
Short answer

CORRECT ANSWER
$944.40

GRADING NOTE Rounding to the nearest cent required. a = P/1.004, r = 1/1.004, n = 360: $P = 180{,}000 \cdot 0.004/(1 - 1.004^{-360}) = 944.40$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q3**

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

A ball is dropped from 10 m and each rebound reaches 60% of the previous height. Find the total vertical distance it has traveled when it hits the ground for the 6th time, to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

419

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

37.67 m

GRADING NOTE Rounding to 0.01 required. $10 + 2(6 + 3.6 + 2.16 + 1.296 + 0.7776) = 10 + 2 \cdot 6(1 - 0.6^5)/0.4 = 37.6672$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

What is the least number of terms of 2 + 6 + 18 + ... whose sum is greater than 10,000? Show the sums that decide it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5
High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

QUESTION ID
S12-M-Q5

TYPE
Short answer

CORRECT ANSWER
9 terms: $S_8 = 6560$ and $S_9 = 19{,}682$

GRADING NOTE Form required: n and the two sums. $S_n = 3^n - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5
High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 1

Identify the first term, ratio, and number of terms • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

For 5 + 15 + 45 + ... (12 terms), give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

422

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

a = 5, r = 3, n = 12

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B1-Q1**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

For $250 + 250(1.005) + \ldots + 250(1.005)^{35}$, give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

423

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 250, r = 1.005, n = 36

GRADING NOTE Form required: all three. Exponents 0 to 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

For 6 + 3.6 + 2.16 + 1.296 + 0.7776, r is

A. 0.4

B. 1.67

C. 0.6

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

424

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.6

GRADING NOTE 3.6/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A list with exponents 0 through 35 has 35 terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

425

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 36 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

For P/1.004 + P/1.004² + ... + P/1.004³⁶⁰, give the first term and the ratio.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

426

QUESTION ID

S12-B1-Q5

TYPE

Short answer

CORRECT ANSWER

a = P/1.004,
r = 1/1.004

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 2

Change a percent to a growth factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

What factor adds 0.5% interest?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

428

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.005

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A height is 60% of the one before. What is the ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

429

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

0.4% per month as a factor is

A. 1.04

B. 1.004

C. 0.004

D. 1.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

430

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.004

GRADING NOTE 1 + 0.004.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

True or false?

Interest of 12% per year is a factor of 0.12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

431

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The growth factor is 1 + 0.12 = 1.12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

A loss of 15% per year is what factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

QUESTION ID

S12-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.85

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 3

Round results of calculations correctly • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Round 9834.0262 to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

434

QUESTION ID

S12-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9834.03

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 37.6672 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

435

QUESTION ID

S12-B3-Q2

TYPE

Short answer

CORRECT ANSWER

37.67

GRADING NOTE Rounding to 0.01 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

True or false?

Rounding 1.005 to 1.01 before raising it to the 36th power gives the same answer to the nearest cent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

436

QUESTION ID

S12-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Early rounding changes the result a lot; round only at the end.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

1.005^{36} is closest to

A. 1.1967

B. 1.18

C. 1.2

D. 1.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

437

QUESTION ID

S12-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1.1967

GRADING NOTE Calculator value 1.19668....

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 3^{12} exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

438

QUESTION ID

S12-B3-Q5

TYPE

Short answer

CORRECT ANSWER

531441

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 1

Find the number of terms with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

How many end-of-month deposits of $250 at 0.5% per month are needed for the balance to reach $20,000? Use $250(1.005^n - 1)/0.005 \geq 20{,}000$.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

440

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

68 deposits

GRADING NOTE Form required: a whole number. $1.005^n \geq 1 + 20{,}000(0.005)/250 = 1.4$, so $n \geq \ln 1.4/\ln 1.005 \approx 67.46$; check: n = 67 gives about $19,838.70 and n = 68 about $20,187.89.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

Solve $2(3^n - 1)/2 = 3^n - 1 = 1{,}000{,}000$ for n exactly, then give the least whole number of terms whose sum is at least 1,000,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

441

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$n = \log_3(1{,}000{,}001)$

≈ 12.58, so 13 terms

GRADING NOTE Form required: the exact value and the whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

True or false?

For a ratio greater than 1, more terms always give a larger sum when a > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

442

QUESTION ID
S12-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each added term is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

The sum $5(2^n - 1)$ first exceeds 5,000 when n =

A. 9

B. 11

C. 10

D. 1000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

443

QUESTION ID

S12-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE $2^n > 1001$; $2^{10} = 1024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

Explain why the number of terms must be rounded up, not to the nearest whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

444

QUESTION ID

S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The sum must reach the target; rounding down would give a sum that is still too small.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 2

Compare two payment plans with the series formula • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Plan A pays $1,000 per month for 24 months. Plan B pays $500 the first month and 8% more each month after that, for 24 months. Which plan pays more in total, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

446

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Plan B: $500(1.08^{24} - 1)/0.08 \approx \$33{,}382.38$ vs $24,000; about $9,382.38 more

GRADING NOTE Form required: the plan and the difference to the nearest cent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

Plan A pays $1,000 per month. Plan B pays $500 the first month and 8% more each month after that. After how many months does Plan B's total first exceed Plan A's total for the same number of months?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

447

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

After 18 months

GRADING NOTE Form required: the month. Compare $500(1.08^n - 1)/0.08$ with $1000n$: $n = 17$ gives $16{,}875.11 < 17{,}000$; $n = 18$ gives $18{,}725.12 > 18{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

True or false?

A geometric plan with ratio greater than 1 eventually pays more in total than any plan with a fixed monthly amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

448

QUESTION ID
S12-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its terms grow without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

Plan C pays $2,000 then 5% less each month. Its total over 24 months is closest to

A. $40,000

B. $24,000

C. $48,000

D. $28,320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

449

QUESTION ID

S12-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $28,320

GRADING NOTE $2000(1 - 0.95^{24})/0.05 \approx 28{,}320$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

Name one assumption in comparing the plans by their totals only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

450

QUESTION ID
S12-F2-Q5

TYPE
Short answer

CORRECT ANSWER
For example: money received later is treated as equal in value to money now (no interest or inflation).

GRADING NOTE Any reasonable assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

Add and subtract polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Combine like terms with the same power	**7.EE.A.1**
BACKWARD 2	Subtract a whole expression	**6.EE.A.3**
BACKWARD 3	Write polynomials in standard form	**HSA-SSE.A.1.a**
FORWARD 1	Reason about the degree of a sum or difference	**HSA-APR.A.1**
FORWARD 2	Add and subtract rational expressions	**HSA-APR.D.7**

SET 13 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Simplify $(4x^3 - 2x^2 + 7) + (-x^3 + 5x - 9)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

453

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3 - 2x^2 + 5x - 2$

GRADING NOTE Form required: standard form (descending powers). $4x^3 - x^3 = 3x^3$; $-2x^2$; $5x$; $7 - 9 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

Simplify $(6x^4 - x^3 + 2x) - (2x^4 + 3x^3 - x^2 + 2x - 1)$.
Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

454

QUESTION ID
S13-M-Q2

TYPE
Short answer

CORRECT ANSWER
$4x^4 - 4x^3 + x^2 + 1$

GRADING NOTE Form required: standard form. The x terms cancel: $2x - 2x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

A company's revenue is $R(x) = -0.01x^3 + 3x^2 + 40x$ dollars and its cost is $C(x) = 0.02x^2 + 15x + 900$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ in standard form.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

455

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-0.01x^3 + 2.98x^2 + 25x - 900$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

Simplify $(x^5 - 3x^3 + x) - (x^5 + 2x^4 - 3x^3 - 7)$. Write the result in standard form and give its degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

456

QUESTION ID
S13-M-Q4

TYPE
Short answer

CORRECT ANSWER
$-2x^4 + x + 7$; degree 4

GRADING NOTE Form required: standard form and the degree. The x^5 and x^3 terms cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

$p(x) = 5x^3 + x^2 - 6$. Find the polynomial $q(x)$ for which $p(x) + q(x) = 2x^3 - x + 4$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

457

QUESTION ID

S13-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-3x^3 - x^2 - x + 10$

GRADING NOTE Form required: standard form. $q = (2x^3 - x + 4) - (5x^3 + x^2 - 6)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 1

Combine like terms with the same power • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify $4x^3 - x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

459

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-2x^2 + 5x - 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

460

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 + 5x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^3 + x^2 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

461

QUESTION ID

S13-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^3 and x^2 are unlike terms; they cannot be combined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

Which pair are like terms?

A. $4x^2$ and $-x^2$

B. $4x^2$ and $4x$

C. x^3 and $3x$

D. x and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

462

QUESTION ID

S13-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2$ and $-x^2$

GRADING NOTE Same variable, same power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $7 - 9 + 2x - 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

463

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 2

Subtract a whole expression • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite $-(2x^4 + 3x^3 - x^2 + 2x - 1)$ without parentheses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

465

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^4 - 3x^3 + x^2 - 2x + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

Simplify (5 − x) − (3 − 4x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

466

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

True or false?

$a - (b - c) = a - b - c.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

467

QUESTION ID

S13-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $a - (b - c) = a - b + c.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

$-(x^2 - 7)$ equals

A. $x^2 + 7$

B. $-x^2 - 7$

C. $-x^2 + 7$

D. $x^2 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

468

QUESTION ID

S13-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^2 + 7$

GRADING NOTE Distribute -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

Simplify 10 − (x + 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

469

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6 − x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 3

Write polynomials in standard form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write $7 + 5x - 2x^3 + x^2$ in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

471

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$-2x^3 + x^2 + 5x + 7$

GRADING NOTE Form required: descending powers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

What is the leading coefficient of $3 - 4x^2 + x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

472

QUESTION ID

S13-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

The degree of $-2x^4 + x + 7$ is

A. 7

B. 1

C. 4

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

473

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE Highest power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

True or false?

The standard form of $2x - x^3$ is $2x - x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

474

QUESTION ID

S13-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Standard form lists powers in descending order: $-x^3 + 2x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

What is the constant term of $6x^3 - 2x + 9$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

475

QUESTION ID

S13-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 1

Reason about the degree of a sum or difference • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

p(x) has degree 4 and q(x) has degree 4. What degrees can p(x) + q(x) have? Give an example where the degree drops.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

477

QUESTION ID
S13-F1-Q1

TYPE
Short answer

CORRECT ANSWER
4 or less (or the zero polynomial); for example $(x^4 + x) + (-x^4 + 1) = x + 1$

GRADING NOTE Any correct example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

For which value of k does
$(kx^3 + 2x - 1) - (5x^3 - x^2 + 3)$ have degree 2?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

478

QUESTION ID
S13-F1-Q2

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form. The x^3 terms cancel when k = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

479

QUESTION ID
S13-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Polynomials are closed under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

p has degree 5 and q has degree 3. The degree of $p - q$ is

A. 5

B. 2

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

480

QUESTION ID
S13-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 5

GRADING NOTE The x^5 term of p cannot cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q4**
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

Explain why the sum of two polynomials is a polynomial.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

481

QUESTION ID
S13-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Adding like terms gives terms that are a coefficient times a whole-number power of x, which is the form of a polynomial.

GRADING NOTE Any equivalent explanation (closure under addition).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 2

Add and subtract rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Write 3/x + 2/(x + 1) as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

483

QUESTION ID
S13-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(5x + 3)/(x(x + 1)); x ≠ 0, −1

GRADING NOTE Form required: one fraction and the exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

Write 1/(x − 2) − 1/(x + 2) as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

484

QUESTION ID
S13-F2-Q2

TYPE
Short answer

CORRECT ANSWER
4/((x − 2)(x + 2)); x ≠ 2, −2

GRADING NOTE Form required: one fraction and the exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

True or false?

Polynomials are closed under division.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

485

QUESTION ID
S13-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For example 1 ÷ x is not a polynomial; rational expressions are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

x/2 + x/3 equals

A. 5x/5

B. x/5

C. 5x/6

D. 2x/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

486

QUESTION ID
S13-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 5x/6

GRADING NOTE Common denominator 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q4**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

Explain how adding rational expressions is like adding fractions of integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

487

QUESTION ID
S13-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Rewrite both over a common denominator, then add the numerators; polynomials play the role of integers.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

Multiply polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Distribute a monomial over a polynomial	**HSA-APR.A.1**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
BACKWARD 3	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Find a coefficient of a product without full expansion	**HSA-APR.A.1**
FORWARD 2	Expand powers of binomials with the Binomial Theorem	**HSA-APR.C.5**

SET 14 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x^2 + 3x - 2)(2x - 5)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

490

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + x^2 - 19x + 10$

GRADING NOTE Form required: standard form. $2x^3 - 5x^2 + 6x^2 - 15x - 4x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

Expand $(x - 2)^3$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

491

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 6x^2 + 12x - 8$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

Multiply $(x^2 - x + 1)(x^2 + x + 1)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

492

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^4 + x^2 + 1$

GRADING NOTE Form required: standard form. $((x^2 + 1) - x)((x^2 + 1) + x) = (x^2 + 1)^2 - x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

Multiply (2x + 1)(x − 3)(x + 4). Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

493

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 3x^2 - 23x - 12$

GRADING NOTE Form required: standard form. $(x - 3)(x + 4) = x^2 + x - 12$; times $(2x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

A box has length x + 2, width x − 1, and height 2x + 3 (in cm). Write its volume as a polynomial in standard form.

V(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

494

QUESTION ID

S14-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 5x^2 - x - 6$

GRADING NOTE Form required: standard form. $(x^2 + x - 2)(2x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 1

Distribute a monomial over a polynomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $2x(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

496

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^3 + 6x^2 - 4x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $-5(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

497

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-5x^2 - 15x + 10$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

True or false?

$x^2(x + 4) = x^3 + 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

498

QUESTION ID

S14-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 \cdot 4 = 4x^2$, so the product is $x^3 + 4x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$-3x(x - 1)$ equals

A. $3x^2 - 3x$

B. $-3x^2 + 3x$

C. $-3x^2 - 3x$

D. $-3x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

499

QUESTION ID

S14-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $-3x^2 + 3x$

GRADING NOTE $-3x \cdot -1 = 3x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $x(x^2 - x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

500

QUESTION ID

S14-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - x^2 + x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $x^2 \cdot x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

502

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x^5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $2x \cdot 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

503

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$6x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$x^2 \cdot x^2 = x^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

504

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$-x \cdot x^3$ equals

A. x^4

B. $-x^3$

C. $-x^4$

D. $-x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

505

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x^4$

GRADING NOTE $-1 \cdot x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x^2)(-4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

506

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-4x^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 3

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x - 3)(x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

508

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 12$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

Multiply $(x + 2)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

509

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + x - 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 2)^2$ equals

A. $x^2 + 4$

B. $x^2 - 4$

C. $x^2 - 4x + 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

510

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2 - 4x + 4$

GRADING NOTE Square of a binomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q3/5

True or false?

$(2x + 1)(x - 3) = 2x^2 - 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

511

QUESTION ID

S14-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The middle terms $-6x + x = -5x$ are missing: $2x^2 - 5x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(2x + 3)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

512

QUESTION ID

S14-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 + x - 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 1

Find a coefficient of a product without full expansion • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find the coefficient of x^2 in $(x^2 + 3x - 2)(4x^2 - x + 5)$ without expanding everything.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

514

QUESTION ID
S14-F1-Q1

TYPE
Short answer

CORRECT ANSWER
−6

GRADING NOTE Any equivalent form. x^2 terms: $x^2 \cdot 5 + 3x \cdot (-x) + (-2) \cdot 4x^2 = 5 - 3 - 8 = -6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

For which k does $(x + k)(x^2 - 3x + 1)$ have no x^2 term?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

515

QUESTION ID
S14-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. x^2 coefficient: $-3 + k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

True or false?

The constant term of a product of polynomials is the product of their constant terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

516

QUESTION ID
S14-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Only constants multiply to a constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

The leading coefficient of $(2x^3 - 1)(5x^2 + x)$ is

A. 7

B. 5

C. 2

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

517

QUESTION ID
S14-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 10

GRADING NOTE $2 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

Find the coefficient of x in (x − 1)(x − 2)(x − 3).

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

518

QUESTION ID
S14-F1-Q5

TYPE
Short answer

CORRECT ANSWER
11

GRADING NOTE Any equivalent form. 1·2 + 1·3 + 2·3 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5
High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Expand $(x + 1)^4$ using Pascal's Triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

520

QUESTION ID
S14-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^4 + 4x^3 + 6x^2 + 4x + 1$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

Expand $(x - 2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

521

QUESTION ID
S14-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x^4 - 8x^3 + 24x^2 - 32x + 16$

GRADING NOTE Form required: standard form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

The coefficients of $(x + y)^5$ are 1, 5, 10, 5, 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

522

QUESTION ID
S14-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Row 5 of Pascal's Triangle has six entries: 1, 5, 10, 10, 5, 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

The coefficient of x^2y^2 in $(x + y)^4$ is

A. 2

B. 6

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

523

QUESTION ID
S14-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 6

GRADING NOTE C(4, 2) = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

Explain why the coefficients of $(x + y)^n$ are symmetric.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

524

QUESTION ID
S14-F2-Q5

TYPE
Short answer

CORRECT ANSWER
C(n, k) = C(n, n − k): choosing k of the n factors to give x is the same as choosing the n − k that give y.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

Find remainders with the Remainder Theorem

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Evaluate a polynomial function	**HSF-IF.A.2**
BACKWARD 2	Powers of negative numbers	**7.NS.A.2**
BACKWARD 3	Dividend = divisor × quotient + remainder	**4.NBT.B.6**
FORWARD 1	Find unknown coefficients from given remainders	**HSA-APR.B.2**
FORWARD 2	Prove the Remainder Theorem	**HSA-APR.B.2**

SET 15 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Use the Remainder Theorem to find the remainder when $p(x) = x^3 - 4x^2 + 2x - 5$ is divided by $x - 3$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

527

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form. $p(3) = 27 - 36 + 6 - 5 = -8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

Use the Remainder Theorem to find the remainder when $p(x) = 2x^4 + x^3 - 3x + 7$ is divided by $x + 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

528

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

37

GRADING NOTE Any equivalent form. Divisor $x - (-2)$: $p(-2) = 32 - 8 + 6 + 7 = 37$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

Use the Remainder Theorem to find the remainder when $p(x) = x^{100} + 3$ is divided by $x + 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

529

QUESTION ID

S15-M-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(-1) = (-1)^{100} + 3 = 1 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

Use the Remainder Theorem to find the remainder when $p(x) = 4x^3 - 6x^2 + x - 1$ is divided by $x - 1/2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

530

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form (−1.5). p(1/2) = 1/2 − 3/2 + 1/2 − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 5x^2 + 4x$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

531

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $p(2) = 16 - 20 + 8 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 1

Evaluate a polynomial function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 - 4x^2 + 2x - 5$. Find $p(1)$.

$p(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

533

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

$p(x) = 2x^2 - x + 3$. Find $p(-2)$.

$p(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

534

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 8 + 2 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

True or false?

If $p(x) = x^2 + 1$, then $p(-3) = -8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

535

QUESTION ID

S15-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 + 1 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

If $p(x) = x^3 - x$, then $p(2) =$

A. 4

B. 6

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

536

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE 8 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = 4x - 1$. Find $p(1/2)$.

$p(1/2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

537

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 2

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2

Evaluate $(-2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

539

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

Evaluate $(-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

540

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(-1)^{100}$ equals

A. -100

B. -1

C. 100

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

541

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE An even power of -1 is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

True or false?

$-2^4 = (-2)^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

542

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2^4 = -16$ but $(-2)^4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

Evaluate $2(-2)^4 + (-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

543

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. $32 - 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 3

Dividend = divisor × quotient + remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Write 47 = 5 × q + r with 0 ≤ r < 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

545

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

47 = 5 × 9 + 2

GRADING NOTE Form required: q = 9 and r = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

A division gives quotient 12 and remainder 3 with divisor 7. What was the dividend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

546

QUESTION ID

S15-B3-Q2

TYPE

Short answer

CORRECT ANSWER

87

GRADING NOTE Any equivalent form. 7 × 12 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

If a division leaves remainder 0, the divisor is a factor of the dividend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

547

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dividend = divisor × quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

$x^3 - 1 = (x - 1)(x^2 + x + 1) + 0$. The remainder on division by $x - 1$ is

A. 0

B. 1

C. $x^2 + x + 1$

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

548

QUESTION ID

S15-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE Read from the identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

$p(x) = (x - 3)(x^2 - x - 1) - 8$. What is the remainder when $p(x)$ is divided by $x - 3$?

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

549

QUESTION ID

S15-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 1

Find unknown coefficients from given remainders • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

When $p(x) = x^3 + kx^2 - 4$ is divided by $x - 2$, the remainder is 12. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

551

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $p(2) = 8 + 4k - 4 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 + ax + b$ leaves remainder 5 on division by $x - 1$ and remainder -1 on division by $x + 1$. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

552

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = 2

GRADING NOTE Form required: both. $1 + a + b = 5$ and $-1 - a + b = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

True or false?

If p(x) leaves remainder 0 on division by x − 4, then p(−4) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

553

QUESTION ID

S15-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The remainder on division by x − 4 is p(4), so p(4) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q3/5

$p(x) = 2x^2 + kx + 1$ leaves remainder 7 on division by $x + 1$. Then $k =$

A. 4

B. -4

C. -6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

554

QUESTION ID

S15-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −4

GRADING NOTE $2 - k + 1 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

For which k is the remainder of $x^4 - kx + 6$ divided by $x + 2$ equal to 0?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

555

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−11

GRADING NOTE Any equivalent form. 16 + 2k + 6 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 2

Prove the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Which argument proves that the remainder when a polynomial p(x) is divided by x − a is p(a)?

A. Write p(x) = (x − a)q(x) + r, where r is a constant because it has degree less than 1. Substituting x = a gives p(a) = 0 · q(a) + r = r.

B. Dividing x^3 by x − 2 leaves remainder 8, and $2^3 = 8$, so the remainder is always p(a).

C. Since x − a = 0 exactly when x = a, dividing p(x) by x − a is the same as replacing x with a, so the remainder must be p(a).

D. Write p(x) = (x − a)q(x) + r. Substituting x = 0 gives p(0) = (0 − a)q(0) + r = −a · q(0) + r, so the remainder r equals p(a) for every a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

557

QUESTION ID
S15-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. Write p(x) = (x − a)q(x) + r, where r is a constant because it has degree less than 1. Substituting x = a gives p(a) = 0 · q(a) + r = r.

GRADING NOTE The division identity with a constant remainder, then x = a makes the factor (a − a) zero. The others check one example, substitute the wrong value (x = 0 gives p(0), not p(a)), or state the result without a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

Why must the remainder on division by x − a be a constant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

558

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The remainder has a lower degree than the divisor x − a, which has degree 1, so it has degree 0 (a constant).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

True or false?

The Remainder Theorem also works for division by 2x − 1 with remainder p(1/2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

559

QUESTION ID

S15-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE p(x) = (2x − 1)q(x) + r gives p(1/2) = r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q3/5

Which step uses the fact that a − a = 0?

A. Graphing p(x)

B. Dividing p(x) by x

C. Factoring p(x)

D. Substituting x = a into p(x) = (x − a)q(x) + r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

560

QUESTION ID

S15-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Substituting x = a into p(x) = (x − a)q(x) + r

GRADING NOTE The product term vanishes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

Explain how the proof shows that p(a) = 0 exactly when x − a is a factor of p(x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

561

QUESTION ID

S15-F2-Q5

TYPE

Short answer

CORRECT ANSWER

p(a) = r, and r = 0 exactly when p(x) = (x − a)q(x), i.e. x − a is a factor.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

Use p(a) = 0 to find factors

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
BACKWARD 2	Factor the quadratic quotient	**HSA-SSE.A.2**
BACKWARD 3	Divide a polynomial by a linear factor	**HSA-APR.A.1**
FORWARD 1	Choose parameters so that a given binomial is a factor	**HSA-APR.B.2**
FORWARD 2	Finish factorizations over the complex numbers	**HSN-CN.C.8**

SET 16 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Show that $x - 2$ is a factor of $p(x) = x^3 - 3x^2 - 4x + 12$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

564

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$p(2) = 0$; $p(x) = (x - 2)(x + 2)(x - 3)$

GRADING NOTE Form required: the check and the complete factorization. $p(2) = 8 - 12 - 8 + 12 = 0$; quotient $x^2 - x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

Show that $x + 1$ is a factor of $p(x) = x^3 + 2x^2 - 5x - 6$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

565

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

$p(-1) = 0$; $p(x) = (x + 1)(x + 3)(x - 2)$

GRADING NOTE Form required: the check and the factorization. Quotient $x^2 + x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

Show that $x + 3$ is a factor of $p(x) = 2x^3 + 5x^2 - 4x - 3$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

566

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

$p(-3) = 0$; $p(x) = (x + 3)(2x + 1)(x - 1)$

GRADING NOTE Form required: the check and the factorization. $p(-3) = -54 + 45 + 12 - 3 = 0$; quotient $2x^2 - x - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

Is $x - 4$ a factor of $p(x) = x^3 - 5x^2 + 2x + 9$? Justify with the Remainder Theorem.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

567

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

No: $p(4) = 1 \neq 0$

GRADING NOTE Form required: the decision and $p(4)$. $64 - 80 + 8 + 9 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-M-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

$p(x) = x^3 - 7x + 6$. Use $p(a) = 0$ to find all the zeros of p, and write p(x) in factored form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

568

QUESTION ID

S16-M-Q5

TYPE

Short answer

CORRECT ANSWER

Zeros 1, 2, −3; p(x) = (x − 1)(x − 2)(x + 3)

GRADING NOTE Form required: all zeros and the factorization. $p(1) = 0$, then the quotient $x^2 + x - 6 = (x - 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 1

Use the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^3 - 3x^2 - 4x + 12$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

570

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

Find the remainder when $x^3 + 2x^2 - 5x - 6$ is divided by $x - 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

571

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

True or false?

Dividing by x + 3 means evaluating at x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

572

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x + 3 = x − (−3): evaluate at −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

The remainder of p(x) ÷ (x + 1) is

A. −p(1)

B. p(−1)

C. p(0)

D. p(1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

573

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. p(−1)

GRADING NOTE x − (−1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

Find $p(-3)$ for $p(x) = 2x^3 + 5x^2 - 4x - 3$.

$p(-3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

574

QUESTION ID

S16-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 2

Factor the quadratic quotient • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

576

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $2x^2 - x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

577

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2x + 1)(x − 1)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

$x^2 + x - 6$ equals

A. $(x - 3)(x + 2)$

B. $(x - 6)(x + 1)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

578

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x + 3)(x − 2)

GRADING NOTE Product −6, sum 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x^2 + 4$ factors over the integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

579

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

580

QUESTION ID

S16-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - 3)(x + 3)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 3

Divide a polynomial by a linear factor • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find q(x) if $x^3 - 3x^2 - 4x + 12 = (x - 2)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

582

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - x - 6$

GRADING NOTE Any equivalent form. Check by multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(x + 1)(x^2 + x - 6)$ to check a division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

583

QUESTION ID

S16-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 2x^2 - 5x - 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

True or false?

If $(x - 1)(x^2 + x - 6) = x^3 - 7x + 6$, then $x^2 + x - 6$ is the quotient of $x^3 - 7x + 6 \div (x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

584

QUESTION ID

S16-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of the quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^3 - 8 \div (x - 2)$ equals

A. $x^2 + 2x + 4$

B. $x^2 + 4$

C. $x^2 - 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

585

QUESTION ID

S16-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 2x + 4$

GRADING NOTE Difference of cubes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

Find q(x) if $2x^3 + 5x^2 - 4x - 3 = (x + 3)\,q(x)$.

q(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

586

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 - x - 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 1

Choose parameters so that a given binomial is a factor • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

ANSWER KEY Reduced copy of the question page, with its answer

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 3x + 9$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

588

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form. $27 + 9k - 9 + 9 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

Find a and b so that $x - 1$ and $x + 2$ are both factors of $x^3 + ax^2 + bx - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

589

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = −1

GRADING NOTE Form required: both. $1 + a + b - 2 = 0$ and $-8 + 4a - 2b - 2 = 0$ give $a + b = 1$ and $2a - b = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

True or false?

If $x - 2$ and $x + 2$ are factors of $p(x)$, then $x^2 + 4$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

590

QUESTION ID

S16-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 2)(x + 2) = x^2 - 4$, not $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

For which k is x + 1 a factor of $x^4 + kx + 3$?

A. 2

B. −4

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

591

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $1 - k + 3 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

Write a cubic with leading coefficient 1 that has $x - 1$, $x + 1$, and $x - 4$ as factors, in standard form.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

592

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^3 - 4x^2 - x + 4$

GRADING NOTE Form required: standard form. $(x^2 - 1)(x - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 2

Finish factorizations over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

ANSWER KEY Reduced copy of the question page, with its answer

$x - 1$ is a factor of $x^3 - 1$. Factor $x^3 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

594

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1/2 - (\sqrt{3}/2)i)(x + 1/2 + (\sqrt{3}/2)i)$

GRADING NOTE Any equivalent form. The quotient $x^2 + x + 1$ has zeros $-1/2 \pm (\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$x + 2$ is a factor of $x^3 + 2x^2 + 9x + 18$. Factor it completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

595

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 2)(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

True or false?

Every factor found with the factor theorem gives a zero of the polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

596

QUESTION ID

S16-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x - a$ is a factor exactly when $p(a) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

$x - 2$ is a factor of $x^3 - 2x^2 + x - 2$. The other zeros are

A. i and −i

B. −2

C. 1 and −1

D. 2i and −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

597

QUESTION ID

S16-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. i and −i

GRADING NOTE Quotient $x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$x - 3$ is a factor of $x^3 - 3x^2 + 4x - 12$. List all complex zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

598

QUESTION ID

S16-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3, 2i, −2i

GRADING NOTE Form required: all three. Quotient $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

Identify zeros and multiplicities from a factorization

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Solve by the zero-product property	**HSA-REI.B.4.b**
BACKWARD 2	Factor to find hidden repeated factors	**HSA-SSE.A.2**
BACKWARD 3	An exponent counts repeated factors	**6.EE.A.1**
FORWARD 1	Build a polynomial from zeros and one more condition	**HSA-APR.B.3**
FORWARD 2	Decide what the zeros determine and what remains free	**HSA-APR.B.3**

SET 17 • HSA-APR.B.3

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $p(x) = (x - 4)^2(x + 1)(2x - 3)$ and give the multiplicity of each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

601

QUESTION ID
S17-M-Q1

TYPE
Short answer

CORRECT ANSWER
4 (multiplicity 2), −1 (multiplicity 1), 3/2 (multiplicity 1)

GRADING NOTE Form required: every zero with its multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

Factor $p(x) = x^3 - 6x^2 + 9x$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

602

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

$p(x) = x(x - 3)^2$; 0 (multiplicity 1), 3 (multiplicity 2)

GRADING NOTE Form required: factorization and zeros with multiplicities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

List the zeros of $p(x) = (x^2 - 9)(x^2 + 2x + 1)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

603

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 (1), −3 (1), −1 (2)

GRADING NOTE Form required: every zero with its multiplicity. $(x - 3)(x + 3)(x + 1)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $p(x) = -2x(x + 5)^3$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

604

QUESTION ID

S17-M-Q4

TYPE

Short answer

CORRECT ANSWER

0 (1), −5 (3)

GRADING NOTE Form required: every zero with its multiplicity. The constant −2 is not a zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

Factor $p(x) = x^4 - 8x^2 + 16$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

605

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

$p(x) = (x - 2)^2(x + 2)^2$; 2 (2), −2 (2)

GRADING NOTE Form required: factorization and zeros with multiplicities. $(x^2 - 4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 1

Solve by the zero-product property • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 4)(x + 1) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

607

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 or −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x(2x - 3) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

608

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0 or 3/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

True or false?

If $(x - 2)^2 = 0$, then $x = 2$ or $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

609

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

The solutions of $(x + 5)(x - 5) = 0$ are

A. 25

B. −5 and 5

C. 0 and 5

D. 5 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

610

QUESTION ID
S17-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. −5 and 5

GRADING NOTE Each factor equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-2x(x + 5) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

611

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0 or −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 2

Factor to find hidden repeated factors • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 6x + 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

613

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

Factor $x^2 + 2x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

614

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

$x^4 - 8x^2 + 16$ equals

A. $(x^2 + 4)^2$

B. $(x^2 - 4)^2$

C. $(x^2 - 8)^2$

D. $(x - 4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

615

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x^2 - 4)^2$

GRADING NOTE $u^2 - 8u + 16$ with $u = x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x^3 - 6x^2 + 9x = x(x + 3)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

616

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x + 3)^2 = x^3 + 6x^2 + 9x$; the correct form is $x(x - 3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

617

QUESTION ID

S17-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 3

An exponent counts repeated factors • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write $(x - 4)^2$ as a product without an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

619

QUESTION ID

S17-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 4)(x − 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many factors of $(x + 5)$ are in $(x + 5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

620

QUESTION ID

S17-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

True or false?

$(x - 2)^2(x - 2) = (x - 2)^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

621

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

In $7(x - 1)^4$, the factor $x - 1$ appears

A. 28 times

B. 4 times

C. 1 time

D. 7 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

622

QUESTION ID

S17-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4 times

GRADING NOTE Exponent 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (x + 1)(x + 1)(x − 3) with an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

623

QUESTION ID

S17-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2(x - 3)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 1

Build a polynomial from zeros and one more condition • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write the cubic polynomial with zeros −1, 2, and 4 (each multiplicity 1) whose graph passes through (0, 16).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

625

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2(x + 1)(x − 2)(x − 4)

GRADING NOTE Any equivalent form. a(1)(−2)(−4) = 8a = 16 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

Write a polynomial of degree 3 with a zero of multiplicity 2 at x = 3, a zero at x = −2, and leading coefficient −1.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

626

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-(x-3)^2(x+2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

True or false?

Zeros alone determine a polynomial completely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

627

QUESTION ID

S17-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero constant multiple has the same zeros; a further condition fixes the scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

A quartic has zeros 1 and −1, each of multiplicity 2, and p(0) = 3. Then p(x) =

A. $-3(x-1)^2(x+1)^2$

B. $3(x^2-1)$

C. $(x-1)^2(x+1)^2$

D. $3(x-1)^2(x+1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

628

QUESTION ID

S17-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $3(x-1)^2(x+1)^2$

GRADING NOTE $a \cdot 1 \cdot 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

A polynomial has zeros 0 (multiplicity 2) and 5 (multiplicity 1), degree 3, and p(1) = −8. Find p(x).

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

629

QUESTION ID

S17-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2(x - 5)$

GRADING NOTE Any equivalent form. a · 1 · (−4) = −8 gives a = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 2

Decide what the zeros determine and what remains free • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

All polynomials with zeros 1 and 3 only (each multiplicity 1) and degree 2 have what form?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

631

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

a(x − 1)(x − 3) with any real a ≠ 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

If the degree is not fixed, give two different polynomials whose only real zeros are 1 and 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

632

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

For example $(x - 1)(x - 3)$ and $(x - 1)^2(x - 3)(x^2 + 1)$

GRADING NOTE Any two valid examples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

True or false?

Two cubic polynomials with the same three zeros (each multiplicity 1) must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

633

QUESTION ID

S17-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They can differ by a constant factor, for example $(x-1)(x-2)(x-3)$ and $2(x-1)(x-2)(x-3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

Which extra information fixes a cubic with zeros −2, 0, 2?

A. That it has degree 3

B. That 0 is a zero

C. Its end behavior only

D. Its value at x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

634

QUESTION ID

S17-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Its value at x = 1

GRADING NOTE One point fixes the scale a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

Explain why knowing only the real zeros of a degree-4 polynomial may not determine it even up to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

635

QUESTION ID

S17-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It may have non-real zeros (an irreducible quadratic factor such as $x^2 + c$) that the real zeros do not show, so different quartics share the same real zeros.

GRADING NOTE Any equivalent explanation with an example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

Construct a rough graph of a polynomial from its zeros

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Read zeros and multiplicities	**HSA-APR.B.3**
BACKWARD 2	End behavior from the leading term	**HSF-IF.B.4**
BACKWARD 3	Find the y-intercept by evaluating at 0	**HSF-IF.A.2**
BACKWARD 4	Sign of a power of a negative or positive number	**7.NS.A.2**
FORWARD 1	Sketch graphs from standard form by factoring first	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its rough graph	**HSA-APR.B.3**

SET 18 • HSA-APR.B.3

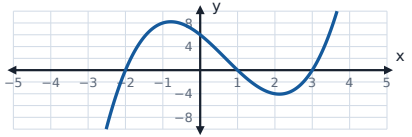
MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

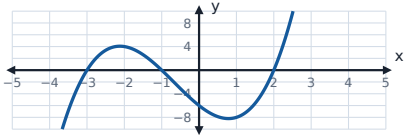
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = (x + 2)(x - 1)(x - 3)$?

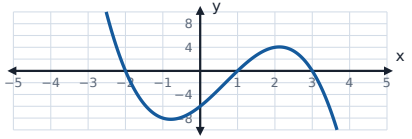
A



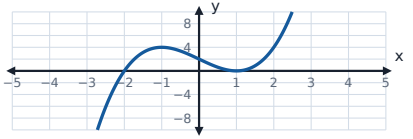
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

638

QUESTION ID

S18-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Zeros −2, 1, 3 (each crossed, multiplicity 1); y-intercept (2)(−1)(−3) = 6; degree 3 with positive leading coefficient, so it falls left and rises right. The others have the zeros with the wrong signs, the opposite end behavior, or a double zero at 1.

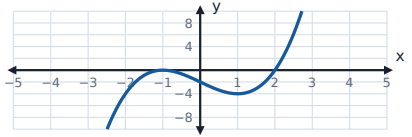
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

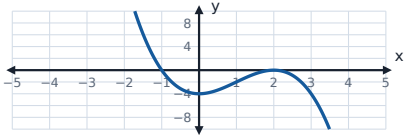
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = -(x + 1)^2(x - 2)$?

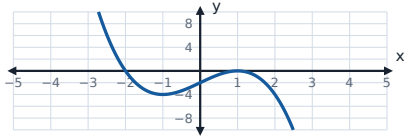
A



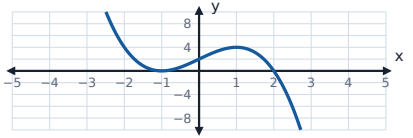
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

639

QUESTION ID

S18-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Double zero at −1 (touches), single zero at 2 (crosses); y-intercept $-(1)^2(-2) = 2$; leading coefficient −1 with odd degree, so it rises left and falls right. The others flip the sign, swap the multiplicities, or use the zeros 1 and −2.

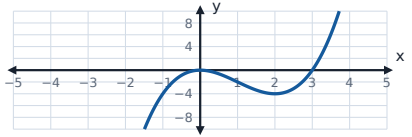
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

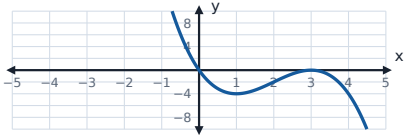
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = x(x - 3)^2$?

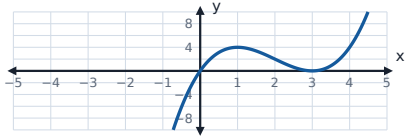
A



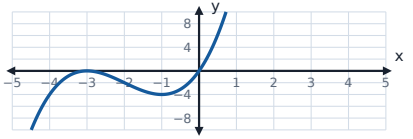
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

640

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Single zero at 0 (crosses), double zero at 3 (touches); y-intercept 0; $y(1) = 4 > 0$; falls left and rises right. The others swap the multiplicities, flip the sign, or put the double zero at -3.

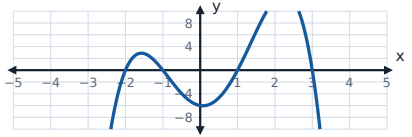
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

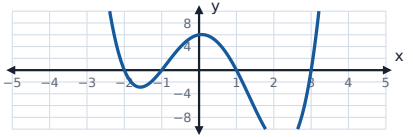
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = (x + 2)(x + 1)(x - 1)(x - 3)$?

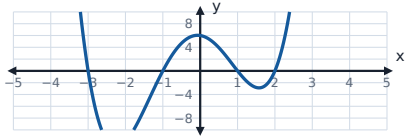
A



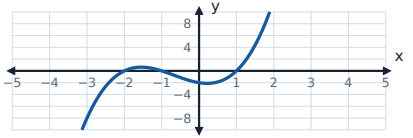
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

641

QUESTION ID

S18-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Four single zeros −2, −1, 1, 3; y-intercept (2)(1)(−1)(−3) = 6; degree 4 with positive leading coefficient, so it rises at both ends. The others are reflected, have zeros with the wrong signs (same y-intercept), or leave out a factor.

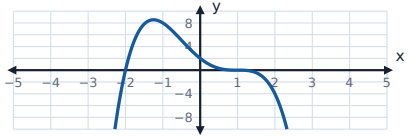
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

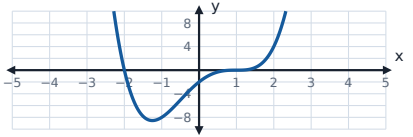
ANSWER KEY Reduced copy of the question page, with its answer

Which graph is a rough graph of $y = -(x - 1)^3(x + 2)$?

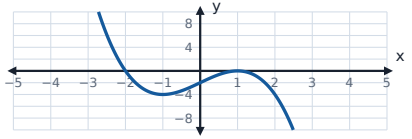
A



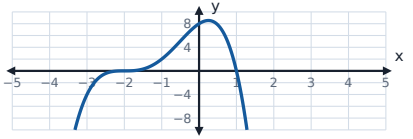
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

642

QUESTION ID

S18-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Zero 1 has multiplicity 3 (crosses with a flattened shape), zero −2 is single; y-intercept $-(-1)^3(2) = 2$; degree 4 with negative leading coefficient, so it falls at both ends. The others flip the sign, use multiplicity 2 at 1, or put the triple zero at −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 1

Read zeros and multiplicities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

644

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−2, 1, 3

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $-(x + 1)^2(x - 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

645

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−1 (2), 2 (1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

True or false?

$x(x - 3)^2$ has a zero of multiplicity 2 at $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

646

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 0$ has multiplicity 1; $x = 3$ has multiplicity 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

The zero $x = 1$ of $-(x - 1)^3(x + 2)$ has multiplicity

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

647

QUESTION ID

S18-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE Exponent 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $(x + 2)(x + 1)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

QUESTION ID

S18-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$-2, -1, 1, 3$

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-B1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 2

End behavior from the leading term • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

As x grows large and positive, what happens to x^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

650

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

It grows large and positive.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

As x grows large and negative, what happens to $-x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

651

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

It becomes large and negative.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

The leading term of $(x + 2)(x - 1)(x - 3)$ is

A. $-x^3$

B. -6

C. $3x$

D. x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

652

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^3

GRADING NOTE $x \cdot x \cdot x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

True or false?

A polynomial of even degree with a positive leading coefficient falls at both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

653

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Like x^2, it rises at both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the leading term of $-(x-1)^3(x+2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

654

QUESTION ID

S18-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$-x^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 3

Find the y-intercept by evaluating at 0 • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find y when x = 0 for y = (x + 2)(x − 1)(x − 3).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

656

QUESTION ID

S18-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 2 · (−1) · (−3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

Find y when x = 0 for $y = -(x + 1)^2(x - 2)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

657

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. −(1)(−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

True or false?

The y-intercept of $x(x - 3)^2$ is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

658

QUESTION ID

S18-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 0 (a factor x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

The y-intercept of $-(x-1)^3(x+2)$ is

A. 0

B. -2

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

659

QUESTION ID

S18-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $-(-1)(2) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

Find y when x = 1 for $y = x(x - 3)^2$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

660

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 4

Sign of a power of a negative or positive number • Grade 7 • The Number System • 7.NS.A.2

Is $(-0.5)^2$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

662

QUESTION ID

S18-B4-Q1

TYPE

Short answer

CORRECT ANSWER

Positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

Is $(-0.5)^3$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

663

QUESTION ID

S18-B4-Q2

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

True or false?

An even power of a negative number is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

664

QUESTION ID

S18-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pairs of negative factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

Near $x = 3$, $(x - 3)^2$ is

A. negative on both sides

B. positive on both sides

C. zero everywhere

D. positive on the left and negative on the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

665

QUESTION ID

S18-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. positive on both sides

GRADING NOTE A square does not change sign at $x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $(x - 1)^3$ change sign at $x = 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

666

QUESTION ID

S18-B4-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: negative for $x < 1$ and positive for $x > 1$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

SET 18 • HSA-APR.B.3

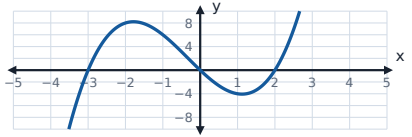
FORWARD 1

Sketch graphs from standard form by factoring first • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

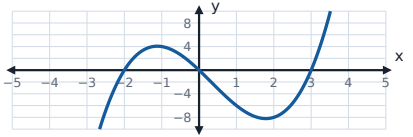
ANSWER KEY Reduced copy of the question page, with its answer

Factor $y = x^3 - x^2 - 6x$. Which graph is a rough graph of it?

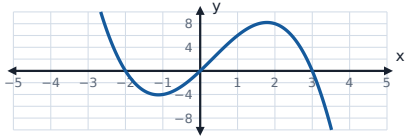
A



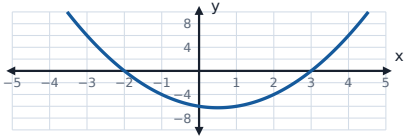
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

668

QUESTION ID

S18-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE $x^3 - x^2 - 6x = x(x - 3)(x + 2)$: crosses at -2, 0, 3; y-intercept 0; falls left and rises right. The others use $x(x + 3)(x - 2)$, the reflection, or drop the factor x.

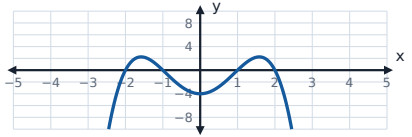
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

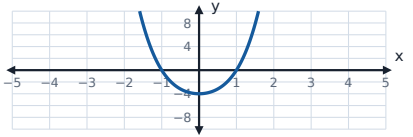
ANSWER KEY Reduced copy of the question page, with its answer

Factor $y = x^4 - 5x^2 + 4$. Which graph is a rough graph of it?

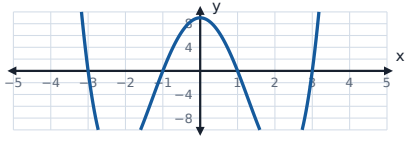
A



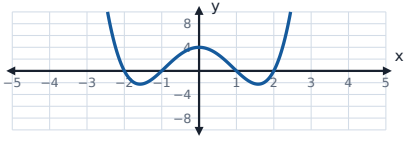
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

669

QUESTION ID

S18-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$: crosses at ±1 and ±2; y-intercept 4; rises at both ends. The others are reflected, factor as $(x^2 - 1)(x^2 + 4)$, or use $(x^2 - 1)(x^2 - 9)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

True or false?

$y = x^3 + 4x$ crosses the x-axis at $x = -2$, 0, and 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

670

QUESTION ID

S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x^2 + 4) = 0$ only at $x = 0$; $x^2 + 4$ has no real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

$y = -x^3 + 9x$ crosses the x-axis at

A. 0 and 9

B. 3 only

C. −3, 0, 3

D. −9, 0, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

671

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −3, 0, 3

GRADING NOTE $-x(x - 3)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

$y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$. Describe its behavior at each zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

672

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Crosses at x = −1; touches (turns) at x = 2.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

FORWARD 2

Write a polynomial from its rough graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A cubic's graph crosses the x-axis at −3 and touches it at 2, and passes through (0, 12). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

674

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 2)^2$

GRADING NOTE Any equivalent form. a(3)(4) = 12 gives a = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

A quartic rises at both ends, touches the x-axis at −1 and at 3, and passes through (0, 18). Write the polynomial.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

675

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$2(x + 1)^2(x - 3)^2$

GRADING NOTE Any equivalent form. a(1)(9) = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

True or false?

A graph that touches the x-axis at x = a without crossing has a zero of even multiplicity at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

676

QUESTION ID

S18-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The factor does not change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

A cubic falls to the right and crosses the x-axis at 0, 1, and 4. Its leading coefficient is

A. zero

B. not determined by the end behavior

C. positive

D. negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

677

QUESTION ID

S18-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. negative

GRADING NOTE A cubic that falls to the right has a negative leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

A polynomial graph crosses at −2, touches at 1, falls to the left and rises to the right, and has y-intercept 4. Give a polynomial of least degree.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

678

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2(x + 2)(x - 1)^2$

GRADING NOTE Any equivalent form. Degree 3 with a > 0; a(2)(1) = 4 gives a = 2, which agrees with the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

Prove polynomial identities

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Expand the square of a binomial	**HSA-APR.A.1**
BACKWARD 2	Recognize a difference of squares	**HSA-SSE.A.2**
BACKWARD 3	What makes an identity	**6.EE.A.4**
FORWARD 1	Prove or disprove proposed identities	**HSA-APR.C.4**
FORWARD 2	Prove identities with the Binomial Theorem	**HSA-APR.C.5**

SET 19 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$?

A. For $x = 2$ and $y = 1$: $25 = 9 + 16$, so the identity is true.

B. Take the square root of both sides: $x^2 + y^2 = (x^2 - y^2) + 2xy$, and the right side simplifies to $x^2 - y^2 + 2xy$, so the identity is true.

C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

D. The right side is $(x^2 - y^2)^2 + (2xy)^2 = x^4 - y^4 + 4x^2y^2$, and the left side is $x^4 + y^4 + 4x^2y^2$, so the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

681

QUESTION ID

S19-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

GRADING NOTE Both sides expand to $x^4 + 2x^2y^2 + y^4$. The others check one example, square $x^2 - y^2$ as $x^4 - y^4$, or take square roots term by term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

Which argument proves the identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$?

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

B. For $a = 1$ and $b = 2$ the left side is $1 + 8 = 9$ and the right side is $3 \cdot (1 - 2 + 4) = 9$, so the identity is true for all a and b.

C. Multiply the first terms and the last terms: $a \cdot a^2 = a^3$ and $b \cdot b^2 = b^3$, so the product is $a^3 + b^3$.

D. The product is $a^3 - 2a^2b + b^3$, which equals $a^3 + b^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

682

QUESTION ID

S19-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

GRADING NOTE All six products, then the cancellations. The others check one example, skip four of the six products, or multiply incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

Which argument proves the identity $(a + b)^2 - (a - b)^2 = 4ab$?

A. For $a = 3$ and $b = 1$: $16 - 4 = 12 = 4 \cdot 3$, so the identity is true.

B. $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab$.

C. $(a + b)^2 - (a - b)^2 = [(a + b) - (a - b)]^2 = (2b)^2 = 4ab$.

D. $(a + b)^2 - (a - b)^2 = (a^2 + b^2) - (a^2 - b^2) = a^2 + b^2 - a^2 + b^2 = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

683

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab.$

GRADING NOTE Subtract the whole second square, changing the sign of every term (or factor as $[(a + b) - (a - b)][(a + b) + (a - b)] = (2b)(2a)$). The others check one example, square binomials as $a^2 + b^2$, or use the false rule $x^2 - y^2 = (x - y)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

Which argument proves the identity $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$?

A. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 + (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

B. For $x = 0$: $4 = 2 \cdot 2$, so the identity is true.

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

D. Multiply the first terms and the last terms: $x^2 \cdot x^2 = x^4$ and $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

684

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

GRADING NOTE Group as a difference of squares (or expand all nine products). The others check one example, use $(u + v)(u - v) = u^2 + v^2$, or skip the middle products.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

Which argument proves the identity
$(n + 1)^3 - n^3 = 3n^2 + 3n + 1$?

A. For n = 1, 2, and 3 the left side is 7, 19, and 37, and so is the right side, so it is true for every n.

B. $(n + 1)^3 = n^3 + 1$, so $(n + 1)^3 - n^3 = 1$, which is $3n^2 + 3n + 1$.

C. $(n + 1)^3 - n^3 = (n + 1 - n)^3 = 1^3 = 1$.

D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

685

QUESTION ID
S19-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

GRADING NOTE Expand $(n + 1)^3$ correctly and subtract. The others check several values (evidence, not a proof) or use the false rules $a^3 - b^3 = (a - b)^3$ and $(n + 1)^3 = n^3 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 1

Expand the square of a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

687

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$a^2 + 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

Expand $(x^2 - y^2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

688

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^4 - 2x^2y^2 + y^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

True or false?

$(a - b)^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

689

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(a - b)^2 = a^2 - 2ab + b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(2xy)^2$ equals

A. $2xy^2$

B. $4xy$

C. $2x^2y^2$

D. $4x^2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

690

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $4x^2y^2$

GRADING NOTE Square each factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

Expand $(n + 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

691

QUESTION ID

S19-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$n^3 + 3n^2 + 3n + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 2

Recognize a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $a^2 - b^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

693

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(a - b)(a + b)$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

Factor $(x^2 + 2)^2 - (2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

694

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 2 - 2x)(x^2 + 2 + 2x)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

True or false?

$u^2 - v^2 = (u - v)(u + v)$ for any expressions u and v.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

695

QUESTION ID

S19-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Structure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

Which is a difference of squares?

A. $x^3 - 1$

B. $x^2 - 3x$

C. $9x^2 - 16$

D. $9x^2 + 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

696

QUESTION ID

S19-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $9x^2 - 16$

GRADING NOTE $(3x)^2 - 4^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

Factor $25 - x^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

697

QUESTION ID

S19-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$(5 - x^2)(5 + x^2)$

GRADING NOTE Any equivalent factored form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 3

What makes an identity • Grade 6 • Expressions & Equations • 6.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Test $(a + b)^2 = a^2 + b^2$ with $a = 1$ and $b = 2$. Is it an identity?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

699

QUESTION ID

S19-B3-Q1

TYPE

Short answer

CORRECT ANSWER

No: 9 ≠ 5

GRADING NOTE Form required: the test and the decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain the difference between an equation and an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

700

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

An identity is true for every value of the variables; an equation may be true only for some values.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

True or false?

Checking an identity with three numbers proves it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

701

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Examples cannot prove a statement about all values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$2(x + 3) = 2x + 6$ is

A. an identity

B. true only for $x = 3$

C. true only for $x = 0$

D. never true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

702

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. an identity

GRADING NOTE Distributive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

Give values that show $a^3 + b^3 = (a + b)^3$ is not an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

703

QUESTION ID

S19-B3-Q5

TYPE

Short answer

CORRECT ANSWER

For example

$a = b = 1$: $2 \neq 8$

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 1

Prove or disprove proposed identities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Is $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ an identity? Which answer is correct?

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

B. Yes, because $a = 1$ and $b = 1$ give $8 = 2 + 6$.

C. No: for $a = 1$ and $b = 2$ the left side is $3^3 = 27$ and the right side is $1 + 8 + 3 \cdot 2 = 15$, so the sides differ.

D. No: $(a + b)^3 = a^3 + b^3$, so the extra term makes the two sides different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

705

QUESTION ID

S19-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

GRADING NOTE True by expansion. The others use $(a + b)^3 = a^3 + b^3$, check one example, or miscompute ($a = 1$, $b = 2$ gives $27 = 1 + 8 + 18$ on both sides).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $a^4 - b^4 = (a - b)^4$ an identity? Which answer is correct?

A. Yes: for a = 1 and b = 0 both sides equal 1.

B. No: for a = 2 and b = 1, the left side is 16 − 1 = 15 and the right side is $1^4 = 1$.

C. Yes: a fourth power applies to each term inside the parentheses, so $(a - b)^4 = a^4 - b^4$ for all a and b.

D. No, because $a^4 - b^4 = (a - b)^2(a + b)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

706

QUESTION ID

S19-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No: for a = 2 and b = 1, the left side is 16 − 1 = 15 and the right side is $1^4 = 1$.

GRADING NOTE One counterexample disproves an identity. The others distribute the exponent, generalize from one example, or use a false factorization ($a^4 - b^4 = (a - b)(a + b)(a^2 + b^2)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

True or false?

$(x - 1)(x^2 + x + 1) = x^3 + 1$ is an identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

707

QUESTION ID

S19-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The product is $x^3 - 1$ (difference of cubes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

Which is an identity?

A. $(x + 2)^2 = x^2 + 4$

B. $(x + 1)^3 = x^3 + 1$

C. $x^2 - 4 = (x - 4)^2$

D. $(x + 2)^2 - (x - 2)^2 = 8x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

708

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $(x + 2)^2 - (x - 2)^2 = 8x$

GRADING NOTE 4ab with a = x, b = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $(x + c)^2 - (x - c)^2 = 12x$ is an identity.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

709

QUESTION ID

S19-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 4cx = 12x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 2

Prove identities with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Which argument uses the Binomial Theorem to prove that the entries in row n of Pascal's Triangle add to 2^n?

A. Row n has n + 1 entries, and each entry is the sum of the two entries above it, so every row has twice as many entries as the row before it and the sum of row n is 2^n.

B. Row 3 is 1, 3, 3, 1, which adds to $8 = 2^3$, so every row n adds to 2^n.

C. Set x = y = 1 in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + ... + C(n, n)y^n$. The left side is 2^n and the right side is C(n, 0) + C(n, 1) + ... + C(n, n), the row sum.

D. Set x = 1 and y = 0 in $(x + y)^n = C(n, 0)x^n + ... + C(n, n)y^n$: the left side is 1 and the right side is C(n, 0) = 1, so each row adds to 2^n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

711

QUESTION ID
S19-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
C. Set x = y = 1 in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + ... + C(n, n)y^n$. The left side is 2^n and the right side is C(n, 0) + C(n, 1) + ... + C(n, n), the row sum.

GRADING NOTE Substituting 1 for x and y makes every term equal to its coefficient. The others check one row, use a substitution that gives 1, or argue from the number of entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

Use the Binomial Theorem with $x = 1$ and $y = -1$ to find $C(5,0) - C(5,1) + C(5,2) - C(5,3) + C(5,4) - C(5,5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

712

QUESTION ID
S19-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form. $(1 - 1)^5 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

True or false?

$(1 + 1)^4 = 1 + 4 + 6 + 4 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

713

QUESTION ID

S19-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

The sum of the coefficients of $(2x + 1)^3$ is

A. 9

B. 8

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

714

QUESTION ID
S19-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 27

GRADING NOTE Set x = 1: 3^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $(1 + 0.1)^3 = 1 + 3(0.1) + 3(0.01) + 0.001$ to find 1.1^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

715

QUESTION ID
S19-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1.331

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

Use polynomial identities to describe numerical relationships

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Substitute values into an identity	**6.EE.A.2**
BACKWARD 2	Check a Pythagorean triple	**8.G.B.6**
BACKWARD 3	Choose a and b to fit a product	**HSA-APR.C.4**
FORWARD 1	Generate a triple with a required side	**HSA-APR.C.4**
FORWARD 2	Connect the triple identity to complex numbers	**HSN-CN.A.2**

SET 20 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ generates Pythagorean triples. Use $x = 3$ and $y = 2$ to find a triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

718

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

5, 12, 13

GRADING NOTE Any order. $x^2 - y^2 = 5$, $2xy = 12$, $x^2 + y^2 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

Use the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ with $x = 5$ and $y = 2$ to find a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

719

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

20, 21, 29

GRADING NOTE Any order. $25 - 4 = 21$, $2 \cdot 5 \cdot 2 = 20$, $25 + 4 = 29$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

Use the identity $(a + b)(a - b) = a^2 - b^2$ to compute 47×53 mentally. Show which a and b you use.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

720

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

a = 50, b = 3:

2500 − 9 = 2491

GRADING NOTE Form required: the choice of a and b and the product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q3**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

Use the identity $(n + 1)^2 - n^2 = 2n + 1$ to write 41 as a difference of two consecutive squares.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

721

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

$41 = 21^2 - 20^2$

GRADING NOTE Form required: the two squares. $2n + 1 = 41$ gives $n = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

Use the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$ to compute $13^3 - 12^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

722

QUESTION ID
S20-M-Q5

TYPE
Short answer

CORRECT ANSWER
469

GRADING NOTE Any equivalent form. 3·144 + 36 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 1

Substitute values into an identity • Grade 6 • Expressions & Equations • 6.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^2 - y^2$ for $x = 3$ and $y = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

724

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2xy for x = 5 and y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

725

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

True or false?

For $n = 12$, $3n^2 + 3n + 1 = 433$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

726

QUESTION ID

S20-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 432 + 36 + 1 = 469.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $a = 50$ and $b = 3$, $a^2 - b^2$ is

A. 2491

B. 2500

C. 2494

D. 2509

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

727

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2491

GRADING NOTE 2500 − 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2n + 1 for n = 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

728

QUESTION ID

S20-B1-Q5

TYPE

Short answer

CORRECT ANSWER

41

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 2

Check a Pythagorean triple • Grade 8 • Geometry • 8.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Do 5, 12, 13 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

730

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: 25 + 144 = 169

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

Do 20, 21, 29 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

731

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes:

400 + 441 = 841

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

True or false?

6, 8, 11 is a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

732

QUESTION ID

S20-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $36 + 64 = 100 \neq 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

Which is a Pythagorean triple?

A. 8, 15, 17

B. 6, 7, 9

C. 2, 3, 4

D. 4, 5, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

733

QUESTION ID

S20-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8, 15, 17

GRADING NOTE 64 + 225 = 289.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

Explain why a triangle with sides 5, 12, 13 is a right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

734

QUESTION ID

S20-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$5^2 + 12^2 = 13^2$, so by the converse of the Pythagorean Theorem it is a right triangle.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 3

Choose a and b to fit a product • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Write 47 and 53 as 50 − b and 50 + b. What is b?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

736

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

Write 98 × 102 as (a − b)(a + b).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

737

QUESTION ID

S20-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(100 − 2)(100 + 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

True or false?

$29 \times 31 = 30^2 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

738

QUESTION ID

S20-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(30 - 1)(30 + 1) = 30^2 - 1 = 899.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

64 × 56 equals

A. $(60 - 4)^2$

B. $64^2 - 56^2$

C. $60^2 - 4^2$

D. $60^2 + 4^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

739

QUESTION ID

S20-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $60^2 - 4^2$

GRADING NOTE a = 60, b = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 98 × 102.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

740

QUESTION ID

S20-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9996

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 1

Generate a triple with a required side • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Use the triple identity to find x and y (whole numbers, x > y) that give a triple with hypotenuse 25 and legs 7 and 24.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

742

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 4, y = 3

GRADING NOTE Any equivalent form. $x^2 + y^2 = 25$, $x^2 - y^2 = 7$, $2xy = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

Find x and y (whole numbers, x > y) for which the identity gives a triple with a leg 2xy = 40 and hypotenuse 41.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

743

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 5, y = 4

GRADING NOTE Any equivalent form. $x^2 - y^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

True or false?

Every choice of whole numbers $x > y > 0$ in the identity gives a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

744

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All three expressions are positive whole numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

Which choice gives the triple 8, 15, 17?

A. x = 4, y = 1

B. x = 5, y = 3

C. x = 4, y = 2

D. x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

745

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x = 4, y = 1

GRADING NOTE 16 − 1 = 15, 8, 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

Explain why $x = 2y$ gives a triple with legs in the ratio $3 : 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

746

QUESTION ID

S20-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x^2 - y^2 = 3y^2$ and $2xy = 4y^2$, so the legs are $3y^2$ and $4y^2$ (ratio 3 : 4).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 2

Connect the triple identity to complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(3 + 2i)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

748

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 12i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

Compute $(x + yi)^2$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

749

QUESTION ID
S20-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$(x^2 - y^2) + 2xyi$

GRADING NOTE Form required: $a + bi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

True or false?

The real and imaginary parts of $(x + yi)^2$ are the legs of the Pythagorean triple from the identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

750

QUESTION ID

S20-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - y^2$ and $2xy$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

$(5 + 2i)^2$ equals

A. 25 + 4i

B. 21 + 20i

C. 21 + 10i

D. 29 + 20i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

751

QUESTION ID

S20-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 21 + 20i

GRADING NOTE 25 − 4 + 20i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

Use $(4 + i)^2$ to find a Pythagorean triple.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

752

QUESTION ID

S20-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15, 8, 17

GRADING NOTE Any order. $(4 + i)^2 = 15 + 8i$; $4^2 + 1^2 = 17$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

Expand with the Binomial Theorem

MAIN	Algebra II main question	**HSA-APR.C.5**
BACKWARD 1	Multiply binomials repeatedly	**HSA-APR.A.1**
BACKWARD 2	Build Pascal's Triangle by a rule	**4.OA.C.5**
BACKWARD 3	Powers of products and of negatives	**8.EE.A.1**
FORWARD 1	Find a specific term or coefficient with a condition	**HSA-APR.C.5**
FORWARD 2	Use binomial coefficients to compute probabilities	**HSS-CP.B.9**

SET 21 • HSA-APR.C.5

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Use the Binomial Theorem to expand $(x + 2)^5$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

755

QUESTION ID
S21-M-Q1

TYPE
Short answer

CORRECT ANSWER
$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$

GRADING NOTE Form required: standard form. Coefficients $C(5, k)\cdot 2^k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q1**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

Use the Binomial Theorem to expand $(2a - b)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

QUESTION ID
S21-M-Q2

TYPE
Short answer

CORRECT ANSWER
$16a^4 - 32a^3b + 24a^2b^2 - 8ab^3 + b^4$

GRADING NOTE Any order of terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

Use the Binomial Theorem to find the coefficient of x^3 in $(x + 3)^7$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

757

QUESTION ID
S21-M-Q3

TYPE
Short answer

CORRECT ANSWER
2835

GRADING NOTE Any equivalent form. $C(7, 3)\cdot 3^4 = 35\cdot 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

Use the Binomial Theorem to find the term containing x^3y^3 in $(x - 2y)^6$.

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

758

QUESTION ID
S21-M-Q4

TYPE
Short answer

CORRECT ANSWER
$-160x^3y^3$

GRADING NOTE Any equivalent form. $C(6, 3)(-2)^3 = 20 \cdot (-8)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

Use the first three terms of the binomial expansion of $(1 + 0.02)^5$ to estimate 1.02^5 to four decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

759

QUESTION ID
S21-M-Q5

TYPE
Short answer

CORRECT ANSWER
1.1040

GRADING NOTE Any value 1.1040 or 1.1041 is accepted (1 + 0.1 + 0.004 = 1.104; the exact value is 1.10408...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-M-Q5**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 1

Multiply binomials repeatedly • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

761

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 4x + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x + 2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

762

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + 6x^2 + 12x + 8$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

True or false?

$(x + y)^3 = x^3 + y^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

763

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$; for $x = y = 1$, $8 \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

The number of terms in the expansion of $(x + y)^5$ is

A. 6

B. 5

C. 10

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

764

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE n + 1 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

Expand $(a - b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

765

QUESTION ID

S21-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 2

Build Pascal's Triangle by a rule • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Row 4 of Pascal's Triangle is 1, 4, 6, 4, 1. Write row 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

767

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1, 5, 10, 10, 5, 1

GRADING NOTE Form required: all six entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

Write row 7 of Pascal's Triangle (row 6 is 1, 6, 15, 20, 15, 6, 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

768

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 7, 21, 35, 35, 21, 7, 1

GRADING NOTE Form required: all entries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

True or false?

Each inner entry of Pascal's Triangle is the sum of the two entries above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

769

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The rule of the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

The fourth entry of row 7 (counting from the first 1) is

A. 20

B. 7

C. 35

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

770

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 35

GRADING NOTE 1, 7, 21, 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the sum of the entries in row 5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

771

QUESTION ID

S21-B2-Q5

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

BACKWARD 3

Powers of products and of negatives • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(2a)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

773

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$16a^4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-2y)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

774

QUESTION ID

S21-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$-8y^3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$(-b)^4 = -b^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

775

QUESTION ID

S21-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An even power of $-b$ is positive: $(-b)^4 = b^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

3^4 equals

A. 27

B. 12

C. 64

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

776

QUESTION ID

S21-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 81

GRADING NOTE $3 \cdot 3 \cdot 3 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(0.02)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

777

QUESTION ID

S21-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.0004

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 1

Find a specific term or coefficient with a condition • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Find the constant term of $(x + 1/x)^6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

779

QUESTION ID
S21-F1-Q1

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE Any equivalent form. Middle term C(6, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q1**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

The coefficient of x^2 in $(1 + kx)^5$ is 90. Find k (k > 0).

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

780

QUESTION ID
S21-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. $C(5, 2)k^2 = 10k^2 = 90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

True or false?

The middle term of $(x + y)^8$ is $56x^4y^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

781

QUESTION ID
S21-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $C(8, 4) = 70$, so the middle term is $70x^4y^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

The term with x^4 in $(x^2 - 1)^5$ is

A. $-5x^4$

B. $-10x^4$

C. $10x^4$

D. $5x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

782

QUESTION ID
S21-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $-10x^4$

GRADING NOTE $C(5, 2)(x^2)^2(-1)^3 = -10x^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

Find the coefficient of x in $(2x - 1)^6$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

783

QUESTION ID
S21-F1-Q5

TYPE
Short answer

CORRECT ANSWER
−12

GRADING NOTE Any equivalent form. $C(6, 1)\cdot 2\cdot(-1)^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 2

Use binomial coefficients to compute probabilities • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is tossed 5 times. Use C(5, k) to find P(exactly 3 heads).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

785

QUESTION ID
S21-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10/32 = 5/16

GRADING NOTE Any equivalent form (0.3125).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

A spinner lands on red with probability 0.3. Find P(exactly 2 reds in 4 spins).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

786

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.2646

GRADING NOTE Any equivalent form. $C(4, 2)(0.3)^2(0.7)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

True or false?

The probabilities of 0, 1, ..., n successes add to 1 because $(p + (1 - p))^n = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

787

QUESTION ID
S21-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Binomial Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For 6 fair coin tosses, P(exactly 3 heads) is

A. 6/64

B. 3/6

C. 15/64

D. 20/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

788

QUESTION ID

S21-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20/64

GRADING NOTE C(6, 3) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

Explain why C(n, k) counts the outcomes with exactly k successes in n trials.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

789

QUESTION ID

S21-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Each outcome is a choice of which k of the n trials are successes; there are C(n, k) such choices.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

Rewrite rational expressions by polynomial division

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Write a division with a remainder as a mixed number	**4.NF.B.3**
BACKWARD 2	Multiply to check a division	**HSA-APR.A.1**
BACKWARD 3	Use the Remainder Theorem to check the remainder	**HSA-APR.B.2**
FORWARD 1	Divide with unknown coefficients and higher-degree divisors	**HSA-APR.D.6**
FORWARD 2	Use division to find slant asymptotes	**HSF-IF.C.7.d**

SET 22 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Which shows $(x^3 + 2x^2 - 5x + 3)/(x - 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^2 + x - 6 + 9/(x - 1)$

B. $x^2 + 3x - 2 + 1/(x + 1)$

C. $x^2 + 3x - 2 - 1/(x - 1)$

D. $x^2 + 3x - 2 + 1/(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

792

QUESTION ID

S22-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 + 3x - 2 + 1/(x - 1)$

GRADING NOTE Synthetic division by 1: 1, 3, −2, remainder 1. The others divide by −1 instead of 1 ($x^2 + x - 6$, remainder 9), change the sign of the remainder, or change the divisor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

Which shows $(2x^3 - 3x^2 + 4x - 7)/(x^2 + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $2x - 3 + (2x + 4)/(x^2 + 1)$

B. $2x - 3 + (2x - 10)/(x^2 + 1)$

C. $2x - 3 + (2x - 4)/(x^2 + 1)$

D. $2x - 3 + (6x - 10)/(x^2 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

793

QUESTION ID

S22-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C.

$2x - 3 + (2x - 4)/(x^2 + 1)$

GRADING NOTE $(x^2 + 1)(2x - 3) = 2x^3 - 3x^2 + 2x - 3$; subtracting from the dividend leaves $2x - 4$ (degree less than 2). The others add instead of subtracting, or make a sign error in one term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

Which shows $(x^4 - 16)/(x - 2)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

B. $x^3 + 2x^2 + 4x + 8 - 32/(x - 2)$

C. $x^3 - 2x^2 + 4x - 8$ (remainder 0)

D. $x^3 + 8$ (remainder 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

794

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

GRADING NOTE Synthetic division by 2 with coefficients 1, 0, 0, 0, −16: 1, 2, 4, 8, remainder 0. The others divide by x + 2, leave out the zero placeholders, or miscompute the remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

Which shows $(6x^2 + x - 4)/(2x + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $6x - 2 - 3/(2x + 1)$

B. $3x - 1 - 4/(2x + 1)$

C. $3x - 1 + 3/(2x + 1)$

D. $3x - 1 - 3/(2x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

795

QUESTION ID

S22-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D.

$3x - 1 - 3/(2x + 1)$

GRADING NOTE $(2x + 1)(3x - 1) = 6x^2 + x - 1$, so the remainder is $-4 - (-1) = -3$. The others change the sign of the remainder, use synthetic division by $-1/2$ without dividing the quotient by 2, or forget to subtract the -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

Which shows $(x^3 - 4x + 5)/(x + 3)$ written in the form $q(x) + r(x)/b(x)$?

A. $x - 7 + 26/(x + 3)$

B. $x^2 - 3x + 5 - 10/(x + 3)$

C. $x^2 - 3x + 5 + 10/(x + 3)$

D. $x^2 + 3x + 5 + 20/(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

796

QUESTION ID

S22-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B.

$x^2 - 3x + 5 - 10/(x + 3)$

GRADING NOTE Synthetic division by -3 with coefficients 1, 0, -4, 5: 1, -3, 5, remainder -10. The others leave out the 0 for x^2, divide by 3 instead of -3, or change the sign of the remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 1

Write a division with a remainder as a mixed number • Grade 4 • Number & Operations–Fractions • 4.NF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Write 23/4 as a mixed number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

798

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5 3/4

GRADING NOTE Any equivalent form (5 + 3/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

Write 47/5 in the form q + r/5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

799

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9 + 2/5

GRADING NOTE Form required: q + r/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

17/3 = 5 + 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800

QUESTION ID

S22-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 17 = 3 · 5 + 2, so 17/3 = 5 + 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5

800

ANSWER KEY Reduced copy of the question page, with its answer

31/7 equals

A. 4 + 1/7

B. 4 + 3/7

C. 3 + 4/7

D. 4 + 3/31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

801

QUESTION ID

S22-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4 + 3/7

GRADING NOTE 31 = 7·4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

Write 100/9 in the form q + r/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

802

QUESTION ID

S22-B1-Q5

TYPE

Short answer

CORRECT ANSWER

11 + 1/9

GRADING NOTE Form required: q + r/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 2

Multiply to check a division • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Compute $(x - 1)(x^2 + 3x - 2) + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5

804

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^3 + 2x^2 - 5x + 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(x^2 + 1)(2x - 3) + (2x - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

805

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$2x^3 - 3x^2 + 4x - 7$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

True or false?

$(2x + 1)(3x - 1) - 3 = 6x^2 + x - 4.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

806

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $6x^2 + x - 1 - 3.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

$(x + 3)(x^2 - 3x + 5)$ equals

A. $x^3 - 4x + 15$

B. $x^3 + 15$

C. $x^3 - 4x + 5$

D. $x^3 + 6x^2 - 4x + 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5

807

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^3 - 4x + 15$

GRADING NOTE $x^3 - 3x^2 + 5x + 3x^2 - 9x + 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B2-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(x - 2)(x^3 + 2x^2 + 4x + 8)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5

808

QUESTION ID

S22-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$x^4 - 16$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 3

Use the Remainder Theorem to check the remainder • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

ANSWER KEY Reduced copy of the question page, with its answer

Find the remainder of $x^3 + 2x^2 - 5x + 3$ divided by $x - 1$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

810

QUESTION ID

S22-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

Find the remainder of $x^3 - 4x + 5$ divided by $x + 3$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5

811

QUESTION ID

S22-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5

True or false?

The remainder of $x^4 - 16$ divided by $x - 2$ is 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

812

QUESTION ID

S22-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^4 - 16 = 0$, so the remainder is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The remainder of $6x^2 + x - 4$ divided by $2x + 1$ is

A. 3

B. -4

C. 1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5

813

QUESTION ID

S22-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −3

GRADING NOTE Evaluate at $x = -1/2$: $1.5 - 0.5 - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the remainder of $2x^3 + x - 1$ divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

814

QUESTION ID

S22-B3-Q5

TYPE

Short answer

CORRECT ANSWER

17

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B3-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 1

Divide with unknown coefficients and higher-degree divisors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

When $x^3 + ax^2 + 2x + 4$ is divided by $x^2 - x + 1$, the remainder is a constant. Find a and the remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

816

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

a = −2; remainder 5

GRADING NOTE Form required: both. Quotient x + (a + 1); remainder (2 + a)x + (3 − a), constant when a = −2, giving 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

Rewrite $(x^4 + x^2 + 1)/(x^2 + x + 1)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

817

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - x + 1$ (remainder 0)

GRADING NOTE Form required: quotient and remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

True or false?

Dividing a degree-5 polynomial by a degree-2 polynomial gives a quotient of degree 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

818

QUESTION ID

S22-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient has degree $5 - 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The remainder of a division by $x^2 + 4$ can have degree at most

A. 0

B. 1

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

819

QUESTION ID

S22-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE Lower than the divisor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

Find the polynomial a(x) for which
$a(x)/(x - 2) = x^2 + 1 + 3/(x - 2)$.

a(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5

820

QUESTION ID
S22-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$x^3 - 2x^2 + x + 1$

GRADING NOTE Any equivalent form. $(x - 2)(x^2 + 1) + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 2

Use division to find slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Use division to find the slant asymptote of $f(x) = (x^2 + 3x + 1)/(x + 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5

822

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = x + 2$

GRADING NOTE Form required: an equation. $(x^2 + 3x + 1)/(x + 1) = x + 2 - 1/(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5

Find the slant asymptote of $f(x) = (2x^2 - x - 4)/(x - 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5

823

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x + 1

GRADING NOTE Form required: an equation. Remainder −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5

True or false?

A rational function with numerator degree exactly one more than the denominator degree has a slant asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

824

QUESTION ID
S22-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The quotient is linear (unless the division is exact, when the graph is the line with a hole).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

The slant asymptote of $y = (x^2 + 1)/x$ is

A. $y = x$

B. $y = 1$

C. $x = 0$

D. $y = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

825

QUESTION ID

S22-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. y = x

GRADING NOTE $x + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

Explain why the graph of $y = x + 2 - 1/(x + 1)$ gets close to $y = x + 2$ for large $|x|$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

826

QUESTION ID

S22-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1/(x + 1) approaches 0 as |x| grows, so y − (x + 2) approaches 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

Rewrite simple rational expressions by inspection

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Split a fraction with a sum in the numerator	**7.EE.A.1**
BACKWARD 2	Rewrite a numerator as a multiple of the denominator plus a cons…	**6.EE.A.3**
BACKWARD 3	Divide by long division when inspection fails	**HSA-APR.D.6**
FORWARD 1	Use the rewritten form to reason about values	**HSA-APR.D.6**
FORWARD 2	Graph rational functions as transformations of y = 1/x	**HSF-BF.B.3**

SET 23 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite (x + 5)/(x + 2) by inspection in the form q + r/(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

829

QUESTION ID
S23-M-Q1

TYPE
Short answer

CORRECT ANSWER
1 + 3/(x + 2)

GRADING NOTE Form required: q + r/(x + 2). x + 5 = (x + 2) + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

Rewrite (2x − 1)/(x − 3) by inspection in the form q + r/(x − 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

830

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 5/(x − 3)

GRADING NOTE Form required. 2x − 1 = 2(x − 3) + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $(x^2 + 1)/x$ by inspection in the form $q(x) + r/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

831

QUESTION ID

S23-M-Q3

TYPE

Short answer

CORRECT ANSWER

x + 1/x

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

Rewrite (3x + 7)/(x + 1) by inspection in the form q + r/(x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

832

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 + 4/(x + 1)

GRADING NOTE Form required. 3x + 7 = 3(x + 1) + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $x^2/(x - 1)$ by inspection in the form $q(x) + r/(x - 1)$. (Hint: $x^2 = (x^2 - 1) + 1$.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5

833

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x + 1 + 1/(x - 1)$

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 1

Split a fraction with a sum in the numerator • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write (x + 3)/x as a sum of two fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

835

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x/x + 3/x = 1 + 3/x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (2(x − 3) + 5)/(x − 3) as a sum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

836

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 5/(x − 3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

True or false?

(a + b)/c = a/c + b/c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

837

QUESTION ID

S23-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division distributes over a sum in the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 + 1)/x$ equals

A. $x + 1/x$

B. $1 + 1/x$

C. $x + 1$

D. $x^2 + 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

838

QUESTION ID

S23-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x + 1/x$

GRADING NOTE $x^2/x + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

True or false?

c/(a + b) = c/a + c/b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

839

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Splitting works only for the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 2

Rewrite a numerator as a multiple of the denominator plus a constant • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that x + 5 = (x + 2) + c.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

841

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $2x - 1 = 2(x - 3) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

842

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

True or false?

$3x + 7 = 3(x + 1) + 7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

843

QUESTION ID

S23-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3(x + 1) + 7 = 3x + 10$; the constant is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

$x^2 - 1$ equals

A. $x(x - 1)$

B. $(x + 1)^2$

C. $(x - 1)^2$

D. $(x - 1)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

844

QUESTION ID

S23-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $(x - 1)(x + 1)$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $4x + 1 = 4(x - 2) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

845

QUESTION ID

S23-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 3

Divide by long division when inspection fails • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Divide x^2 by $x - 1$ with long division. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

847

QUESTION ID

S23-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Quotient x + 1, remainder 1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-B3-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

Divide 3x + 7 by x + 1. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

848

QUESTION ID

S23-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Quotient 3, remainder 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

True or false?

Long division and inspection give the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

849

QUESTION ID

S23-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The form q + r/b is unique.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The remainder of $(2x - 1) \div (x - 3)$ is

A. -5

B. 2

C. -1

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

850

QUESTION ID

S23-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 5

GRADING NOTE $2 \cdot 3 - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

Divide x + 5 by x + 2. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5

851

QUESTION ID

S23-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Quotient 1, remainder 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-B3-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 1

Use the rewritten form to reason about values • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

For which integers x is (x + 5)/(x + 2) an integer? Use 1 + 3/(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5

853

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = −5, −3, −1, 1

GRADING NOTE Form required: all four. x + 2 must divide 3: ±1, ±3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why (2x − 1)/(x − 3) can never equal 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

854

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

It equals 2 + 5/(x − 3), and 5/(x − 3) is never 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

$(3x + 7)/(x + 1) > 3$ for every $x > -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

855

QUESTION ID

S23-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4/(x + 1) > 0$ when $x > -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

As x grows large, (x + 5)/(x + 2) approaches

A. 3

B. 0

C. 5/2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

856

QUESTION ID

S23-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE 3/(x + 2) → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For $x > 0$, what is the least value of $x + 1/x$, and where does it occur?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5

857

QUESTION ID

S23-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2, at $x = 1$

GRADING NOTE Form required: value and location. $x + 1/x - 2 = (x - 1)^2/x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 2

Graph rational functions as transformations of y = 1/x • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Describe how the graph of $y = 1 + 3/(x + 2)$ is obtained from $y = 3/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

859

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Shift left 2 and up 1.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the asymptotes of $y = 2 + 5/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5

860

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 and y = 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

y = (3x + 7)/(x + 1) has horizontal asymptote y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

861

QUESTION ID

S23-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (3x + 7)/(x + 1) = 3 + 4/(x + 1), so the asymptote is y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

$y = (x + 5)/(x + 2)$ is a translation of $y = 3/x$ by

A. left 2, up 1

B. right 2, up 1

C. left 2, up 5

D. left 5, up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5

862

QUESTION ID

S23-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. left 2, up 1

GRADING NOTE $1 + 3/(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5

Write (4x − 3)/(x − 1) as a transformation of k/x and give its asymptotes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5

863

QUESTION ID

S23-F2-Q5

TYPE

Short answer

CORRECT ANSWER

4 + 1/(x − 1); x = 1 and y = 4

GRADING NOTE Form required: form and both asymptotes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5

Multiply and divide rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Multiply and divide fractions	**6.NS.A.1**
BACKWARD 2	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 3	Division by zero is undefined	**7.NS.A.2.b**
FORWARD 1	Simplify complex fractions	**HSA-APR.D.7**
FORWARD 2	Explain closure of rational expressions	**HSA-APR.D.7**

SET 24 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x^2 - 9)/(x^2 + x - 6) \cdot (x - 2)/(x - 3)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

866

QUESTION ID
S24-M-Q1

TYPE
Short answer

CORRECT ANSWER
1; $x \neq -3, 2, 3$

GRADING NOTE Form required: simplified result and all exclusions from the original denominators. $(x - 3)(x + 3)/((x + 3)(x - 2)) \cdot (x - 2)/(x - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

Simplify $(x + 2)/(3x^2) \div (x^2 - 4)/x$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

867

QUESTION ID
S24-M-Q2

TYPE
Short answer

CORRECT ANSWER
$1/(3x(x - 2))$; $x \neq -2, 0, 2$

GRADING NOTE Form required: result and all exclusions, including the zeros of the divisor $x^2 - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

Simplify $(2x - 6)/(x + 1) \cdot (x + 1)/(x^2 - 9)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q3/5

868

QUESTION ID
S24-M-Q3

TYPE
Short answer

CORRECT ANSWER
$2/(x + 3)$; $x \neq -3, -1, 3$

GRADING NOTE Form required: result and all exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q3**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q3/5

Simplify (3x + 3)/(x − 4) ÷ (x + 1)/(x − 4). State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

869

QUESTION ID
S24-M-Q4

TYPE
Short answer

CORRECT ANSWER
3; x ≠ −1, 4

GRADING NOTE Form required: result and all exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

Simplify $(5x^2 + 5x)/(x^2 - 4) \div (x + 1)/(x - 2)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

870

QUESTION ID
S24-M-Q5

TYPE
Short answer

CORRECT ANSWER
$5x/(x + 2)$; $x \neq -2, -1, 2$

GRADING NOTE Form required: result and all exclusions. $5x(x + 1)/((x - 2)(x + 2)) \cdot (x - 2)/(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 1

Multiply and divide fractions • Grade 6 • The Number System • 6.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2/3 · 9/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

872

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5/6 ÷ 10/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

873

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

True or false?

$a/b \div c/d = a/b \cdot c/d$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

874

QUESTION ID

S24-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Dividing multiplies by the reciprocal d/c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

3/8 ÷ 3/4 equals

A. 9/32

B. 2

C. 3/4

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

875

QUESTION ID

S24-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/2

GRADING NOTE 3/8 · 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

Compute 14/15 · 5/7.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

876

QUESTION ID

S24-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 2

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

878

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x − 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

Factor $5x^2 + 5x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

879

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5x(x + 1)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

880

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x − 2)(x + 2)

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$2x - 6 = 2(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

881

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Common factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

882

QUESTION ID

S24-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 3

Division by zero is undefined • Grade 7 • The Number System • 7.NS.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

For which x is 1/(x − 4) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

884

QUESTION ID

S24-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $(x + 2)/(x^2 - 9)$ undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

885

QUESTION ID

S24-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3 or −3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

True or false?

After canceling $(x - 3)$, $x = 3$ is allowed in the original expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

886

QUESTION ID

S24-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The original expression is undefined at $x = 3$, so it stays excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Dividing by (x + 1)/(x − 2) is impossible when

A. x = −1 only

B. x = 0

C. x = −1 or x = 2

D. x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

887

QUESTION ID

S24-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x = −1 or x = 2

GRADING NOTE Divisor undefined or equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

Why is $x = 0$ excluded from $(x + 2)/(3x^2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

888

QUESTION ID

S24-B3-Q5

TYPE

Short answer

CORRECT ANSWER

The denominator $3x^2$ is 0 when $x = 0$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 1

Simplify complex fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (1/x + 1)/(1/x − 1). State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

890

QUESTION ID
S24-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(1 + x)/(1 − x); x ≠ 0, 1

GRADING NOTE Form required: result and exclusions. Multiply top and bottom by x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

Simplify $(1 - 4/x^2)/(1 + 2/x)$. State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

891

QUESTION ID
S24-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 2)/x;\ x \neq 0, -2$

GRADING NOTE Form required: result and exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

True or false?

$(a/b)/(c/d) = ad/(bc)$ when $b, c, d \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

892

QUESTION ID
S24-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Multiply by the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

(x/2)/(x/4) for x ≠ 0 equals

A. 2

B. 1/2

C. $x^2/8$

D. $8/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

893

QUESTION ID
S24-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 2

GRADING NOTE x/2 · 4/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

Simplify (x − 1/x)/(x + 1). State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

894

QUESTION ID
S24-F1-Q5

TYPE
Short answer

CORRECT ANSWER
(x − 1)/x; x ≠ 0, −1

GRADING NOTE Form required: result and exclusions. $(x^2 - 1)/(x(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q5**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 2

Explain closure of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Explain why the product of two rational expressions is a rational expression.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

896

QUESTION ID
S24-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(a/b)(c/d) = ac/(bd), and ac and bd are polynomials (bd not the zero polynomial).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

Explain why division by a nonzero rational expression gives a rational expression.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

897

QUESTION ID
S24-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(a/b) ÷ (c/d) = ad/(bc), a ratio of polynomials when c is not the zero polynomial.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

True or false?

Rational expressions are closed under division by the zero expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

898

QUESTION ID

S24-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

Which system is closed under addition, subtraction, multiplication, and division by nonzero elements?

A. Whole numbers

B. Rational expressions

C. Integers

D. Polynomials

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

899

QUESTION ID
S24-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Rational expressions

GRADING NOTE Like the rational numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q4**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

Give an example showing polynomials are not closed under division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

900

QUESTION ID
S24-F2-Q5

TYPE
Short answer

CORRECT ANSWER
x ÷ (x + 1) = x/(x + 1), which is not a polynomial

GRADING NOTE Any valid example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q5**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

Add and subtract rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Add fractions with unlike denominators	**5.NF.A.1**
BACKWARD 2	Find a least common denominator by factoring	**HSA-SSE.A.2**
BACKWARD 3	Subtract a whole numerator	**6.EE.A.3**
FORWARD 1	Decompose into partial fractions	**HSA-APR.D.7**
FORWARD 2	Solve rational equations	**HSA-REI.A.2**

SET 25 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x + 1)/(x^2 + x) + 1/x$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

903

QUESTION ID
S25-M-Q1

TYPE
Short answer

CORRECT ANSWER
$2/x$; $x \neq 0, -1$

GRADING NOTE Form required: result and all exclusions. $(x + 1)/(x(x + 1)) = 1/x$, but $x = -1$ stays excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

Write 2/(x + 1) + 3/(x − 2) as one rational expression.
State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

904

QUESTION ID
S25-M-Q2

TYPE
Short answer

CORRECT ANSWER
(5x − 1)/((x + 1)(x − 2)); x ≠ −1, 2

GRADING NOTE Form required: one fraction and exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify x/(x − 3) − 3/(x − 3). State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q3/5

905

QUESTION ID
S25-M-Q3

TYPE
Short answer

CORRECT ANSWER
1; x ≠ 3

GRADING NOTE Form required: result and exclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q3/5

Write $1/(x^2 - 1) - 1/(x - 1)$ as one rational expression. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

906

QUESTION ID
S25-M-Q4

TYPE
Short answer

CORRECT ANSWER
$-x/((x - 1)(x + 1))$; $x \neq -1, 1$

GRADING NOTE Form required: one fraction and exclusions. $(1 - (x + 1))/((x - 1)(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

Write $4/(x^2 - 4x) - 1/x$ as one rational expression. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

907

QUESTION ID
S25-M-Q5

TYPE
Short answer

CORRECT ANSWER
$(8 - x)/(x(x - 4))$; $x \neq 0, 4$

GRADING NOTE Form required: one fraction and exclusions. $(4 - (x - 4))/(x(x - 4))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q5**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 1

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2/3 + 3/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

909

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

17/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5/6 − 1/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

910

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

True or false?

1/2 + 1/3 = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

911

QUESTION ID

S25-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

1/4 + 1/6 equals

A. 5/12

B. 2/10

C. 2/12

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

912

QUESTION ID

S25-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5/12

GRADING NOTE 3/12 + 2/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 3/10 + 2/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

913

QUESTION ID

S25-B1-Q5

TYPE

Short answer

CORRECT ANSWER

13/30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 2

Find a least common denominator by factoring • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + x$ and $x^2 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

915

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x(x + 1); x(x − 4)

GRADING NOTE Form required: both factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

What is the least common denominator of $1/(x^2 - 1)$ and $1/(x - 1)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

916

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

The LCD of $\frac{1}{x}$ and $\frac{1}{x(x-4)}$ is

A. $x - 4$

B. $x^2(x - 4)$

C. $x(x - 4)$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

917

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x(x - 4)$

GRADING NOTE Use each factor to its highest power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

True or false?

The LCD of $\frac{2}{x+1}$ and $\frac{3}{x-2}$ is $x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The denominators have no common factor; the LCD is $(x + 1)(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5

What is the LCD of $1/(x^2 + x)$ and $1/x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

919

QUESTION ID

S25-B2-Q5

TYPE

Short answer

CORRECT ANSWER

x(x + 1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-B2-Q5**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 3

Subtract a whole numerator • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 4 − (x − 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

921

QUESTION ID

S25-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8 − x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 1 − (x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

922

QUESTION ID

S25-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$3 - (x - 3) = -x.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5

923

QUESTION ID

S25-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3 - x + 3 = 6 - x.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

2(x − 2) − 3(x + 1) equals

A. 5x − 7

B. −x + 1

C. −x − 7

D. −x − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

924

QUESTION ID

S25-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −x − 7

GRADING NOTE 2x − 4 − 3x − 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify x − (3 − x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

925

QUESTION ID

S25-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2x − 3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 1

Decompose into partial fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Find A and B so that
(5x + 1)/((x − 1)(x + 2)) = A/(x − 1) + B/(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

927

QUESTION ID
S25-F1-Q1

TYPE
Short answer

CORRECT ANSWER
A = 2, B = 3

GRADING NOTE Form required: both. 5x + 1 = A(x + 2) + B(x − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find A and B so that 1/(x(x + 1)) = A/x + B/(x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

928

QUESTION ID
S25-F1-Q2

TYPE
Short answer

CORRECT ANSWER
A = 1, B = −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

True or false?

$1/(x(x + 1)) = 1/x + 1/(x + 1)$ for $x \neq 0, -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

929

QUESTION ID
S25-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $1/x + 1/(x + 1) = (2x + 1)/(x(x + 1))$; the difference $1/x - 1/(x + 1)$ gives $1/(x(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

$4/(x^2 - 4)$ equals

A. $2/(x - 2) - 2/(x + 2)$

B. $1/(x - 2) + 1/(x + 2)$

C. $4/(x - 2) - 4/(x + 2)$

D. $1/(x - 2) - 1/(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

930

QUESTION ID
S25-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D.
$1/(x - 2) - 1/(x + 2)$

GRADING NOTE Combine to check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $1/(n(n+1)) = 1/n - 1/(n+1)$ to find $1/(1\cdot2) + 1/(2\cdot3) + 1/(3\cdot4) + 1/(4\cdot5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

931

QUESTION ID
S25-F1-Q5

TYPE
Short answer

CORRECT ANSWER
4/5

GRADING NOTE Any equivalent form. Telescoping: $1 - 1/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q5**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 2

Solve rational equations • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve 2/(x + 1) + 3/(x − 2) = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

933

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/5

GRADING NOTE Any equivalent form. 5x − 1 = 0; not an excluded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

Solve x/(x − 3) − 3/(x − 3) = 1. Explain your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

934

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Every real x except 3 (the left side simplifies to 1 for x ≠ 3)

GRADING NOTE Form required: the solution set with the exclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

True or false?

A solution of a rational equation that makes a denominator 0 must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

935

QUESTION ID

S25-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

The solution of 1/x + 1/(2x) = 3 is

A. x = 1/2

B. x = 2

C. x = 3/2

D. no solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

936

QUESTION ID

S25-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x = 1/2

GRADING NOTE 3/(2x) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

Solve $4/(x^2 - 4x) - 1/x = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

937

QUESTION ID

S25-F2-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. $(8 - x)/(x(x - 4)) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

Create and solve rational, radical, and exponential equations from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Express rates of work and unit rates	**6.RP.A.3.b**
BACKWARD 2	Write exponential growth models	**HSF-LE.A.2**
BACKWARD 3	Solve linear equations after clearing a denominator	**HSA-REI.B.3**
FORWARD 1	Create equations whose algebraic solutions must be checked aga.	**HSA-CED.A.1**
FORWARD 2	Solve exponential context equations with logarithms	**HSF-LE.A.4**

SET 26 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Ana can paint a room in 4 hours. Working together, Ana and Ben paint it in 2.4 hours. Write an equation for Ben's time b (in hours) working alone, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5

940

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/4 + 1/b = 1/2.4;
b = 6 hours

GRADING NOTE Form required: the equation and the solution with units. 1/b = 5/12 − 3/12 = 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5

A car's speed s (in mph) before braking is estimated from its skid length d (in feet) by $s = \sqrt{(22.5d)}$. Write and solve an equation for the skid length of a car going 45 mph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

941

QUESTION ID
S26-M-Q2

TYPE
Short answer

CORRECT ANSWER
$45 = \sqrt{(22.5d)}$;
d = 90 feet

GRADING NOTE Form required: the equation and the solution with units. 2025 = 22.5d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A town of 1,200 people doubles in population every 5 years. Write an equation for the time t (in years) when the population is 9,600, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

942

QUESTION ID

S26-M-Q3

TYPE

Short answer

CORRECT ANSWER

$1200 \cdot 2^{t/5} = 9600$; $t = 15$ years

GRADING NOTE Form required: the equation and the solution with units. $2^{t/5} = 8 = 2^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A printer costs $500 plus $4 per shirt. Write an equation for the number of shirts x that makes the average cost per shirt $9, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

943

QUESTION ID

S26-M-Q4

TYPE

Short answer

CORRECT ANSWER

(500 + 4x)/x = 9;
x = 100 shirts

GRADING NOTE Form required: the equation and the solution. 500 = 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

A tank holds 10 L of a 20% salt solution. Write an equation for the liters x of pure water to add to make an 8% solution, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

944

QUESTION ID

S26-M-Q5

TYPE

Short answer

CORRECT ANSWER

2/(10 + x) = 0.08;
x = 15 L

GRADING NOTE Form required: the equation and the solution with units. The salt stays 2 L; 10 + x = 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 1

Express rates of work and unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana paints a room in 4 hours. What fraction of the room does she paint per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

946

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

A pump fills a pool in 6 hours. What fraction does it fill in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

947

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

True or false?

If two people work together, their times to finish the job add.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

948

QUESTION ID

S26-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Their rates (jobs per hour) add; working together takes less time, not more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

A machine makes 300 parts in 5 hours. Its rate is

A. 1500 parts per hour

B. 300 parts per hour

C. 60 parts per hour

D. 5/300 parts per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

949

QUESTION ID

S26-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 60 parts per hour

GRADING NOTE 300/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

A job takes 2.4 hours together. What fraction is done per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

950

QUESTION ID

S26-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/2.4 = 5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 2

Write exponential growth models • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A population of 1,200 doubles every 5 years. Write the population after t years.

P(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

952

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$1200 \cdot 2^{t/5}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-B2-Q1**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A $500 deposit grows 3% per year. Write its value after t years.

V(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

953

QUESTION ID
S26-B2-Q2

TYPE
Short answer

CORRECT ANSWER
$500(1.03)^t$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

True or false?

A quantity that doubles every 5 years is multiplied by 2^5 after 20 years.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

954

QUESTION ID

S26-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20/5 = 4 doublings, a factor of $2^4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

A quantity triples every year starting at 7. After t years it is

A. $7 + 3t$

B. $21t$

C. $3 \cdot 7^t$

D. $7 \cdot 3^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

955

QUESTION ID

S26-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $7 \cdot 3^t$

GRADING NOTE Repeated multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

How many doublings happen in 15 years if the doubling time is 5 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

956

QUESTION ID
S26-B2-Q5

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 3

Solve linear equations after clearing a denominator • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 500 + 4x = 9x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

958

QUESTION ID

S26-B3-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2 = 0.08(10 + x).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

959

QUESTION ID

S26-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

True or false?

To solve (500 + 4x)/x = 9 we may multiply both sides by x when x ≠ 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q3/5

960

QUESTION ID

S26-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiplying by a nonzero quantity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

2025 = 22.5d gives d =

A. 2002.5

B. 45

C. 9

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

961

QUESTION ID

S26-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 90

GRADING NOTE 2025/22.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5x/12 = 1/2.4 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

962

QUESTION ID

S26-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 1

Create equations whose algebraic solutions must be checked against the context • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A boat travels 24 km downstream and 24 km back in 5 hours. The current is 2 km/h. Write an equation for the boat's still-water speed b and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5

964

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

24/(b + 2) + 24/(b − 2) = 5; b = 10 km/h (reject b = −0.4)

GRADING NOTE Form required: equation, solution, and rejection. $5b^2 - 48b - 20 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5

The length of a rectangle is 3 more than its width w, and its diagonal is 15. Write an equation and find w.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5

965

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$w^2 + (w + 3)^2 = 225$;
$w = 9$ (reject -12)

GRADING NOTE Form required: equation and solution with the rejection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5

True or false?

A negative solution of an equation for a time must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

966

QUESTION ID

S26-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Time in the context is nonnegative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

A model gives t = −3 or t = 4 for the time (in seconds) of an event. The answer is

A. neither

B. both

C. 4 seconds

D. −3 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

967

QUESTION ID

S26-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4 seconds

GRADING NOTE Reject the negative time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe fills a tank in x hours and a second pipe takes 2 hours longer. Together they take 2.4 hours. Write an equation and find x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

968

QUESTION ID

S26-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1/x + 1/(x + 2) = 1/2.4; x = 4 hours (reject −1.2)

GRADING NOTE Form required: equation and the solution. $2.4(2x + 2) = x(x + 2)$ gives $x^2 - 2.8x - 4.8 = 0$, so x = 4 or −1.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 2

Solve exponential context equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The town of 1,200 doubles every 5 years. Write and solve an equation for when the population reaches 5,000. Give t to the nearest tenth of a year. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

970

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$1200 \cdot 2^{t/5} = 5000$; $t = 5 \log_2(25/6) \approx 10.3$ years

GRADING NOTE Form required: the equation and t (10.29 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

Solve $500(1.04)^t = 800$ for t. Give an exact answer and a decimal to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

971

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

t = log(1.6)/log(1.04) ≈ 11.98

GRADING NOTE Form required: exact form and decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

True or false?

$2^{t/5} = 8$ can be solved without logarithms, but $2^{t/5} = 5$ cannot be solved exactly without them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5

972

QUESTION ID

S26-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 8 is a power of 2; 5 is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5

The exact solution of $3^x = 20$ is

A. 20/3

B. log(20/3)

C. log 20/log 3

D. 3 log 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

973

QUESTION ID

S26-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. log 20/log 3

GRADING NOTE Take logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

Explain why log(1.6)/log(1.04) is the time for an amount growing 4% per year to grow by 60%.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

QUESTION ID

S26-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$(1.04)^t = 1.6$ is solved by t = log 1.6/log 1.04.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Create and solve nonlinear inequalities from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Sign of a product on intervals	**7.NS.A.2**
BACKWARD 3	Reverse the inequality when multiplying by a negative	**HSA-REI.B.3**
FORWARD 1	Solve rational inequalities with sign charts and excluded values	**HSA-CED.A.1**
FORWARD 2	Write systems of constraints and judge viability	**HSA-CED.A.3**

SET 27 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 64t$ feet after t seconds. Write and solve an inequality for the times when the ball is at least 48 feet high.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

977

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-16t^2 + 64t \geq 48$;
$1 \leq t \leq 3$ seconds

GRADING NOTE Form required: inequality and solution interval. $t^2 - 4t + 3 \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

A company sells x items at a price of 100 − 2x dollars each. Write and solve an inequality for the values of x that give revenue of at least $1,200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

978

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x(100 - 2x) \geq 1200$; $20 \leq x \leq 30$

GRADING NOTE Form required: inequality and solution. $(x - 20)(x - 30) \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

The average cost of making x posters is (300 + 2x)/x dollars. Write and solve an inequality for the numbers of posters (x > 0) with an average cost of at most $5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

979

QUESTION ID
S27-M-Q3

TYPE
Short answer

CORRECT ANSWER
(300 + 2x)/x ≤ 5; x ≥ 100

GRADING NOTE Form required: inequality and solution. With x > 0, 300 + 2x ≤ 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Write and solve an inequality for the times when C is at least 4 mg/L.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5

980

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

$20t/(t^2 + 4) \geq 4$;

$1 \leq t \leq 4$ hours

GRADING NOTE Form required: inequality and interval. $t^2 + 4 > 0$, so $20t \geq 4t^2 + 16$, $t^2 - 5t + 4 \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5

$500 is invested at 4% interest compounded yearly. Write an inequality for the number of whole years t until the value is at least $800, and solve it. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

981

QUESTION ID

S27-M-Q5

TYPE

Short answer

CORRECT ANSWER

$500(1.04)^t \geq 800$; $t \geq 11.98$, so 12 years

GRADING NOTE Form required: the inequality and the whole-year answer. Tables or logarithms both accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $t^2 - 4t + 3 = 0$.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

983

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 or 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 50x + 600 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

984

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20 or 30

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solutions of $t^2 - 5t + 4 = 0$ are -1 and -4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

985

QUESTION ID

S27-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(t - 1)(t - 4) = 0$ gives $t = 1$ or $t = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

The solutions of $x^2 - x - 12 = 0$ are

A. 4 and −3

B. 6 and −2

C. 12 and −1

D. −4 and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q4/5

986

QUESTION ID

S27-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 and −3

GRADING NOTE $(x - 4)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2x^2 - 100x + 1200 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

987

QUESTION ID

S27-B1-Q5

TYPE

Short answer

CORRECT ANSWER

20 or 30

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 2

Sign of a product on intervals • Grade 7 • The Number System • 7.NS.A.2

Is (t − 1)(t − 3) positive or negative when t = 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

989

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Negative:

(1)(−1) = −1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $(t - 1)(t - 3)$ positive or negative when $t = 5$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

990

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Positive: (4)(2) = 8

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

True or false?

A product of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

991

QUESTION ID

S27-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 20)(x - 30) \leq 0$ when

A. $x \leq 20$ or $x \geq 30$

B. $x \leq 20$

C. $20 \leq x \leq 30$

D. $x \geq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

992

QUESTION ID

S27-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $20 \leq x \leq 30$

GRADING NOTE Factors have opposite signs between the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For which t is $(t - 1)(t - 4) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

993

QUESTION ID

S27-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$1 < t < 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 3

Reverse the inequality when multiplying by a negative • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-16t^2 + 64t - 48 \geq 0$ after dividing both sides by -16.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

995

QUESTION ID

S27-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$t^2 - 4t + 3 \leq 0$

GRADING NOTE Form required: the reversed inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-3x \geq 12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

996

QUESTION ID

S27-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x \leq -4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying both sides of an inequality by x is always allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

997

QUESTION ID
S27-B3-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only when the sign of x is known.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $x > 0$, then $(300 + 2x)/x \leq 5$ is equivalent to

A. $300 + 2x \leq 5x$

B. $300 + 2x \geq 5x$

C. $2x \leq 5$

D. $300 \leq 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q4/5

998

QUESTION ID

S27-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 300 + 2x ≤ 5x

GRADING NOTE Multiplying by positive x keeps the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2 - x < 7$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

999

QUESTION ID

S27-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x > -5$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 1

Solve rational inequalities with sign charts and excluded values • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 3)/(x + 1) \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5

1001

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$-1 < x \leq 3$

GRADING NOTE Form required: the interval; −1 is excluded and 3 is included.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x + 2)/(x - 4) > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5

1002

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x < -2$ or $x > 4$

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying $(x - 3)/(x + 1) \leq 0$ by $x + 1$ gives $x - 3 \leq 0$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

1003

QUESTION ID
S27-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The direction depends on the sign of $x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

The solution set of $1/x > 2$ is

A. $x > 1/2$

B. $x < 1/2$

C. $x > 0$

D. $0 < x < 1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

1004

QUESTION ID

S27-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $0 < x < 1/2$

GRADING NOTE For $x > 0$, $1 > 2x$; for $x < 0$ the left side is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

A model's average cost is (400 + 3x)/x dollars. For how many items x > 0 is the average cost under $7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5

1005

QUESTION ID
S27-F1-Q5

TYPE
Short answer

CORRECT ANSWER
x > 100

GRADING NOTE Any equivalent form. 400 + 3x < 7x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 2

Write systems of constraints and judge viability • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

The ball's height must be at least 48 ft and the time at most 2.5 s. Write the constraints and find the viable times.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5

1007

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-16t^2 + 64t \geq 48$ and $0 \leq t \leq 2.5$; viable times $1 \leq t \leq 2.5$

GRADING NOTE Form required: both constraints and the combined interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5

Revenue $x(100 - 2x)$ must be at least \$1,200 and at most 24 items can be made. Which x are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5

1008

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$20 \le x \le 24$ (whole numbers)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5

True or false?

A value that solves the inequality may still not be viable in the context.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

1009

QUESTION ID
S27-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE For example a non-whole number of items.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

Revenue $x(100 - 2x) \geq 1200$ with x a whole number is satisfied by

A. x = 19

B. x = 25

C. x = 31

D. x = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

1010

QUESTION ID

S27-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = 25

GRADING NOTE $20 \leq x \leq 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

Explain why 12 years, not 11.98 years, answers the compound-interest question.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

1011

QUESTION ID
S27-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Interest is added once a year, so the value only reaches $800 at the end of a whole year.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

Create and graph nonlinear two-variable models

MAIN	Algebra II main question	**HSA-CED.A.2**
BACKWARD 1	Make a table of values from an equation	**6.EE.C.9**
BACKWARD 2	Choose scales and label axes	**HSN-Q.A.1**
BACKWARD 3	Write a growth or decay model from a percent	**HSF-LE.A.2**
FORWARD 1	Create equations in more than two variables and graph a member	**HSA-CED.A.2**
FORWARD 2	Interpret key features of the created models	**HSF-IF.B.4**

SET 28 • HSA-CED.A.2

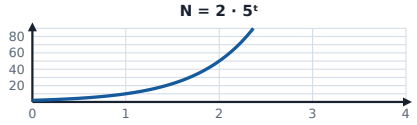
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

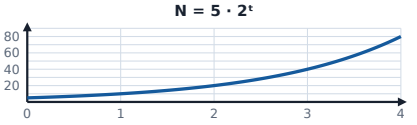
ANSWER KEY Reduced copy of the question page, with its answer

A rabbit population starts at 5 and doubles every year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 4$ (t in years, N rabbits)?

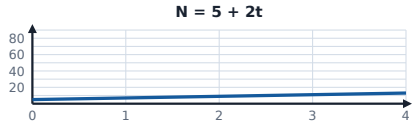
A



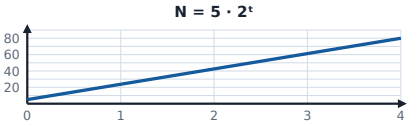
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

1014

QUESTION ID

S28-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Doubling every year from 5 gives $N = 5 \cdot 2^t$, an exponential curve through (0, 5), (1, 10), (2, 20), (3, 40), (4, 80). The others use the wrong model ($2 \cdot 5^t$ or $5 + 2t$) or pair the right equation with a straight line, which is not its graph.

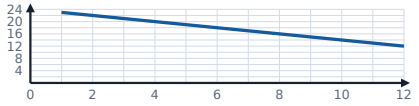
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

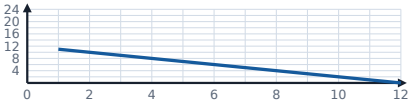
ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has area 24 square units. Which graph shows a correct equation for the situation with a correct graph of that equation for $1 \le L \le 12$ (length L, width w)?

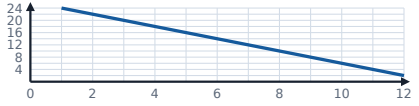
A



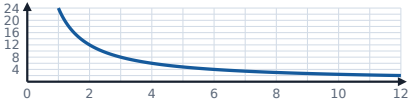
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

1015

QUESTION ID

S28-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

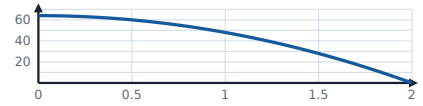
GRADING NOTE Area = Lw = 24 gives w = 24/L, a decreasing curve that flattens and approaches the L-axis, through (1, 24), (2, 12), (4, 6), (12, 2). The others use w = 24 − L, w = 12 − L (a perimeter idea), or draw a straight line for 24/L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

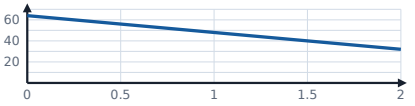
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

A stone is dropped from 64 feet. Its height after t seconds is 64 minus 16 times the square of t. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 2$ (h in feet)?

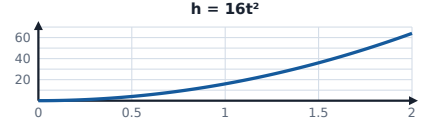
A



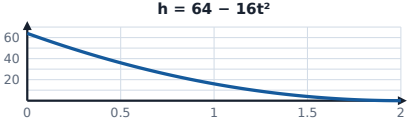
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

1016

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $h = 64 - 16t^2$: the height falls slowly at first and then faster, through (0, 64), (0.5, 60), (1, 48), (1.5, 28), (2, 0). The others use $64 - 16t$ (not squared), $16t^2$ (the distance fallen), or draw a curve that does not match $64 - 16t^2$ (it gives 16 at t = 1 instead of 48).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

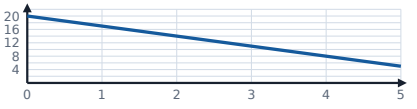
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

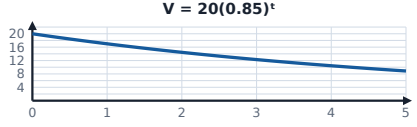
A $20,000 car loses 15% of its value each year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 5$ (V in thousands of dollars, t in years)?

A

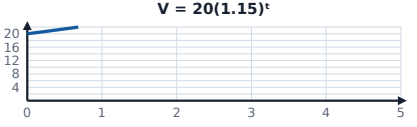
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

1017

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Keeping 85% each year gives $V = 20(0.85)^t$, through (1, 17), (2, 14.45), (5, 8.87). The others keep only 15% each year, subtract a fixed $3,000 each year (15% of the first value only), or grow 15% each year.

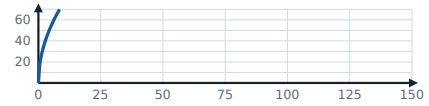
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

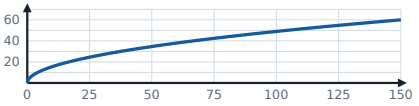
ANSWER KEY Reduced copy of the question page, with its answer

A car's speed s (mph) is the square root of 24 times its skid length d (ft). Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le d \le 150$?

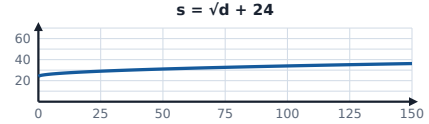
A



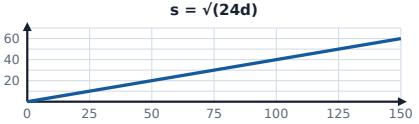
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

1018

QUESTION ID

S28-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE s = √(24d): an increasing curve that flattens, through (6, 12), (24, 24), (96, 48), (150, 60). The others use 24√d or √d + 24, or draw a straight line for √(24d) (it gives 24 at d = 60, not 37.9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 1

Make a table of values from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

For $N = 200 \cdot 2^t$, find N when t = 0, 1, 2, 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

1020

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

200, 400, 800, 1600

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For w = 24/L, find w when L = 2, 3, 4, 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

1021

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12, 8, 6, 4

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

True or false?

For $h = 64 - 16t^2$, $h = 48$ when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

1022

QUESTION ID

S28-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $h(2) = 64 - 64 = 0$; $h = 48$ when $t = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For s = √(24d), s when d = 96 is

A. 96

B. 48

C. 2304

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

1023

QUESTION ID

S28-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 48

GRADING NOTE √2304.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For $V = 20000(0.85)^t$, find V when t = 1 and t = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

1024

QUESTION ID

S28-B1-Q5

TYPE

Short answer

CORRECT ANSWER

17000 and 14450

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 2

Choose scales and label axes • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Values of N go from 200 to 3,200. Give a sensible step for the vertical axis on a grid with about 7 lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

1026

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

500

GRADING NOTE Any sensible step (for example 500).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which quantity goes on the horizontal axis when V depends on t?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

1027

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

t (time in years)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

True or false?

Axis labels should include units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

1028

QUESTION ID

S28-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Units make the graph readable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The best vertical scale for values from 0 to 64 is

A. steps of 0.5

B. steps of 1 from 0 to 10

C. steps of 100 from 0 to 1000

D. steps of 10 from 0 to 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

1029

QUESTION ID
S28-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. steps of 10 from 0 to 70

GRADING NOTE Fits the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the horizontal range for $0 \le d \le 150$ with a step of 10. How many steps is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

1030

QUESTION ID

S28-B2-Q5

TYPE

Short answer

CORRECT ANSWER

15 steps

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 3

Write a growth or decay model from a percent • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the value of $20,000 after losing 15% per year for t years.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

1032

QUESTION ID

S28-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$20000(0.85)^t$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the growth factor for 4% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5

1033

QUESTION ID

S28-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1.04

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B3-Q2**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5

True or false?

Losing 15% each year means multiplying by 0.15 each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

1034

QUESTION ID

S28-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply by $1 - 0.15 = 0.85$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A quantity doubles every hour starting at 200. After t hours it is

A. $2 \cdot 200^t$

B. $200 \cdot 2^t$

C. 200 + 2t

D. 400t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

1035

QUESTION ID
S28-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $200 \cdot 2^t$

GRADING NOTE Doubling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the decay factor for 30% decay?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5

1036

QUESTION ID

S28-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B3-Q5**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 1

Create equations in more than two variables and graph a member of the family • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

The current I (amps) through a resistor is the voltage V divided by the resistance R. Write the equation, and write I as a function of R when V = 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

1038

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

I = V/R; I = 12/R

GRADING NOTE Form required: both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The volume of a cylinder is $V = \pi r^2 h$. Write h in terms of r when $V = 100\pi$, and describe its graph for $r > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5

1039

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$h = 100/r^2$; decreasing, approaching 0 as r grows

GRADING NOTE Form required: the equation and a description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5

True or false?

Fixing one variable of a three-variable equation gives a two-variable equation that can be graphed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

1040

QUESTION ID
S28-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A family of curves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $A = P(1.05)^t$ with $P = 1000$, the graph of A against t is

A. a line

B. an increasing exponential curve

C. a parabola

D. a decreasing curve

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

1041

QUESTION ID
S28-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. an increasing exponential curve

GRADING NOTE Growth factor 1.05 > 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For the area formula $A = \pi r^2$, which graph (A against r) is correct for $r \geq 0$: a line, a half parabola, or an exponential?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

1042

QUESTION ID

S28-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Half of a parabola (from the origin, increasing)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 2

Interpret key features of the created models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

For $N = 200 \cdot 2^t$, what does the N-intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

1044

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

200 bacteria at the start (t = 0).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For $h = 64 - 16t^2$, when does the stone hit the ground, and what is the t-intercept's meaning?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

1045

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

t = 2 s; the time the height is 0

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

True or false?

The graph of $w = 24/L$ never reaches the L-axis because a rectangle of area 24 cannot have width 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

1046

QUESTION ID
S28-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $w > 0$ for every $L > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $V = 20000(0.85)^t$, the graph

A. decreases and approaches 0

B. increases

C. is a line

D. decreases to 0 at t = 6.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5

1047

QUESTION ID
S28-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. decreases and approaches 0

GRADING NOTE Decay.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe the rate of change of $s = \sqrt{(24d)}$ as d grows.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

1048

QUESTION ID
S28-F2-Q5

TYPE
Short answer

CORRECT ANSWER
It stays positive but decreases (the graph flattens).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

Represent nonlinear constraints and judge viable options

MAIN	Algebra II main question	**HSA-CED.A.3**
BACKWARD 1	Represent constraints with a linear system	**HSA-CED.A.3**
BACKWARD 2	Test whether a point satisfies an inequality	**6.EE.B.5**
BACKWARD 3	Solve quadratic inequalities	**HSA-CED.A.1**
FORWARD 1	Optimize within nonlinear constraints	**HSA-CED.A.3**
FORWARD 2	Locate constraint boundaries as graph intersections	**HSA-REI.D.11**

SET 29 • HSA-CED.A.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A garden has length L and width W (feet). The fencing for L + W is 30 ft and the area must be at least 200 ft². Write the constraints, find the viable widths, and decide whether W = 8 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5

1051

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

L + W = 30 and LW ≥ 200; 10 ≤ W ≤ 20; W = 8 is not viable

GRADING NOTE Form required: constraints, the interval, and the decision. W(30 − W) ≥ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5

A box has a square base of side x and a volume of 32 in^3; both x and the height h must be at most 4 in. Write the constraints, find the viable x, and decide whether x = 3 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

1052

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2h = 32$, $x \le 4$, $h \le 4$; $2\sqrt{2} \le x \le 4$ (about 2.83 to 4); x = 3 is viable ($h \approx 3.56$)

GRADING NOTE Form required: constraints, the range, and the decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A plan invests $1,000 at 5% compounded yearly and needs at least $1,500 within 10 years. Write the constraint and decide whether the plan is viable. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

1053

QUESTION ID
S29-M-Q3

TYPE
Short answer

CORRECT ANSWER

$1000(1.05)^t \geq 1500$ for some $t \leq 10$; viable: $1000(1.05)^{10} \approx 1628.89$

GRADING NOTE Form required: the constraint and the decision with evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Attendance at a show is 500 − 10p people when tickets cost p dollars. Revenue must be at least $6,000. Write the constraints, find the viable prices, and decide whether $35 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

1054

QUESTION ID
S29-M-Q4

TYPE
Short answer

CORRECT ANSWER
p(500 − 10p) ≥ 6000, p ≥ 0; 20 ≤ p ≤ 30; $35 is not viable

GRADING NOTE Form required: constraints, interval, decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular pen with sides x and y (m) needs area at least 50 m² and perimeter at most 40 m. Write the system and decide whether (5, 10) and (4, 12) are viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

1055

QUESTION ID
S29-M-Q5

TYPE
Short answer

CORRECT ANSWER
xy ≥ 50 and 2x + 2y ≤ 40; (5, 10) is viable (area 50, perimeter 30); (4, 12) is not (area 48)

GRADING NOTE Form required: the system and both decisions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 1

Represent constraints with a linear system • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write a constraint for "the total of L and W is 30".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

1057

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

L + W = 30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a constraint for "at most 40 m of fence for a rectangle with sides x and y".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

1058

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2x + 2y ≤ 40

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At least 200" is written ≤ 200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

1059

QUESTION ID

S29-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At least 200 means 200 or more: ≥ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

1059

ANSWER KEY Reduced copy of the question page, with its answer

"The height h must be at most 4" is

A. $h \leq 4$

B. $h \geq 4$

C. $h < 4$

D. $h = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

1060

QUESTION ID

S29-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $h \leq 4$

GRADING NOTE At most: less than or equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (12, 20) a solution of $x + y \leq 30$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

1061

QUESTION ID

S29-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No: 32 > 30

GRADING NOTE Form required: the decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 2

Test whether a point satisfies an inequality • Grade 6 • Expressions & Equations • 6.EE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Does W = 8 satisfy W(30 − W) ≥ 200?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

1063

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

No: 8 · 22 = 176

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does p = 25 satisfy p(500 − 10p) ≥ 6000?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

1064

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 25 · 250 = 6250

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(4, 12) satisfies $xy \geq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

1065

QUESTION ID

S29-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $48 < 50$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which x satisfies $x^2 \geq 8$?

A. 3

B. 2

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

1066

QUESTION ID

S29-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE $9 \geq 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Does t = 10 satisfy $1000(1.05)^t \geq 1500$? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

1067

QUESTION ID

S29-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: about 1628.89

GRADING NOTE Form required: decision and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 3

Solve quadratic inequalities • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve $W^2 - 30W + 200 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

1069

QUESTION ID

S29-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$10 \leq W \leq 20$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $p^2 - 50p + 600 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

1070

QUESTION ID

S29-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$20 \leq p \leq 30$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 \geq 8$ with $x > 0$ means $x \geq 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

1071

QUESTION ID

S29-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x \geq \sqrt{8} = 2\sqrt{2} \approx 2.83$; for example $x = 3$ works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

The solution of $(x - 2)(x - 6) \leq 0$ is

A. $x = 2$ or $x = 6$

B. $2 \leq x \leq 6$

C. $x \leq 2$ or $x \geq 6$

D. $x \geq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

1072

QUESTION ID

S29-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2 \leq x \leq 6$

GRADING NOTE Between the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $32/x^2 \leq 4$ for $x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

1073

QUESTION ID

S29-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 2\sqrt{2}$

GRADING NOTE Any equivalent form ($x \geq 2.83$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 1

Optimize within nonlinear constraints • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A rectangular garden has length L and width W (feet) with L + W = 30. Which width gives the greatest area, and what is that area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

1075

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

W = 15; 225 ft²

GRADING NOTE Form required: both. A = W(30 − W) has vertex at W = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which ticket price p gives the greatest revenue p(500 − 10p), and what is that revenue?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5

1076

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

p = $25; $6,250

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5

True or false?

A maximum found from the model must still satisfy every other constraint to be a viable choice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

1077

QUESTION ID
S29-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Check all constraints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

For the pen with $xy \geq 50$ and $2x + 2y \leq 40$, which is viable?

A. (10, 10)

B. (15, 10)

C. (2, 20)

D. (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

1078

QUESTION ID

S29-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (10, 10)

GRADING NOTE (10, 10): area 100, perimeter 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

Among whole-number widths, list all viable garden widths when the area must be at least 216 ft².

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5

1079

QUESTION ID

S29-F1-Q5

TYPE

Short answer

CORRECT ANSWER

12, 13, 14, 15, 16, 17, 18

GRADING NOTE Form required: all. $W^2 - 30W + 216 \le 0$ gives $12 \le W \le 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 2

Locate constraint boundaries as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Explain how the viable widths $10 \le W \le 20$ can be read from the graphs of $y = W(30 - W)$ and $y = 200$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

1081

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The graphs intersect at W = 10 and W = 20; the parabola is above the line between them.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where do y = p(500 − 10p) and y = 6000 intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5

1082

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

p = 20 and p = 30

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5

True or false?

The x-coordinates where $y = 32/x^2$ meets $y = 4$ solve $32/x^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

1083

QUESTION ID
S29-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Intersection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 1000(1.05)^t$ and $y = 1500$ meet at $t \approx$

A. 15

B. 8.3

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

1084

QUESTION ID
S29-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 8.3

GRADING NOTE log 1.5/log 1.05 ≈ 8.31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

Explain why the region above y = 200 and below the parabola shows the viable widths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

1085

QUESTION ID

S29-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Those W give area W(30 − W) greater than or equal to 200.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

Rearrange rational, radical, and exponential formulas

MAIN	Algebra II main question	**HSA-CED.A.4**
BACKWARD 1	Rearrange linear formulas	**HSA-CED.A.4**
BACKWARD 2	Undo squares and square roots	**8.EE.A.2**
BACKWARD 3	Clear fractions in an equation	**HSA-REI.B.3**
FORWARD 1	Rearrange formulas where conditions decide the result	**HSA-CED.A.4**
FORWARD 2	Rearrange exponential formulas with logarithms	**HSF-LE.A.4**

SET 30 • HSA-CED.A.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The lens formula is 1/f = 1/a + 1/b. Solve for b and state the condition needed.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

1088

QUESTION ID
S30-M-Q1

TYPE
Short answer

CORRECT ANSWER
af/(a − f), a ≠ f

GRADING NOTE Form required: the formula and the condition. 1/b = (a − f)/(af).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pendulum's period is $T = 2\pi\sqrt{(L/g)}$. Solve for L.

L = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5

1089

QUESTION ID

S30-M-Q2

TYPE

Short answer

CORRECT ANSWER

$gT^2/(4\pi^2)$

GRADING NOTE Any equivalent form. Square both sides of $T/(2\pi) = \sqrt{(L/g)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An investment grows by $A = P(1 + r)^t$. Solve for r.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

1090

QUESTION ID

S30-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(A/P)^{1/t} - 1$

GRADING NOTE Any equivalent form (the t-th root of A/P minus 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The speed of a falling object is $v = \sqrt{(2gh)}$. Solve for h and state the condition on v.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

1091

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

$v^2/(2g)$, with $v \geq 0$

GRADING NOTE Form required: formula and condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In a circuit, I = V/(R + r). Solve for R and state the condition needed.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5

1092

QUESTION ID

S30-M-Q5

TYPE

Short answer

CORRECT ANSWER

V/I − r, I ≠ 0

GRADING NOTE Form required: formula and condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 1

Rearrange linear formulas • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2L + 2W for W.

W = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

1094

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(P − 2L)/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve d = rt for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

1095

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

d/r

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving V = IR for R gives R = VI.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

1096

QUESTION ID

S30-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide both sides by I: R = V/I.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

Solving y = mx + b for x gives

A. (y − b)/m

B. y/m − b

C. (y + b)/m

D. m(y − b)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

1097

QUESTION ID

S30-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (y − b)/m

GRADING NOTE Subtract b, divide by m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve A = (1/2)bh for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

1098

QUESTION ID

S30-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2A/b

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 2

Undo squares and square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 49$ for $x > 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

1100

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{y} = 6$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

1101

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If √(L/g) = k, then L/g = √k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5

1102

QUESTION ID

S30-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Square both sides: L/g = k².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(\sqrt{(2gh)})^2$ equals

A. $2g\sqrt{h}$

B. $2gh$

C. $\sqrt{(2gh)}$

D. $4g^2h^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

1103

QUESTION ID

S30-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2gh

GRADING NOTE Squaring undoes the root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve T/(2π) = √m for m.

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

1104

QUESTION ID

S30-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$T^2/(4\pi^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 3

Clear fractions in an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 1/2 + 1/x = 1 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

1106

QUESTION ID

S30-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Combine 1/a − 1/f into one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

1107

QUESTION ID

S30-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(f − a)/(af)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $1/b = (a - f)/(af)$, then $b = af/(a - f)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

1108

QUESTION ID

S30-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving 3 = 12/(R + 2) for R gives

A. 36

B. 4

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

1109

QUESTION ID

S30-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2

GRADING NOTE R + 2 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve I = V/(R + r) for R + r.

R + r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

1110

QUESTION ID

S30-B3-Q5

TYPE

Short answer

CORRECT ANSWER

V/I

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B3-Q5**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 1

Rearrange formulas where conditions decide the result • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = \pi r^2$ for r, given r is a length.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5

1112

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(A/\pi)}$

GRADING NOTE Form required: the positive root only ($r \geq 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = (x + 1)/(x - 2)$ for x and state the condition on y.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5

1113

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(2y + 1)/(y - 1)$, $y \neq 1$

GRADING NOTE Form required: formula and condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5

True or false?

Solving $s = \sqrt{(24d)}$ for d gives $d = s^2/24$ for every real s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

1114

QUESTION ID
S30-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only $s \geq 0$ comes from the original formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

Solving $E = (1/2)mv^2$ for v ($v \geq 0$) gives

A. $\sqrt{(2E/m)}$

B. $(2E/m)^2$

C. $\sqrt{(E/2m)}$

D. $2E/m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5

1115

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $\sqrt{(2E/m)}$

GRADING NOTE Multiply by 2/m and take the positive root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5

Solve $P = k/V^2$ for V ($V > 0$).

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5

1116

QUESTION ID

S30-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(k/P)}$

GRADING NOTE Form required: positive root with $k/P > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 2

Rearrange exponential formulas with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = Pe^{rt}$ for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

1118

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

ln(A/P)/r

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = P(1 + r)^t$ for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

1119

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

log(A/P)/log(1 + r)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

True or false?

Solving $N = N_0 \cdot 2^{t/d}$ for t gives $t = d \cdot \log_2(N/N_0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q3/5

1120

QUESTION ID

S30-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take log base 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving $y = 5 \cdot 10^x$ for x gives

A. y/50

B. log(y/5)

C. $10^{y/5}$

D. log y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

1121

QUESTION ID

S30-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. log(y/5)

GRADING NOTE $10^x = y/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use t = log(A/P)/log(1 + r) to find t for A = 2000, P = 1000, r = 0.05. A calculator may be used.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

1122

QUESTION ID

S30-F2-Q5

TYPE

Short answer

CORRECT ANSWER

14.21

GRADING NOTE Any value 14.2 to 14.21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

Solve rational equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Solve proportions	**7.RP.A.2**
BACKWARD 2	Find a common denominator of rational expressions	**HSA-APR.D.7**
BACKWARD 3	Values that make a denominator zero	**7.NS.A.2.b**
FORWARD 1	Explain and construct extraneous solutions	**HSA-REI.A.2**
FORWARD 2	See rational equations as graph intersections	**HSA-REI.D.11**

SET 31 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3/(x − 1) = 6/(x + 2). Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

1125

QUESTION ID
S31-M-Q1

TYPE
Short answer

CORRECT ANSWER
x = 4; none extraneous

GRADING NOTE Form required: solution and the check. 3x + 6 = 6x − 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

Solve x/(x − 2) + 1 = 2/(x − 2). Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

1126

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

No solution; x = 2 is extraneous

GRADING NOTE Form required: the conclusion and the extraneous value. 2x − 2 = 2 gives x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-M-Q2**

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 1/x + 1/(x + 2) = 4/3. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

1127

QUESTION ID

S31-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = 1 or x = −3/2; none extraneous

GRADING NOTE Form required: both solutions. $2x^2 + x - 3 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2/(x - 3) = 9/(x - 3)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

1128

QUESTION ID

S31-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = −3; x = 3 is extraneous

GRADING NOTE Form required: solution and extraneous value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

Solve $2/(x + 1) - 1/(x - 1) = 2/(x^2 - 1)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

1129

QUESTION ID

S31-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = 5; none extraneous

GRADING NOTE Form required: solution and check. $2(x - 1) - (x + 1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 1

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3/x = 6/10.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

1131

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 4/7 = x/21.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

1132

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

True or false?

$a/b = c/d$ implies $ac = bd$ when $b, d \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q3/5

1133

QUESTION ID

S31-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Cross-multiplying gives $ad = bc$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

2/5 = 6/x gives x =

A. 12

B. 2.4

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

1134

QUESTION ID

S31-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

GRADING NOTE 2x = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3/(x - 1) = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

1135

QUESTION ID

S31-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 2

Find a common denominator of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

What is the LCD of 1/x and 1/(x + 2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

1137

QUESTION ID
S31-B2-Q1

TYPE
Short answer

CORRECT ANSWER
x(x + 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

What is the LCD of $2/(x + 1)$, $1/(x - 1)$, and $2/(x^2 - 1)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

1138

QUESTION ID
S31-B2-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 1)(x - 1)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The LCD of $1/(x-2)$ and $2/(x-2)$ is $(x-2)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

1139

QUESTION ID
S31-B2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is $x-2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiplying $1/x + 1/(x + 2) = 4/3$ by $3x(x + 2)$ gives

A. $3 + 3 = 4$

B. $3(x + 2) + 3x = 4$

C. $3(x + 2) + 3x = 4x(x + 2)$

D. $x + 2 + x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

1140

QUESTION ID

S31-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C.

$3(x + 2) + 3x = 4x(x + 2)$

GRADING NOTE Each term times the LCD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1/x + 1/(x + 2) as one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

1141

QUESTION ID
S31-B2-Q5

TYPE
Short answer

CORRECT ANSWER
(2x + 2)/(x(x + 2))

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 3

Values that make a denominator zero • Grade 7 • The Number System • 7.NS.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

Which x-values make x/(x − 2) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

1143

QUESTION ID

S31-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which x-values make $2/(x^2 - 1)$ undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

1144

QUESTION ID

S31-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1 or −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

True or false?

A rational equation cannot have a solution that makes any denominator zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

1145

QUESTION ID

S31-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The expression is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

Which value is excluded from 1/x + 1/(x + 2) = 4/3?

A. 2

B. 1

C. 4/3

D. −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

1146

QUESTION ID

S31-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −2

GRADING NOTE x + 2 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the excluded values of $x^2/(x - 3) = 9/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

1147

QUESTION ID

S31-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 1

Explain and construct extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Explain how multiplying both sides of $x/(x-2) + 1 = 2/(x-2)$ by $x-2$ creates an extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

1149

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Multiplying by $x-2$ is not reversible when $x-2=0$, so the new equation is true at $x=2$ although the original is undefined there.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

Write a rational equation that has the extraneous solution x = 4 and no actual solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

1150

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

For example

x/(x − 4) = 4/(x − 4)

GRADING NOTE Any valid example with a check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

True or false?

Every rational equation has at least one extraneous solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

1151

QUESTION ID

S31-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3/(x − 1) = 6/(x + 2) has none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

For which a does a/(x − 1) = 2/(x − 1) have infinitely many solutions?

A. a = 2

B. no value

C. a = 0

D. a = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

1152

QUESTION ID

S31-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. a = 2

GRADING NOTE Then every x ≠ 1 works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve (x + 1)/(x − 1) = 2/(x − 1) + 1 and explain the result.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

1153

QUESTION ID
S31-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Every x ≠ 1 is a solution (the equation is an identity on its domain).

GRADING NOTE Any equivalent answer. (x + 1)/(x − 1) − 2/(x − 1) = (x − 1)/(x − 1) = 1 for x ≠ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 2

See rational equations as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Explain why $x/(x - 2) + 1 = 2/(x - 2)$ has no solution using the graphs of $y = x/(x - 2) + 1$ and $y = 2/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

1155

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

For $x \neq 2$ the left side minus the right side is $(2x - 4)/(x - 2) = 2$, so the graphs never meet; at $x = 2$ neither is defined.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

Where do y = 3/(x − 1) and y = 6/(x + 2) intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5

1156

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 1)

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5

True or false?

An extraneous solution corresponds to an intersection point of the two graphs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

1157

QUESTION ID

S31-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Neither graph is defined there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = x^2/(x - 3)$ and $y = 9/(x - 3)$ intersect at

A. (3, 3/2)

B. (−3, 3/2)

C. (−3, −3/2)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5

1158

QUESTION ID

S31-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, −3/2)

GRADING NOTE $9/(-6) = -3/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graph or table to approximate the solution of $1/x = x - 1$ with $x > 0$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5

1159

QUESTION ID
S31-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1.62

GRADING NOTE Any value 1.61 to 1.62 ($(1 + \sqrt{5})/2 \approx 1.618$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5

Solve radical equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Square and cube both sides; principal roots	**8.EE.A.2**
BACKWARD 2	Solve the resulting quadratic	**HSA-REI.B.4.b**
BACKWARD 3	Check a candidate in the original equation	**6.EE.B.5**
FORWARD 1	Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$	**HSA-REI.A.2**
FORWARD 2	Explain extraneous solutions with graphs of radical functions	**HSF-IF.C.7.b**

SET 32 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 7) = x − 5. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

1162

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 9; x = 2 is extraneous

GRADING NOTE Form required: solution and extraneous value. $x^2 - 11x + 18 = 0$; $\sqrt{9} = 3 \neq -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

Solve √(2x − 1) = 3. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

1163

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = 5; none extraneous

GRADING NOTE Form required: solution and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(x + 3) + 3 = x. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

1164

QUESTION ID

S32-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = 6; x = 1 is extraneous

GRADING NOTE Form required: solution and extraneous value. $x^2 - 7x + 6 = 0$; $\sqrt{4} + 3 = 5 \neq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

Solve $\sqrt[3]{(x - 2)} = -3$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

1165

QUESTION ID

S32-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = −25; none extraneous

GRADING NOTE Form required: solution. Cubing is reversible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q4**

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(x − 1) + 5 = 2. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

1166

QUESTION ID

S32-M-Q5

TYPE

Short answer

CORRECT ANSWER

No solution; x = 10 is extraneous

GRADING NOTE Form required: conclusion. √(x − 1) = −3 is impossible; squaring gives x = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 1

Square and cube both sides; principal roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{y} = 3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

1168

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt[3]{y} = -3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

1169

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−27

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{9} = 3$ or -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5

1170

QUESTION ID

S32-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The principal square root is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{y} = -2$ has

A. $y = 2$

B. no solution

C. $y = 4$

D. $y = -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

1171

QUESTION ID

S32-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. no solution

GRADING NOTE A principal square root is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Square both sides of $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

1172

QUESTION ID

S32-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x + 7 = x^2 - 10x + 25$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 2

Solve the resulting quadratic • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 11x + 18 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

1174

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 or 9

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 7x + 6 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

1175

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 or 6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)^2 = x^2 + 25.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

1176

QUESTION ID

S32-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 10x + 25.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 6x + 9 = x + 3$ gives

A. $x^2 - 7x + 12 = 0$

B. $x^2 - 5x + 12 = 0$

C. $x^2 - 6x + 6 = 0$

D. $x^2 - 7x + 6 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

1177

QUESTION ID

S32-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - 7x + 6 = 0$

GRADING NOTE Subtract x + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 4x = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

1178

QUESTION ID

S32-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0 or 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 3

Check a candidate in the original equation • Grade 6 • Expressions & Equations • 6.EE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = 2$ in $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

1180

QUESTION ID

S32-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{9} = 3$ but $2 - 5 = -3$; not a solution

GRADING NOTE Form required: the check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

Check x = 6 in √(x + 3) + 3 = x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

1181

QUESTION ID

S32-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√9 + 3 = 6; a solution

GRADING NOTE Form required: the check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

True or false?

$x = 4$ satisfies $\sqrt{(2x - 1)} = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

1182

QUESTION ID

S32-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(8 - 1)} = \sqrt{7} \neq 3$; the solution is $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value satisfies √(x + 3) + 3 = x?

A. 1

B. 0

C. 6

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

1183

QUESTION ID

S32-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Check each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = 10$ in $\sqrt{(x - 1)} + 5 = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

1184

QUESTION ID

S32-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3 + 5 = 8 ≠ 2; not a solution

GRADING NOTE Form required: check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 1

Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$ • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 5)} - \sqrt{x} = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

1186

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $\sqrt{(x + 5)} = 1 + \sqrt{x}$ gives $\sqrt{x} = 2$; check $\sqrt{9} - 2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^{3/2} = 8$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

1187

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $x = 8^{2/3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{(x^2)} = 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

1188

QUESTION ID

S32-F1-Q3

TYPE

Short answer

CORRECT ANSWER

5 or −5

GRADING NOTE Form required: both. $\sqrt{(x^2)} = |x|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q4/5

1189

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$; for example $x = -3$ gives 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q4/5

The solution set of $\sqrt{(x^2)} = x$ is

A. $x \geq 0$

B. $x \leq 0$

C. $x = 0$ only

D. all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

1190

QUESTION ID

S32-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. x ≥ 0

GRADING NOTE $|x| = x$ exactly when $x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 2

Explain extraneous solutions with graphs of radical functions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = \sqrt{(x + 7)}$ and $y = x - 5$ meet at one point. Give it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

1192

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(9, 4)

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

The extraneous value x = 2 is where y = x − 5 meets which other graph?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

1193

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = −√(x + 7)

GRADING NOTE Any equivalent answer. Squaring cannot tell √ from −√.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

True or false?

The graph of $y = \sqrt{(x + 7)}$ starts at (7, 0) and increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

1194

QUESTION ID

S32-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 7 = 0$ at $x = -7$, so it starts at $(-7, 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $y = \sqrt{(x - 1)} + 5$ lies

A. below $y = 2$

B. below the x-axis

C. at or below $y = 5$

D. at or above $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5

1195

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. at or above y = 5

GRADING NOTE $\sqrt{} \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the graphs to explain why $\sqrt{(x - 1)} + 5 = 2$ has no solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

1196

QUESTION ID

S32-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$y = \sqrt{(x - 1)} + 5$ is always at least 5, so it never meets $y = 2$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

Explain why intersections of nonlinear graphs solve f(x) = g(x)

MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	A graph is the set of its solutions	**HSA-REI.D.10**
BACKWARD 2	Solve a linear system by graphing	**8.EE.C.8.a**
BACKWARD 3	Evaluate nonlinear functions at an input	**HSF-IF.A.2**
FORWARD 1	Relate intersections to zeros of f - g	**HSA-REI.D.11**
FORWARD 2	Find all solutions of a trigonometric equation on an interval from a…	**HSF-TF.B.7**

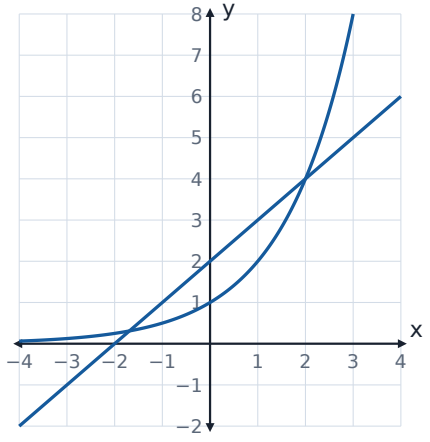
SET 33 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 2^x$ and $y = x + 2$ are shown. They meet at $x = 2$ and at a negative x. Explain why $x = 2$ solves $2^x = x + 2$, and verify it.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

1199

QUESTION ID
S33-M-Q1

TYPE
Short answer

CORRECT ANSWER
At an intersection both functions have the same y-value for the same x, so $2^x = x + 2$ there; $2^2 = 4 = 2 + 2$.

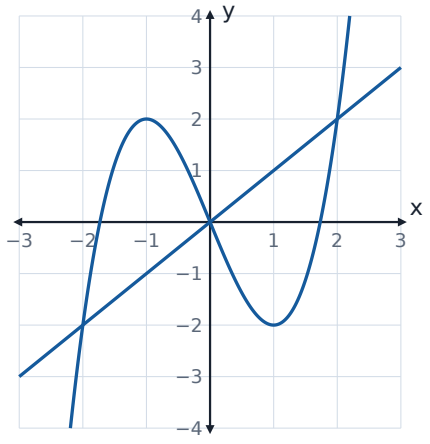
GRADING NOTE Form required: the explanation and the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

The graphs of $y = x^3 - 3x$ and $y = x$ are shown. Use them to list every solution of $x^3 - 3x = x$, and explain why they are solutions.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S33-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

1200

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = −2, 0, 2: at these x-values the two outputs are equal (each point lies on both graphs).

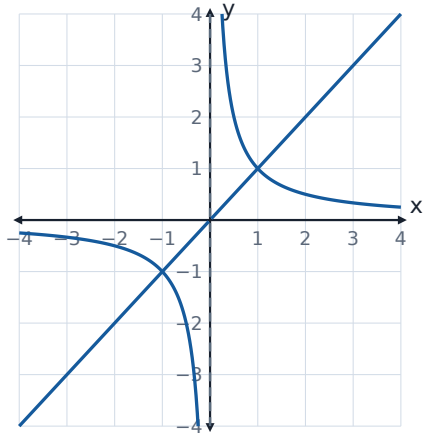
GRADING NOTE Form required: all three solutions and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S33-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 1/x$ and $y = x$ are shown. List every solution of $1/x = x$ and explain using the graphs.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

1201

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = −1 and x = 1: the graphs meet at (−1, −1) and (1, 1), where 1/x and x are equal.

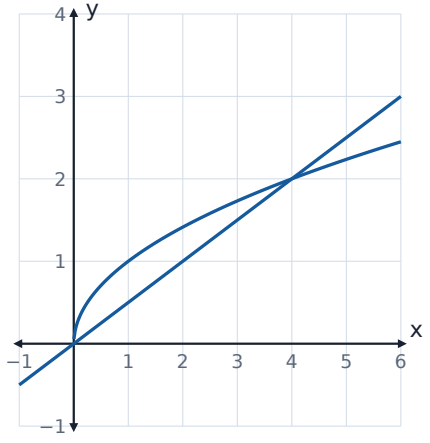
GRADING NOTE Form required: both solutions and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

The graphs of $y = \sqrt{x}$ and $y = x/2$ are shown. List every solution of $\sqrt{x} = x/2$ and verify each.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5

1202

QUESTION ID

S33-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x = 0$ and $x = 4$: $\sqrt{0} = 0 = 0/2$ and $\sqrt{4} = 2 = 4/2$.

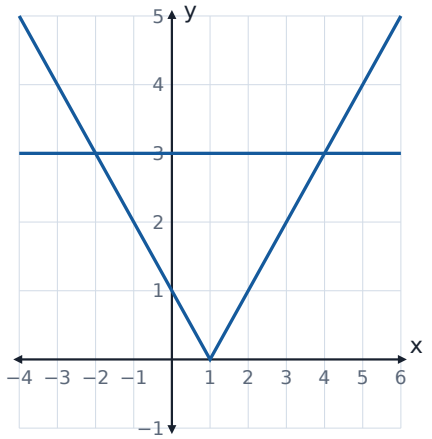
GRADING NOTE Form required: both solutions and the checks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5

The graphs of $y = |x - 1|$ and $y = 3$ are shown. List every solution of $|x - 1| = 3$ and explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

1203

QUESTION ID

S33-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = −2 and x = 4: the V-shaped graph is at height 3 there.

GRADING NOTE Form required: both solutions and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 1

A graph is the set of its solutions • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 4) on the graph of $y = 2^x$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

1205

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: $2^2 = 4$

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (3, 6) on the graph of y = x + 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

1206

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No: 3 + 2 = 5

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-B1-Q2**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every point on the graph of y = f(x) has the form (x, f(x)).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

1207

QUESTION ID
S33-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition of the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on $y = x^3 - 3x$?

A. (0, 3)

B. (2, 6)

C. (2, 2)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

1208

QUESTION ID

S33-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 2)

GRADING NOTE $8 - 6 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (−1, −1) on the graph of y = 1/x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

1209

QUESTION ID

S33-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: 1/(−1) = −1

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 2

Solve a linear system by graphing • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

The lines y = x + 1 and y = −x + 5 meet at (2, 3).
Explain why x = 2 solves x + 1 = −x + 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

1211

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

At x = 2 both equal 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where do y = 2x and y = 6 meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

1212

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 6)

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Two lines that do not intersect give an equation with no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

1213

QUESTION ID

S33-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 3$ and $y = x - 1$ meet at $x =$

A. 4

B. 3

C. -1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

1214

QUESTION ID

S33-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $x - 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve x + 1 = −x + 5 algebraically.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

1215

QUESTION ID

S33-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 3

Evaluate nonlinear functions at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find 2^x and $x + 2$ when $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

1217

QUESTION ID

S33-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4 and 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $x^3 - 3x$ when $x = -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

1218

QUESTION ID

S33-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{9} = 9/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

1219

QUESTION ID

S33-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{9} = 3$, but $9/2 = 4.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$|x - 1|$ when $x = -2$ is

A. 3

B. 1

C. -1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

1220

QUESTION ID

S33-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE $|-3|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 1/x when x = −1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

1221

QUESTION ID

S33-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 1

Relate intersections to zeros of f − g • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the solutions of $x^3 - 3x = x$ are the zeros of $h(x) = x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

1223

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(x) = g(x) exactly when f(x) − g(x) = 0, and f − g = x³ − 4x.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

Find the zeros of $h(x) = x^3 - 4x$ algebraically.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5

1224

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−2, 0, 2

GRADING NOTE Form required: all three. x(x − 2)(x + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5

True or false?

If y = f(x) and y = g(x) meet at x = a, then the graph of y = f(x) − g(x) crosses or touches the x-axis at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

1225

QUESTION ID

S33-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(a) − g(a) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $\sqrt{x} = x/2$ are the zeros of

A. $\sqrt{x} + x/2$

B. $\sqrt{x} - x/2$

C. $\sqrt{x} \cdot x/2$

D. $x/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q4/5

1226

QUESTION ID
S33-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $\sqrt{x} - x/2$

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q4/5

Explain why $2^x = x + 2$ has exactly two solutions using $h(x) = 2^x - x - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5

1227

QUESTION ID

S33-F1-Q5

TYPE

Short answer

CORRECT ANSWER

h changes sign twice (h(−2) > 0, h(0) < 0, h(3) > 0) and h is decreasing then increasing, so it has exactly two zeros.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 2

Find all solutions of a trigonometric equation on an interval from a graph • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

On $0 \le x < 2\pi$, where does $y = \sin x$ meet $y = 1/2$? Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5

1229

QUESTION ID
S33-F2-Q1

TYPE
Short answer

CORRECT ANSWER
π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

On $0 \le x < 2\pi$, where does $y = \cos x$ meet $y = 0$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5

1230

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

π/2 and 3π/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5

True or false?

sin x = 1/2 has exactly two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

1231

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has two solutions in each period, so infinitely many real solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

How many solutions does $\sin x = 0.3$ have on $0 \le x < 2\pi$?

A. 1

B. 0

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5

1232

QUESTION ID

S33-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE The horizontal line crosses twice in one period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why sin x = 2 has no solution using the graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

1233

GRADING NOTE Any equivalent explanation.

QUESTION ID

S33-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The graph of y = sin x stays between −1 and 1, so it never meets y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

Approximate solutions of f(x) = g(x)

MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	Read solutions of linear-quadratic systems from graphs	**HSA-REI.D.11**
BACKWARD 2	Evaluate exponential and logarithmic expressions with a calculat…	**HSF-IF.A.2**
BACKWARD 3	Round to a place value	**5.NBT.A.4**
FORWARD 1	Guarantee the accuracy of an approximation	**HSA-REI.D.11**
FORWARD 2	Compare an approximation with an exact logarithmic solution	**HSF-LE.A.4**

SET 34 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Use a table or successive approximation to solve $2^x = 5 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5

1236

QUESTION ID
S34-M-Q1

TYPE
Short answer

CORRECT ANSWER
1.72

GRADING NOTE Accept 1.71 to 1.72 (exact root 1.7156...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = 7 - x$ to the nearest hundredth with a table or successive approximation. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5

1237

QUESTION ID
S34-M-Q2

TYPE
Short answer

CORRECT ANSWER
1.74

GRADING NOTE Accept 1.73 to 1.74 (root 1.7392...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve log x = 2 − x (log base 10) to the nearest hundredth. A calculator may be used.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5

1238

QUESTION ID
S34-M-Q3

TYPE
Short answer

CORRECT ANSWER
1.76

GRADING NOTE Accept 1.75 to 1.76 (root 1.7556...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5

The table shows $f(x) = 3(0.8)^x$ and $g(x) = x$. Between which two table values is the solution of $f(x) = g(x)$? Estimate it to the nearest tenth.

x	0	0.5	1	1.5	2	2.5
f(x)	3	2.683	2.4	2.147	1.92	1.717
g(x)	0	0.5	1	1.5	2	2.5

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

1239

QUESTION ID

S34-M-Q4

TYPE

Short answer

CORRECT ANSWER

Between 1.5 and 2; x ≈ 1.9

GRADING NOTE Form required: the interval and the estimate (root 1.944...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graph or table to find the negative solution of $2^x = x + 2$ to the nearest tenth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5

1240

QUESTION ID
S34-M-Q5

TYPE
Short answer

CORRECT ANSWER
−1.7

GRADING NOTE Any equivalent form (root −1.690...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 1

Read solutions of linear-quadratic systems from graphs • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Where do $y = x^2$ and $y = 4$ meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

1242

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = -2$ and $x = 2$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Read the intersection of $y = x^2$ and $y = x + 2$ from a graph or table.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

1243

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 and x = 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solutions of f(x) = g(x) can be approximated from a table where f − g changes sign.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

1244

QUESTION ID
S34-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Sign change between rows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) - g(1) = -2$ and $f(2) - g(2) = 1$. A solution lies

A. below 1

B. at 1.5 exactly

C. above 2

D. between 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

1245

QUESTION ID

S34-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. between 1 and 2

GRADING NOTE Sign change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = 3$. Between which consecutive integers is the positive solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

1246

QUESTION ID

S34-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1 and 2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 2

Evaluate exponential and logarithmic expressions with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find $2^{1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

1248

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find log 1.75 (base 10) to the nearest thousandth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

1249

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.243

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1.7^3 = 5.1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

1250

QUESTION ID

S34-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1.7 \cdot 1.7 \cdot 1.7 = 4.913$ (5.1 is $3 \cdot 1.7$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3(0.8)^2$ equals

A. 0.192

B. 4.8

C. 1.92

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

1251

QUESTION ID

S34-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1.92

GRADING NOTE $3 \cdot 0.64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $2^{-1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

1252

QUESTION ID

S34-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.31

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 3

Round to a place value • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Round 1.7156 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

1254

QUESTION ID

S34-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.72

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 1.944 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

1255

QUESTION ID

S34-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1.9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

1.755 rounded to the nearest hundredth is 1.76.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

1256

QUESTION ID

S34-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Round half up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

−1.690 to the nearest tenth is

A. −1.6

B. −1.69

C. −2

D. −1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

1257

QUESTION ID

S34-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −1.7

GRADING NOTE Nearest tenth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 1.7392 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

1258

QUESTION ID

S34-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1.74

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 1

Guarantee the accuracy of an approximation • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

$h(x) = 2^x + x - 5$. Find h(1.71) and h(1.72) and explain why a solution lies between them. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5

1260

QUESTION ID
S34-F1-Q1

TYPE
Short answer

CORRECT ANSWER
h(1.71) ≈ −0.018 and h(1.72) ≈ 0.014; h changes sign, so h has a zero between them

GRADING NOTE Form required: both values and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5

Use one step of bisection on [1, 2] for $x^3 + x - 7 = 0$. Which half contains the solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

1261

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

[1.5, 2], since h(1.5) = −2.125 < 0 and h(2) = 3 > 0

GRADING NOTE Form required: the half and the evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

True or false?

After 4 bisection steps starting from an interval of length 1, the interval has length 1/8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

1262

QUESTION ID

S34-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each step halves the length: $(1/2)^4 = 1/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

How many bisection steps from [1, 2] guarantee an interval shorter than 0.01?

A. 5

B. 100

C. 10

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

1263

QUESTION ID

S34-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE $2^7 = 128 > 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why checking h(1.715) and h(1.725) decides whether the solution rounds to 1.72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

1264

QUESTION ID

S34-F1-Q5

TYPE

Short answer

CORRECT ANSWER

If h changes sign between 1.715 and 1.725, the zero is in that interval, and every number there rounds to 1.72.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 2

Compare an approximation with an exact logarithmic solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^x = 5$ exactly, then give a decimal to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

1266

QUESTION ID
S34-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$\log_2 5 = \log 5/\log 2 \approx 2.32$

GRADING NOTE Form required: exact and decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

Solve $3(0.8)^x = 1$ exactly and to the nearest hundredth.
A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

1267

QUESTION ID
S34-F2-Q2

TYPE
Short answer

CORRECT ANSWER
log(1/3)/log(0.8) ≈ 4.92

GRADING NOTE Form required: exact and decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

True or false?

$2^x = 5 - x$ can be solved exactly with one logarithm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

1268

QUESTION ID

S34-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x appears both in an exponent and outside it; approximation is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation can be solved exactly with logarithms?

A. $\log x = 2 - x$

B. $2^x = x + 2$

C. $5(1.2)^x = 30$

D. $3(0.8)^x = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q4/5

1269

QUESTION ID

S34-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $5(1.2)^x = 30$

GRADING NOTE Only one exponential term and no other x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why approximation methods are needed for $2^x = x + 2$ but not for $2^x = 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

1270

QUESTION ID

S34-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2^x = 4$ isolates the exponential, so $x = \log_2 4 = 2$ exactly; $2^x = x + 2$ mixes an exponential and a linear term, which no algebra step separates.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSF-TF.A.1 • TRIGONOMETRIC FUNCTIONS

Use radian measure as arc length on the unit circle

MAIN	Algebra II main question	**HSF-TF.A.1**
BACKWARD 1	Circumference of a circle	**7.G.B.4**
BACKWARD 2	Arc length is proportional to the angle	**HSG-C.B.5**
BACKWARD 3	Degrees as fractions of a turn	**4.MD.C.5**
FORWARD 1	Measure negative angles and angles beyond one turn	**HSF-TF.A.1**
FORWARD 2	Read sine and cosine of radian angles from the unit circle	**HSF-TF.A.2**

SET 35 • HSF-TF.A.1

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

On the unit circle, a central angle subtends an arc of length 2.5. What is the radian measure of the angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

1273

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.5 radians: on the unit circle the radian measure equals the length of the subtended arc.

GRADING NOTE Form required: the measure and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 150° to radians. Give an exact answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

1274

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

5π/6

GRADING NOTE Form required: exact. 150 · π/180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 7π/4 radians to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

1275

QUESTION ID

S35-M-Q3

TYPE

Short answer

CORRECT ANSWER

315°

GRADING NOTE Any equivalent form. (7/4)·180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle of 2π/3 radians is drawn in a circle of radius 6 cm. Find the length of the arc it subtends.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

1276

QUESTION ID
S35-M-Q4

TYPE
Short answer

CORRECT ANSWER
4π cm (about 12.57 cm)

GRADING NOTE Form required: exact or decimal with units. s = rθ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why one full turn measures 2π radians.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

1277

QUESTION ID

S35-M-Q5

TYPE

Short answer

CORRECT ANSWER

A full turn subtends the whole unit circle, whose circumference is 2π · 1 = 2π.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 1

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the circumference of a circle of radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

1279

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the circumference of a circle of radius 6 cm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

1280

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12π cm

GRADING NOTE Any equivalent form with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

True or false?

The circumference of a circle is πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

1281

QUESTION ID

S35-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE πr^2 is the area; the circumference is $\pi d = 2\pi r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

Half of the unit circle has length

A. 2π

B. 1

C. π

D. π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

1282

QUESTION ID

S35-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π

GRADING NOTE 2π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What length is one quarter of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

1283

QUESTION ID

S35-B1-Q5

TYPE

Short answer

CORRECT ANSWER

π/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 2

Arc length is proportional to the angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A 60° angle subtends what fraction of a full circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5

1285

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q1**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the arc length of a 90° angle in a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

1286

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q2**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

True or false?

Doubling the radius doubles the arc length for the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

1287

QUESTION ID

S35-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Arc length is proportional to the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

The arc length for angle θ (radians) in a circle of radius r is

A. θ/r

B. $r^2\theta$

C. $2\pi\theta$

D. rθ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

1288

QUESTION ID

S35-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. rθ

GRADING NOTE Definition of radian measure scaled by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of the circle is an angle of π/3 radians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5

1289

QUESTION ID

S35-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form. (π/3)/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q5**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 3

Degrees as fractions of a turn • Grade 4 • Measurement & Data • 4.MD.C.5

ANSWER KEY Reduced copy of the question page, with its answer

How many degrees are in a full turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

1291

QUESTION ID

S35-B3-Q1

TYPE

Short answer

CORRECT ANSWER

360°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a turn is 150°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

1292

QUESTION ID

S35-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

True or false?

A 315° angle is 3/4 of a turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

1293

QUESTION ID

S35-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 315/360 = 7/8; 3/4 of a turn is 270°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A quarter turn is

A. 90°

B. 180°

C. 45°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

1294

QUESTION ID

S35-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

GRADING NOTE 360/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

How many degrees are in 2/3 of a turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

1295

QUESTION ID

S35-B3-Q5

TYPE

Short answer

CORRECT ANSWER

240°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 1

Measure negative angles and angles beyond one turn • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Give a positive angle less than 2π that is coterminal with $-\pi/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

1297

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$5\pi/3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the angle between 0 and 2π that is coterminal with 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

1298

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

True or false?

An arc of length 10 on the unit circle goes more than once around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

1299

QUESTION ID

S35-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10 > 2\pi \approx 6.28$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

An angle of $-\pi/2$ radians is traced

A. a quarter turn clockwise

B. one full turn

C. half a turn

D. a quarter turn counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

1300

QUESTION ID

S35-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. a quarter turn clockwise

GRADING NOTE Negative means clockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A point travels 10 units counterclockwise around the unit circle from (1, 0). How far past one full turn is it? Give a decimal to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

1301

QUESTION ID

S35-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$10 - 2\pi \approx 3.72$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 2

Read sine and cosine of radian angles from the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

What point of the unit circle is at angle π? Give cos π and sin π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

1303

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, 0); cos π = −1, sin π = 0

GRADING NOTE Form required: the point and both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find cos(3π/2) and sin(3π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

1304

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0 and −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

1305

QUESTION ID

S35-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Back at (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin(π/2) equals

A. −1

B. 0

C. π/2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

1306

QUESTION ID

S35-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE Point (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cos θ and sin θ make sense for θ = 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

1307

QUESTION ID
S35-F2-Q5

TYPE
Short answer

CORRECT ANSWER
13π/6 ends at the same point as π/6, so cos and sin take the values at that point (√3/2, 1/2).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Find exact trigonometric values beyond the first quadrant

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Special right triangle ratios	**HSG-SRT.C.6**
BACKWARD 2	Reference angles for radian measures	**HSF-TF.A.1**
BACKWARD 3	Signs of coordinates by quadrant	**6.NS.C.6.b**
FORWARD 1	Values for angles beyond 2π and solving for angles	**HSF-TF.A.2**
FORWARD 2	Verify the Pythagorean identity at exact values	**HSF-TF.C.8**

SET 36 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find sin(5π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

1310

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE Form required: exact. Reference angle π/6, quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find cos(4π/3) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

1311

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Form required: exact. Reference angle π/3, quadrant III.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find tan(7π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

1312

QUESTION ID

S36-M-Q3

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Form required: exact. Point (√2/2, −√2/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find $\sin(-\pi/2)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

1313

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Form required: exact. Point (0, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

Use the unit circle to find cos(11π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

1314

QUESTION ID

S36-M-Q5

TYPE

Short answer

CORRECT ANSWER

√3/2

GRADING NOTE Form required: exact. Reference angle π/6, quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 1

Special right triangle ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In a 30°−60°−90° triangle with hypotenuse 1, give the side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2 (opposite 30°) and √3/2 (opposite 60°)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find sin 45° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5

True or false?

cos 60° = √3/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S36-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE cos 60° = 1/2; √3/2 is sin 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

sin 60° equals

A. √2/2

B. √3/2

C. √3

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S36-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. √3/2

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find tan 30° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S36-B1-Q5

TYPE

Short answer

CORRECT ANSWER

√3/3

GRADING NOTE Any equivalent form (1/√3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 2

Reference angles for radian measures • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the reference angle of 5π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the reference angle of 4π/3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

True or false?

The reference angle of 7π/4 is 3π/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S36-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2\pi - 7\pi/4 = \pi/4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which quadrant contains 11π/6?

A. I

B. IV

C. III

D. II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S36-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. IV

GRADING NOTE Between 3π/2 and 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 4π/3 to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S36-B2-Q5

TYPE

Short answer

CORRECT ANSWER

240°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q5**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 3

Signs of coordinates by quadrant • Grade 6 • The Number System • 6.NS.C.6.b

ANSWER KEY Reduced copy of the question page, with its answer

In which quadrant are both coordinates negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

1328

QUESTION ID

S36-B3-Q1

TYPE

Short answer

CORRECT ANSWER

III

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the signs of the coordinates in quadrant II?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

1329

QUESTION ID

S36-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x negative, y positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

True or false?

A point in quadrant IV has a positive y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

1330

QUESTION ID

S36-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Quadrant IV points have $x > 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

1330

The point $(-\sqrt{3}/2, 1/2)$ is in quadrant

A. I

B. II

C. III

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

1331

QUESTION ID

S36-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. II

GRADING NOTE (−, +).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the signs of x and y for a point in quadrant IV.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

1332

QUESTION ID

S36-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x positive, y negative

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 1

Values for angles beyond 2π and solving for angles • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(13π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

1334

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find cos(−3π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

1335

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−√2/2

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every θ with $0 \le \theta < 2\pi$ and $\cos\theta = -1/2$.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

1336

QUESTION ID

S36-F1-Q3

TYPE

Short answer

CORRECT ANSWER

2π/3 and 4π/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

tan(π/2) is undefined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

1337

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE cos(π/2) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which angle has sin θ = −√3/2 and cos θ = 1/2?

A. 5π/3

B. 4π/3

C. 2π/3

D. 7π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

1338

QUESTION ID

S36-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 5π/3

GRADING NOTE Quadrant IV, reference π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 2

Verify the Pythagorean identity at exact values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Verify $\sin^2\theta + \cos^2\theta = 1$ for $\theta = 5\pi/6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

1340

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(1/2)^2 + (-\sqrt{3}/2)^2$
$= 1/4 + 3/4 = 1$

GRADING NOTE Form required: the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Verify $\sin^2\theta + \cos^2\theta = 1$ for $\theta = 7\pi/4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

1341

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/2 + 1/2 = 1

GRADING NOTE Form required: the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^2\theta + \cos^2\theta = 1$ is true only for first-quadrant angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

1342

QUESTION ID

S36-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every point (cos θ, sin θ) is on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If cos θ = −1/2 and θ is in quadrant III, then sin θ =

A. √3/2

B. −√3/2

C. 1/2

D. −1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

1343

QUESTION ID

S36-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −√3/2

GRADING NOTE $\sin^2 = 3/4$, negative in III.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the values found from the unit circle always satisfy $\sin^2\theta + \cos^2\theta = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

1344

QUESTION ID

S36-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(cos θ, sin θ) lies on $x^2 + y^2 = 1$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

SET 37 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Explain the extension of sine and cosine to all real numbers

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Right-triangle definitions of sine and cosine	**HSG-SRT.C.6**
BACKWARD 2	Points on the unit circle	**HSG-GPE.A.1**
BACKWARD 3	Radian measure as directed arc length	**HSF-TF.A.1**
FORWARD 1	Explain where tangent is defined and how values repeat	**HSF-TF.A.2**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 37 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which statement correctly explains how the unit circle defines cos t and sin t for every real number t?

A. Start at (0, 1) and travel t units clockwise; cos t is the y-coordinate and sin t is the x-coordinate of the endpoint.

B. Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if t > 0 and clockwise if t < 0; cos t and sin t are the x- and y-coordinates of the endpoint.

C. Travel t units counterclockwise from (1, 0); for t < 0 the values are undefined.

D. Draw a right triangle with an acute angle of t radians; cos t and sin t are the ratios adjacent/hypotenuse and opposite/hypotenuse, so the definition works only for t between 0 and π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

1347

QUESTION ID

S37-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if t > 0 and clockwise if t < 0; cos t and sin t are the x- and y-coordinates of the endpoint.

GRADING NOTE Arc length |t| from (1, 0), direction by the sign of t, coordinates of the endpoint. The others use the right-triangle definition (only acute angles), the wrong start and coordinates, or leave negative t undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which explanation shows why sin(t + 2π) = sin t for every real number t?

A. For t = π/2 both sides equal 1, so they are always equal.

B. Sine distributes over addition, so sin(t + 2π) = sin t + sin 2π, and sin 2π = 0 because the point at 2π is (1, 0); therefore sin(t + 2π) = sin t.

C. Adding 2π to t reflects the endpoint on the unit circle over the x-axis, and a reflection over the x-axis keeps the y-coordinate, so sin(t + 2π) = sin t.

D. The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and t + 2π end at the same point, which has the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

1348

QUESTION ID
S37-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and t + 2π end at the same point, which has the same y-coordinate.

GRADING NOTE One full turn returns to the same point. The others distribute sine over a sum (false in general), check one value, or describe the wrong motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

Why is cos t < 0 for π/2 < t < π?

A. For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.

B. For these t the endpoint is in quadrant II, where y-coordinates are negative, and cos t is the y-coordinate.

C. cos(2π/3) = −1/2, so cos t is negative on this interval.

D. On the unit circle cos t decreases steadily as t increases from 0, so once t passes π/2, where cos t = 0, the value of cos t is negative for every larger t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5

1349

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.

GRADING NOTE Quadrant II and the x-coordinate. The others use the wrong coordinate and sign, a false claim (cos t increases again after π and is positive after 3π/2), or one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5

Why is $-1 \le \sin t \le 1$ for every real number t?

A. $\sin(\pi/2) = 1$ and $\sin(3\pi/2) = -1$, so every value lies between them.

B. In a right triangle the hypotenuse is longer than either leg, so opposite/hypotenuse is less than 1, which means $\sin t < 1$ for every real number t.

C. sin t is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.

D. sin t is an arc length on the unit circle, and those arcs are at most 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q4/5

1350

QUESTION ID

S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. sin t is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.

GRADING NOTE Coordinates of points on the unit circle satisfy $x^2 + y^2 = 1$. The others only cover acute angles (and wrongly exclude sin t = 1), misstate the definition, or reason from two values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q4/5

How does the unit-circle definition give sin(−π/6)?

A. Travel π/6 clockwise from (1, 0) to (1/2, −√3/2), so sin(−π/6) = −√3/2.

B. Negative inputs are not on the unit circle, so sin(−π/6) is undefined.

C. Travel π/6 clockwise from (1, 0) to the point (√3/2, −1/2), so sin(−π/6) = −1/2.

D. Travel π/6 counterclockwise from (1, 0) to the point (√3/2, 1/2), so sin(−π/6) = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

1351

QUESTION ID

S37-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Travel π/6 clockwise from (1, 0) to the point (√3/2, −1/2), so sin(−π/6) = −1/2.

GRADING NOTE A negative t means clockwise travel; the endpoint is (√3/2, −1/2) and the sine is its y-coordinate. The others go the wrong way, use the point for π/3, or leave negative t undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 1

Right-triangle definitions of sine and cosine • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Define sin A in a right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5

1353

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

opposite/hypotenuse

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle with hypotenuse 1, what are the legs next to and opposite an acute angle θ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5

1354

QUESTION ID
S37-B1-Q2

TYPE
Short answer

CORRECT ANSWER
cos θ (adjacent) and sin θ (opposite)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5

True or false?

The right-triangle definition works for θ = 120°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

1355

QUESTION ID

S37-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A right triangle has no 120° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

In a right triangle with hypotenuse 1, the leg opposite θ is

A. sin θ

B. tan θ

C. 1

D. cos θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5

1356

QUESTION ID
S37-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. sin θ

GRADING NOTE opposite/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5

Why does the right-triangle definition fail for negative angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

1357

QUESTION ID

S37-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Triangle angles are positive (between 0° and 90° for the acute angles).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 2

Points on the unit circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the circle with center (0, 0) and radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

1359

QUESTION ID
S37-B2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^2 + y^2 = 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

Is (−1/2, √3/2) on the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

1360

QUESTION ID
S37-B2-Q2

TYPE
Short answer

CORRECT ANSWER
Yes: 1/4 + 3/4 = 1

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

True or false?

Every point on the unit circle has $-1 \le x \le 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

1361

QUESTION ID

S37-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 \le 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

Which point is on the unit circle?

A. (0.9, 0.9)

B. (0.5, 0.5)

C. (1, 1)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5

1362

QUESTION ID
S37-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. (0.6, −0.8)

GRADING NOTE 0.36 + 0.64 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q4**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A point on the unit circle has x = 0. Give both possible points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

1363

QUESTION ID
S37-B2-Q5

TYPE
Short answer

CORRECT ANSWER
(0, 1) and (0, −1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 3

Radian measure as directed arc length • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

What arc length on the unit circle does t = π/6 cut off?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

1365

QUESTION ID

S37-B3-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which direction does t = −π/6 travel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

1366

QUESTION ID

S37-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Clockwise

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

True or false?

t = π returns to the starting point (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

1367

QUESTION ID

S37-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE t = π is half a turn and ends at (−1, 0); t = 2π returns to (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc of length π from (1, 0) ends at

A. (0, 1)

B. (0, −1)

C. (1, 0)

D. (−1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

1368

QUESTION ID

S37-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (−1, 0)

GRADING NOTE Half a turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give a negative t that ends at (0, −1).

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

1369

QUESTION ID

S37-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−π/2

GRADING NOTE Any valid value (−π/2 + 2πk).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 1

Explain where tangent is defined and how values repeat • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Why is tan t undefined at t = π/2 + kπ (k an integer)?

A. At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.

B. tan t is undefined whenever sin t = 1.

C. At these t the endpoint on the unit circle is (1, 0) or (−1, 0), so sin t = 0 and tan t = sin t/cos t would have to divide by 0.

D. tan t = cos t/sin t, and sin t = 0 at these t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

1371

QUESTION ID
S37-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.

GRADING NOTE The endpoints are (0, ±1), where the x-coordinate cos t is 0. The others use the wrong points, invert the ratio, or give a rule that misses t = 3π/2 (where sin t = −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain and range of y = sin t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

1372

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Domain all real numbers; range −1 ≤ y ≤ 1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

cos t and sin t can both equal 0 for the same t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

1373

QUESTION ID

S37-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (0, 0) is not on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The unit-circle definition agrees with the triangle definition for

A. $t < 0$ only

B. $\pi/2 < t < \pi$ only

C. all t

D. $0 < t < \pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

1374

QUESTION ID

S37-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $0 < t < \pi/2$

GRADING NOTE Acute angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cos t = cos(−t) using the unit circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

1375

QUESTION ID

S37-F1-Q5

TYPE

Short answer

CORRECT ANSWER

t and −t end at points that are reflections over the x-axis, which have the same x-coordinate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle argument shows that sin(−t) = −sin t?

A. sin(−π/2) = −1 = −sin(π/2), so the rule holds for every t.

B. The endpoints for t and −t are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.

C. The endpoints for t and −t are the same point, so their sines are opposites.

D. The endpoints for t and −t are always reflections of each other over the y-axis, so their y-coordinates (the sines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

1377

QUESTION ID

S37-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. The endpoints for t and −t are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.

GRADING NOTE Equal arcs in opposite directions from (1, 0) give mirror images over the x-axis. The others name the wrong reflection (it keeps y), check one value, or describe the points wrongly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is sine an even function, an odd function, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

1378

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Odd

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

True or false?

tan(t + π) = tan t for every t where it is defined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

1379

QUESTION ID

S37-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The endpoints of t and t + π are opposite points; the ratio y/x is unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The period of y = tan t is

A. π

B. 4π

C. π/2

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

1380

QUESTION ID

S37-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. π

GRADING NOTE Opposite points give equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

Explain why cos is an even function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

1381

QUESTION ID

S37-F2-Q5

TYPE

Short answer

CORRECT ANSWER

cos(−t) = cos t because reflection over the x-axis keeps the x-coordinate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Write sinusoidal models with given amplitude, period, and midline

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Amplitude and midline from the maximum and minimum	**7.NS.A.3**
BACKWARD 2	Period and frequency	**6.RP.A.2**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Write models from data or with a phase shift	**HSF-TF.B.5**
FORWARD 2	Solve for times in sinusoidal models	**HSF-TF.B.7**

SET 38 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write a cosine function with amplitude 3, period 2π, midline y = 1, and a maximum at x = 0.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

1384

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

3cos x + 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 4 minutes. A rider starts at the bottom. Write the rider's height h after t minutes.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

1385

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

25 − 20cos(πt/2)

GRADING NOTE Any equivalent form. Amplitude 20, midline 25, period 4 so B = 2π/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Water depth at a harbor varies between 1 ft (low tide) and 9 ft (high tide) with a period of 12 hours. High tide is at t = 0. Write the depth d after t hours.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

1386

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

4cos(πt/6) + 5

GRADING NOTE Any equivalent form. Midline (9 + 1)/2, amplitude (9 − 1)/2, B = 2π/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

A town's average temperature is lowest, 10°C, in month t = 0 and highest, 30°C, at t = 6, with a 12-month period. Write a model T(t).

T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

1387

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

20 − 10cos(πt/6)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

Write a sine function with amplitude 2, period π, and midline y = −1 that is at the midline and increasing at x = 0.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

1388

QUESTION ID
S38-M-Q5

TYPE
Short answer

CORRECT ANSWER
2sin(2x) − 1

GRADING NOTE Any equivalent form. B = 2π/π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 1

Amplitude and midline from the maximum and minimum • Grade 7 • The Number System • 7.NS.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find (9 + 1)/2 and (9 − 1)/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

1390

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5 and 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

A quantity varies between 10 and 30. Find the midline value and amplitude.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

1391

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Midline 20; amplitude 10

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

True or false?

The amplitude is the maximum minus the minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

1392

QUESTION ID

S38-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

For max 45 and min 5, the midline is

A. y = 20

B. y = 50

C. y = 25

D. y = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

1393

QUESTION ID

S38-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. y = 25

GRADING NOTE (45 + 5)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m with center 25 m high. Give its maximum and minimum heights.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

1394

QUESTION ID

S38-B1-Q5

TYPE

Short answer

CORRECT ANSWER

45 m and 5 m

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 2

Period and frequency • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A wheel turns once every 4 minutes. How many turns per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

1396

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

A spring oscillates 3 times per second. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

1397

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/3 s

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

True or false?

Frequency is equal to the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

1398

QUESTION ID

S38-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Frequency is the reciprocal: f = 1/period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A period of 12 hours means a frequency of

A. 12 cycles per hour

B. 2 cycles per hour

C. 1/2 cycle per hour

D. 1/12 cycle per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

1399

QUESTION ID

S38-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/12 cycle per hour

GRADING NOTE Reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

y = sin(Bx) has period 2π/B. Find B for period 4.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

1400

QUESTION ID

S38-B2-Q5

TYPE

Short answer

CORRECT ANSWER

π/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Describe how y = 3cos x + 1 is obtained from y = cos x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

1402

QUESTION ID

S38-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Vertical stretch by 3, then shift up 1

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe y = −cos x compared with y = cos x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

1403

QUESTION ID

S38-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Reflection over the x-axis

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

True or false?

y = cos x + 5 has a maximum of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

1404

QUESTION ID

S38-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The maximum of cos x is 1, so the maximum is 6; 5 is the midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

$y = -20\cos x$ starts (at $x = 0$) at

A. its minimum

B. its maximum

C. the midline going up

D. the midline going down

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q4/5

1405

GRADING NOTE −20.

QUESTION ID

S38-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. its minimum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q4/5

Write y = sin x shifted down 1 and stretched vertically by 2.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

1406

QUESTION ID

S38-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2sin x − 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 1

Write models from data or with a phase shift • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A rider's height is 3 m at t = 0 (bottom), 27 m at the top after 15 s, and the motion repeats every 30 s. Write h(t).

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

1408

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15 − 12cos(πt/15)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

Sunrise times vary between 5:30 and 7:30 (as 5.5 and 7.5 hours) with period 365 days, latest at day 0. Write a model.

S = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

1409

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6.5 + cos(2πt/365)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

True or false?

$y = 4\sin(x - \pi/2)$ is the same as $y = 4\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S38-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q3/5

1410

QUESTION ID

S38-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin(x - \pi/2) = -\cos x$, so $4\sin(x - \pi/2) = -4\cos x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S38-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q3/5

A sine model with period 10, amplitude 3, midline 7, and a maximum at t = 2.5 is

A. 7 + 3sin(10t)

B. 7 + 3sin(πt/5)

C. 3 + 7sin(πt/5)

D. 7 + 3cos(πt/5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

1411

QUESTION ID

S38-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7 + 3sin(πt/5)

GRADING NOTE Quarter period 2.5 gives the maximum of sine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A model has a maximum of 18 at t = 3 and the next minimum of 6 at t = 7. Write a cosine model.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

1412

QUESTION ID

S38-F1-Q5

TYPE

Short answer

CORRECT ANSWER

12 + 6cos(π(t − 3)/4)

GRADING NOTE Any equivalent form. Period 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 2

Solve for times in sinusoidal models • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

For h = 25 − 20cos(πt/2), find the first time t > 0 when the rider is at 35 m. A calculator may be used.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

1414

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4/3 min (about 1.33 min)

GRADING NOTE Any equivalent form. cos(πt/2) = −1/2 gives πt/2 = 2π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For $d = 4\cos(\pi t/6) + 5$, find all times in $0 \le t < 12$ when the depth is 5 ft.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5

1415

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 and 9 hours

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5

True or false?

A sinusoidal model usually reaches a given value between its min and max twice per period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

1416

QUESTION ID

S38-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Once going up and once going down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For T = 20 − 10cos(πt/6), T = 25 first when

A. t = 2

B. t = 4

C. t = 3

D. t = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

1417

QUESTION ID

S38-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. t = 4

GRADING NOTE cos(πt/6) = −1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

Explain why there are two solutions per period when the value is strictly between the minimum and maximum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

1418

QUESTION ID

S38-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The graph passes through that height once while rising and once while falling in each period.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

SET 39 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Interpret the parameters of sinusoidal models

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Build the model from amplitude, period, and midline	**HSF-TF.B.5**
BACKWARD 2	Evaluate a sinusoidal model	**HSF-TF.A.2**
BACKWARD 3	Convert rates between units	**6.RP.A.3.d**
FORWARD 1	Predict how changing a parameter changes the situation	**HSF-TF.B.5**
FORWARD 2	Interpret periodic features from graphs and tables	**HSF-IF.B.4**

SET 39 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ meters after t seconds. Give the maximum height, the minimum height, and the time for one revolution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

1421

QUESTION ID

S39-M-Q1

TYPE

Short answer

CORRECT ANSWER

22 m, 2 m, 30 s

GRADING NOTE Form required: all three with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

Hours of daylight are $D(t) = 12 + 3\sin(2\pi(t - 80)/365)$, t in days. Interpret 12, 3, and 365.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

1422

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

12: average daylight (midline, hours); 3: the most daylight varies from 12 (amplitude, hours); 365: days in one cycle (period)

GRADING NOTE Form required: all three interpretations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

A model of blood pressure is P(t) = 100 + 20sin(2.5πt), t in seconds. Find the period and the number of heartbeats per minute.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

1423

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

Period 0.8 s; 75 beats per minute

GRADING NOTE Form required: both. 2π/(2.5π) = 0.8; 60/0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A spring's position is $y = -4\cos(3\pi t)$ cm, t in seconds. Give the amplitude, period, and frequency.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

1424

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 cm; 2/3 s; 1.5 cycles per second

GRADING NOTE Form required: all three with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

Two sounds are y = sin(880πt) and y = sin(440πt), t in seconds. Give the frequency of each in cycles per second and say which is higher.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

1425

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

440 and 220 cycles per second; the first is higher

GRADING NOTE Form required: both frequencies and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 1

Build the model from amplitude, period, and midline • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

What is the amplitude of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

1427

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the midline of y = 12 + 10sin(πt/15)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

1428

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

True or false?

The period of $y = \sin(Bt)$ is $B/(2\pi)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

1429

QUESTION ID

S39-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is $2\pi/B$ (for $B > 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The period of $y = \sin(\pi t/15)$ is

A. 30

B. 15

C. $\pi/15$

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

1430

QUESTION ID

S39-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 30

GRADING NOTE $2\pi/(\pi/15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

Write a model with amplitude 4, period 2/3, and midline 0.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

1431

QUESTION ID

S39-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4sin(3πt)

GRADING NOTE Any equivalent form (cos also accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 2

Evaluate a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 12 + 10sin(π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

1433

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 12 + 10sin(3π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

1434

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

True or false?

$-4\cos(0) = 0.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

1435

QUESTION ID

S39-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE cos 0 = 1, so −4cos 0 = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

100 + 20sin(π/2) equals

A. 100

B. 20

C. 120

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

1436

QUESTION ID

S39-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 120

GRADING NOTE sin(π/2) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q4**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 12 + 3sin(0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

1437

QUESTION ID

S39-B2-Q5

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 3

Convert rates between units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

A cycle lasts 0.8 s. How many cycles are in 60 s?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

1439

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

75

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

1.5 cycles per second is how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

1440

QUESTION ID

S39-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

True or false?

A period of 30 s means 30 revolutions per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

1441

QUESTION ID

S39-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 60/30 = 2 revolutions per minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

440 cycles per second is how many cycles per millisecond?

A. 4.4

B. 44

C. 440

D. 0.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

1442

QUESTION ID

S39-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.44

GRADING NOTE 1000 ms per second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A tide has period 12 hours. How many cycles in 3 days?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

1443

QUESTION ID

S39-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 1

Predict how changing a parameter changes the situation • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel rider's height is h(t) = 12 + 10sin(πt/15) m after t seconds. The wheel is replaced by one of the same size that turns twice as fast. Rewrite h(t).

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

1445

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12 + 10sin(2πt/15)

GRADING NOTE Any equivalent form. Period 15 s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

Daylight in one city is D(t) = 12 + 3sin(2π(t − 80)/365) hours. A city farther from the equator has daylight between 8 and 16 hours with the same period and timing. Which parameter changes, and to what?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

1446

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The amplitude changes to 4

GRADING NOTE Form required: parameter and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Changing the midline changes the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

1447

QUESTION ID

S39-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The midline only shifts the graph vertically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

Multiplying B by 3 makes the period

A. 3 units longer

B. unchanged

C. three times as long

D. one third as long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

1448

QUESTION ID

S39-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. one third as long

GRADING NOTE 2π/(3B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A heart rate rises from 75 to 100 beats per minute. What is the new period, in seconds?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

1449

QUESTION ID
S39-F1-Q5

TYPE
Short answer

CORRECT ANSWER
0.6 s

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 2

Interpret periodic features from graphs and tables • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A table of a tide's depth shows maxima at t = 2 and t = 14 hours. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5

1451

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12 hours

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5

A graph of a temperature model has a high of 30 and a low of 10. What are the midline and amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

1452

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Midline 20; amplitude 10

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

True or false?

Consecutive minima of a periodic function are one period apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

1453

QUESTION ID

S39-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Repeating pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

On which interval does $h(t) = 12 + 10\sin(\pi t/15)$ increase?

A. $0 < t < 15$

B. $0 < t < 7.5$

C. $15 < t < 30$

D. $7.5 < t < 22.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

1454

QUESTION ID

S39-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 < t < 7.5

GRADING NOTE Rises to the maximum at t = 7.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

Explain how to read the frequency of a periodic graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

1455

QUESTION ID

S39-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Find the length of one cycle (the period) and take its reciprocal (cycles per unit).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Prove the Pythagorean identity

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	Coordinates of the point at angle θ	**HSF-TF.A.2**
BACKWARD 3	Expand and simplify algebraic squares	**HSA-SSE.A.2**
FORWARD 1	Prove identities derived from the Pythagorean identity	**HSF-TF.C.8**
FORWARD 2	Use the angle-addition formulas	**HSF-TF.C.9**

SET 40 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Which argument proves $\sin^2\theta + \cos^2\theta = 1$ for every angle θ, using the unit circle?

A. On the unit circle $\sin\theta + \cos\theta = 1$ for every angle θ, because the coordinates of a point add to the radius; squaring both sides gives $\sin^2\theta + \cos^2\theta = 1$.

B. The point at angle θ is $(\sin\theta, \cos\theta)$, so $\sin\theta + \cos\theta = 1$.

C. For $\theta = 30°$: $(1/2)^2 + (\sqrt{3}/2)^2 = 1$, so the identity holds.

D. The point at angle θ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

1458

QUESTION ID

S40-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. The point at angle θ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.

GRADING NOTE The coordinates of the point satisfy the circle equation. The others check one angle or rely on the false statement $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

A right triangle has legs a and b, hypotenuse c, and acute angle θ opposite a. Which argument proves $\sin^2\theta + \cos^2\theta = 1$?

A. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a + b)^2/c^2$, and $(a + b)^2 = c^2$ by the Pythagorean Theorem, so the sum is 1.

B. $\sin\theta = b/c$ and $\cos\theta = a/c$, so $\sin\theta + \cos\theta = (a + b)/c = 1$.

C. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.

D. In a 3-4-5 triangle, $(3/5)^2 + (4/5)^2 = 1$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

1459

QUESTION ID

S40-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.

GRADING NOTE Both ratios and $a^2 + b^2 = c^2$. The others swap the ratios and use $a + b = c$, square a sum incorrectly, or check one triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

Which argument proves $1 + \tan^2\theta = 1/\cos^2\theta$ wherever $\cos\theta \neq 0$?

A. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$ to get $1 + \cos^2\theta/\sin^2\theta = 1/\sin^2\theta$; since $\cos^2\theta/\sin^2\theta = \tan^2\theta$, this gives $1 + \tan^2\theta = 1/\cos^2\theta$.

B. For $\theta = 0$: $1 + 0 = 1/1$, so the identity holds.

C. Square $\tan\theta = \sin\theta/\cos\theta$ to get $\tan^2\theta = \sin^2\theta/\cos^2\theta = 1/\cos^2\theta$.

D. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

1460

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.

GRADING NOTE Divide the Pythagorean identity by $\cos^2\theta$. The others divide by $\sin^2\theta$ and then treat $\cos^2\theta/\sin^2\theta$ as $\tan^2\theta$ (it is $1/\tan^2\theta$), replace $\sin^2\theta$ by 1, or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

θ is in quadrant II, and its point on the unit circle is (x, y) with $x < 0 < y$. Which argument proves $\sin^2\theta + \cos^2\theta = 1$ with a reference triangle?

A. The reference triangle has legs x and y, so $x + y = 1$, which means $\cos\theta + \sin\theta = 1$.

B. The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.

C. The legs of the reference triangle are x and y, and x is negative in quadrant II, so $x^2 + y^2 = -1$ there, which means $\cos^2\theta + \sin^2\theta = -1$ instead of 1.

D. The identity was proved for acute angles, so it is true in quadrant II with no further work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

1461

QUESTION ID

S40-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.

GRADING NOTE Side lengths are $|x|$ and $|y|$; squares remove the signs. The others add the legs, mishandle the negative coordinate, or skip the argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$?

A. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.

B. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta = 1$, so the term $2\sin\theta\cos\theta$ is extra.

C. For $\theta = 0$: $(0 + 1)^2 = 1 + 0$, so the identity holds.

D. $(\sin\theta + \cos\theta)^2 = 1^2 = 1$ because $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

1462

QUESTION ID
S40-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.

GRADING NOTE Expand the square, then use the Pythagorean identity. The others drop the middle term, check one angle, or use the false statement $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 3 and 4. Find the hypotenuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

1464

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

A right triangle has hypotenuse 1 and a leg 0.6. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

1465

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, $a^2 + b^2 = c^2$ where c is the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

1466

QUESTION ID

S40-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

Legs 5 and 12 give hypotenuse

A. 60

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

1467

QUESTION ID

S40-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13

GRADING NOTE √169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

Legs a and b and hypotenuse c: write c^2 in terms of a and b.

c^2 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

1468

QUESTION ID

S40-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 2

Coordinates of the point at angle θ • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give the coordinates of the point on the unit circle at angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

1470

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(cos θ, sin θ)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3) and sin(2π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

1471

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2 and √3/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

True or false?

sin θ is the x-coordinate of the point at angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

1472

QUESTION ID

S40-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

The point at angle π/4 is

A. (1/2, √3/2)

B. (1, 1)

C. (√3/2, 1/2)

D. (√2/2, √2/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

1473

QUESTION ID

S40-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (√2/2, √2/2)

GRADING NOTE 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the radius of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

1474

QUESTION ID

S40-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 3

Expand and simplify algebraic squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

1476

QUESTION ID

S40-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$a^2 + 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(a/c)^2 + (b/c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

1477

QUESTION ID

S40-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$(a^2 + b^2)/c^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

True or false?

$\sin^2\theta$ means $\sin(\theta^2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

1478

QUESTION ID

S40-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin^2\theta$ means $(\sin\theta)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

Dividing $x^2 + y^2 = 1$ by x^2 gives

A. $1 + y^2/x^2 = 1/x^2$

B. $1 + y^2 = 1/x$

C. $y^2 = 1$

D. $x^2 + y^2 = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

1479

QUESTION ID

S40-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $1 + y^2/x^2 = 1/x^2$

GRADING NOTE Divide each term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(s + c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

1480

QUESTION ID

S40-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$s^2 + 2sc + c^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 1

Prove identities derived from the Pythagorean identity • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$?

A. Divide both sides by $\sin^2\theta - \cos^2\theta$ to get $\sin^2\theta + \cos^2\theta = 1$, which is the Pythagorean identity, so the identity is true.

B. $\sin^4\theta - \cos^4\theta = (\sin\theta - \cos\theta)^4$, which equals $\sin^2\theta - \cos^2\theta$.

C. For $\theta = \pi/4$ both sides are 0, so the identity holds.

D. $\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

1482

QUESTION ID

S40-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D.
$\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta.$

GRADING NOTE Factor as a difference of squares and use the Pythagorean identity. The others assume the identity (and divide by an expression that can be 0), use the false rule $a^4 - b^4 = (a - b)^4$, or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

Which argument proves $(1 - \cos\theta)(1 + \cos\theta) = \sin^2\theta$?

A. For $\theta = \pi/2$: $(1 - 0)(1 + 0) = 1 = \sin^2(\pi/2)$, so the identity holds.

B. $(1 - \cos\theta)(1 + \cos\theta) = 1 - 2\cos\theta + \cos^2\theta$, and by the Pythagorean identity this equals $\sin^2\theta$.

C. $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.

D. $(1 - \cos\theta)(1 + \cos\theta) = 1 + \cos^2\theta$, which equals $\sin^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

1483

QUESTION ID

S40-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.

GRADING NOTE Multiply (difference of squares), then use the identity. The others multiply incorrectly or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin \theta + \cos \theta = 1$ for every θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

1484

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\theta = \pi/4$ gives $\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equals $1 - \sin^2\theta$?

A. $\tan^2\theta$

B. $1 - \cos^2\theta$

C. $\cos\theta$

D. $\cos^2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

1485

QUESTION ID
S40-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $\cos^2\theta$

GRADING NOTE Rearranged identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$?

A. Replace $\sin^2\theta$ with $1 - \cos^2\theta$:
$\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.

B. Add $\sin^2\theta$ to both sides to get $\cos^2\theta = 2\cos^2\theta - 1 + \sin^2\theta$, which is always true.

C. For $\theta = 0$: $1 - 0 = 2 - 1$, so the identity holds.

D. Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - 1 - \cos^2\theta = -1$, so the identity is false for every θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

1486

QUESTION ID
S40-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.

GRADING NOTE Substitute $\sin^2\theta = 1 - \cos^2\theta$ and subtract the whole expression. The others drop the parentheses, check one angle, or end with an unproved statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 2

Use the angle-addition formulas • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

Use sin(a + b) = sin a cos b + cos a sin b to find sin(75°) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5

1488

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(√6 + √2)/4

GRADING NOTE Any equivalent form. a = 45°, b = 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5

Use cos(a + b) = cos a cos b − sin a sin b with a = b = θ to write cos 2θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

1489

QUESTION ID
S40-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$\cos^2\theta - \sin^2\theta$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

True or false?

$\cos 2\theta = 1 - 2\sin^2\theta$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

1490

QUESTION ID

S40-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Use the Pythagorean identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

sin 2θ equals

A. $\sin^2\theta - \cos^2\theta$

B. 2sin θ

C. 2sin θ cos θ

D. sin θ + cos θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5

1491

QUESTION ID

S40-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2sin θ cos θ

GRADING NOTE a = b = θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$ to write $(\sin\theta + \cos\theta)^2$ in terms of $\sin 2\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

1492

QUESTION ID
S40-F2-Q5

TYPE
Short answer

CORRECT ANSWER
$1 + \sin 2\theta$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q5**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

SET 41 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Use the Pythagorean identity to find trigonometric values

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	Signs of sine, cosine, and tangent by quadrant	**HSF-TF.A.2**
BACKWARD 2	Square roots of fractions	**8.EE.A.2**
BACKWARD 3	Solve $x^2 = c$	**HSA-REI.B.4.b**
FORWARD 1	Find values when the quadrant is not given, and combine values	**HSF-TF.C.8**
FORWARD 2	Find the angles with inverse trigonometric functions	**HSF-TF.B.7**

SET 41 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 3/5 and θ is in quadrant II. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

1495

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

cos θ = −4/5, tan θ = −3/4

GRADING NOTE Form required: both exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

cos θ = −5/13 and θ is in quadrant III. Find sin θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

1496

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

sin θ = −12/13, tan θ = 12/5

GRADING NOTE Form required: both exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = −2 and θ is in quadrant IV. Find sin θ and cos θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

1497

QUESTION ID

S41-M-Q3

TYPE

Short answer

CORRECT ANSWER

sin θ = −2√5/5, cos θ = √5/5

GRADING NOTE Form required: both exact (−2/√5 and 1/√5 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

sin θ = −1/3 and θ is in quadrant IV. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

1498

QUESTION ID

S41-M-Q4

TYPE

Short answer

CORRECT ANSWER

cos θ = 2√2/3, tan θ = −√2/4

GRADING NOTE Form required: both exact (−1/(2√2) accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 0.6 and sin θ < 0. Find sin θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

1499

QUESTION ID

S41-M-Q5

TYPE

Short answer

CORRECT ANSWER

sin θ = −0.8, tan θ = −4/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 1

Signs of sine, cosine, and tangent by quadrant • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In quadrant II, which of sin, cos, tan are positive?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

1501

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

sin only

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrant III, what is the sign of tan θ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

1502

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In quadrant IV, $\cos\theta < 0$ and $\sin\theta > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

1503

QUESTION ID

S41-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Quadrant IV has $x > 0$ and $y < 0$, so $\cos\theta > 0$ and $\sin\theta < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $\sin\theta < 0$ and $\cos\theta > 0$, θ is in quadrant

A. I

B. II

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

1504

QUESTION ID

S41-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. IV

GRADING NOTE (+, −).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

If tan θ < 0 and sin θ > 0, which quadrant is θ in?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

1505

QUESTION ID

S41-B1-Q5

TYPE

Short answer

CORRECT ANSWER

II

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 2

Square roots of fractions • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find √(16/25).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

1507

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find √(8/9) in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

1508

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2√2/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

√(144/169) = 12/169.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

1509

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Take the root of both: √144/√169 = 12/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

√(1/5) equals

A. 5

B. √5

C. 1/25

D. √5/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

1510

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √5/5

GRADING NOTE Rationalize.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find √0.64.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

1511

QUESTION ID

S41-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 3

Solve $x^2 = c$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 16/25$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

1513

QUESTION ID

S41-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4/5 or −4/5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 1 - 9/25$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

1514

QUESTION ID

S41-B3-Q2

TYPE

Short answer

CORRECT ANSWER

±4/5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 = 0.36$ has only the solution 0.6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q3/5

1515

QUESTION ID

S41-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Also −0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 + (3/5)^2 = 1$ gives x =

A. ±2/5

B. 2/5

C. 4/5

D. ±4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

1516

QUESTION ID

S41-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ±4/5

GRADING NOTE $x^2 = 16/25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5x^2 = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

1517

QUESTION ID

S41-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$\pm\sqrt{5}/5$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 1

Find values when the quadrant is not given, and combine values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 3/5. List both possible values of cos θ and the quadrants they go with.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

1519

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4/5 (quadrant I) or −4/5 (quadrant II)

GRADING NOTE Form required: both with quadrants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = −5/13 and θ is in quadrant III. Find sin θ · cos θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

1520

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60/169

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Knowing tan θ = 1 determines sin θ uniquely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

1521

QUESTION ID

S41-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE θ can be in quadrant I or III, so sin θ = ±√2/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $\sin\theta = 0.8$ and $\tan\theta < 0$, then $\cos\theta =$

A. -0.8

B. 0.2

C. 0.6

D. -0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

1522

QUESTION ID

S41-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −0.6

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $1 + \tan^2\theta = 1/\cos^2\theta$ to find $\cos\theta$ if $\tan\theta = 3$ and θ is in quadrant III.

$\cos\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

1523

QUESTION ID
S41-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$-\sqrt{10}/10$

GRADING NOTE Any equivalent form ($-1/\sqrt{10}$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 2

Find the angles with inverse trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

$\sin\theta = 3/5$ with $0 \le \theta < 2\pi$. Find both values of θ to the nearest thousandth. A calculator may be used.

$\theta \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5

1525

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.644 and 2.498

GRADING NOTE Form required: both. arcsin(0.6) and π − arcsin(0.6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = −5/13 and θ is in quadrant III. Find θ to the nearest thousandth (0 ≤ θ < 2π). A calculator may be used.

θ ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5

1526

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4.32

GRADING NOTE Accept 4.318 to 4.320. 2π − arccos(−5/13).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5

True or false?

arcsin gives only angles between −π/2 and π/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

1527

QUESTION ID

S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

The quadrant II angle with sin θ = 3/5 is

A. π + arcsin(3/5)

B. π − arcsin(3/5)

C. arcsin(3/5)

D. −arcsin(3/5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5

1528

QUESTION ID

S41-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. π − arcsin(3/5)

GRADING NOTE Reflect over the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5

Explain why the identity finds cos θ but the inverse function is needed to find θ itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

1529

QUESTION ID

S41-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The identity relates values to each other; finding the angle from a value requires undoing sine (with arcsin and the quadrant).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of polynomial and rational models

MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret key features of linear and quadratic models	**HSF-IF.B.4**
BACKWARD 2	Describe increasing and decreasing behavior	**8.F.B.5**
BACKWARD 3	Zeros and sign from factors	**HSA-APR.B.3**
FORWARD 1	Sketch a model from a verbal description of key features	**HSF-IF.B.4**
FORWARD 2	Compare key features across representations	**HSF-IF.C.9**

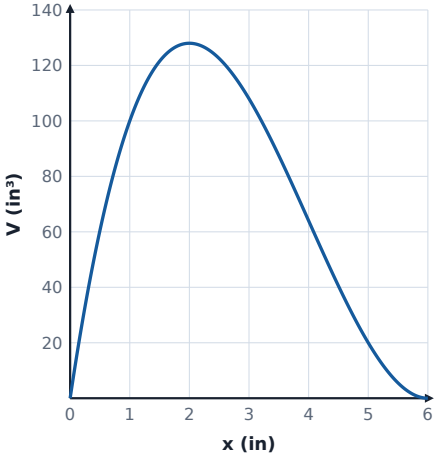
SET 42 • HSF-IF.B.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 12-in square sheet by cutting squares of side x from the corners. Its volume is $V(x) = x(12 - 2x)^2$. Use the graph to give the x that maximizes V, the maximum volume, and the interval where V increases.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

1532

QUESTION ID
S42-M-Q1

TYPE
Short answer

CORRECT ANSWER
x = 2 in; V = 128 in³; increases for 0 < x < 2

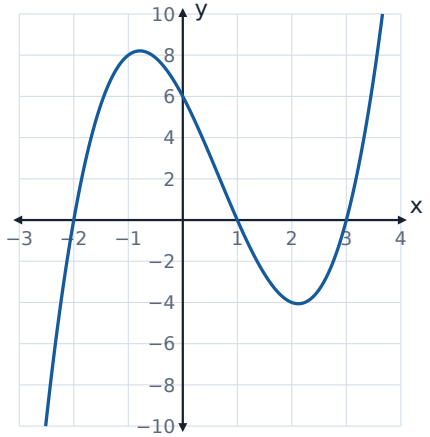
GRADING NOTE Form required: all three. Values read from the graph within ±0.2 in and ±5 in³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = (x + 2)(x - 1)(x - 3)$ is shown. Give the intervals where $f(x) > 0$ and the intervals where $f(x) < 0$.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

1533

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

$f > 0$ on $-2 < x < 1$ and $x > 3$; $f < 0$ on $x < -2$ and $1 < x < 3$

GRADING NOTE Form required: all intervals.

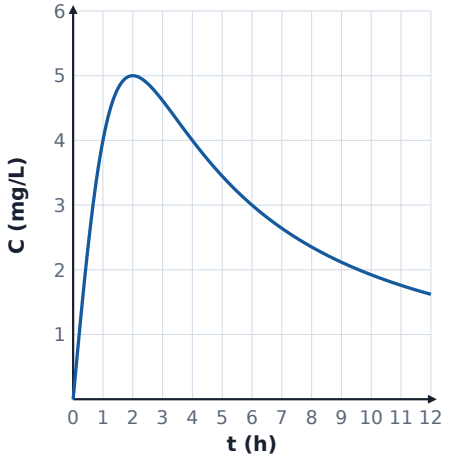
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Use the graph to give the maximum concentration, when it occurs, and the end behavior. Interpret the end behavior.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

1534

QUESTION ID
S42-M-Q3

TYPE
Short answer

CORRECT ANSWER
Maximum 5 mg/L at t = 2 h; C → 0 as t grows: the drug is eventually eliminated

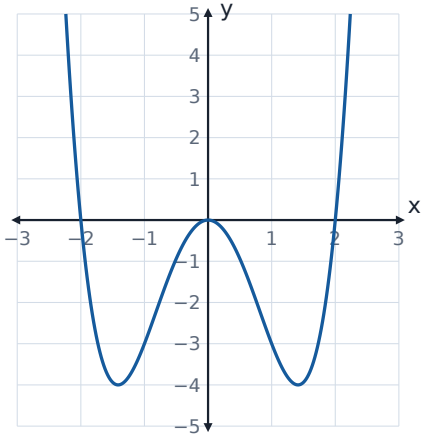
GRADING NOTE Form required: maximum, time, end behavior and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = x^4 - 4x^2$ is shown. Give its x-intercepts and its relative maximum and minimum points.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

1535

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

x-intercepts −2, 0, 2; relative maximum (0, 0); relative minima at about (−1.41, −4) and (1.41, −4)

GRADING NOTE Form required: intercepts and all extrema (x within ±0.1).

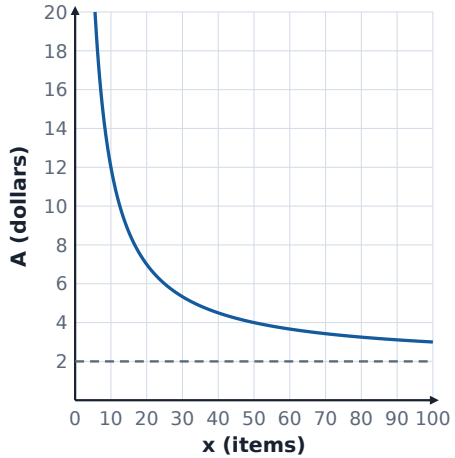
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The average cost of making x items is A(x) = (100 + 2x)/x dollars. Use the graph to describe the end behavior and interpret it.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

1536

QUESTION ID
S42-M-Q5

TYPE
Short answer

CORRECT ANSWER
As x grows, A(x) approaches $2 (horizontal asymptote y = 2): the average cost approaches the $2 cost of each item as the $100 fixed cost is spread out

GRADING NOTE Form required: the asymptote value and the interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 1

Interpret key features of linear and quadratic models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 32t + 4$. When is the maximum height, and what is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

1538

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

t = 1; 20 ft

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What does the y-intercept of C = 25 + 3x mean if C is the cost of x rides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

1539

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A fixed cost of $25

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

True or false?

A parabola that opens down has a minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q3/5

1540

QUESTION ID

S42-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its vertex is the highest point: a maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 32t + 4$ increases for

A. all t

B. $0 < t < 1$

C. $t > 1$

D. $1 < t < 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

1541

QUESTION ID

S42-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 < t < 1

GRADING NOTE Before the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Where is $y = (x - 2)(x - 6)$ negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

1542

QUESTION ID

S42-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$2 < x < 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 2

Describe increasing and decreasing behavior • Grade 8 • Functions • 8.F.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A graph rises from x = 0 to x = 2 and then falls.
Describe it in words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

1544

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Increasing on 0 < x < 2, decreasing after 2

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

Sketch-free: describe the graph of the concentration that rises then levels toward 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

1545

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Increases to a peak, then decreases and approaches 0

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

True or false?

A highest point between rising and falling parts is a relative maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5

1546

QUESTION ID

S42-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5

A function that falls on $1 < x < 3$ is

A. constant there

B. decreasing there

C. undefined there

D. increasing there

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

1547

QUESTION ID

S42-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. decreasing there

GRADING NOTE Falling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

What is a relative minimum?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

1548

QUESTION ID

S42-B2-Q5

TYPE

Short answer

CORRECT ANSWER

A point lower than all nearby points of the graph

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 3

Zeros and sign from factors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

1550

QUESTION ID

S42-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−2, 1, 3

GRADING NOTE Form required: all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

List the zeros of $x^4 - 4x^2 = x^2(x - 2)(x + 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5

1551

QUESTION ID

S42-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0 (2), 2 (1), −2 (1)

GRADING NOTE Form required: all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5

True or false?

$V(x) = x(12 - 2x)^2$ has zeros 0 and 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

1552

QUESTION ID

S42-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $12 - 2x = 0$ at $x = 6$, so the zeros are 0 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$20t/(t^2 + 4) = 0$ when

A. never

B. $t = -2$

C. $t = 2$

D. $t = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q4/5

1553

QUESTION ID

S42-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. t = 0

GRADING NOTE Numerator 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $(x + 2)(x - 1)(x - 3)$ positive or negative at $x = 2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5

1554

QUESTION ID

S42-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Negative: (4)(1)(−1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSF-IF.B.4

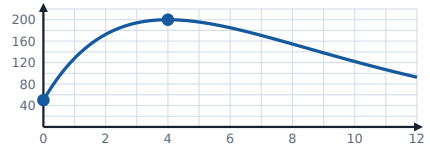
FORWARD 1

Sketch a model from a verbal description of key features • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

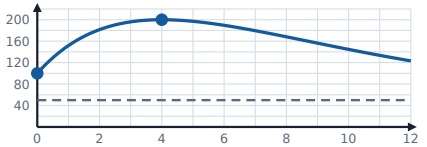
ANSWER KEY Reduced copy of the question page, with its answer

Which graph could show a population that is 50 at t = 0, rises to a maximum of 200 at t = 4, and then decreases toward a level of 100 as t grows?

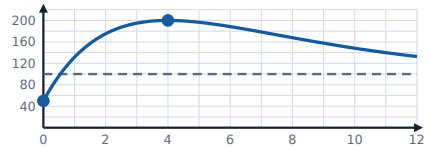
A



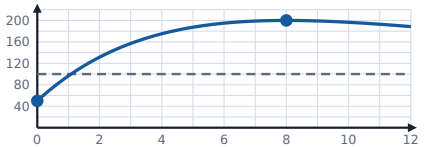
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

1556

QUESTION ID

S42-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE The graph must start at (0, 50), peak at (4, 200), and level off toward y = 100. The others level off toward 0, start at 100 and level off toward 50, or peak at t = 8.

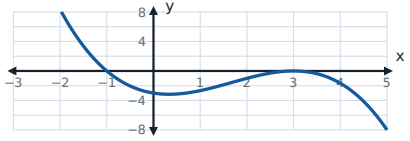
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

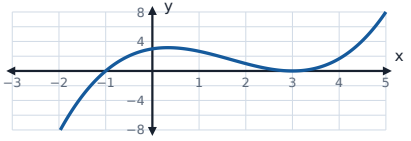
ANSWER KEY Reduced copy of the question page, with its answer

Which graph could show a cubic with zeros at −1 and 3 only, a negative y-intercept, falling to the right, and touching the x-axis at 3?

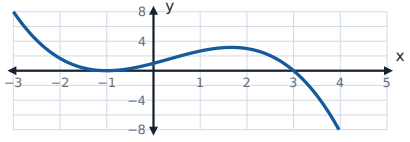
A



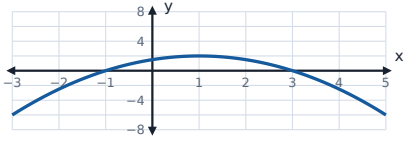
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

1557

QUESTION ID

S42-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Touching at 3 and falling to the right means the graph is below the axis on both sides of 3, so it is negative between −1 and 3 and the y-intercept is negative; for example $y = -(x + 1)(x - 3)^2/3$. The others rise to the right, touch at −1 instead of 3, or are a parabola (not a cubic).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

A graph can have a relative maximum that is not the absolute maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

1558

QUESTION ID
S42-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Cubics, for example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

A model rises to a peak and approaches 0 as t grows. Which formula fits?

A. $20t/(t^2 + 4)$

B. $20t$

C. t^2

D. $20 - t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5

1559

QUESTION ID
S42-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $20t/(t^2 + 4)$

GRADING NOTE Rational with a peak.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a polynomial with zeros 0 and 6 and a relative maximum in between, whose values are positive between the zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

1560

QUESTION ID

S42-F1-Q5

TYPE

Short answer

CORRECT ANSWER

For example

$f(x) = x(6 - x)$

GRADING NOTE Any valid example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

FORWARD 2

Compare key features across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x(12 - 2x)^2$ and g is a quadratic with vertex (3, 120). Which has the greater maximum on $0 < x < 6$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5

1562

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

f: 128 > 120

GRADING NOTE Form required: answer and values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$C(t) = 20t/(t^2 + 4)$ and a table of $D(t)$ gives $D(1) = 3$, $D(2) = 6$, $D(3) = 4$. Which has the greater peak among these values, and when?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

1563

QUESTION ID
S42-F2-Q2

TYPE
Short answer

CORRECT ANSWER
D: 6 at t = 2 (C peaks at 5)

GRADING NOTE Form required: function, value, time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

True or false?

Two functions with the same zeros have the same relative extrema.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5

1564

QUESTION ID

S42-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Scale changes the extrema values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5

Which has a horizontal asymptote?

A. $(x + 2)(x - 1)(x - 3)$

B. $x^4 - 4x^2$

C. $x(12 - 2x)^2$

D. $(100 + 2x)/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

1565

QUESTION ID

S42-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (100 + 2x)/x

GRADING NOTE Rational with equal degrees.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to compare end behavior of a polynomial given by formula and a rational function given by a graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

1566

QUESTION ID
S42-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Use the leading term of the polynomial and read the rational graph's asymptote for large |x|.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret periodic models from graphs and tables

MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret maximums and minimums on a graph	**HSF-IF.B.4**
BACKWARD 2	Read values from a table	**6.EE.C.9**
BACKWARD 3	Amplitude, period, and midline of a sinusoidal model	**HSF-TF.B.5**
FORWARD 1	Predict with periodicity	**HSF-IF.B.4**
FORWARD 2	Write a model from periodic features	**HSF-TF.B.5**

SET 43 • HSF-IF.B.4

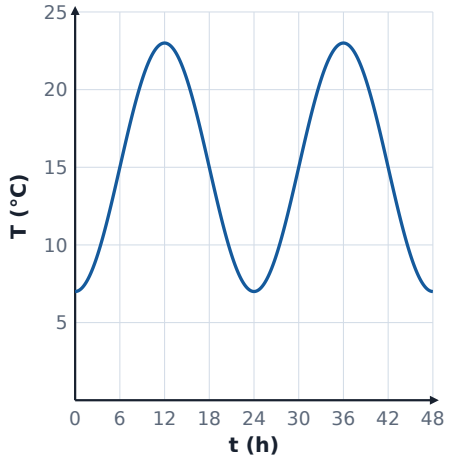
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a city's temperature T (°C) t hours after midnight on two days. Give the period, the maximum temperature, and the first time it occurs.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

1569

QUESTION ID
S43-M-Q1

TYPE
Short answer

CORRECT ANSWER
Period 24 h; maximum 23°C at t = 12 (noon)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

The table shows the water depth d (ft) at a harbor every 2 hours. Give the period, the depth at low tide, and the first time of low tide.

t (h)	0	2	4	6	8	10	12	14	16	18
d (ft)	9	7	3	1	3	7	9	7	3	1

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

1570

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

Period 12 h; low tide 1 ft at t = 6 h

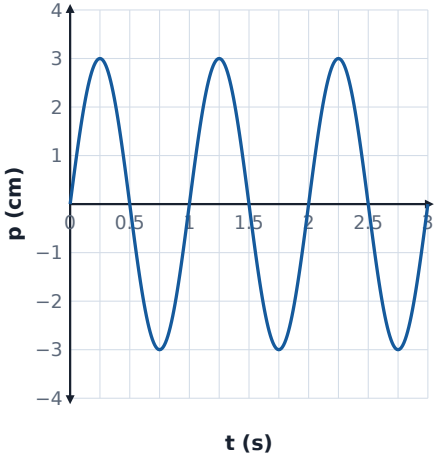
GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

A piston's position p (cm) is graphed for $0 \le t \le 3$ seconds. Give the amplitude, the period, and the intervals of the first cycle where p increases.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

1571

QUESTION ID
S43-M-Q3

TYPE
Short answer

CORRECT ANSWER
Amplitude 3 cm; period 1 s; increasing on $0 < t < 0.25$ and $0.75 < t < 1$

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows hours of daylight D in month m (m = 0 is January). Give the maximum, the minimum, the period, and the months in which daylight increases.

m	0	1	2	3	4	5	6	7	8	9	10	11	12
D (h)	9	9.4	10.5	12	13.5	14.6	15	14.6	13.5	12	10.5	9.4	9

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

1572

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

Maximum 15 h (m = 6), minimum 9 h (m = 0 and 12), period 12 months, increasing from m = 0 to m = 6

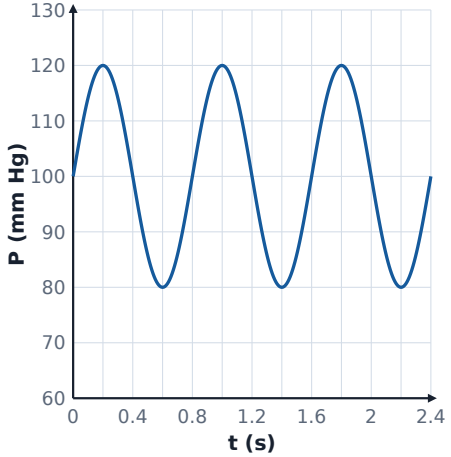
GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows blood pressure P (mm Hg) over 2.4 seconds. Give the maximum and minimum pressures and the number of heartbeats per minute.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

1573

QUESTION ID
S43-M-Q5

TYPE
Short answer

CORRECT ANSWER
Maximum 120, minimum 80; period 0.8 s, so 75 beats per minute

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 1

Interpret maximums and minimums on a graph • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A graph of height peaks at 45 m. What does the peak mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

1575

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The greatest height reached is 45 m

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

A graph is lowest at (6, 1). Interpret it if it shows depth (ft) against time (h).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

1576

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The least depth, 1 ft, at 6 hours

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

True or false?

The minimum value of a graph is its lowest y-value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

1577

QUESTION ID

S43-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

A relative maximum on a graph is

A. the last point

B. any x-intercept

C. a point higher than nearby points

D. the y-intercept

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

1578

QUESTION ID

S43-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a point higher than nearby points

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

A graph rises on $0 < x < 2$. What does this mean for a temperature graph?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

1579

QUESTION ID

S43-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The temperature is increasing during that time

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 2

Read values from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. What is d when t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

1581

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 ft

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. At which listed times is d = 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

1582

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0 and 12

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

True or false?

A repeated pattern in a table suggests a periodic relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

1583

QUESTION ID

S43-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same values recur.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Hours of daylight in months m = 0, 3, 6, 12 are 9, 12, 15, 9. Daylight of 15 h occurs at

A. m = 12

B. m = 6

C. m = 0

D. m = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

1584

QUESTION ID

S43-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. m = 6

GRADING NOTE Read.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depth is 7 ft at t = 2 h and 3 ft at t = 4 h. By how much does the depth d change from t = 2 to t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

1585

QUESTION ID

S43-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−4 ft

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 3

Amplitude, period, and midline of a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the amplitude and midline of $T = 15 - 8\cos(\pi t/12)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

1587

QUESTION ID

S43-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8 and y = 15

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of $p = 3\sin(2\pi t)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

1588

QUESTION ID

S43-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

True or false?

A wave between 80 and 120 has amplitude 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

1589

QUESTION ID

S43-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Amplitude is half the difference: (120 − 80)/2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A wave with maximum 9 and minimum 1 has midline

A. y = 8

B. y = 5

C. y = 10

D. y = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

1590

QUESTION ID

S43-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 5

GRADING NOTE (9 + 1)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A period of 0.8 s means how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

1591

QUESTION ID

S43-B3-Q5

TYPE

Short answer

CORRECT ANSWER

75

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 1

Predict with periodicity • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

Use the temperature model $T(t) = 15 - 8\cos(\pi t/12)$ to find T at t = 100 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

1593

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

11°C

GRADING NOTE Any equivalent form. $100 = 4 \cdot 24 + 4$; $T(4) = 15 - 8\cos(\pi/3) = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depths repeat every 12 hours, and the depth is 1 ft at t = 6 h. Predict the depth at t = 30 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5

1594

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1 ft

GRADING NOTE Any equivalent form. 30 = 24 + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

If f has period 12, then f(50) = f(10).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

1595

QUESTION ID

S43-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $50 = 4 \cdot 12 + 2$, so f(50) = f(2); f(10) need not equal it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A piston's position is $p = 3\sin(2\pi t)$ cm, with period 1 s. At $t = 7.25$ s, p is

A. 0

B. 3

C. −3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

1596

QUESTION ID

S43-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE Same as $t = 0.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Daylight follows a 12-month cycle with a maximum of 15 h at m = 6 (m = 0 is January). In which month after m = 12 does daylight next reach 15 h?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

1597

QUESTION ID

S43-F1-Q5

TYPE

Short answer

CORRECT ANSWER

m = 18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 2

Write a model from periodic features • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depth varies between 9 ft (at t = 0) and 1 ft (at t = 6 h) and repeats every 12 hours. Write a cosine model for the depth d.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

1599

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 4cos(πt/6)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Daylight varies between 9 h (month m = 0) and 15 h (m = 6) and repeats every 12 months. Write a model for daylight D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q2/5

1600

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$12 - 3\cos(\pi m/6)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q2/5

True or false?

A temperature that is 7°C at t = 0, 23°C at t = 12 h, and repeats every 24 hours is modeled by $15 - 8\cos(\pi t/12)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q3/5

1601

QUESTION ID

S43-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Minimum 7 at t = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q3/5

A blood pressure that starts at 100 mm Hg and rises, varies between 80 and 120, and repeats every 0.8 s is modeled by

A. 100 + 20sin(0.8t)

B. 20 + 100sin(2.5πt)

C. 120sin(2.5πt)

D. 100 + 20sin(2.5πt)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q4/5

1602

QUESTION ID
S43-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D.
100 + 20sin(2.5πt)

GRADING NOTE Period 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A piston's position starts at 0 cm and increases, varies between −3 and 3 cm, and repeats every 1 s. Write a model for p.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q5/5

1603

QUESTION ID

S43-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$3\sin(2\pi t)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

State domains of radical, rational, logarithmic, and polynomial models

MAIN	Algebra II main question	**HSF-IF.B.5**
BACKWARD 1	Domains of linear, quadratic, and exponential models	**HSF-IF.B.5**
BACKWARD 2	Division by zero	**7.NS.A.2.b**
BACKWARD 3	Solve linear inequalities	**HSA-REI.B.3**
FORWARD 1	Domains defined by nonlinear conditions	**HSF-IF.B.5**
FORWARD 2	Domains and ranges of inverse functions	**HSF-BF.B.4.a**

SET 44 • HSF-IF.B.5

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A car's speed is $s(d) = \sqrt{(22.5d)}$ mph for a skid of d feet. Give the domain of the function and the domain that makes sense in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q1/5

1606

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

Function: $d \geq 0$; context: $d \geq 0$ (real skid lengths, up to a realistic maximum)

GRADING NOTE Form required: both. $d < 0$ gives no real root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q1/5

The average cost of x shirts is A(x) = (500 + 4x)/x. Give the domain of the function and the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q2/5

1607

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

Function: all real x ≠ 0; context: whole numbers x ≥ 1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q2/5

pH = −log[H⁺], where [H⁺] is the hydrogen ion concentration. What is the domain?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q3/5

1608

QUESTION ID

S44-M-Q3

TYPE

Short answer

CORRECT ANSWER

$[H^+] > 0$

GRADING NOTE Form required: the inequality. Logarithms need positive inputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A box from a 12-in sheet has volume $V(x) = x(12 - 2x)^2$. Give the domain of the polynomial and the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q4/5

1609

QUESTION ID
S44-M-Q4

TYPE
Short answer

CORRECT ANSWER
Polynomial: all real numbers; context: $0 < x < 6$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $f(x) = \sqrt{(x - 2)}/(x - 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q5/5

1610

QUESTION ID

S44-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 2$ and $x \neq 5$

GRADING NOTE Form required: both conditions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 1

Domains of linear, quadratic, and exponential models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ models height until the ball lands. Give the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q1/5

1612

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$0 \le t \le 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

C(n) = 2n + 5 is the cost of n tickets. Give the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q2/5

1613

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Whole numbers

n ≥ 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q2/5

True or false?

The domain of $f(x) = 2^x$ is $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q3/5

1614

QUESTION ID

S44-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2^x is defined for every real x; $y > 0$ is its range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A model for the number of people uses a domain of

A. fractions only

B. all real numbers

C. whole numbers

D. negative numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q4/5

1615

QUESTION ID

S44-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. whole numbers

GRADING NOTE Count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q4/5

Why might a model's domain be smaller than the function's domain?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q5/5

1616

QUESTION ID

S44-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The context restricts inputs (for example to nonnegative or whole numbers).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 2

Division by zero • Grade 7 • The Number System • 7.NS.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

For which x is 1/(x − 5) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q1/5

1618

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is (500 + 4x)/x undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q2/5

1619

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q2/5

True or false?

3/0 = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q3/5

1620

QUESTION ID

S44-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)/(x^2 - 9)$ is undefined at

A. $x = 0$

B. $x = 9$

C. $x = -1$

D. $x = \pm 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q4/5

1621

GRADING NOTE $x^2 = 9$.

QUESTION ID

S44-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x = \pm 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 0/5 defined? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q5/5

1622

QUESTION ID

S44-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes, 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 3

Solve linear inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x - 2 \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q1/5

1624

QUESTION ID

S44-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $12 - 2x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q2/5

1625

QUESTION ID

S44-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x < 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

22.5d ≥ 0 means d ≥ 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q3/5

1626

QUESTION ID

S44-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by a positive number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3 - x \geq 0$ means

A. $x \geq 3$

B. $x \leq 3$

C. $x \geq -3$

D. $x \leq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q4/5

1627

QUESTION ID

S44-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x \leq 3$

GRADING NOTE Add x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 8 > 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q5/5

1628

QUESTION ID

S44-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x > −4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 1

Domains defined by nonlinear conditions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $f(x) = \sqrt{(9 - x^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q1/5

1630

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$-3 \leq x \leq 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $g(x) = \log(x^2 - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q2/5

1631

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x < -2$ or $x > 2$

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $h(x) = 1/(x^2 - x - 6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q3/5

1632

QUESTION ID

S44-F1-Q3

TYPE

Short answer

CORRECT ANSWER

All real x except −2 and 3

GRADING NOTE Form required: both exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The domain of $\sqrt{(x^2)}$ is $x \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q4/5

1633

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 \geq 0$ for every x, so $\sqrt{(x^2)}$ is defined for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log(x - 1) + \sqrt{(5 - x)}$ is

A. $1 < x \leq 5$

B. $x \leq 5$

C. $1 \leq x \leq 5$

D. $x > 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q5/5

1634

QUESTION ID

S44-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $1 < x \leq 5$

GRADING NOTE Both conditions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 2

Domains and ranges of inverse functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{(x - 2)}$ has domain $x \geq 2$ and range $y \geq 0$. Write f^{-1} and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q1/5

1636

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = x^2 + 2$ with domain $x \geq 0$

GRADING NOTE Form required: formula and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$ has range $y > 0$. What is the domain of $f^{-1}(x) = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q2/5

1637

QUESTION ID
S44-F2-Q2

TYPE
Short answer

CORRECT ANSWER
x > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q2/5

True or false?

The domain of an inverse function is the range of the original function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q3/5

1638

QUESTION ID
S44-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Inputs and outputs swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 1/x + 3 (x ≠ 0) has an inverse with domain

A. x ≠ 0

B. x > 3

C. all real x

D. x ≠ 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q4/5

1639

QUESTION ID

S44-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. x ≠ 3

GRADING NOTE The range of f excludes 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the inverse of $s(d) = \sqrt{(22.5d)}$ needs the restriction $s \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q5/5

1640

QUESTION ID

S44-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The range of s is $s \geq 0$, so $d = s^2/22.5$ is the inverse only for $s \geq 0$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate and interpret average rates of change of nonlinear models

MAIN	Algebra II main question	**HSF-IF.B.6**
BACKWARD 1	Average rate of change of simpler functions	**HSF-IF.B.6**
BACKWARD 2	Slope between two points	**8.F.B.4**
BACKWARD 3	Evaluate nonlinear functions	**HSF-IF.A.2**
FORWARD 1	Compare average rates on different intervals	**HSF-IF.B.6**
FORWARD 2	Compare rates of change across representations	**HSF-IF.C.9**

SET 45 • HSF-IF.B.6

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

A culture has $N(t) = 200 \cdot 2^t$ bacteria after t hours. Find the average rate of change from t = 1 to t = 3 and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q1/5

1643

QUESTION ID
S45-M-Q1

TYPE
Short answer

CORRECT ANSWER
600 bacteria per hour: on average the culture grew by 600 bacteria each hour from hour 1 to hour 3

GRADING NOTE Form required: value with units and interpretation. (1600 − 400)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q1/5

A stone's height is $h(t) = 64 - 16t^2$ ft. Find the average rate of change from $t = 0.5$ to $t = 1.5$ s and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q2/5

1644

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

−32 ft/s: the stone fell an average of 32 feet per second

GRADING NOTE Form required: value with units and interpretation. (28 − 60)/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q2/5

The table shows a car's value. Find the average rate of change from t = 2 to t = 4 years and interpret it.

t (years)	0	1	2	3	4	5
V ($)	20000	17000	14450	12282.50	10440.06	8874.11

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q3/5

1645

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

About −$2,004.97 per year: the value dropped about $2,005 per year on average

GRADING NOTE Form required: value with units and interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q3/5

Speed from skid length is s(d) = √(24d) mph. Find the average rate of change from d = 24 to d = 96 ft and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q4/5

1646

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/3 mph per foot: on average each extra foot of skid adds 1/3 mph of speed

GRADING NOTE Form required: value with units and interpretation. (48 − 24)/72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Temperature is $T(t) = 15 - 8\cos(\pi t/12)$ °C after t hours. Find the average rate of change from t = 0 to t = 6 and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q5/5

1647

QUESTION ID
S45-M-Q5

TYPE
Short answer

CORRECT ANSWER
4/3 °C per hour: on average the temperature rose about 1.33°C per hour

GRADING NOTE Form required: value with units and interpretation. (15 − 7)/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 1

Average rate of change of simpler functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$. Find the average rate of change from $x = 1$ to $x = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q1/5

1649

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(2) = 10 and f(5) = 1. Find the average rate of change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q2/5

1650

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q2/5

True or false?

The average rate of change from a to b is (f(b) − f(a))/(b − a).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q3/5

1651

QUESTION ID

S45-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q3/5

The average rate of change of f(x) = 3x + 1 on any interval is

A. it varies

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q4/5

1652

QUESTION ID

S45-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the units of a rate if f is in dollars and x is in years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q5/5

1653

QUESTION ID

S45-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Dollars per year

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 2

Slope between two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope between (1, 400) and (3, 1600).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q1/5

1655

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

600

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope between (0.5, 60) and (1.5, 28).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q2/5

1656

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q2/5

True or false?

A negative slope means the output increases as the input increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q3/5

1657

QUESTION ID

S45-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A negative slope means the output decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The slope between (24, 24) and (96, 48) is

A. 72/24

B. 1/3

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q4/5

1658

QUESTION ID

S45-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/3

GRADING NOTE 24/72 = 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope between (0, 7) and (6, 15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q5/5

1659

QUESTION ID

S45-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 3

Evaluate nonlinear functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find N(3) for $N(t) = 200 \cdot 2^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q1/5

1661

QUESTION ID

S45-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1600

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find h(1.5) for $h(t) = 64 - 16t^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q2/5

1662

QUESTION ID

S45-B3-Q2

TYPE

Short answer

CORRECT ANSWER

28

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(24 \cdot 96)} = 2304$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q3/5

1663

QUESTION ID

S45-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $24 \cdot 96 = 2304$ and $\sqrt{2304} = 48$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

15 − 8cos(π/2) equals

A. 15

B. 7

C. 23

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q4/5

1664

QUESTION ID

S45-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 15

GRADING NOTE cos(π/2) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find T(0) for T(t) = 15 − 8cos(πt/12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q5/5

1665

QUESTION ID

S45-B3-Q5

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 1

Compare average rates on different intervals • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

For $N(t) = 200 \cdot 2^t$, compare the average rates on [0, 1] and [3, 4]. What does the comparison show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q1/5

1667

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

200 and 1600 bacteria per hour: the growth speeds up

GRADING NOTE Form required: both rates and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For s(d) = √(24d), compare the average rates on [0, 24] and [96, 150].

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q2/5

1668

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1 and 2/9 ≈ 0.22 mph per foot: the rate decreases

GRADING NOTE Form required: both and the trend.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q2/5

True or false?

For a function with period 24, the average rate of change over [0, 24] is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q3/5

1669

QUESTION ID

S45-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(24) = f(0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = x^3, the average rate on [−1, 1] is

A. 3

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q4/5

1670

QUESTION ID

S45-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE (1 − (−1))/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q4/5

Find an interval of length 6 on which $T(t) = 15 - 8\cos(\pi t/12)$ has average rate of change $-4/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q5/5

1671

QUESTION ID

S45-F1-Q5

TYPE

Short answer

CORRECT ANSWER

[12, 18]

GRADING NOTE Any valid interval (for example [12, 18]).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 2

Compare rates of change across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(t) = 200 \cdot 2^t$, and g is given by $g(1) = 500$, $g(3) = 1500$. Which has the greater average rate on [1, 3]?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q1/5

1673

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

f: 600 > 500

GRADING NOTE Form required: answer and both rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q1/5

A graph of a cooling liquid falls from 90°C at t = 0 to 50°C at t = 20 min. Compare with T(t) = 90 − 1.5t on the same interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q2/5

1674

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Graph: −2°C/min; formula: −1.5°C/min; the graph cools faster on average

GRADING NOTE Form required: both rates and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q2/5

True or false?

A function with a greater average rate on one interval must be greater everywhere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q3/5

1675

QUESTION ID

S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rates and values are different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q3/5

A table shows h(0) = 0, h(2) = 40; the formula $k(t) = 5t^2$ gives

A. no rate for k

B. a greater rate for k

C. a greater rate for h

D. equal average rates on [0, 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q4/5

1676

QUESTION ID

S45-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a greater rate for h

GRADING NOTE h: 40/2 = 20; k: (20 − 0)/2 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q4/5

Explain how to compare rates when one function is a graph and one is a formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q5/5

1677

QUESTION ID

S45-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Read the two endpoint values from the graph, evaluate the formula at the same inputs, and compute both quotients.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions with transformations

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table	**HSF-IF.C.7.b**
BACKWARD 2	Translations and reflections of graphs	**HSF-BF.B.3**
BACKWARD 3	Square and cube roots of numbers	**8.EE.A.2**
FORWARD 1	Graph radical functions with horizontal stretches and reflections	**HSF-IF.C.7.b**
FORWARD 2	Graph a radical function and its inverse	**HSF-BF.B.4.a**

SET 46 • HSF-IF.C.7.b

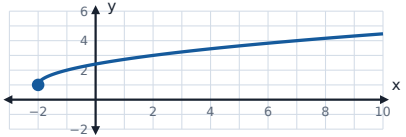
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

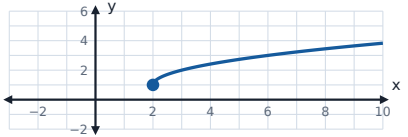
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \sqrt{(x - 2)} + 1$?

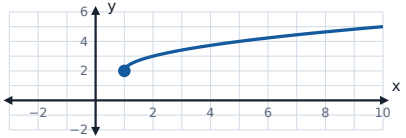
A



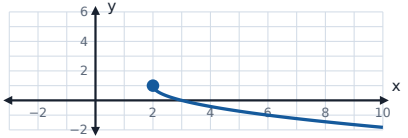
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q1/5

1680

QUESTION ID

S46-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE The graph starts at (2, 1) (right 2 and up 1 from $y = \sqrt{x}$) and rises through (3, 2) and (6, 3); domain $x \geq 2$, range $y \geq 1$. The others show $\sqrt{(x + 2)} + 1$ (shifted the wrong way), $\sqrt{(x - 1)} + 2$ (shifts swapped), and $-\sqrt{(x - 2)} + 1$ (reflected).

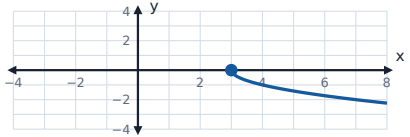
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q1/5

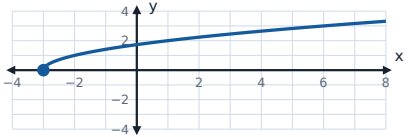
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -\sqrt{(x + 3)}$?

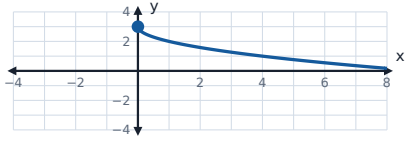
A



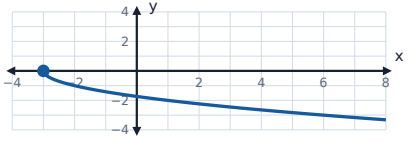
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q2/5

1681

QUESTION ID

S46-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Left 3, then reflected over the x-axis: starts at (−3, 0) and falls through (−2, −1) and (1, −2); domain $x \geq -3$, range $y \leq 0$. The others show $-\sqrt{(x - 3)}$, $\sqrt{(x + 3)}$, and $-\sqrt{x} + 3$.

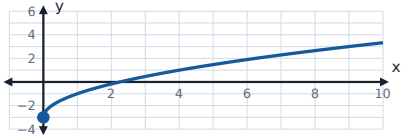
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q2/5

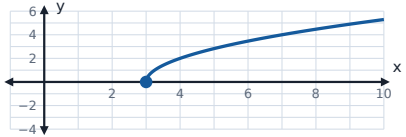
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2\sqrt{x} - 3$?

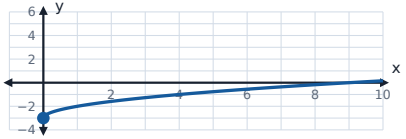
A



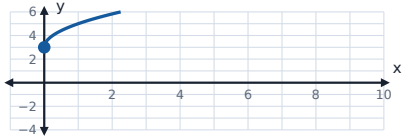
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q3/5

1682

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Vertical stretch by 2, then down 3: starts at (0, −3) and passes (1, −1), (4, 1), (9, 3); domain $x \geq 0$, range $y \geq -3$. The others show $2\sqrt{(x-3)}$, $\sqrt{x} - 3$ (no stretch), and $2\sqrt{x} + 3$.

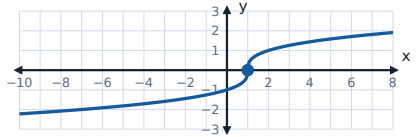
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q3/5

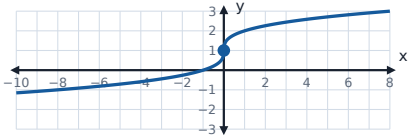
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \sqrt[3]{(x + 1)}$?

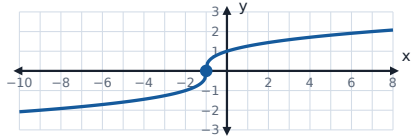
A



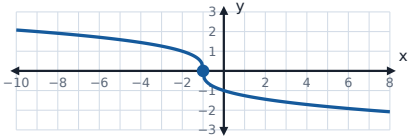
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q4/5

1683

QUESTION ID

S46-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Left 1: increasing, centered at (−1, 0), through (−9, −2), (−2, −1), (0, 1), (7, 2); domain and range all real numbers. The others show $\sqrt[3]{(x - 1)}$, $\sqrt[3]{x} + 1$, and $-\sqrt[3]{(x + 1)}$.

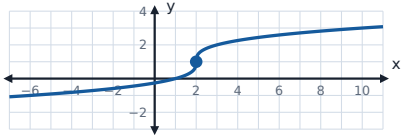
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q4/5

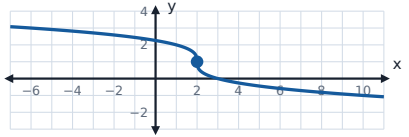
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -\sqrt[3]{(x - 2)} + 1$?

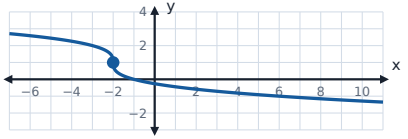
A



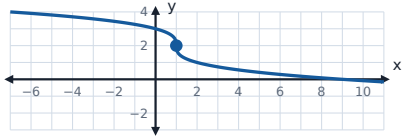
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q5/5

1684

QUESTION ID

S46-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Right 2, reflected, up 1: decreasing, centered at (2, 1), through (1, 2), (3, 0), (10, −1), (−6, 3); domain and range all real numbers. The others show $\sqrt[3]{(x - 2)} + 1$, $-\sqrt[3]{(x + 2)} + 1$, and $-\sqrt[3]{(x - 1)} + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 1

Graph y = √x and y = ∛x from a table • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Complete the table for $y = \sqrt{x}$ at $x = 0, 1, 4, 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q1/5

1686

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 1, 2, 3

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the table for $y = \sqrt[3]{x}$ at $x = -8, -1, 0, 1, 8$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q2/5

1687

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2, −1, 0, 1, 2

GRADING NOTE Form required: all five.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q2**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q2/5

True or false?

The graph of $y = \sqrt{x}$ starts at the origin and decreases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q3/5

1688

QUESTION ID

S46-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{x}$ increases as x increases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $y = \sqrt[3]{x}$ is

A. $x > 0$

B. all real numbers

C. $x \leq 0$

D. $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q4/5

1689

QUESTION ID

S46-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. all real numbers

GRADING NOTE Cube roots of negatives exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $y = \sqrt{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q5/5

1690

QUESTION ID

S46-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$y \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 2

Translations and reflections of graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How is $y = \sqrt{(x - 2)} + 1$ obtained from $y = \sqrt{x}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q1/5

1692

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Right 2, up 1

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How is $y = -\sqrt{(x + 3)}$ obtained from $y = \sqrt{x}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q2/5

1693

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Left 3, then reflect over the x-axis

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 2\sqrt{x}$ is a vertical stretch of $y = \sqrt{x}$ by 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q3/5

1694

QUESTION ID

S46-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outputs doubled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q3/5

The point (0, 0) of $y = \sqrt{x}$ moves to which point for $y = \sqrt{(x - 2)} + 1$?

A. (2, 1)

B. (1, 2)

C. (−2, 1)

D. (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q4/5

1695

QUESTION ID

S46-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 1)

GRADING NOTE Right 2, up 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Where does (1, 1) of $y = \sqrt[3]{x}$ go under $y = \sqrt[3]{(x + 1)}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q5/5

1696

QUESTION ID

S46-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 3

Square and cube roots of numbers • Grade 8 • Expressions & Equations • 8.EE.A.2

Find $\sqrt{9}$ and $\sqrt[3]{(-8)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q1/5

1698

QUESTION ID

S46-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3 and −2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $\sqrt[3]{27}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q2/5

1699

QUESTION ID

S46-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

√(−4) is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q3/5

1700

QUESTION ID

S46-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No real square root of a negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt[3]{(-1)}$ equals

A. i

B. undefined

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q4/5

1701

QUESTION ID

S46-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −1

GRADING NOTE $(-1)^3 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find √16 − 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q5/5

1702

QUESTION ID

S46-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSF-IF.C.7.b

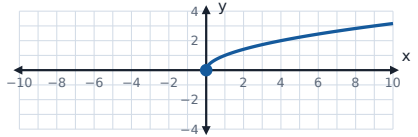
FORWARD 1

Graph radical functions with horizontal stretches and reflections • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

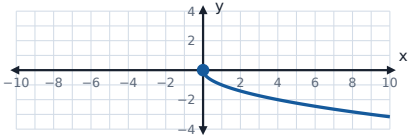
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = √(−x)?

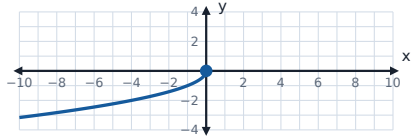
A



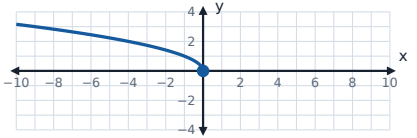
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q1/5

1704

QUESTION ID

S46-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Reflection of y = √x over the y-axis: starts at (0, 0) and rises to the left through (−1, 1), (−4, 2), (−9, 3); domain x ≤ 0. The others show √x, −√x, and −√(−x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that √(4x) = 2√x for x ≥ 0 and explain what this says about the two graphs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q2/5

1705

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

√(4x) = √4 · √x = 2√x, so the horizontal compression by 1/4 equals the vertical stretch by 2.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = \sqrt[3]{(-x)}$ is the same as $y = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q3/5

1706

QUESTION ID

S46-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt[3]{(-x)} = -\sqrt[3]{x}$; at $x = 8$ they give -2 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $y = \sqrt{(3 - x)}$ is

A. all reals

B. $x \geq 3$

C. $x \leq 3$

D. $x \geq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q4/5

1707

QUESTION ID

S46-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x \leq 3$

GRADING NOTE $3 - x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a square root function whose graph starts at (4, −2) and goes up to the left.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q5/5

1708

QUESTION ID
S46-F1-Q5

TYPE
Short answer

CORRECT ANSWER
√(4 − x) − 2 (for example)

GRADING NOTE Any valid function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

FORWARD 2

Graph a radical function and its inverse • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = \sqrt{(x - 2)} + 1$ and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q1/5

1710

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = (x - 1)^2 + 2$ for $x \geq 1$

GRADING NOTE Form required: formula and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $g(x) = \sqrt[3]{(x + 1)}$.

$g^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q2/5

1711

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of an inverse is the reflection of the graph over y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q3/5

1712

QUESTION ID
S46-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Inputs and outputs swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (6, 3) on $f(x) = \sqrt{(x - 2)} + 1$ gives which point on f^{-1}?

A. (6, 3)

B. (3, 6)

C. (−6, −3)

D. (3, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q4/5

1713

QUESTION ID

S46-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 6)

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the inverse of $f(x) = \sqrt{(x - 2)} + 1$ needs the domain $x \geq 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q5/5

1714

QUESTION ID

S46-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The range of f is $y \geq 1$, which becomes the domain of the inverse; without it $(x - 1)^2 + 2$ is not one-to-one.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and absolute value functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate and graph simple piecewise functions	**HSF-IF.C.7.b**
BACKWARD 2	Absolute value as distance	**6.NS.C.7**
BACKWARD 3	Open and closed endpoints	**6.EE.B.8**
FORWARD 1	Write piecewise functions from graphs and contexts	**HSF-IF.C.7.b**
FORWARD 2	Describe absolute value graphs as transformations, and identify…	**HSF-BF.B.3**

SET 47 • HSF-IF.C.7.b

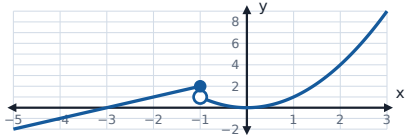
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

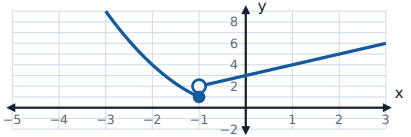
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows f(x) = x + 3 for x < −1, and f(x) = x² for x ≥ −1?

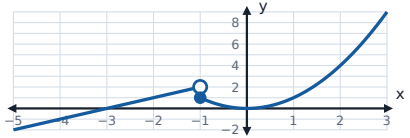
A



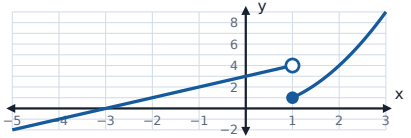
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q1/5

1717

QUESTION ID

S47-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE The line y = x + 3 is used to the left of −1 and ends with an open circle at (−1, 2) (x = −1 is not included); the parabola y = x² starts with a closed dot at (−1, 1). The others swap the circles, swap the pieces, or break at x = 1.

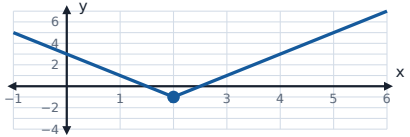
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q1/5

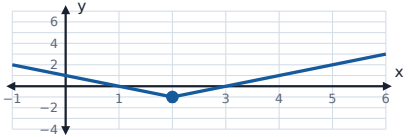
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = |2x - 4| - 1$?

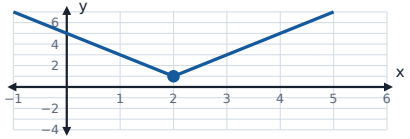
A



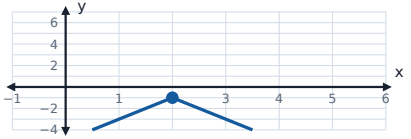
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q2/5

1718

QUESTION ID

S47-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $|2x - 4| = 2|x - 2|$: vertex $(2, -1)$, slopes ± 2, through $(0, 3)$, $(1, 1)$, $(3, 1)$, $(4, 3)$. The others show $|x - 2| - 1$ (slopes ± 1), $|2x - 4| + 1$, and $-|2x - 4| - 1$.

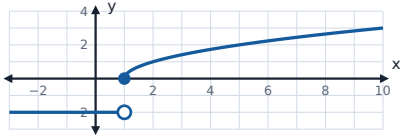
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q2/5

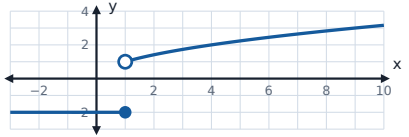
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows f(x) = −2 for x ≤ 1, and f(x) = √(x − 1) for x > 1?

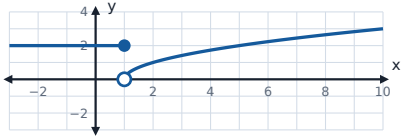
A



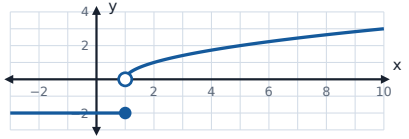
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q3/5

1719

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE The constant piece includes x = 1 (closed dot at (1, −2)); the square-root piece starts at x = 1 but does not include it (open circle at (1, 0)) and passes (2, 1), (5, 2), (10, 3). The others swap the circles, use √x, or use y = 2.

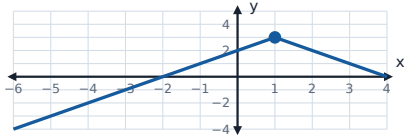
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q3/5

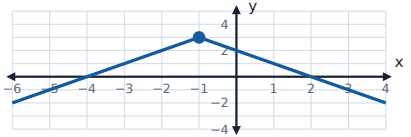
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -|x + 1| + 3$?

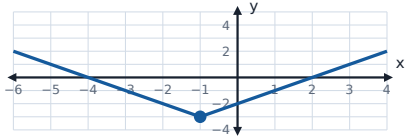
A



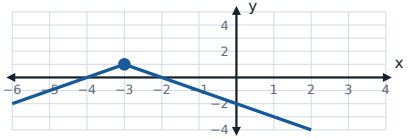
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q4/5

1720

QUESTION ID

S47-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Left 1, reflected, up 3: vertex (−1, 3), opening down, through (−4, 0), (0, 2), (2, 0). The others show $-|x - 1| + 3$, $|x + 1| - 3$, and $-|x + 3| + 1$.

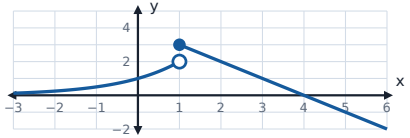
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q4/5

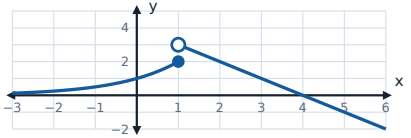
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $f(x) = 2^x$ for $x < 1$, and $f(x) = 4 - x$ for $x \geq 1$?

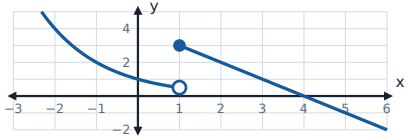
A



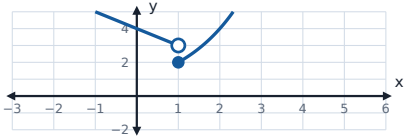
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q5/5

1721

QUESTION ID

S47-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE For $x < 1$ the graph is $y = 2^x$ (through (0, 1), approaching $y = 0$ to the left) with an open circle at (1, 2); for $x \geq 1$ it is $y = 4 - x$ from a closed dot at (1, 3) through (4, 0). The others swap the circles, use $(1/2)^x$, or swap the pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 1

Evaluate and graph simple piecewise functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(-3)$ and $f(5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q1/5

1723

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2 and 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(0)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q2/5

1724

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q2/5

True or false?

A piecewise function uses different rules on different parts of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q3/5

1725

QUESTION ID

S47-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x + 3$ $(x < -1)$, x^2 $(x \geq -1)$, $f(-1)$ is

A. 3

B. 2

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q4/5

1726

QUESTION ID

S47-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE Use the second rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 3$ for $x < -1$, and $f(x) = x^2$ for $x \geq -1$. Find $f(-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q5/5

1727

QUESTION ID

S47-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 2

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Find |−5| and |3|.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q1/5

1729

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 and 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $|2(1) - 4| - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q2/5

1730

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q2/5

True or false?

|x| can be negative for negative x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q3/5

1731

QUESTION ID

S47-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE |x| is a distance, so it is never negative: |−3| = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q3/5

$|x + 1| = 0$ when

A. $x = 0$

B. $x = -1$

C. $x = 1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q4/5

1732

QUESTION ID

S47-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = −1

GRADING NOTE Vertex input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $-|0 + 1| + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q5/5

1733

QUESTION ID

S47-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 3

Open and closed endpoints • Grade 6 • Expressions & Equations • 6.EE.B.8

ANSWER KEY Reduced copy of the question page, with its answer

On a number line, is $x < -1$ drawn with an open or a closed circle at -1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q1/5

1735

QUESTION ID

S47-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Open

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $x \geq 1$ drawn with an open or a closed circle at 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q2/5

1736

QUESTION ID

S47-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Closed

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q2/5

True or false?

An open circle means the endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q3/5

1737

QUESTION ID

S47-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

On a number line, $x \leq 1$ is drawn with

A. an open circle and shading right

B. a closed circle at 1 and shading right

C. an open circle at 1 and shading left

D. a closed circle at 1 and shading left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q4/5

1738

QUESTION ID

S47-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a closed circle at 1 and shading left

GRADING NOTE Included endpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inequality for "all numbers greater than 1".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q5/5

1739

QUESTION ID

S47-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x > 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 1

Write piecewise functions from graphs and contexts • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

A shipping company charges $5 for packages up to 2 lb and $5 plus $1.50 per pound over 2 lb. Write the cost C(w).

C(w) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q1/5

1741

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5 for $0 < w \leq 2$; $5 + 1.5(w - 2)$ for $w > 2$

GRADING NOTE Any equivalent form with correct endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q1/5

Write |x − 3| as a piecewise function without absolute value bars.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q2/5

1742

QUESTION ID
S47-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3 − x for x < 3; x − 3 for x ≥ 3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A piecewise function can have a jump and still be a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q3/5

1743

QUESTION ID

S47-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each input still has one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which piecewise function is continuous at x = 1?

A. 2^x for $x < 1$, $4 - x$ for $x \geq 1$

B. x^2 for $x < 1$, $2x - 1$ for $x \geq 1$

C. 2 for $x < 1$, 3 for $x \geq 1$

D. x for $x < 1$, $x + 1$ for $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q4/5

1744

QUESTION ID
S47-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. x^2 for $x < 1$, $2x - 1$ for $x \geq 1$

GRADING NOTE Both pieces give 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find k so that $f(x) = kx$ for $x < 2$ and $f(x) = x^2$ for $x \geq 2$ has no jump at 2.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q5/5

1745

QUESTION ID

S47-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 2k = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 2

Describe absolute value graphs as transformations, and identify symmetry • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Describe y = |2x − 4| − 1 as a transformation of y = |x|.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q1/5

1747

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 2|x − 2| − 1: right 2, vertical stretch by 2, down 1

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q1/5

Write an absolute value function with vertex (−1, 3) that opens down and passes through (2, 0).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q2/5

1748

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−|x + 1| + 3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

y = |x| is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q3/5

1749

QUESTION ID

S47-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE |−x| = |x|, so it is even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q3/5

$y = |x - 2|$ is symmetric about

A. the origin

B. $y = 2$

C. $x = 0$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q4/5

1750

QUESTION ID

S47-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2

GRADING NOTE Vertex at 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = -|x + 1| + 3$ an even function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q5/5

1751

QUESTION ID

S47-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No: f(1) = 1 but f(−1) = 3

GRADING NOTE Form required: decision and evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c • INTERPRETING FUNCTIONS

Graph polynomial functions showing zeros and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.c**
BACKWARD 1	Factor out common factors and special products	**HSA-SSE.A.2**
BACKWARD 2	Crossing and touching from multiplicity	**HSA-APR.B.3**
BACKWARD 3	Evaluate polynomials at test points	**HSF-IF.A.2**
FORWARD 1	Locate relative extrema of polynomial graphs	**HSF-IF.C.7.c**
FORWARD 2	Interpret polynomial graphs in context	**HSF-IF.B.4**

SET 48 • HSF-IF.C.7.c

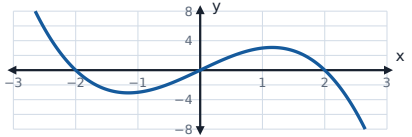
MAIN

Algebra II main question • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

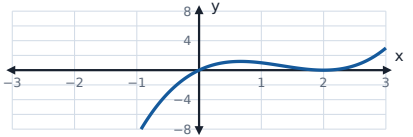
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = x^3 - 4x$?

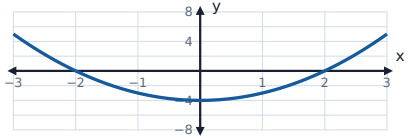
A



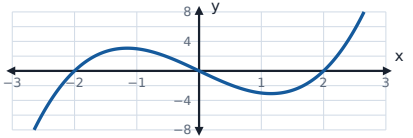
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q1/5

1754

QUESTION ID

S48-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE $x^3 - 4x = x(x - 2)(x + 2)$: crosses at -2, 0, 2; $(-1, 3)$ and $(1, -3)$; falls left, rises right (odd degree, positive leading coefficient). The others show $-x^3 + 4x$, $x(x - 2)^2$, and $x^2 - 4$.

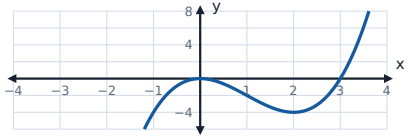
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q1/5

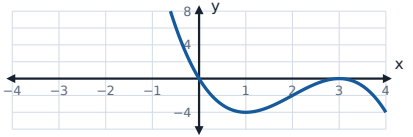
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -x^3 + 3x^2$?

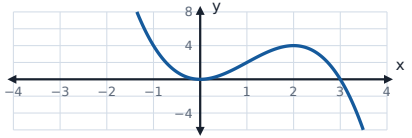
A



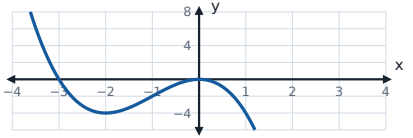
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q2/5

1755

QUESTION ID

S48-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $-x^3 + 3x^2 = -x^2(x - 3)$: touches at 0 (double zero), crosses at 3, (2, 4), rises left and falls right. The others show $x^3 - 3x^2$, $-x(x - 3)^2$, and $-x^2(x + 3)$.

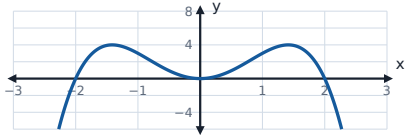
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q2/5

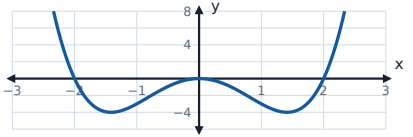
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = x^4 - 4x^2$?

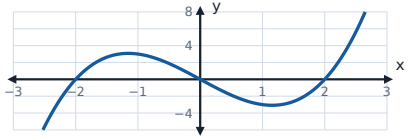
A



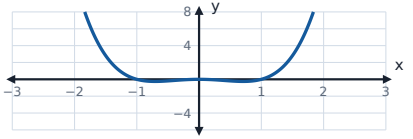
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q3/5

1756

QUESTION ID

S48-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE $x^4 - 4x^2 = x^2(x - 2)(x + 2)$: touches at 0, crosses at ±2, (±1, −3), rises at both ends (even degree, positive leading coefficient). The others show $-x^4 + 4x^2$, $x^3 - 4x$, and $x^4 - x^2$.

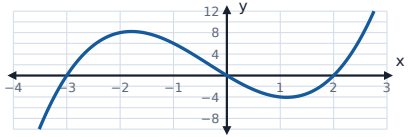
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q3/5

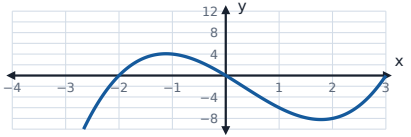
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = x^3 + x^2 - 6x$?

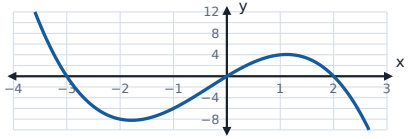
A



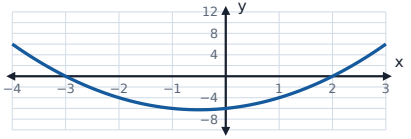
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q4/5

1757

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $x^3 + x^2 - 6x = x(x + 3)(x - 2)$: crosses at -3, 0, 2; $(-2, 8)$ and $(1, -4)$; falls left, rises right. The others show $x(x - 3)(x + 2)$, its reflection $-x(x + 3)(x - 2)$, and $x^2 + x - 6$.

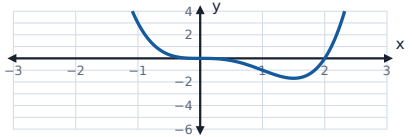
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q4/5

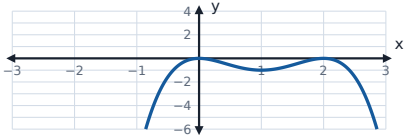
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -x^4 + 2x^3$?

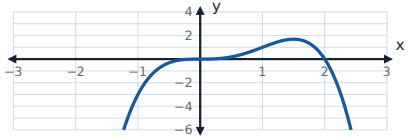
A



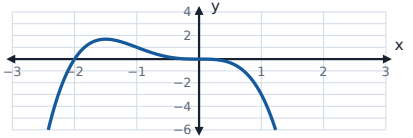
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q5/5

1758

QUESTION ID

S48-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $-x^4 + 2x^3 = -x^3(x - 2)$: crosses at 0 with a flattened shape (multiplicity 3) and at 2; (1, 1); falls at both ends (even degree, negative leading coefficient). The others show $x^4 - 2x^3$, $-x^2(x - 2)^2$, and $-x^3(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 1

Factor out common factors and special products • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q1/5

1760

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x(x − 2)(x + 2)

GRADING NOTE Form required: complete.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $-x^3 + 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q2/5

1761

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-x^2(x - 3)$

GRADING NOTE Any equivalent complete factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q2/5

True or false?

$x^4 - 4x^2 = x^2(x^2 - 4) = x^2(x - 2)(x + 2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q3/5

1762

QUESTION ID

S48-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Common factor, then difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q3/5

$x^3 + x^2 - 6x$ equals

A. $(x + 3)(x - 2)$

B. $x(x + 3)(x - 2)$

C. $x(x - 3)(x + 2)$

D. $x(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q4/5

1763

QUESTION ID

S48-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x(x + 3)(x - 2)$

GRADING NOTE $x^2 + x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $-x^4 + 2x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q5/5

1764

QUESTION ID

S48-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$-x^3(x - 2)$

GRADING NOTE Any equivalent complete factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 2

Crossing and touching from multiplicity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Does $y = x^2(x - 3)$ cross or touch the x-axis at 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q1/5

1766

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Touches

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $y = x^3(x - 2)$ cross at 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q2/5

1767

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, with a flattened shape (odd multiplicity 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A zero of odd multiplicity is a crossing point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q3/5

1768

QUESTION ID

S48-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sign change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $y = (x + 2)^2(x - 1)$ at $x = -2$

A. crosses

B. touches

C. has an asymptote

D. is undefined

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q4/5

1769

QUESTION ID

S48-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. touches

GRADING NOTE Even multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the end behavior of $y = -x^4 + 2x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q5/5

1770

QUESTION ID

S48-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Falls at both ends

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 3

Evaluate polynomials at test points • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^3 - 4x$ at $x = -1$ and $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q1/5

1772

QUESTION ID

S48-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3 and −3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-x^3 + 3x^2$ at $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q2/5

1773

QUESTION ID

S48-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q2/5

True or false?

$x^4 - 4x^2$ at $x = 1$ equals 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S48-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q3/5

1774

QUESTION ID

S48-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S48-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^3 + x^2 - 6x$ at $x = -2$ equals

A. 8

B. 0

C. -8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q4/5

1775

QUESTION ID

S48-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8

GRADING NOTE $-8 + 4 + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-x^4 + 2x^3$ at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q5/5

1776

QUESTION ID

S48-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 1

Locate relative extrema of polynomial graphs • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Use a table with steps of 0.1 (or technology) to estimate the relative maximum of $y = x^3 - 4x$ to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q1/5

1778

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About (−1.2, 3.1)

GRADING NOTE Accept x from −1.2 to −1.1 and y about 3.08. Exact x = −2/√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use technology or a table to estimate the relative minimum of $y = x^3 + x^2 - 6x$ on $0 < x < 2$ to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q2/5

1779

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About (1.1, −4.1)

GRADING NOTE Accept x from 1.1 to 1.2 and y from −4.1 to −4.0. Exact $x = (-1 + \sqrt{19})/3 \approx 1.12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 4 always has exactly 3 turning points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S48-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q3/5

1780

QUESTION ID

S48-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has at most 3; $y = x^4$ has only one turning point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S48-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many turning points can $y = x^4 - 4x^2$ have?

A. 1

B. 4

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q4/5

1781

QUESTION ID

S48-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE It has 3: two minima and one maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the graph of $y = x^3 - 4x$ has a turning point between 0 and 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q5/5

1782

QUESTION ID

S48-F1-Q5

TYPE

Short answer

CORRECT ANSWER

It is 0 at x = 0 and x = 2 and negative between them, so it must turn back up.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 2

Interpret polynomial graphs in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A box's volume is $V(x) = x(12 - 2x)^2$. Which part of the polynomial graph is meaningful, and what is the maximum volume there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q1/5

1784

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0 < x < 6; maximum 128 in³ at x = 2

GRADING NOTE Form required: interval and maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A company's profit (thousands of dollars) is $P(x) = -x^3 + 9x^2 - 15x$ for x hundred items, $0 \le x \le 8$. When is the profit positive?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q2/5

1785

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

For about $2.2 < x < 6.8$ (exactly between $(9 - \sqrt{21})/2 \approx 2.21$ and $(9 + \sqrt{21})/2 \approx 6.79$)

GRADING NOTE Form required: the interval. Endpoints accepted within ±0.05. $P(x) = -x(x^2 - 9x + 15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

A zero of a profit model is a break-even point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q3/5

1786

QUESTION ID

S48-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Profit 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A population model $P(t) = -t^3 + 6t^2 + 15t + 100$ rises then falls. The relative maximum means

A. the largest population before decline

B. the starting population

C. the end behavior

D. the time the population is 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q4/5

1787

QUESTION ID

S48-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the largest population before decline

GRADING NOTE Turning point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q4/5

Why does the end behavior of $V(x) = x(12 - 2x)^2$ not matter for the box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q5/5

1788

QUESTION ID

S48-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The context only uses $0 < x < 6$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions with intercepts, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph basic exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Zero and negative exponents	**8.EE.A.1**
BACKWARD 3	Vertical and horizontal shifts and reflections	**HSF-BF.B.3**
FORWARD 1	Compare and describe exponential graphs	**HSF-IF.C.7.e**
FORWARD 2	Graph the inverse of an exponential function	**HSF-BF.B.4.a**

SET 49 • HSF-IF.C.7.e

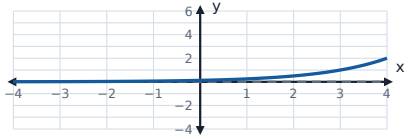
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

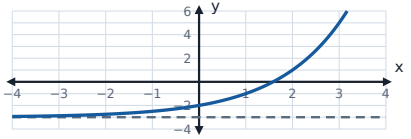
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2^x - 3$?

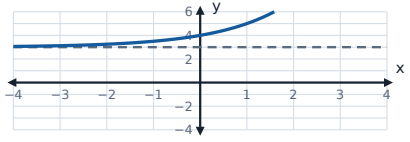
A



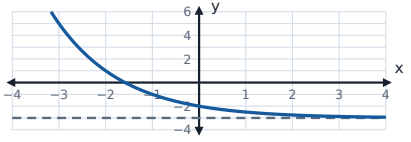
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q1/5

1791

QUESTION ID

S49-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Down 3 from $y = 2^x$: asymptote $y = -3$, y-intercept $(0, -2)$, x-intercept $\log_2 3 \approx 1.58$, increasing. The others show 2^{x-3}, $2^x + 3$, and $(1/2)^x - 3$.

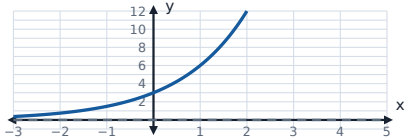
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q1/5

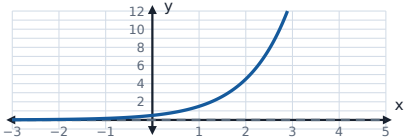
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 3(0.5)^x$?

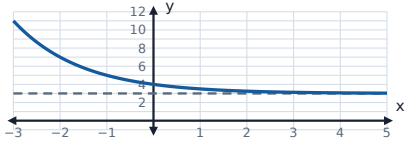
A



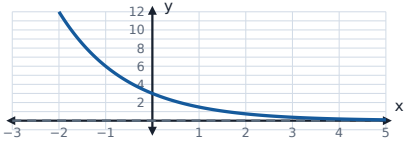
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q2/5

1792

QUESTION ID

S49-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Initial value 3, factor 0.5: y-intercept (0, 3), (1, 1.5), (−1, 6); decreasing; asymptote $y = 0$ ($y \to 0$ as $x \to \infty$). The others show $3(2)^x$, $0.5(3)^x$, and $(0.5)^x + 3$.

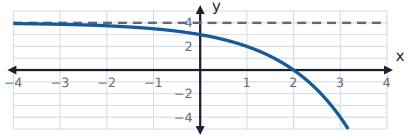
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q2/5

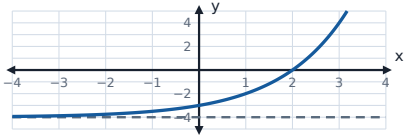
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -2^x + 4$?

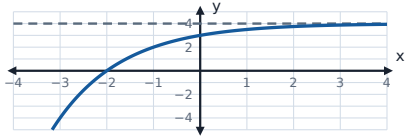
A



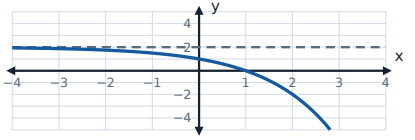
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q3/5

1793

QUESTION ID

S49-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Reflect $y = 2^x$ over the x-axis and move up 4: asymptote $y = 4$ (approached to the left), y-intercept (0, 3), x-intercept (2, 0), decreasing. The others show $2^x - 4$, $4 - 2^{-x}$, and $-2^x + 2$.

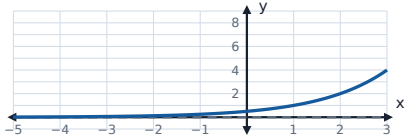
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q3/5

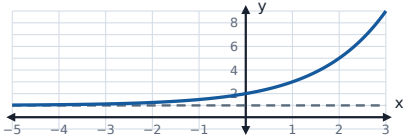
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2^{x+1}$?

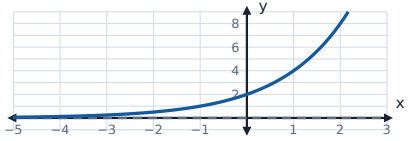
A



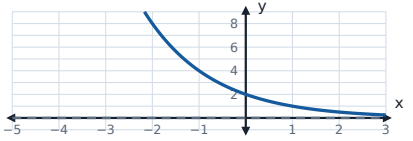
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q4/5

1794

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Left 1 from $y = 2^x$ (or $2 \cdot 2^x$): asymptote $y = 0$, y-intercept $(0, 2)$, through $(-1, 1)$, $(1, 4)$, $(2, 8)$. The others show 2^{x-1}, $2^x + 1$ (same y-intercept, asymptote $y = 1$), and 2^{1-x}.

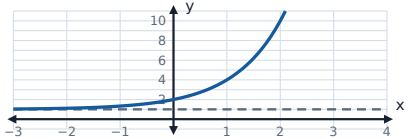
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q4/5

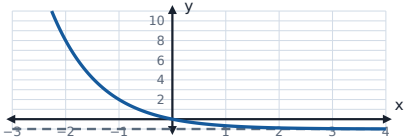
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (1/3)^x + 1$?

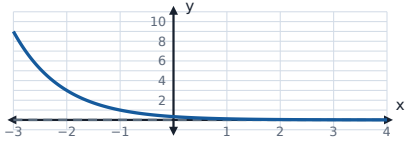
A



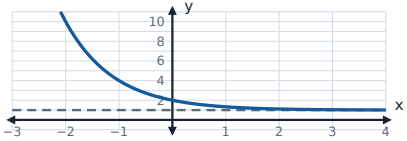
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q5/5

1795

QUESTION ID

S49-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Up 1 from $y = (1/3)^x$: asymptote $y = 1$, y-intercept $(0, 2)$, through $(-1, 4)$ and $(-2, 10)$, decreasing toward 1. The others show $3^x + 1$, $(1/3)^x - 1$, and $(1/3)^{x+1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 1

Graph basic exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Complete the table for $y = 2^x$ at $x = -2, -1, 0, 1, 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q1/5

1797

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4, 1/2, 1, 2, 4

GRADING NOTE Form required: all five.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 3(0.5)^x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q2/5

1798

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q2/5

True or false?

$y = 2^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q3/5

1799

QUESTION ID

S49-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^x \to 0$ as $x \to -\infty$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = (0.5)^x$ is

A. linear

B. increasing

C. constant

D. decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q4/5

1800

QUESTION ID

S49-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. decreasing

GRADING NOTE Base less than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $y = 2^x$ have an x-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q5/5

1801

QUESTION ID

S49-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No: $2^x > 0$ for every x

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 2

Zero and negative exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2^{-2}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q1/5

1803

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(1/3)^{-2}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q2/5

1804

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$b^0 = 0$ for $b \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q3/5

1805

QUESTION ID

S49-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero base to the 0 power is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3(0.5)^{-1}$ equals

A. 6

B. 3

C. -1.5

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q4/5

1806

QUESTION ID

S49-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE $0.5^{-1} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2^{-1} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q5/5

1807

QUESTION ID

S49-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 3

Vertical and horizontal shifts and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How does $y = 2^x - 3$ relate to $y = 2^x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q1/5

1809

QUESTION ID

S49-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Shifted down 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How does $y = 2^{x+1}$ relate to $y = 2^{x}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q2/5

1810

QUESTION ID

S49-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Shifted left 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A vertical shift moves the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q3/5

1811

QUESTION ID

S49-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y = 0 becomes y = k.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q3/5

$y = -2^x + 4$ has asymptote

A. $y = 4$

B. $y = 0$

C. $y = -4$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q4/5

1812

QUESTION ID

S49-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4

GRADING NOTE Reflect, then up 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the asymptote of $y = (1/3)^x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q5/5

1813

QUESTION ID

S49-B3-Q5

TYPE

Short answer

CORRECT ANSWER

y = 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 1

Compare and describe exponential graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the graphs of $y = 2^x$ and $y = (1/2)^x$ are reflections of each other over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q1/5

1815

QUESTION ID
S49-F1-Q1

TYPE
Short answer

CORRECT ANSWER
$(1/2)^x = 2^{-x}$, which replaces x by −x.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that $2^{x+1} = 2 \cdot 2^x$ and explain what this says about two transformations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q2/5

1816

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$2^{x+1} = 2^x \cdot 2^1$: a left shift by 1 equals a vertical stretch by 2.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q2/5

True or false?

$y = 5 \cdot 3^x$ and $y = 3^x$ have different horizontal asymptotes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q3/5

1817

QUESTION ID

S49-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A vertical stretch keeps the asymptote y = 0 for both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q3/5

Which exponential graph passes through (0, 4) and (1, 2)?

A. $y = 2^x + 3$

B. $y = 4 \cdot 2^x$

C. $y = 0.5 \cdot 4^x$

D. $y = 4(0.5)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q4/5

1818

QUESTION ID

S49-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 4(0.5)^x$

GRADING NOTE a = 4, b = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an exponential function with asymptote $y = -1$ and y-intercept 2 that is increasing.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q5/5

1819

QUESTION ID
S49-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$3 \cdot 2^x - 1$ (for example)

GRADING NOTE Any valid function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 2

Graph the inverse of an exponential function • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $f(x) = 2^x$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q1/5

1821

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2 x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (3, 8) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q2/5

1822

QUESTION ID
S49-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(8, 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q2/5

True or false?

The inverse of $y = 2^x$ has the vertical asymptote $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q3/5

1823

QUESTION ID
S49-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Reflection of $y = 0$ over $y = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q3/5

The inverse of $y = 2^x - 3$ is

A. $y = \log_2(x - 3)$

B. $y = \log_2 x - 3$

C. $y = 2^{x+3}$

D. $y = \log_2(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q4/5

1824

QUESTION ID

S49-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = \log_2(x + 3)$

GRADING NOTE $x = 2^y - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q4/5

Give the domain and range of $y = \log_2 x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q5/5

1825

QUESTION ID

S49-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Domain x > 0; range all real numbers

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph logarithmic functions with intercept, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Find inverses by swapping inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 3	Shift and reflect graphs	**HSF-BF.B.3**
FORWARD 1	Graph logarithms with reflections that change the domain	**HSF-IF.C.7.e**
FORWARD 2	Solve equations with logarithms	**HSF-LE.A.4**

SET 50 • HSF-IF.C.7.e

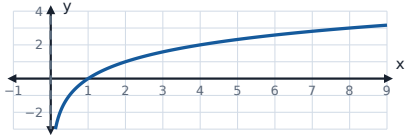
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

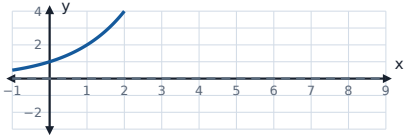
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2 x$?

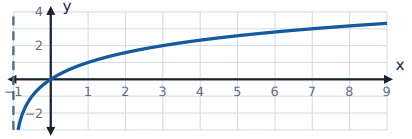
A



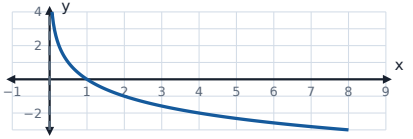
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q1/5

1828

QUESTION ID

S50-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Asymptote $x = 0$, x-intercept (1, 0), through (2, 1), (4, 2), (8, 3); domain $x > 0$; increases slowly. The others show 2^x (the inverse), $\log_2(x + 1)$, and $-\log_2 x$.

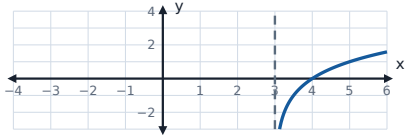
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q1/5

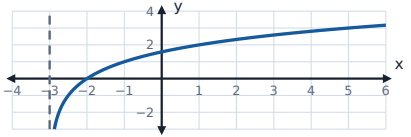
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2(x + 3)$?

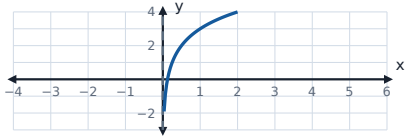
A



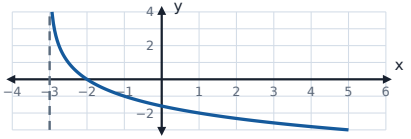
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q2/5

1829

QUESTION ID

S50-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Left 3 from $y = \log_2 x$: asymptote $x = -3$, x-intercept $(-2, 0)$, through $(-1, 1)$, $(1, 2)$, $(5, 3)$; domain $x > -3$. The others show $\log_2(x - 3)$, $\log_2 x + 3$, and $-\log_2(x + 3)$.

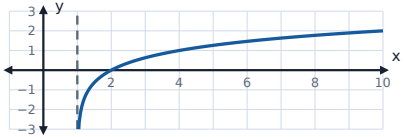
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q2/5

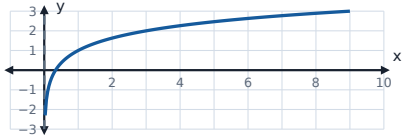
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_3 x - 1$?

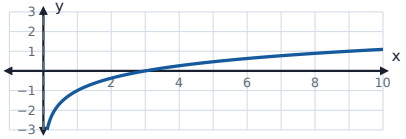
A



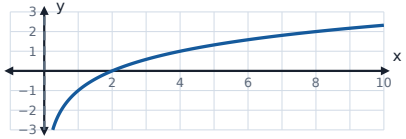
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q3/5

1830

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Down 1 from $y = \log_3 x$: asymptote $x = 0$, x-intercept $(3, 0)$, through $(1, -1)$ and $(9, 1)$; domain $x > 0$. The others show $\log_3(x - 1)$, $\log_3 x + 1$, and $\log_2 x - 1$ (wrong base: it passes $(2, 0)$).

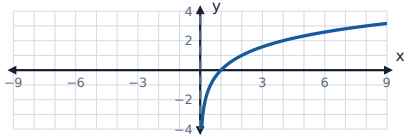
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q3/5

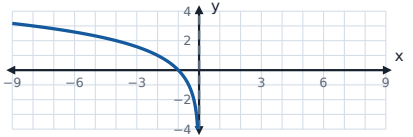
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -\log_2 x$?

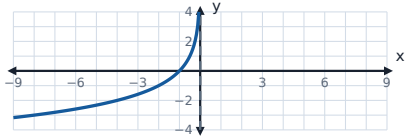
A



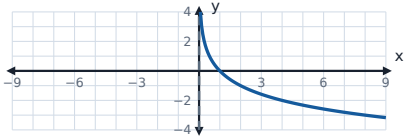
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q4/5

1831

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Reflection of $y = \log_2 x$ over the x-axis: asymptote $x = 0$, x-intercept $(1, 0)$, through $(2, -1)$, $(4, -2)$, $(8, -3)$; decreasing; domain $x > 0$. The others show $\log_2 x$, $\log_2(-x)$, and $-\log_2(-x)$.

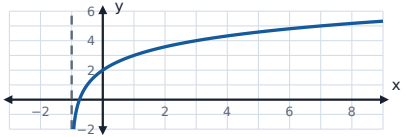
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q4/5

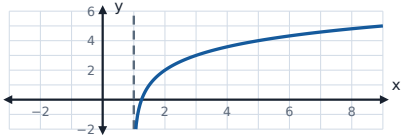
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2(x - 1) + 2$?

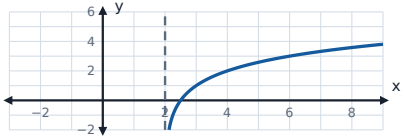
A



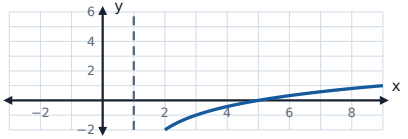
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q5/5

1832

QUESTION ID

S50-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Right 1 and up 2: asymptote $x = 1$, through (2, 2), (3, 3), (5, 4), x-intercept (1.25, 0); domain $x > 1$. The others show $\log_2(x + 1) + 2$, $\log_2(x - 2) + 1$, and $\log_2(x - 1) - 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 1

Graph exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Give three points on $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q1/5

1834

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

For example (0, 1), (1, 2), (2, 4)

GRADING NOTE Any three correct points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the asymptote of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q2/5

1835

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q2/5

True or false?

$y = 3^x$ passes through (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q3/5

1836

QUESTION ID

S50-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^3 = 27$, not 1; the graph passes through (1, 3) and (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 2^x$ passes through

A. (2, 1)

B. (−1, −2)

C. (1/2, −1)

D. (−1, 1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q4/5

1837

QUESTION ID

S50-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (−1, 1/2)

GRADING NOTE 2^{-1}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $y = 3^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q5/5

1838

QUESTION ID

S50-B1-Q5

TYPE

Short answer

CORRECT ANSWER

y > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 2

Find inverses by swapping inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

(2, 4) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

1840

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $y = 3^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

1841

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$y = \log_3 x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Swapping x and y in $y = 2^x$ gives $x = 2^y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

1842

QUESTION ID
S50-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Inverse relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

Which point is on $y = \log_2 x$?

A. (3, 8)

B. (2, 4)

C. (8, 3)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

1843

QUESTION ID

S50-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (8, 3)

GRADING NOTE $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\log_2 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

1844

QUESTION ID

S50-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 3

Shift and reflect graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How does $y = \log_2(x + 3)$ relate to $y = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q1/5

1846

QUESTION ID

S50-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Shifted left 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How does $y = -\log_2 x$ relate to $y = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q2/5

1847

QUESTION ID

S50-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Reflected over the x-axis

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

A horizontal shift moves the vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q3/5

1848

QUESTION ID

S50-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE x = 0 becomes x = h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = \log_2(x - 1) + 2$ has asymptote

A. $x = -1$

B. $y = 2$

C. $x = 1$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q4/5

1849

QUESTION ID

S50-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x = 1

GRADING NOTE Right 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Where does (1, 0) of $y = \log_3 x$ move under $y = \log_3 x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q5/5

1850

QUESTION ID

S50-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(1, −1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSF-IF.C.7.e

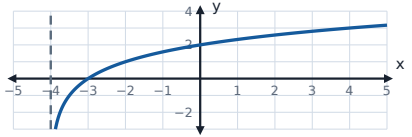
FORWARD 1

Graph logarithms with reflections that change the domain • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

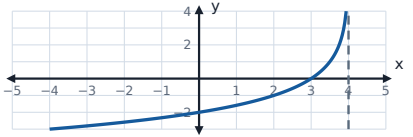
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2(4 - x)$?

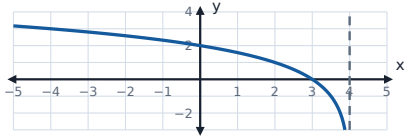
A



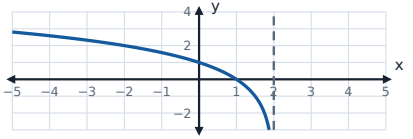
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q1/5

1852

QUESTION ID

S50-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $4 - x > 0$ gives domain $x < 4$ and asymptote $x = 4$; x-intercept (3, 0); through (2, 1), (0, 2), (−4, 3); decreasing. The others show $\log_2(x + 4)$, $-\log_2(4 - x)$, and $\log_2(2 - x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $y = \log(x^2 - 1)$ and explain the graph has two branches.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q2/5

1853

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x < -1$ or $x > 1$: $x^2 - 1 > 0$ on two separate intervals

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q2/5

True or false?

$y = \log_2(2x)$ is $y = \log_2 x$ shifted up 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q3/5

1854

QUESTION ID

S50-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\log_2 2 + \log_2 x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The x-intercept of $y = \log_2(x + 3)$ is

A. -2

B. 1

C. -3

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q4/5

1855

QUESTION ID

S50-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −2

GRADING NOTE $x + 3 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x-intercept of $y = \log_2(x - 1) + 2$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q5/5

1856

QUESTION ID

S50-F1-Q5

TYPE

Short answer

CORRECT ANSWER

5/4

GRADING NOTE Any equivalent form. $x - 1 = 2^{-2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

FORWARD 2

Solve equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 12$ exactly and to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

1858

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2 12 = \log 12/\log 2 \approx 3.58$

GRADING NOTE Form required: exact and decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5 \cdot 10^{x} = 40$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

1859

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

log 8

GRADING NOTE Any equivalent form (≈ 0.903).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

True or false?

$\log_2 12$ is between 4 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

1860

QUESTION ID

S50-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^3 = 8 < 12 < 16 = 2^4$, so $\log_2 12$ is between 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

The solution of $3^x = 7$ is

A. log 7/log 3

B. log 3/log 7

C. 7/3

D. log 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

1861

QUESTION ID

S50-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. log 7/log 3

GRADING NOTE Change of base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the graph of $y = \log_2 x$ to estimate $\log_2 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

1862

QUESTION ID

S50-F2-Q5

TYPE

Short answer

CORRECT ANSWER

About 2.6

GRADING NOTE Accept 2.5 to 2.6 (exact 2.585).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q5**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph trigonometric functions showing period, midline, and amplitude

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Unit-circle values at quarter turns	**HSF-TF.A.2**
BACKWARD 2	Read amplitude, period, and midline from a formula	**HSF-TF.B.5**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Graph shifted sinusoids and the tangent function	**HSF-IF.C.7.e**
FORWARD 2	Horizontal stretches f(kx) of trigonometric graphs	**HSF-BF.B.3**

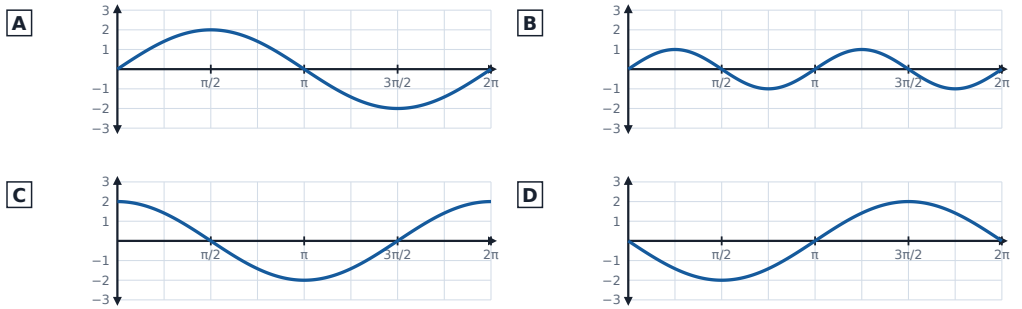
SET 51 • HSF-IF.C.7.e

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = 2sin x on 0 ≤ x ≤ 2π?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q1/5

1865

QUESTION ID

S51-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

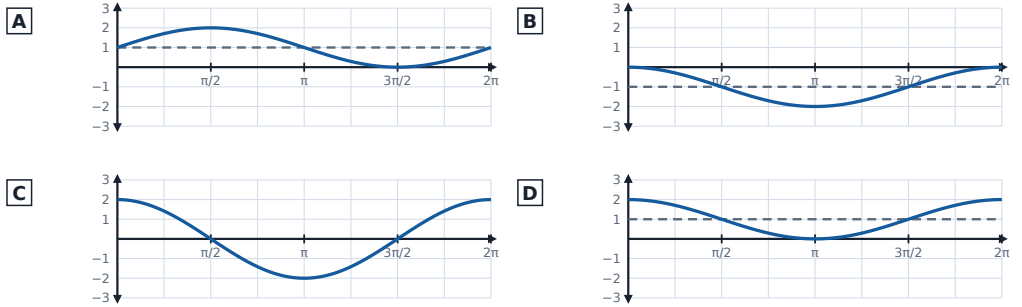
GRADING NOTE Amplitude 2, period 2π, midline y = 0: (0, 0), (π/2, 2), (π, 0), (3π/2, −2), (2π, 0). The others show sin(2x), 2cos x, and −2sin x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = cos x + 1 on 0 ≤ x ≤ 2π?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q2/5

1866

QUESTION ID

S51-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

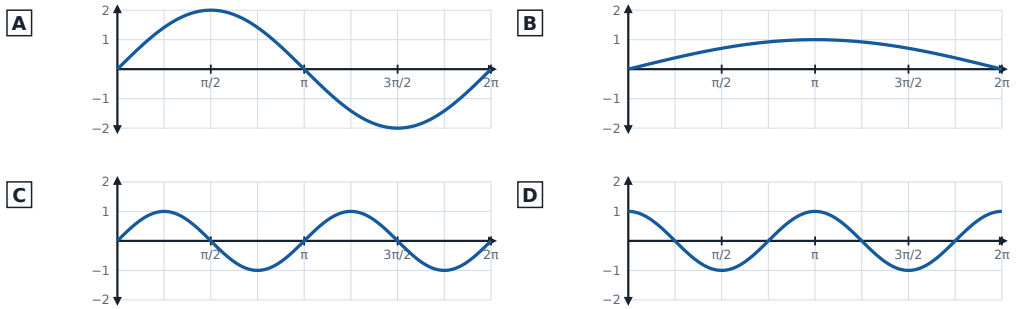
GRADING NOTE Midline y = 1, amplitude 1, period 2π: (0, 2), (π/2, 1), (π, 0), (3π/2, 1), (2π, 2). The others show sin x + 1, cos x − 1, and 2cos x (same maximum, wrong minimum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = sin(2x) on 0 ≤ x ≤ 2π?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q3/5

1867

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

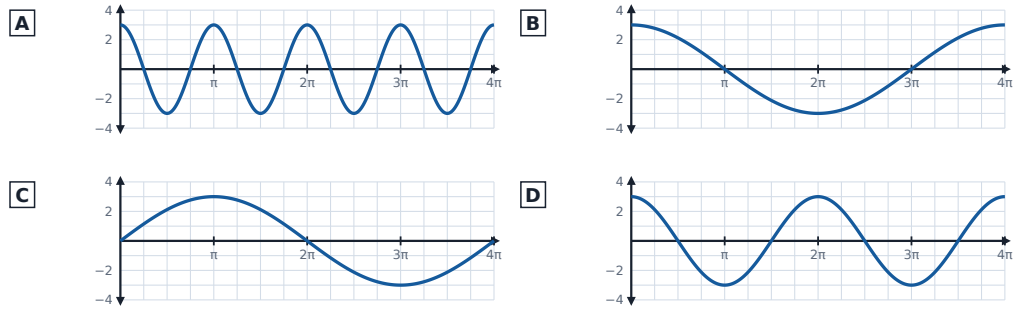
GRADING NOTE Period 2π/2 = π, amplitude 1, midline y = 0: two cycles on [0, 2π], starting at (0, 0) and rising to (π/4, 1). The others show 2sin x, sin(x/2), and cos(2x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = 3cos(x/2) on 0 ≤ x ≤ 4π?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q4/5

1868

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Period 2π/(1/2) = 4π, amplitude 3, midline y = 0: (0, 3), (π, 0), (2π, −3), (3π, 0), (4π, 3). The others show 3cos(2x), 3sin(x/2), and 3cos x.

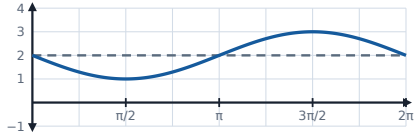
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q4/5

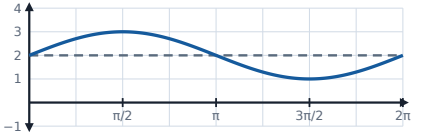
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = −sin x + 2 on 0 ≤ x ≤ 2π?

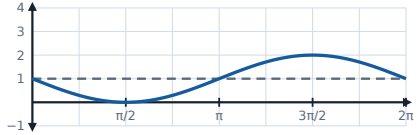
A



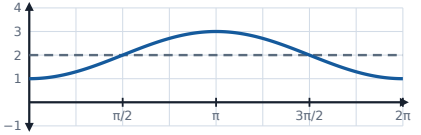
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q5/5

1869

QUESTION ID

S51-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Midline y = 2, amplitude 1, reflected: (0, 2), (π/2, 1), (π, 2), (3π/2, 3), (2π, 2). The others show sin x + 2, −sin x + 1, and −cos x + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 1

Unit-circle values at quarter turns • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give sin x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q1/5

1871

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 1, 0, −1, 0

GRADING NOTE Form required: all five.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q2/5

1872

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1, 0, −1, 0, 1

GRADING NOTE Form required: all five.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(3\pi/2) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q3/5

1873

QUESTION ID

S51-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The point at $3\pi/2$ is $(0, -1)$, so $\sin(3\pi/2) = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos(π) equals

A. π

B. 1

C. −1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q4/5

1874

QUESTION ID

S51-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −1

GRADING NOTE Point (−1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(π/4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q5/5

1875

QUESTION ID

S51-B1-Q5

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 2

Read amplitude, period, and midline from a formula • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the amplitude and period of y = 3cos(x/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q1/5

1877

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 and 4π

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the midline of $y = -\sin x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q2/5

1878

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = sin(2x) has period 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q3/5

1879

QUESTION ID

S51-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is 2π/2 = π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

y = 2sin x has amplitude

A. 1

B. 2

C. π

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q4/5

1880

QUESTION ID

S51-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE |A|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of y = cos(3x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q5/5

1881

QUESTION ID

S51-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Describe y = cos x + 1 compared with y = cos x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q1/5

1883

QUESTION ID

S51-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Up 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe y = −sin x compared with y = sin x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q2/5

1884

QUESTION ID

S51-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Reflected over the x-axis

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 2sin x has the same zeros as y = sin x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q3/5

1885

QUESTION ID

S51-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical stretches keep zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The maximum of $y = -\sin x + 2$ is

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q4/5

1886

QUESTION ID

S51-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE $-(-1) + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the minimum of y = cos x + 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q5/5

1887

QUESTION ID

S51-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSF-IF.C.7.e

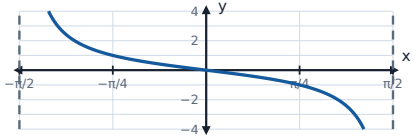
FORWARD 1

Graph shifted sinusoids and the tangent function • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

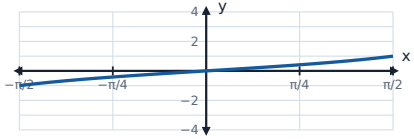
Which graph shows $y = \tan x$ for $-\pi/2 < x < \pi/2$?

A

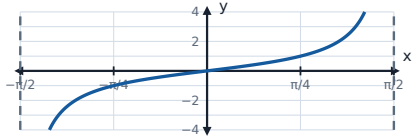


B

C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q1/5

1889

QUESTION ID

S51-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Asymptotes $x = \pm\pi/2$ ($\cos x = 0$ there); increasing through $(-\pi/4, -1)$, $(0, 0)$, $(\pi/4, 1)$. The others show $-\tan x$, $\tan(2x)$, and $\tan(x/2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q1/5

Describe the graph of $y = \sin(x - \pi/2)$ compared with $y = \sin x$, and name a simpler formula for it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q2/5

1890

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Shifted right π/2; it equals −cos x

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The period of $y = \tan x$ is 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q3/5

1891

QUESTION ID

S51-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\tan(x + \pi) = \tan x$, so the period is π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The first maximum of $y = \cos(x - \pi/3)$ for $x \geq 0$ is at

A. $x = 2\pi/3$

B. $x = \pi/3$

C. $x = 0$

D. $x = -\pi/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q4/5

1892

QUESTION ID
S51-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $x = \pi/3$

GRADING NOTE Shift right π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the x-values of the maximum points of $y = \sin(x + \pi/4)$ on $0 \leq x < 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q5/5

1893

QUESTION ID

S51-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = \pi/4$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

FORWARD 2

Horizontal stretches f(kx) of trigonometric graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How is y = sin(2x) obtained from y = sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q1/5

1895

QUESTION ID
S51-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Horizontal compression by a factor of 1/2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How is y = cos(x/2) obtained from y = cos x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q2/5

1896

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Horizontal stretch by a factor of 2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q2/5

True or false?

Replacing x by kx (k > 1) shortens the period by the factor k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q3/5

1897

QUESTION ID
S51-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Period 2π/k.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

y = sin(−x) is the same as

A. cos x

B. sin x

C. −cos x

D. −sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q4/5

1898

QUESTION ID

S51-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −sin x

GRADING NOTE Sine is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For which k does y = cos(kx) have period 6π?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q5/5

1899

QUESTION ID

S51-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form (or −1/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Rewrite exponential functions to reveal rates over other time units

MAIN	Algebra II main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Powers of powers and rational exponents	**HSN-RN.A.1**
BACKWARD 3	Percent change and growth factors	**7.RP.A.3**
FORWARD 1	Compare growth offers by rewriting to the same time unit	**HSF-IF.C.8.b**
FORWARD 2	Find doubling times and half-lives with logarithms	**HSF-LE.A.4**

SET 52 • HSF-IF.C.8.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

$y = 100(1.08)^t$ with t in years. Rewrite it with a monthly growth factor and give the monthly percent rate to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q1/5

1902

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

$y = 100(1.08^{1/12})^{12t} \approx 100(1.006434)^{12t}$; about 0.64% per month

GRADING NOTE Form required: rewritten form and rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 500(0.97)^t$ with t in years. Rewrite it per decade and give the percent decrease per decade to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q2/5

1903

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y = 500(0.97^{10})^{t/10} \approx 500(0.7374)^{t/10}$; about 26.26% decrease per decade

GRADING NOTE Form required: rewritten form and rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$Q = 50(2)^{t/3}$ with t in hours. Rewrite it with an hourly growth factor and give the hourly percent rate to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q3/5

1904

QUESTION ID
S52-M-Q3

TYPE
Short answer

CORRECT ANSWER
$Q = 50(2^{1/3})^t \approx 50(1.2599)^t$; about 25.99% per hour

GRADING NOTE Form required: rewritten form and rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$A = 1000(1.005)^{12t}$ with t in years. Rewrite it with an annual growth factor and give the annual percent rate to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q4/5

1905

QUESTION ID
S52-M-Q4

TYPE
Short answer

CORRECT ANSWER
$A = 1000(1.005^{12})^t \approx 1000(1.061678)^t$; about 6.17% per year

GRADING NOTE Form required: rewritten form and rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$N = 800(0.5)^{t/10}$ with t in years. Rewrite it with a yearly factor and give the yearly percent decrease to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q5/5

1906

QUESTION ID
S52-M-Q5

TYPE
Short answer

CORRECT ANSWER
$N = 800(0.5^{1/10})^t \approx 800(0.933033)^t$; about 6.70% decrease per year

GRADING NOTE Form required: rewritten form and rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 1

Read growth rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = 100(1.08)^t$ growth or decay, and at what percent per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q1/5

1908

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Growth, 8% per year

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = 500(0.97)^t$ growth or decay, and at what percent per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q2/5

1909

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Decay, 3% per year

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 2(1.25)^t$ grows 125% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q3/5

1910

QUESTION ID

S52-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The factor 1.25 means 25% growth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 40(0.9)^t$ decreases by

A. 9% per unit

B. 10% per unit

C. 90% per unit

D. 0.9% per unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q4/5

1911

QUESTION ID

S52-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 10% per unit

GRADING NOTE $1 - 0.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the growth factor for 6.17% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q5/5

1912

QUESTION ID

S52-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1.0617

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 2

Powers of powers and rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite 1.08^{t} as a power of $1.08^{1/12}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q1/5

1914

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(1.08^{1/12})^{12t}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2^{t/3}$ as $(2^{1/3})^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q2/5

1915

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2^{1/3})^t$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B2-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q2/5

True or false?

$(b^m)^n = b^{mn}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q3/5

1916

QUESTION ID

S52-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

1.005^{12t} equals

A. $(1.005^{12})^{t}$

B. $(1.005^{t})/12$

C. $12 \cdot 1.005^{t}$

D. $1.005^{12} + t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q4/5

1917

QUESTION ID

S52-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $(1.005^{12})^{t}$

GRADING NOTE Power of a power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $8^{1/3}$ as a whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q5/5

1918

QUESTION ID

S52-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 3

Percent change and growth factors • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write the growth factor for a 25.99% increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q1/5

1920

QUESTION ID

S52-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.2599

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A factor 0.7374 means what percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q2/5

1921

QUESTION ID

S52-B3-Q2

TYPE

Short answer

CORRECT ANSWER

26.26%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q2/5

True or false?

A factor 1.006434 means about 0.64% growth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q3/5

1922

QUESTION ID

S52-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3% decrease corresponds to the factor

A. 0.03

B. 0.97

C. 0.7

D. 1.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q4/5

1923

QUESTION ID

S52-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.97

GRADING NOTE 1 − 0.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the percent change for a factor of 0.933.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q5/5

1924

QUESTION ID

S52-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6.7% decrease

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 1

Compare growth offers by rewriting to the same time unit • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Account A grows 0.5% per month. Account B grows 6.1% per year. Which grows faster? Rewrite to compare. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q1/5

1926

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A: $1.005^{12} \approx 1.0617$, about 6.17% per year > 6.1%, so A grows faster

GRADING NOTE Form required: comparison on a common unit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A culture doubles every 3 hours; another triples every 5 hours. Which grows faster per hour? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q2/5

1927

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$2^{1/3} \approx 1.260$ vs $3^{1/5} \approx 1.246$; the first grows faster

GRADING NOTE Form required: both hourly factors and the decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q2/5

True or false?

Growing 1% per month is the same as growing 12% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q3/5

1928

QUESTION ID

S52-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1.01^{12} \approx 1.1268$, about 12.68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the greatest annual growth?

A. 4% per half year

B. 0.6% per month

C. 8% per year

D. 2% per quarter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q4/5

1929

QUESTION ID

S52-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2% per quarter

GRADING NOTE $1.02^4 \approx 1.0824$; $1.006^{12} \approx 1.0744$; $1.04^2 = 1.0816$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A half-life of 10 years and a decay of 7% per year: which decays faster? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q5/5

1930

QUESTION ID
S52-F1-Q5

TYPE
Short answer

CORRECT ANSWER
7% per year (0.93 < 0.933 per year)

GRADING NOTE Form required: decision and factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 2

Find doubling times and half-lives with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the doubling time of $y = 100(1.08)^t$, to the nearest hundredth of a year. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q1/5

1932

QUESTION ID
S52-F2-Q1

TYPE
Short answer

CORRECT ANSWER
9.01 years

GRADING NOTE Any equivalent form. log 2/log 1.08.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the half-life of $y = 500(0.97)^t$ to the nearest hundredth of a year. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q2/5

1933

QUESTION ID
S52-F2-Q2

TYPE
Short answer

CORRECT ANSWER
22.76 years

GRADING NOTE Any equivalent form. log 0.5/log 0.97.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q2/5

True or false?

The doubling time of $y = 50(2)^{t/3}$ is 6 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q3/5

1934

QUESTION ID

S52-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The exponent t/3 increases by 1 every 3 hours, which doubles y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q3/5

The doubling time of a quantity growing 6.17% per year is about

A. 16.2 years

B. 11.6 years

C. 6.17 years

D. 3.1 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q4/5

1935

QUESTION ID

S52-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 11.6 years

GRADING NOTE log 2/log 1.0617 ≈ 11.58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why rewriting $y = 100(1.08)^t$ as $100 \cdot 2^{t/9.01}$ shows the doubling time.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q5/5

1936

QUESTION ID
S52-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The exponent t/9.01 increases by 1 every 9.01 years, which multiplies y by 2.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of Algebra II functions in different representations

MAIN	Algebra II main question	**HSF-IF.C.9**
BACKWARD 1	Compare linear, quadratic, and exponential functions	**HSF-IF.C.9**
BACKWARD 2	Recognize a pattern in a table	**8.F.A.3**
BACKWARD 3	Domains, asymptotes, and periods of Algebra II families	**HSF-IF.B.5**
FORWARD 1	Compare rates of change and long-run behavior across represent…	**HSF-IF.C.9**
FORWARD 2	Build a difference function to compare	**HSF-BF.A.1.b**

SET 53 • HSF-IF.C.9

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3(1.1)^x$. The table shows g. Which has the greater y-intercept, and which is greater at x = 20? A calculator may be used.

x	0	1	2	3	4
g(x)	10	15	20	25	30

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q1/5

1939

QUESTION ID

S53-M-Q1

TYPE

Short answer

CORRECT ANSWER

g has the greater y-intercept (10 > 3); at x = 20, f(20) ≈ 20.18 < g(20) = 110, so g is greater

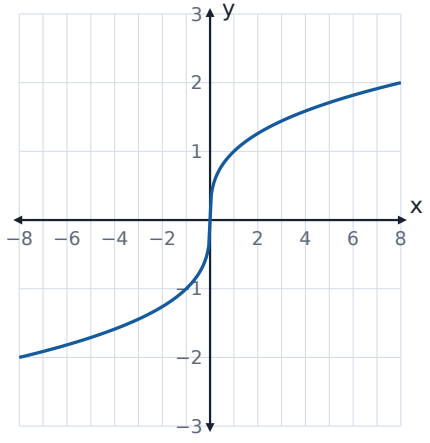
GRADING NOTE Form required: both comparisons with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{(x - 1)}$. The graph of g is shown. Compare their domains.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q2/5

1940

QUESTION ID

S53-M-Q2

TYPE

Short answer

CORRECT ANSWER

f: x ≥ 1; g: all real numbers (g is a cube root graph)

GRADING NOTE Form required: both domains.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q2/5

h(t) = 12 + 10sin(πt/15) gives one Ferris wheel's height (m, t in s). A second wheel has radius 8 m, center 10 m high, and turns once every 40 s. Compare the maximum heights and the periods.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q3/5

1941

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

Maximum heights 22 m and 18 m; periods 30 s and 40 s; the first wheel is higher and faster

GRADING NOTE Form required: both comparisons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$. The table shows g. Which is greater at $x = 9$?

x	1	3	9	27
g(x)	0	1	2	3

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q4/5

1942

QUESTION ID

S53-M-Q4

TYPE

Short answer

CORRECT ANSWER

f(9) ≈ 3.17 > g(9) = 2

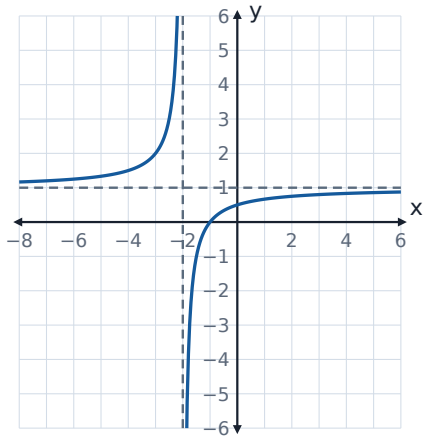
GRADING NOTE Form required: answer with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = (2x + 1)/(x − 3). The graph of g shows the horizontal asymptote y = 1 and the vertical asymptote x = −2. Compare the asymptotes.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q5/5

1943

QUESTION ID

S53-M-Q5

TYPE

Short answer

CORRECT ANSWER

f: horizontal y = 2, vertical x = 3; g: horizontal y = 1, vertical x = −2

GRADING NOTE Form required: both pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 1

Compare linear, quadratic, and exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 2x + 1 and g has table g(0) = 3, g(1) = 4. Which has the greater y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q1/5

1945

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

g (3 > 1)

GRADING NOTE Form required: answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = 2^x$. Which is greater at $x = 10$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q2/5

1946

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

g: 1024 > 100

GRADING NOTE Form required: answer with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q2/5

True or false?

An increasing exponential eventually exceeds any linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q3/5

1947

QUESTION ID

S53-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Algebra I result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q3/5

$f(x) = -(x - 1)^2 + 5$ and g with maximum 4 in a graph. The greater maximum is

A. f: 5

B. cannot tell

C. equal

D. g: 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q4/5

1948

QUESTION ID

S53-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. f: 5

GRADING NOTE Vertex form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table rises by 5 each step. What kind of function is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q5/5

1949

QUESTION ID

S53-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Linear

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 2

Recognize a pattern in a table • Grade 8 • Functions • 8.F.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Is the table x: 0, 1, 2; y: 10, 15, 20 linear? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q1/5

1951

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: constant difference 5

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is the table x: 1, 3, 9, 27; y: 0, 1, 2, 3 linear? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q2/5

1952

QUESTION ID

S53-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No: equal output steps need inputs multiplied by 3 (logarithmic)

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q2/5

True or false?

A table with constant ratios of outputs is exponential.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q3/5

1953

QUESTION ID

S53-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Repeated multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which table is exponential?

A. 1, 4, 9, 16

B. 1, 2, 4, 8

C. 2, 2, 2, 2

D. 1, 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q4/5

1954

GRADING NOTE Ratio 2.

QUESTION ID

S53-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1, 2, 4, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the slope of a linear table with outputs 10, 15, 20 at x = 0, 1, 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q5/5

1955

QUESTION ID

S53-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 3

Domains, asymptotes, and periods of Algebra II families • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{(x - 1)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q1/5

1957

QUESTION ID

S53-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the asymptotes of (2x + 1)/(x − 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q2/5

1958

QUESTION ID

S53-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 and y = 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q2/5

True or false?

The period of $12 + 10\sin(\pi t/15)$ is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q3/5

1959

QUESTION ID

S53-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2\pi/(\pi/15) = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log_2 x$ is

A. all reals

B. $x \geq 0$

C. $x > 0$

D. $x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q4/5

1960

QUESTION ID

S53-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x > 0

GRADING NOTE Positive inputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheel of radius 8 m with center 10 m high has what maximum height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q5/5

1961

QUESTION ID

S53-B3-Q5

TYPE

Short answer

CORRECT ANSWER

18 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 1

Compare rates of change and long-run behavior across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3(1.1)^x$ and $g(x) = 10 + 5x$. Estimate the first whole number x at which f exceeds g. A calculator may be used. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q1/5

1963

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

46

GRADING NOTE Any equivalent form. $f(45) \approx 218.7 < g(45) = 235$; $f(46) \approx 240.6 > g(46) = 240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$, and g has g(1) = 0, g(3) = 1, g(9) = 2. Compare the average rates of change of f and g on $1 \le x \le 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q2/5

1964

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

f: (3.17 − 0)/8 ≈ 0.40; g: 2/8 = 0.25; f is greater

GRADING NOTE Form required: both rates and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q2/5

True or false?

A function with a larger y-intercept is always larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q3/5

1965

QUESTION ID

S53-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE g starts higher but f eventually passes it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which grows fastest for large x?

A. $\sqrt{(x - 1)}$

B. $3(1.1)^x$

C. $5x + 10$

D. $\log_2 x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q4/5

1966

QUESTION ID

S53-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $3(1.1)^x$

GRADING NOTE Exponential.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $g(x) = \log_3 x$ stays below $f(x) = \log_2 x$ for $x > 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q5/5

1967

QUESTION ID

S53-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\log_3 x = \log_2 x / \log_2 3$ and $\log_2 3 > 1$, so g is a smaller positive multiple of f.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 2

Build a difference function to compare • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Write $d(x) = f(x) - g(x)$ for $f(x) = 3(1.1)^x$ and $g(x) = 5x + 10$.

$d(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q1/5

1969

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$3(1.1)^x - 5x - 10$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain what the sign of d(x) tells you.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q2/5

1970

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

d(x) > 0 means f is greater; d(x) < 0 means g is greater.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q2/5

True or false?

A zero of d(x) = f(x) − g(x) is an x where f(x) = g(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q3/5

1971

QUESTION ID
S53-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Equal outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$, $h_1(0) - h_2(0)$ is

A. 2

B. 22

C. −2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q4/5

1972

QUESTION ID

S53-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE 12 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two Ferris wheel heights are $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$. Write $h_1(t) - h_2(t)$ as a function of t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q5/5

1973

QUESTION ID

S53-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$h_1(t) - h_2(t) = 2 + 10\sin(\pi t/15) - 8\sin(\pi t/20)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q5/5

Build models by combining Algebra II function types

MAIN	Algebra II main question	**HSF-BF.A.1.b**
BACKWARD 1	Combine linear and quadratic functions	**HSF-BF.A.1.b**
BACKWARD 2	Multiply and subtract polynomials	**HSA-APR.A.1**
BACKWARD 3	Evaluate exponential and trigonometric expressions	**HSF-IF.A.2**
FORWARD 1	Interpret the parts of a combined model	**HSF-BF.A.1.b**
FORWARD 2	Compose functions in context	**HSF-BF.A.1.c**

SET 54 • HSF-BF.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Revenue is $R(x) = x(50 - 0.5x)$ and cost is $C(x) = 200 + 10x$ dollars for x items. Write the profit $P(x)$ in standard form and find $P(30)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q1/5

1976

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

$P(x) = -0.5x^2 + 40x - 200$; $P(30) = 550$

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The cost of x posters is C(x) = 1200 + 3x dollars. Write the average cost per poster A(x) and find A(100).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q2/5

1977

QUESTION ID
S54-M-Q2

TYPE
Short answer

CORRECT ANSWER
A(x) = (1200 + 3x)/x; A(100) = 15

GRADING NOTE Form required: function and value with units ($15 per poster).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Write the food per person and find it at t = 10 to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q3/5

1978

QUESTION ID
S54-M-Q3

TYPE
Short answer

CORRECT ANSWER
$F(t)/P(t) = (6000 + 100t)/(5000(1.02)^t)$; about 1.15 units per person

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A spring's amplitude is 10 cm and shrinks by 10% each second, while it oscillates once per second. Write y(t) as an exponential times a cosine and find y(1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q4/5

1979

QUESTION ID

S54-M-Q4

TYPE

Short answer

CORRECT ANSWER

$y(t) = 10(0.9)^t \cos(2\pi t)$; $y(1) = 9$

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Coffee starts 70°C above a 20°C room, and the difference shrinks 5% each minute. Write its temperature T(t) as a sum of functions and find T(10) to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q5/5

1980

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

$T(t) = 20 + 70(0.95)^t$; $T(10) \approx 61.91°C$

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 1

Combine linear and quadratic functions • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

R(x) = 8x and C(x) = 3x + 50. Write P = R − C.

P(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q1/5

1982

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5x − 50

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = x + 2 and g(x) = x − 2. Write f · g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q2/5

1983

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Profit is revenue minus cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q3/5

1984

QUESTION ID

S54-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Average cost is

A. total cost times items

B. total cost divided by the number of items

C. cost minus revenue

D. revenue divided by cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q4/5

1985

QUESTION ID
S54-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. total cost divided by the number of items

GRADING NOTE Per item.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x$ and $g(x) = x^2 + 1$. Find $(f + g)(3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q5/5

1986

QUESTION ID

S54-B1-Q5

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 2

Multiply and subtract polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand x(50 − 0.5x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q1/5

1988

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$50x - 0.5x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(50x - 0.5x^2) - (200 + 10x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q2/5

1989

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-0.5x^2 + 40x - 200$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-(200 + 10x) = -200 + 10x.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q3/5

1990

QUESTION ID

S54-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minus sign applies to both terms: $-200 - 10x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(3x + 1) − (x − 4) equals

A. 2x + 5

B. 4x − 3

C. 2x − 3

D. 2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q4/5

1991

QUESTION ID

S54-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2x + 5

GRADING NOTE Subtract each term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $2(x^2 - 3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q5/5

1992

QUESTION ID

S54-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$2x^2 - 6x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 3

Evaluate exponential and trigonometric expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $70(0.95)^{10}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q1/5

1994

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

41.91

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $5000(1.02)^{10}$ to the nearest whole number. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q2/5

1995

QUESTION ID
S54-B3-Q2

TYPE
Short answer

CORRECT ANSWER
6095

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q3/5

1996

QUESTION ID

S54-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Full turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$10(0.9)^2$ equals

A. 9

B. 18

C. 81

D. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q4/5

1997

QUESTION ID

S54-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 8.1

GRADING NOTE $10 \cdot 0.81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\cos(2\pi \cdot 0.5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q5/5

1998

QUESTION ID

S54-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 1

Interpret the parts of a combined model • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

In $T(t) = 20 + 70(0.95)^t$, what do 20 and $70(0.95)^t$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q1/5

2000

QUESTION ID
S54-F1-Q1

TYPE
Short answer

CORRECT ANSWER
20: room temperature; $70(0.95)^t$: how far the coffee is above room temperature

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In $y(t) = 10(0.9)^t \cos(2\pi t)$, which factor controls the size of the swings and which controls their timing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q2/5

2001

QUESTION ID
S54-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$10(0.9)^t$ controls the size; $\cos(2\pi t)$ the timing

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In A(x) = (1200 + 3x)/x = 1200/x + 3, the term 1200/x is the fixed cost spread over x posters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q3/5

2002

QUESTION ID

S54-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Split the fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

As t grows, $T(t) = 20 + 70(0.95)^t$ approaches

A. 20

B. 0

C. 90

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q4/5

2003

QUESTION ID

S54-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20

GRADING NOTE The exponential part → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why food per person F(t)/P(t) eventually decreases.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q5/5

2004

QUESTION ID
S54-F1-Q5

TYPE
Short answer

CORRECT ANSWER
P grows exponentially while F grows linearly, so the denominator eventually grows much faster.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 2

Compose functions in context • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

A balloon's radius is r(t) = 2t cm after t seconds and its volume is V(r) = (4/3)πr³. Write V(r(t)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q1/5

2006

QUESTION ID
S54-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(32/3)πt³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A store takes 20% off with d(p) = 0.8p and then adds a $5 fee with f(p) = p + 5. Write f(d(p)) and find it for p = 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q2/5

2007

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8p + 5; $45

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

f(g(x)) and g(f(x)) are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q3/5

2008

QUESTION ID

S54-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Order matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(x) = x^2$ and $g(x) = x + 1$, then $f(g(2)) =$

A. 4

B. 9

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q4/5

2009

QUESTION ID

S54-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 9

GRADING NOTE f(3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how composition differs from multiplying two functions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q5/5

2010

QUESTION ID

S54-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Composition feeds the output of one function in as the input of the other; multiplication multiplies the two outputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q5/5

Identify vertical shifts, vertical stretches, and horizontal shifts of Algebra II functions

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Shifts and stretches of linear and quadratic graphs	**HSF-BF.B.3**
BACKWARD 2	Translate points in the plane	**8.G.A.3**
BACKWARD 3	Key points of parent functions	**HSF-IF.C.7.e**
FORWARD 1	Write transformed functions from descriptions and order of transfo..	**HSF-BF.B.3**
FORWARD 2	Read sinusoid parameters as transformations	**HSF-TF.B.5**

SET 55 • HSF-BF.B.3

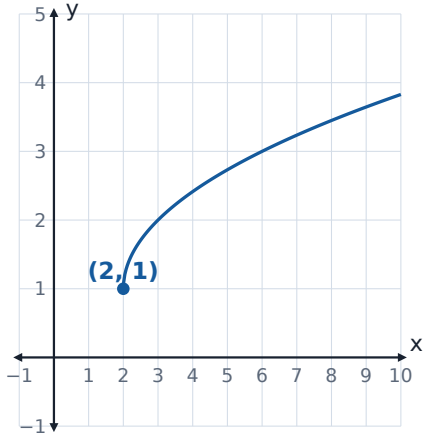
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{x}$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q1/5

2013

QUESTION ID

S55-M-Q1

TYPE

Short answer

CORRECT ANSWER

h = 2, k = 1

GRADING NOTE Form required: both. The starting point moves from (0, 0) to (2, 1).

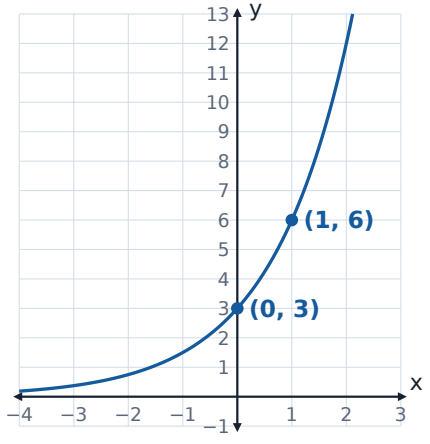
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$. The graph of $g(x) = a \cdot f(x)$ is shown. Find a.

a = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q2/5

2014

QUESTION ID
S55-M-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. g(0) = a.

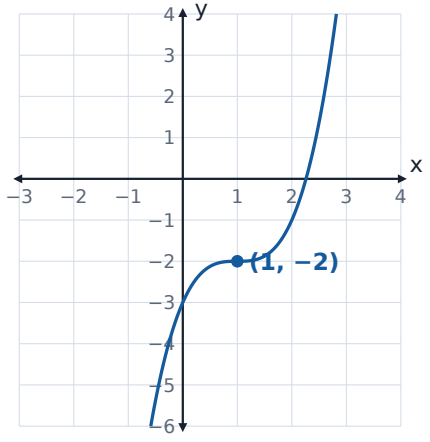
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q2/5

2014

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^3$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q3/5

2015

QUESTION ID

S55-M-Q3

TYPE

Short answer

CORRECT ANSWER

h = 1, k = −2

GRADING NOTE Form required: both.

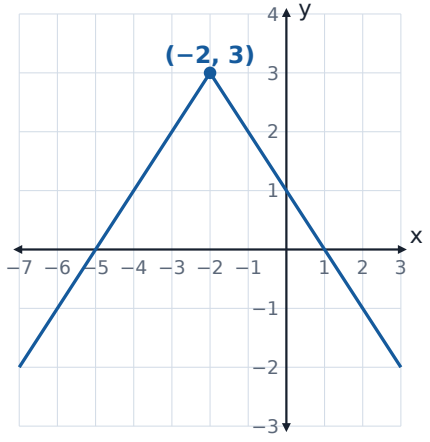
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q3/5

2015

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = |x|. The graph of g(x) = a·f(x − h) + k is shown. Find a, h, and k.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q4/5

2016

QUESTION ID

S55-M-Q4

TYPE

Short answer

CORRECT ANSWER

a = −1, h = −2, k = 3

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

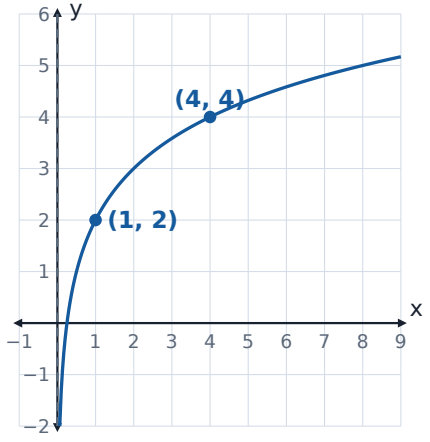
High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q4/5

2016

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$. The graph of $g(x) = f(x) + k$ is shown. Find k.

$k =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q5/5

2017

QUESTION ID
S55-M-Q5

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. f(1) = 0 becomes 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q5/5

2017

SET 55 • HSF-BF.B.3

BACKWARD 1

Shifts and stretches of linear and quadratic graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How is $y = (x - 3)^2 + 1$ obtained from $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q1/5

2019

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Right 3, up 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q1/5

2019

ANSWER KEY Reduced copy of the question page, with its answer

How is $y = 2x^2$ obtained from $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q2/5

2020

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertical stretch by 2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q2/5

2020

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = f(x + 4) is the graph of f shifted right 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q3/5

2021

QUESTION ID

S55-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Left 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The vertex of $y = (x + 2)^2 - 5$ is

A. (2, 5)

B. (2, −5)

C. (−2, −5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q4/5

2022

QUESTION ID

S55-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (−2, −5)

GRADING NOTE $h = -2$, $k = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q4/5

2022

ANSWER KEY Reduced copy of the question page, with its answer

Write $y = x^2$ shifted left 1 and down 4.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q5/5

2023

QUESTION ID

S55-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2 - 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q5/5

2023

SET 55 • HSF-BF.B.3

BACKWARD 2

Translate points in the plane • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Translate (0, 0) right 2 and up 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q1/5

2025

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q1/5

2025

ANSWER KEY Reduced copy of the question page, with its answer

Translate (1, 1) left 2 and up 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q2/5

2026

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, 4)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q2/5

2026

True or false?

Translating (0, 0) down 2 gives (−2, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q3/5

2027

QUESTION ID

S55-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Down 2 changes y: (0, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x, y) \to (x - 2, y + 3)$ moves $(0, 0)$ to

A. $(-2, 3)$

B. $(-2, -3)$

C. $(2, 3)$

D. $(3, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q4/5

2028

QUESTION ID

S55-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (−2, 3)

GRADING NOTE Apply the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which translation takes (0, 0) to (1, −2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q5/5

2029

QUESTION ID

S55-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Right 1, down 2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 3

Key points of parent functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Give the point of $y = \sqrt{x}$ at $x = 0$ and at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q1/5

2031

QUESTION ID

S55-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 0) and (1, 1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the y-intercept of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q2/5

2032

QUESTION ID

S55-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q2/5

True or false?

The graph of $y = \log_2 x$ passes through (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q3/5

2033

QUESTION ID

S55-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\log_2 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q3/5

The vertex of y = |x| is

A. (1, 0)

B. (1, 1)

C. (0, 1)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q4/5

2034

QUESTION ID

S55-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0)

GRADING NOTE Parent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the center (point of symmetry) of $y = x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q5/5

2035

QUESTION ID
S55-B3-Q5

TYPE
Short answer

CORRECT ANSWER
(0, 0)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 1

Write transformed functions from descriptions and order of transformations • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Write g, the graph of $f(x) = 2^x$ stretched vertically by 3, then shifted down 4.

g(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q1/5

2037

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3 \cdot 2^x - 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q1/5

Write g, the graph of f(x) = √x shifted down 4, then stretched vertically by 3.

g(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q2/5

2038

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3√x − 12

GRADING NOTE Any equivalent form. 3(√x − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q2/5

True or false?

A vertical stretch then a shift can give a different graph than the same shift then the stretch.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q3/5

2039

QUESTION ID
S55-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE See the two items above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 2f(x) + 1$ moves the point (3, 4) of f to

A. (6, 5)

B. (4, 9)

C. (3, 9)

D. (3, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q4/5

2040

QUESTION ID

S55-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (3, 9)

GRADING NOTE $2 \cdot 4 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

g(x) = f(x + 2) − 3 and f has asymptote x = 0. Where is g's vertical asymptote?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q5/5

2041

QUESTION ID

S55-F1-Q5

TYPE

Short answer

CORRECT ANSWER

x = −2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 2

Read sinusoid parameters as transformations • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Describe y = 3sin x + 2 as transformations of y = sin x and give its maximum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q1/5

2043

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Vertical stretch by 3, up 2; maximum 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the function for y = cos x stretched vertically by 4 and shifted down 1.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q2/5

2044

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4cos x − 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = A sin x + D, A is the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q3/5

2045

QUESTION ID

S55-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE D gives the midline y = D; |A| is the amplitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = -2\cos x + 5$ has amplitude and midline

A. 5 and $y = 2$

B. -2 and $y = 5$

C. 2 and $y = -5$

D. 2 and $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q4/5

2046

QUESTION ID

S55-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2 and $y = 5$

GRADING NOTE |A|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A model is y = A sin x + D with maximum 9 and minimum 1. Find A (A > 0) and D.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q5/5

2047

QUESTION ID

S55-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A = 4, D = 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q5/5

Horizontal stretches, compressions, and reflections f(kx)

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Reflections and dilations of points	**8.G.A.3**
BACKWARD 2	Solve kx = c for x	**6.EE.B.7**
BACKWARD 3	Period of y = sin(Bx)	**HSF-TF.B.5**
FORWARD 1	Combine horizontal stretches with shifts	**HSF-BF.B.3**
FORWARD 2	Graph exponential and trigonometric functions with horizontal cha…	**HSF-IF.C.7.e**

SET 56 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √x passes through (4, 2). Describe the graph of g(x) = √(4x) compared with f, and give the point of g with y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q1/5

2050

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

Horizontal compression by a factor of 1/4; (1, 2)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = sin x has period 2π. Describe g(x) = sin(3x) compared with f and give its period.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q2/5

2051

QUESTION ID
S56-M-Q2

TYPE
Short answer

CORRECT ANSWER
Horizontal compression by a factor of 1/3; period 2π/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe the graph of $g(x) = 2^{-x}$ compared with $f(x) = 2^{x}$, and give the point of g that comes from (1, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q3/5

2052

QUESTION ID
S56-M-Q3

TYPE
Short answer

CORRECT ANSWER
Reflection over the y-axis; (−1, 2)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q3/5

f passes through (−2, 1), (0, 3), and (4, −1). List the corresponding points of g(x) = f(2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q4/5

2053

QUESTION ID

S56-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−1, 1), (0, 3), (2, −1)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q4/5

f passes through (2, 5). Which point must be on the graph of g(x) = f(−x/2)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q5/5

2054

QUESTION ID

S56-M-Q5

TYPE

Short answer

CORRECT ANSWER

(−4, 5):

g(−4) = f(2) = 5

GRADING NOTE Form required: point and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 1

Reflections and dilations of points • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Reflect (1, 2) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q1/5

2056

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply the x-coordinate of (4, −1) by 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q2/5

2057

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q2/5

True or false?

Reflecting over the y-axis changes the sign of the x-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S56-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q3/5

2058

QUESTION ID

S56-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S56-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q3/5

(x, y) → (x/3, y) applied to (6, 2) gives

A. (18, 2)

B. (6, 2/3)

C. (2, 2)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q4/5

2059

QUESTION ID

S56-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 2)

GRADING NOTE Divide x by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Apply (x, y) → (−2x, y) to (2, 5).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q5/5

2060

QUESTION ID

S56-B1-Q5

TYPE

Short answer

CORRECT ANSWER

(−4, 5)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 2

Solve kx = c for x • Grade 6 • Expressions & Equations • 6.EE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q1/5

2062

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-x/2 = 2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q2/5

2063

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If 4x = 4, then x = 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q3/5

2064

QUESTION ID

S56-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by 4: x = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3x = 2\pi$ gives $x =$

A. $2\pi/3$

B. $2\pi - 3$

C. $\pi/3$

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q4/5

2065

QUESTION ID

S56-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2π/3

GRADING NOTE Divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-x = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q5/5

2066

QUESTION ID

S56-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 3

Period of y = sin(Bx) • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of y = sin(2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q1/5

2068

QUESTION ID

S56-B3-Q1

TYPE

Short answer

CORRECT ANSWER

π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of y = cos(x/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q2/5

2069

QUESTION ID

S56-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For y = sin(Bx) with B > 0, a larger B gives a longer period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q3/5

2070

QUESTION ID
S56-B3-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The period 2π/B gets shorter as B grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

y = sin(4x) has period

A. 8π

B. 4π

C. π/4

D. π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q4/5

2071

QUESTION ID

S56-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. π/2

GRADING NOTE 2π/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which B gives period 6?

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q5/5

2072

QUESTION ID

S56-B3-Q5

TYPE

Short answer

CORRECT ANSWER

π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 1

Combine horizontal stretches with shifts • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Rewrite g(x) = f(2x − 6) in the form f(k(x − h)) and describe the transformation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q1/5

2074

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(2(x − 3)): compress horizontally by 1/2, then shift right 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q1/5

f passes through (4, 1). Which point is on $g(x) = f(2x - 6)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q2/5

2075

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 1)

GRADING NOTE Any equivalent answer. $2x - 6 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q2/5

True or false?

f(2x − 6) is f(2x) shifted right 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q3/5

2076

QUESTION ID

S56-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted right 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

y = √(9x) equals

A. √x + 9

B. 9√x

C. 3√x

D. √x/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q4/5

2077

QUESTION ID
S56-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 3√x

GRADING NOTE Horizontal compression equals vertical stretch for √.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write g, the graph of $f(x) = 2^x$ reflected over the y-axis and then shifted right 1.

g(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q5/5

2078

QUESTION ID

S56-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$2^{-(x-1)} = 2^{1-x}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 2

Graph exponential and trigonometric functions with horizontal changes • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Give the y-intercept, asymptote, and two points of $y = 2^{-x} + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q1/5

2080

QUESTION ID
S56-F2-Q1

TYPE
Short answer

CORRECT ANSWER
y-intercept 2; asymptote y = 1; points (−1, 3), (1, 1.5)

GRADING NOTE Form required: all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the maxima of $y = \cos(2x)$ on $0 \le x \le 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q2/5

2081

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 0, π, 2π (value 1)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 3^{2x} = 9^{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q3/5

2082

QUESTION ID
S56-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Power of a power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = \sin(x/2)$ completes one cycle on

A. $0 \le x \le \pi$

B. $0 \le x \le 2\pi$

C. $0 \le x \le \pi/2$

D. $0 \le x \le 4\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q4/5

2083

QUESTION ID
S56-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $0 \le x \le 4\pi$

GRADING NOTE Period 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $y = 2^{2x}$ is a horizontal compression of $y = 2^{x}$ and also an exponential with base 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q5/5

2084

QUESTION ID
S56-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Replacing x by 2x compresses horizontally; $2^{2x} = (2^2)^x = 4^x$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q5/5

Recognize even and odd functions

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Signs of powers of negatives	**7.NS.A.2**
BACKWARD 2	Reflect points over the axes and the origin	**6.NS.C.6.b**
BACKWARD 3	Substitute -x into an expression	**HSF-IF.A.2**
FORWARD 1	Reason about sums and products of even and odd functions	**HSF-BF.B.3**
FORWARD 2	Explain trigonometric symmetry with the unit circle	**HSF-TF.A.4**

SET 57 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Is $f(x) = x^4 - 3x^2$ even, odd, or neither? Show $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q1/5

2087

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

Even: $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$

GRADING NOTE Form required: decision and the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $f(x) = x^3 - x$ even, odd, or neither? Show $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q2/5

2088

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

Odd: $f(-x) = -x^3 + x = -f(x)$

GRADING NOTE Form required: decision and the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $f(x) = x^2 + x$ even, odd, or neither? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q3/5

2089

QUESTION ID

S57-M-Q3

TYPE

Short answer

CORRECT ANSWER

Neither: f(1) = 2 and f(−1) = 0, so f(−1) ≠ f(1) and f(−1) ≠ −f(1)

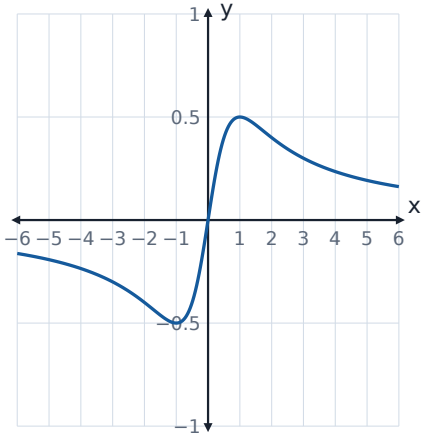
GRADING NOTE Form required: decision and a counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = x/(x^2 + 1)$ is shown. Is f even, odd, or neither? Explain from the graph.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q4/5

2090

QUESTION ID
S57-M-Q4

TYPE
Short answer

CORRECT ANSWER
Odd: the graph is symmetric about the origin

GRADING NOTE Form required: decision and the symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is f(x) = cos x + x² even, odd, or neither? Show f(−x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q5/5

2091

QUESTION ID

S57-M-Q5

TYPE

Short answer

CORRECT ANSWER

Even:

f(−x) = cos(−x) + (−x)² = cos x + x²

GRADING NOTE Form required: decision and the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 1

Signs of powers of negatives • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-x)^4$ and $(-x)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q1/5

2093

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x^4 and $-x^3$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q2/5

2094

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x^2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q2/5

True or false?

$(-x)^5 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q3/5

2095

QUESTION ID

S57-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An odd power keeps the sign: $(-x)^5 = -x^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q3/5

2095

ANSWER KEY Reduced copy of the question page, with its answer

$(-2)^2 - (-2)$ equals

A. -6

B. 2

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q4/5

2096

QUESTION ID

S57-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE 4 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q4/5

2096

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-(-x)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q5/5

2097

QUESTION ID

S57-B1-Q5

TYPE

Short answer

CORRECT ANSWER

x^3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 2

Reflect points over the axes and the origin • Grade 6 • The Number System • 6.NS.C.6.b

ANSWER KEY Reduced copy of the question page, with its answer

Reflect (2, 3) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q1/5

2099

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (2, 3) 180° about the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q2/5

2100

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q2/5

True or false?

A graph symmetric about the y-axis contains (−a, b) whenever it contains (a, b).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q3/5

2101

QUESTION ID

S57-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Even symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Symmetry about the origin pairs (a, b) with

A. (−a, b)

B. (−a, −b)

C. (a, −b)

D. (b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q4/5

2102

QUESTION ID

S57-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−a, −b)

GRADING NOTE 180° rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph is symmetric about the origin and contains (1, 0.5). Name another point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q5/5

2103

QUESTION ID

S57-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(−1, −0.5)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 3

Substitute −x into an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^2 + x$, find $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q1/5

2105

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^3 - x$, find $f(-2)$ and $f(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q2/5

2106

QUESTION ID

S57-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−6 and 6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q2/5

True or false?

For f(x) = 3x, f(−x) = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S57-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q3/5

2107

QUESTION ID

S57-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(−x) = 3(−x) = −3x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S57-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^4 - 3x^2$, $f(-1)$ is

A. 2

B. 4

C. -2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q4/5

2108

GRADING NOTE 1 − 3.

QUESTION ID

S57-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = 1/x, find f(−x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q5/5

2109

QUESTION ID

S57-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1/x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 1

Reason about sums and products of even and odd functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Which argument proves that the sum of two odd functions f and g is odd?

A. x^3 and x^5 are odd and $x^3 + x^5$ is odd, so every sum of two odd functions is odd.

B. $(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f(x) + g(x)) = -(f + g)(x)$.

C. $(f + g)(-x) = f(x) + g(x) = (f + g)(x)$, so the sum is odd.

D. $(f + g)(-x) = -f(-x) - g(-x) = -(f + g)(-x)$, so the sum is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q1/5

2111

QUESTION ID

S57-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f(x) + g(x)) = -(f + g)(x)$.

GRADING NOTE Use $f(-x) = -f(x)$ and $g(-x) = -g(x)$, then factor out -1. The others use the even-function rule (and would show even), check one example, or misapply the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q1/5

Which argument proves that the product of two odd functions f and g is even?

A. $(fg)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = (fg)(x)$.

B. $x \cdot x^3 = x^4$ is even, so every product of two odd functions is even.

C. Odd times odd is odd for whole numbers, so the product of two odd functions is odd.

D. $(fg)(-x) = f(-x)g(-x) = -f(x)g(x)$, so the product is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q2/5

2112

QUESTION ID

S57-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A.

$(fg)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = (fg)(x)$.

GRADING NOTE The two negative signs multiply to +1. The others drop one negative sign, check one example, or confuse odd functions with odd numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q2/5

True or false?

The only function that is both even and odd is f(x) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q3/5

2113

QUESTION ID

S57-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(x) = f(−x) = −f(x) gives f(x) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = |x| + x^3$ is

A. even

B. both

C. neither

D. odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q4/5

2114

QUESTION ID

S57-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. neither

GRADING NOTE $f(1) = 2$, $f(-1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q4/5

Is f(x) = x sin x even, odd, or neither? Explain using the fact that a product of two odd functions is even.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q5/5

2115

QUESTION ID

S57-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Even: x and sin x are both odd, so their product is even.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 2

Explain trigonometric symmetry with the unit circle • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to explain why cos(−x) = cos x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q1/5

2117

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The points at angles x and −x are reflections over the x-axis and have the same x-coordinate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is tan even, odd, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q2/5

2118

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Odd: tan(−x) = sin(−x)/cos(−x) = −sin x/cos x = −tan x

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q2/5

True or false?

sin is an even function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q3/5

2119

QUESTION ID

S57-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE sin(−x) = −sin x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is an even function?

A. cos x

B. tan x

C. sin x + cos x

D. sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q4/5

2120

QUESTION ID

S57-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos x

GRADING NOTE Reflection over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use symmetry to find $\sin(-\pi/6)$ from $\sin(\pi/6) = 1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q5/5

2121

QUESTION ID

S57-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$-1/2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q5/5

Write inverses of radical, rational, cubic, exponential, and logarithmic functions

MAIN	Algebra II main question	**HSF-BF.B.4.a**
BACKWARD 1	Inverses of linear functions	**HSF-BF.B.4.a**
BACKWARD 2	Rearrange formulas with roots, powers, and fractions	**HSA-CED.A.4**
BACKWARD 3	Domain and range	**HSF-IF.B.5**
FORWARD 1	Inverses that are their own inverse, and checking by points	**HSF-BF.B.4.a**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 58 • HSF-BF.B.4.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = (x - 1)^3 + 2$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q1/5

2124

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt[3]{(x - 2)} + 1$

GRADING NOTE Any equivalent form. Domain all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = 2x/(x - 3)$, $x \neq 3$, and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q2/5

2125

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = 3x/(x - 2)$, $x \neq 2$

GRADING NOTE Form required: formula and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q2/5

Find the inverse of $f(x) = \sqrt{(x + 4)}$, $x \geq -4$, and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q3/5

2126

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = x^2 - 4, x \geq 0$

GRADING NOTE Form required: formula and domain (the range of f).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = 5 \cdot 2^x$ and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q4/5

2127

QUESTION ID

S58-M-Q4

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = \log_2(x/5)$, $x > 0$

GRADING NOTE Form required: formula and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = \log_3(x - 1)$ and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q5/5

2128

QUESTION ID

S58-M-Q5

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = 3^x + 1$, all real x

GRADING NOTE Form required: formula and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 1

Inverses of linear functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of f(x) = 3x − 6.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

2130

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 6)/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 5 = y for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

2131

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(y − 5)/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

True or false?

The inverse undoes the function: $f^{-1}(f(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

2132

QUESTION ID

S58-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The inverse of f(x) = x/4 + 1 is

A. 4(x − 1)

B. (x − 1)/4

C. 4x − 1

D. x/4 − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

2133

QUESTION ID

S58-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4(x − 1)

GRADING NOTE Undo in reverse order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(2) = 7$, what is $f^{-1}(7)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

2134

QUESTION ID

S58-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 2

Rearrange formulas with roots, powers, and fractions • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = \sqrt{(x + 4)}$ for x ($y \geq 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q1/5

2136

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$y^2 - 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = 2x/(x - 3)$ for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q2/5

2137

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3y/(y − 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $y = (x - 1)^3 + 2$ for x gives $x = \sqrt[3]{(y - 2)} + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q3/5

2138

QUESTION ID

S58-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Undo each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving $y = x^3$ for x gives

A. y/3

B. $\sqrt[3]{y}$

C. $\sqrt{y}$

D. y^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q4/5

2139

QUESTION ID

S58-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $\sqrt[3]{y}$

GRADING NOTE Cube root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve y = x/(x + 1) for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q5/5

2140

QUESTION ID

S58-B2-Q5

TYPE

Short answer

CORRECT ANSWER

y/(1 − y)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 3

Domain and range • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of f(x) = √(x + 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q1/5

2142

QUESTION ID

S58-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = 5 \cdot 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q2/5

2143

QUESTION ID

S58-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q2/5

True or false?

The range of f(x) = 2x/(x − 3) excludes y = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q3/5

2144

QUESTION ID

S58-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(9) = 18/6 = 3; the excluded value is y = 2 (the horizontal asymptote).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q3/5

The domain of $\log_3(x - 1)$ is

A. $x > 1$

B. all reals

C. $x > 0$

D. $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q4/5

2145

QUESTION ID

S58-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x > 1

GRADING NOTE $x - 1 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = \log_3(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q5/5

2146

QUESTION ID

S58-B3-Q5

TYPE

Short answer

CORRECT ANSWER

All real numbers

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 1

Inverses that are their own inverse, and checking by points • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of f(x) = (x + 1)/(x − 1). What do you notice?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q1/5

2148

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = (x + 1)/(x - 1)$: f is its own inverse

GRADING NOTE Form required: formula and observation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 5 \cdot 2^x$ passes through (1, 10), and
$f^{-1}(x) = \log_2(x/5)$. Check the inverse with this point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q2/5

2149

QUESTION ID
S58-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$\log_2(10/5) = \log_2 2 = 1$

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q2/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q3/5

2150

QUESTION ID

S58-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only one-to-one functions; $f(x) = x^2$ on all reals does not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is its own inverse?

A. $f(x) = x^2$

B. $f(x) = x + 1$

C. $f(x) = 2x$

D. $f(x) = 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q4/5

2151

QUESTION ID

S58-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. f(x) = 1/x

GRADING NOTE $1/(1/x) = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the horizontal line test on the graph of $f(x) = x^3 - x$ to decide whether f has an inverse function. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q5/5

2152

QUESTION ID

S58-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A horizontal line such as y = 0 meets the graph three times, so f is not one-to-one and has no inverse function.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Verify that $f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$ are inverses by computing $f(g(x))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q1/5

2154

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$f(g(x)) = (\sqrt[3]{(x - 2)})^3 + 2 = x$

GRADING NOTE Form required: the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$. Compute $g(f(x))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q2/5

2155

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt[3]{((x - 1)^3)} + 1 = x$

GRADING NOTE Form required: the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q2/5

True or false?

To verify inverses, both f(g(x)) = x and g(f(x)) = x must hold on the domains.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q3/5

2156

QUESTION ID

S58-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two-sided.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = \sqrt{(x + 4)}$ and $g(x) = x^2 - 4$ $(x \geq 0)$, $g(f(x)) =$

A. x for all x

B. $|x|$

C. x for $x \geq -4$

D. $x^2 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q4/5

2157

QUESTION ID

S58-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x for x ≥ −4

GRADING NOTE $(\sqrt{(x + 4)})^2 - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $g(x) = x^2 - 4$ without a domain restriction is not the inverse of $f(x) = \sqrt{(x + 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q5/5

2158

QUESTION ID
S58-F2-Q5

TYPE
Short answer

CORRECT ANSWER
$f(g(x)) = \sqrt{(x^2)} = |x|$, which is not x when $x < 0$.

GRADING NOTE Any equivalent explanation (uses $\sqrt{(x^2)} = |x|$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q5/5

Write exact logarithmic solutions of exponential equations

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Isolate the exponential expression	**HSA-REI.B.3**
BACKWARD 2	Powers of 2, 3, and 10	**8.EE.A.1**
BACKWARD 3	Logarithms as the inverse of exponentials	**HSF-BF.B.4.a**
FORWARD 1	Solve equations with different bases or variables on both sides	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship to solve logarithmic equations	**HSF-BF.B.5**

SET 59 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3 \cdot 2^{5t} = 48$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q1/5

2161

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form. $2^{5t} = 16 = 2^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2 \cdot 10^{3t} = 9$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q2/5

2162

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

log(4.5)/3

GRADING NOTE Form required: exact (≈ 0.2177).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-M-Q2**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5e^{2t} = 30$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q3/5

2163

QUESTION ID

S59-M-Q3

TYPE

Short answer

CORRECT ANSWER

(ln 6)/2

GRADING NOTE Form required: exact (≈ 0.8959).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $100(1.05)^t = 300$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q4/5

2164

QUESTION ID

S59-M-Q4

TYPE

Short answer

CORRECT ANSWER

log 3/log 1.05

GRADING NOTE Form required: exact (any base; ≈ 22.52).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $4 \cdot 3^{t/2} = 20$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q5/5

2165

QUESTION ID

S59-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2 \log_3 5$

GRADING NOTE Form required: exact (2 log 5/log 3 accepted; ≈ 2.93).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-M-Q5**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 1

Isolate the exponential expression • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3y = 48 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q1/5

2167

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5y = 30 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q2/5

2168

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q2/5

True or false?

From $2 \cdot 10^{3t} = 9$, we get $10^{3t} = 4.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q3/5

2169

QUESTION ID

S59-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

From $4 \cdot 3^{t/2} = 20$, $3^{t/2}$ equals

A. 80

B. 5

C. 1/5

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q4/5

2170

QUESTION ID

S59-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE Divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 100y = 300 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q5/5

2171

QUESTION ID

S59-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 2

Powers of 2, 3, and 10 • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 16 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q1/5

2173

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2^4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1000 as a power of 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q2/5

2174

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q2/5

True or false?

$3^4 = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q3/5

2175

QUESTION ID

S59-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81$; 12 is $3 \cdot 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$2^{5t} = 2^4$ gives

A. t = 4

B. 5t = 4

C. 5t = 16

D. t = 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q4/5

2176

QUESTION ID

S59-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5t = 4

GRADING NOTE Equal exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1/8 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q5/5

2177

QUESTION ID

S59-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2^{-3}

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 3

Logarithms as the inverse of exponentials • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $10^x = 4.5$ as a logarithm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q1/5

2179

QUESTION ID

S59-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \log 4.5$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $e^u = 6$ as a logarithm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q2/5

2180

QUESTION ID

S59-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$u = \ln 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q2/5

True or false?

$\log_2 8 = 4$ because $2^4 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q3/5

2181

QUESTION ID

S59-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^4 = 16$; $\log_2 8 = 3$ because $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3^y = 5$ means y =

A. $\log_3 5$

B. 5/3

C. $\log_5 3$

D. 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q4/5

2182

QUESTION ID
S59-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $\log_3 5$

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\log_2 32$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q5/5

2183

QUESTION ID
S59-B3-Q5

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 1

Solve equations with different bases or variables on both sides • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^x = 3^{x-1}$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q1/5

2185

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

ln 3/(ln 3 − ln 2)

GRADING NOTE Form required: exact (any base; ≈ 2.71).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5^{2x} = 7$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q2/5

2186

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

log 7/(2 log 5)

GRADING NOTE Form required: exact (≈ 0.605).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

log(ab) = log a + log b for positive a and b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q3/5

2187

QUESTION ID

S59-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Product property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The exact solution of $e^{x+1} = 10$ is

A. 10/e

B. ln 10/1

C. ln 9

D. ln 10 − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q4/5

2188

QUESTION ID

S59-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ln 10 − 1

GRADING NOTE x + 1 = ln 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3 \cdot 4^x = 2 \cdot 5^x$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q5/5

2189

QUESTION ID

S59-F1-Q5

TYPE

Short answer

CORRECT ANSWER

log(3/2)/log(5/4)

GRADING NOTE Form required: exact (≈ 1.817).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 2

Use the inverse relationship to solve logarithmic equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2(x + 3) = 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q1/5

2191

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

29

GRADING NOTE Any equivalent form. x + 3 = 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve log x + log(x − 3) = 1 and check for extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q2/5

2192

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 5; x = −2 is extraneous

GRADING NOTE Form required: solution and rejection. x(x − 3) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q2/5

True or false?

$\log_b(b^x) = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q3/5

2193

QUESTION ID

S59-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\ln(e^7)$ equals

A. ln 7

B. 1

C. 7

D. e^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q4/5

2194

QUESTION ID

S59-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 7

GRADING NOTE Inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ln(2x) = 3 exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q5/5

2195

QUESTION ID

S59-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$e^3/2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q5/5

Evaluate logarithmic solutions of exponential models with technology

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Write exact logarithmic solutions	**HSF-LE.A.4**
BACKWARD 2	Evaluate logarithms with a calculator	**HSF-IF.A.2**
BACKWARD 3	Choose an appropriate level of accuracy	**HSN-Q.A.3**
FORWARD 1	Analyze how parameters change the solution	**HSF-LE.A.4**
FORWARD 2	Rewrite models to show doubling time or half-life	**HSF-IF.C.8.b**

SET 60 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$2,500 earns 3.5% interest compounded yearly. Solve $2500(1.035)^t = 4000$ and interpret the answer. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q1/5

2198

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

t = log 1.6/log 1.035 ≈ 13.66; the balance first reaches $4,000 after 14 yearly compoundings

GRADING NOTE Form required: value and interpretation (13.66 years; 14 whole years).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Carbon-14 has a half-life of 5730 years. Solve $80(0.5)^{t/5730} = 30$ to the nearest year and interpret it. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q2/5

2199

QUESTION ID
S60-M-Q2

TYPE
Short answer

CORRECT ANSWER
t ≈ 8108 years: a sample with 30 of its original 80 units of carbon-14 is about 8,108 years old

GRADING NOTE Form required: value and interpretation (8100 to 8110 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Coffee cools by $T(t) = 20 + 70(0.95)^t$ °C after t minutes. When is it 50°C? Give minutes to the nearest hundredth. A calculator may be used.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q3/5

2200

QUESTION ID
S60-M-Q3

TYPE
Short answer

CORRECT ANSWER
16.52 minutes

GRADING NOTE Any equivalent form. $0.95^t = 3/7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A city of 1.2 million grows by $P(t) = 1.2e^{0.015t}$ million people. When does it reach 1.5 million? Give years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q4/5

2201

QUESTION ID
S60-M-Q4

TYPE
Short answer

CORRECT ANSWER
14.88 years

GRADING NOTE Any equivalent form. ln 1.25/0.015.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A dose leaves $200(0.8)^t$ mg of a drug after t hours. After which whole hour is the amount first below 50 mg? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q5/5

2202

QUESTION ID
S60-M-Q5

TYPE
Short answer

CORRECT ANSWER
After 7 hours
(t > 6.21)

GRADING NOTE Form required: the whole hour and the evidence. $0.8^t < 0.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 1

Write exact logarithmic solutions • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $1.035^t = 1.6$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q1/5

2204

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

log 1.6/log 1.035

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^{0.015t} = 1.25$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q2/5

2205

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

ln 1.25/0.015

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q2/5

True or false?

$0.8^t = 0.25$ gives t = log 0.25/log 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q3/5

2206

QUESTION ID

S60-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take logs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(0.5)^{t/5730} = 0.375$ gives t =

A. 0.375 · 5730

B. 5730/log 0.375

C. log 0.375/5730

D. 5730 log 0.375/log 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q4/5

2207

QUESTION ID

S60-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 5730 log 0.375/log 0.5

GRADING NOTE Multiply by 5730.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $0.95^t = 3/7$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q5/5

2208

QUESTION ID

S60-B1-Q5

TYPE

Short answer

CORRECT ANSWER

log(3/7)/log 0.95

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 2

Evaluate logarithms with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate log 1.6 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q1/5

2210

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.2041

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q1/5

2210

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate ln 1.25 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q2/5

2211

QUESTION ID
S60-B2-Q2

TYPE
Short answer

CORRECT ANSWER
0.2231

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

log 0.25 is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q3/5

2212

QUESTION ID

S60-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.25 < 1, so log 0.25 < 0 (about −0.602).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

log 1.6/log 1.035 is about

A. 0.20

B. 46.0

C. 1.55

D. 13.66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q4/5

2213

QUESTION ID

S60-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13.66

GRADING NOTE 0.2041/0.01494.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate log 0.25/log 0.8 to two decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q5/5

2214

QUESTION ID
S60-B2-Q5

TYPE
Short answer

CORRECT ANSWER
6.21

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 3

Choose an appropriate level of accuracy • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A model gives 13.66 years for yearly interest. What whole-year answer fits the context?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q1/5

2216

QUESTION ID

S60-B3-Q1

TYPE

Short answer

CORRECT ANSWER

14 years

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 8108.16 years to the nearest hundred years.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q2/5

2217

QUESTION ID

S60-B3-Q2

TYPE

Short answer

CORRECT ANSWER

8100 years

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q2/5

True or false?

A time found from a model should be reported with more decimal places than the data justify.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q3/5

2218

QUESTION ID

S60-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Precision should match the context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug amount first drops below 50 mg at t = 6.21 h. If it is checked every hour, it is first below 50 mg at

A. 6.21 h

B. 6 h

C. 7 h

D. 5 h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q4/5

2219

QUESTION ID

S60-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 7 h

GRADING NOTE Next whole hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why might 14.88 years be reported as "about 15 years"?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q5/5

2220

QUESTION ID

S60-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Population models are approximate, so a rounded value is more honest.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q5/5

2220

SET 60 • HSF-LE.A.4

FORWARD 1

Analyze how parameters change the solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

If the interest rate changes from 3.5% to 5%, how long does it take $2,500 to reach $4,000? Give years to the nearest hundredth. A calculator may be used.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q1/5

2222

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9.63 years

GRADING NOTE Any equivalent form. log 1.6/log 1.05.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the doubling time at 6% yearly interest and compare it with the Rule of 72 estimate 72/6. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q2/5

2223

QUESTION ID
S60-F1-Q2

TYPE
Short answer

CORRECT ANSWER
log 2/log 1.06 ≈ 11.90 years; Rule of 72 gives 12 years (close)

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Doubling the initial amount in $a(b)^t = d$ halves the time t needed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q3/5

2224

QUESTION ID

S60-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It reduces t by one doubling time, not by half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In $200(0.8)^t = 50$, replacing 200 by 400 changes t by

A. doubling it

B. halving it

C. adding log 2/log(1/0.8) ≈ 3.11 hours

D. nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q4/5

2225

QUESTION ID

S60-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. adding log 2/log(1/0.8) ≈ 3.11 hours

GRADING NOTE $400(0.8)^t = 50$ needs $0.8^t = 1/8$ instead of 1/4: t grows by log(1/2)/log 0.8 ≈ 3.11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

At 3.5% yearly interest, explain why $2,500 takes as long to grow to $4,000 as $5,000 takes to grow to $8,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q5/5

2226

QUESTION ID

S60-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Both need the growth factor 1.6, and t = log 1.6/log 1.035 depends only on that ratio.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 2

Rewrite models to show doubling time or half-life • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2500(1.035)^t$ in the form $2500 \cdot 2^{t/D}$ and give D to the nearest hundredth. A calculator may be used.

D ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q1/5

2228

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

20.15 years

GRADING NOTE Any equivalent form. D = log 2/log 1.035.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q1/5

2228

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $200(0.8)^t$ in the form $200(0.5)^{t/H}$ and give H to the nearest hundredth. A calculator may be used.

H ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q2/5

2229

QUESTION ID
S60-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3.11 hours

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$80(0.5)^{t/5730}$ shows the half-life 5730 directly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q3/5

2230

QUESTION ID
S60-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Exponent t/5730.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$1.2e^{0.015t}$ has doubling time about

A. 46.2 years

B. 66.7 years

C. 15 years

D. 0.015 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q4/5

2231

QUESTION ID

S60-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 46.2 years

GRADING NOTE In 2/0.015.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the half-life form makes it easy to answer "how much remains after 3 half-lives".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q5/5

2232

QUESTION ID
S60-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Each half-life multiplies by 0.5, so 3 half-lives leave $(0.5)^3 = 1/8$ of the amount.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate population percentages with a normal distribution

MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Interpret mean and standard deviation	**HSS-ID.A.2**
BACKWARD 2	Describe the shape of a distribution	**HSS-ID.A.1**
BACKWARD 3	Percents of a whole	**6.RP.A.3.c**
FORWARD 1	Find values from percentiles of a normal model	**HSS-ID.A.4**
FORWARD 2	Use a normal model for a margin of error	**HSS-IC.B.4**

SET 61 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent between 59 and 69 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q1/5

2235

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

95%

GRADING NOTE Any equivalent form. 59 and 69 are 2 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent of women taller than 66.5 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q2/5

2236

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

16%

GRADING NOTE Any equivalent form. (100 − 68)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Test scores are approximately normal with mean 500 and standard deviation 100. Estimate the percent of scores between 400 and 700. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q3/5

2237

QUESTION ID

S61-M-Q3

TYPE

Short answer

CORRECT ANSWER

81.5%

GRADING NOTE Any equivalent form. 34 + 34 + 13.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q3**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is approximately normal with mean 40 h and standard deviation 4 h. Use the table to estimate the percent of batteries lasting less than 46 hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q4/5

2238

QUESTION ID

S61-M-Q4

TYPE

Short answer

CORRECT ANSWER

93.32%

GRADING NOTE Any equivalent form. z = 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Cereal box weights are approximately normal with mean 16.0 oz and standard deviation 0.2 oz. Use the table to estimate the percent between 15.8 and 16.3 oz.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q5/5

2239

QUESTION ID

S61-M-Q5

TYPE

Short answer

CORRECT ANSWER

77.45%

GRADING NOTE Any equivalent form. z from −1 to 1.5: 0.9332 − 0.1587.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q5**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 1

Interpret mean and standard deviation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A data set has mean 64 and standard deviation 2.5.
What value is one standard deviation above the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q1/5

2241

QUESTION ID
S61-B1-Q1

TYPE
Short answer

CORRECT ANSWER
66.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is two standard deviations below a mean of 500 with standard deviation 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q2/5

2242

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

300

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q2/5

True or false?

A larger standard deviation means the data are more spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q3/5

2243

QUESTION ID

S61-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

46 is how many standard deviations above a mean of 40 when the standard deviation is 4?

A. 0.67

B. 4

C. 6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q4/5

2244

QUESTION ID

S61-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1.5

GRADING NOTE (46 − 40)/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (15.8 − 16.0)/0.2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q5/5

2245

QUESTION ID

S61-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B1-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 2

Describe the shape of a distribution • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Describe a histogram that is highest in the middle and falls off evenly on both sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q1/5

2247

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric, mound-shaped (bell-shaped)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where are the mean and median of a symmetric distribution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q2/5

2248

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

About the same, at the center

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q2/5

True or false?

A normal curve is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q3/5

2249

QUESTION ID

S61-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Bell shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bell-shaped histogram is called

A. skewed right

B. uniform

C. mound-shaped and symmetric

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q4/5

2250

QUESTION ID

S61-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. mound-shaped and symmetric

GRADING NOTE Shape vocabulary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In a symmetric distribution, what percent of the data are below the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q5/5

2251

QUESTION ID
S61-B2-Q5

TYPE
Short answer

CORRECT ANSWER
50%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B2-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 3

Percents of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 95% of 2,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q1/5

2253

QUESTION ID

S61-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1900

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 34% + 13.5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q2/5

2254

QUESTION ID

S61-B3-Q2

TYPE

Short answer

CORRECT ANSWER

47.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q2/5

True or false?

Half of 68% is 36%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q3/5

2255

QUESTION ID

S61-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 68/2 = 34.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

100% − 95% split equally into two tails gives each tail

A. 5%

B. 47.5%

C. 0.5%

D. 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q4/5

2256

QUESTION ID

S61-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2.5%

GRADING NOTE 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.7745 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q5/5

2257

QUESTION ID

S61-B3-Q5

TYPE

Short answer

CORRECT ANSWER

77.45%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 1

Find values from percentiles of a normal model • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Above what height are the tallest 2.5% of women (mean 64 in, SD 2.5 in)? Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

2259

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

69 inches

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the table: what battery life is the 84th percentile (mean 40 h, SD 4 h)?

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

2260

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

44 hours

GRADING NOTE Any equivalent form. z = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

True or false?

In a normal model, the median equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

2261

QUESTION ID

S61-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q3**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The 16th percentile of scores with mean 500 and SD 100 is

A. 484

B. 600

C. 340

D. 400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

2262

QUESTION ID

S61-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 400

GRADING NOTE One SD below.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cereal box below 15.7 oz is rejected. Use the table to estimate the percent rejected (mean 16.0, SD 0.2).

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

2263

QUESTION ID

S61-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6.68%

GRADING NOTE Any equivalent form. z = −1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q5**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 2

Use a normal model for a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Simulated sample proportions are approximately normal with mean 0.58 and SD 0.049. Within what interval do about 95% of them fall?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q1/5

2265

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.482 to 0.678

GRADING NOTE Any equivalent form. ±2 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simulated sample proportions have standard deviation 0.049. What margin of error does this give for the population proportion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q2/5

2266

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 0.098 (about 10 percentage points)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q2/5

True or false?

The margin of error is about two standard deviations of the simulated statistic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q3/5

2267

QUESTION ID

S61-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 95% range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q3/5

If the SD of simulated means is 0.2 hours, the margin of error is about

A. 0.1 hours

B. 0.4 hours

C. 2 hours

D. 0.2 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q4/5

2268

QUESTION ID

S61-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.4 hours

GRADING NOTE $2 \cdot 0.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q4/5

Explain why the normal model is useful for simulated sample proportions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q5/5

2269

QUESTION ID

S61-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Their distribution is approximately mound-shaped and symmetric, so the 95% range is about the mean ± 2 SD.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Recognize data sets for which a normal model is not appropriate

MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Describe skewed, symmetric, bimodal, and uniform shapes	**HSS-ID.A.3**
BACKWARD 2	Measures of center and spread for a data set	**6.SP.A.2**
BACKWARD 3	Groups combined in one data set	**HSS-ID.A.1**
FORWARD 1	Check a normal model with the 68-95-99.7 rule	**HSS-ID.A.4**
FORWARD 2	Decide whether a probability model fits observed data	**HSS-IC.A.2**

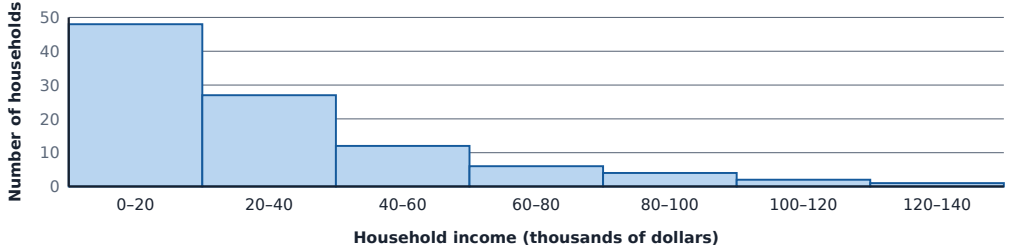
SET 62 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows household incomes in a town. Is a normal model appropriate? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q1/5

2272

QUESTION ID

S62-M-Q1

TYPE

Short answer

CORRECT ANSWER

No: the distribution is strongly skewed right, not symmetric and mound-shaped

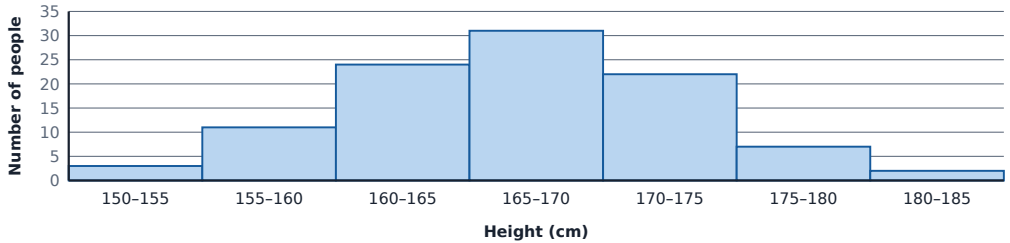
GRADING NOTE Form required: decision and the shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows heights of adults in a club. Is a normal model appropriate? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q2/5

2273

QUESTION ID

S62-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: roughly symmetric and mound-shaped with a single peak

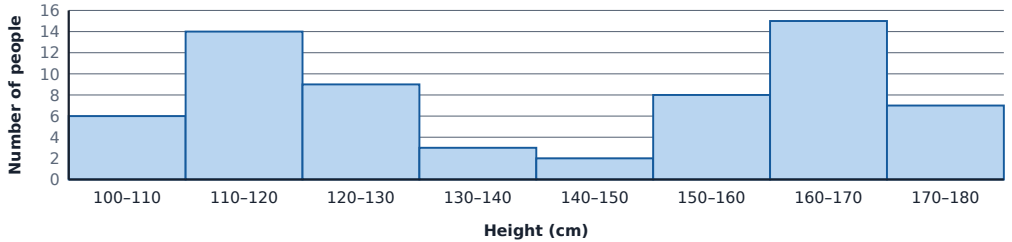
GRADING NOTE Form required: decision and the shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows heights of all people at a school event (children and adults). Is a normal model appropriate? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q3/5

2274

QUESTION ID
S62-M-Q3

TYPE
Short answer

CORRECT ANSWER
No: the distribution has two peaks (bimodal), from two different groups

GRADING NOTE Form required: decision and the shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Waiting times at a clinic have mean 4 minutes, standard deviation 5 minutes, and minimum 0. Is a normal model appropriate? Justify using the numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q4/5

2275

QUESTION ID

S62-M-Q4

TYPE

Short answer

CORRECT ANSWER

No: a normal model would put about 16% of times below 4 − 5 = −1 minute, which is impossible, so the data must be skewed right

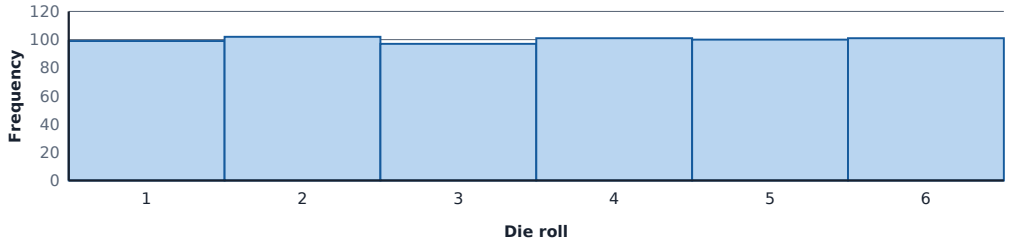
GRADING NOTE Form required: decision and the numerical reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows the results of 600 rolls of a fair die. Is a normal model appropriate? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q5/5

2276

QUESTION ID

S62-M-Q5

TYPE

Short answer

CORRECT ANSWER

No: the distribution is roughly uniform (flat), not mound-shaped

GRADING NOTE Form required: decision and the shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 1

Describe skewed, symmetric, bimodal, and uniform shapes • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A histogram has a long tail to the right. Name its shape.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q1/5

2278

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Skewed right

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q1/5

A histogram has two separate peaks. Name its shape.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q2/5

2279

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Bimodal

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q2/5

True or false?

In a right-skewed distribution, the mean is usually less than the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q3/5

2280

QUESTION ID

S62-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The long right tail pulls the mean above the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram with bars all about the same height is

A. skewed left

B. uniform

C. bimodal

D. normal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q4/5

2281

QUESTION ID
S62-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. uniform

GRADING NOTE Flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is more resistant to a long tail: the mean or the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q5/5

2282

QUESTION ID

S62-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The median

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 2

Measures of center and spread for a data set • Grade 6 • Statistics & Probability • 6.SP.A.2

What does the standard deviation (or another spread measure) describe?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q1/5

2284

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

How spread out the data are around the center

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Data are all at least 0 with mean 4. Can many values be far below 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q2/5

2285

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No: values cannot go below 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q2/5

2285

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A distribution can be described by its center, spread, and shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q3/5

2286

QUESTION ID

S62-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Three features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a measure of center?

A. median

B. range

C. standard deviation

D. IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q4/5

2287

QUESTION ID

S62-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. median

GRADING NOTE Center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give one value two standard deviations below a mean of 4 with SD 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q5/5

2288

QUESTION ID

S62-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 3

Groups combined in one data set • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Two groups with different centers are combined. What might the histogram show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q1/5

2290

QUESTION ID

S62-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Two peaks

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a variable whose histogram for all students in a K-12 school might have several peaks.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q2/5

2291

QUESTION ID

S62-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Height (for example)

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B3-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q2/5

True or false?

A histogram of one group can look different from a histogram of two groups combined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q3/5

2292

QUESTION ID

S62-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mixtures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bimodal histogram suggests

A. an error in counting

B. a small standard deviation

C. two groups in the data

D. a normal model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q4/5

2293

QUESTION ID

S62-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. two groups in the data

GRADING NOTE Two populations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q4/5

How could you get a mound-shaped distribution from the school-event heights?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q5/5

2294

QUESTION ID

S62-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Separate the children and the adults into two data sets

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B3-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 1

Check a normal model with the 68-95-99.7 rule • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

In 1,000 measurements, 70% are within 1 SD of the mean and 96% within 2 SDs. Is a normal model reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q1/5

2296

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: the percents are close to 68% and 95%

GRADING NOTE Form required: decision and comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q1/5

In another data set, 85% of values are within 1 SD and 91% within 2 SDs. Is a normal model reasonable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q2/5

2297

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No: 85% is far above 68% and 91% is below 95%

GRADING NOTE Form required: decision and comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q2/5

True or false?

If a data set has an extreme outlier, the normal model may still fit the rest but the mean and SD are distorted.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q3/5

2298

QUESTION ID

S62-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outliers affect mean and SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q3**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q3/5

About what percent of normal data lies more than 3 SDs from the mean?

A. 5%

B. 3%

C. 0.3%

D. 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q4/5

2299

QUESTION ID

S62-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.3%

GRADING NOTE 100 − 99.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q4/5

A data set of 200 values has 25 values more than 2 SDs from the mean. Compare with a normal model.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q5/5

2300

QUESTION ID

S62-F1-Q5

TYPE

Short answer

CORRECT ANSWER

12.5% vs about 5%: too many extreme values for a normal model

GRADING NOTE Form required: both percents and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q5/5

2300

SET 62 • HSS-ID.A.4

FORWARD 2

Decide whether a probability model fits observed data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A spinner is claimed to land on each of 4 colors equally often. In 400 spins the counts are 98, 105, 97, 100. Are the results consistent with the claim?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q1/5

2302

QUESTION ID

S62-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: each count is close to the expected 100

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q1/5

A model says a coin is fair. In 50 flips there were 34 heads. In 200 simulated runs of 50 fair flips, 3 had 34 or more heads. Is the model consistent with the data?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q2/5

2303

QUESTION ID

S62-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No: 3/200 = 1.5% is very unusual, so the data are evidence against a fair coin

GRADING NOTE Form required: decision and the proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q2/5

True or false?

A model is rejected only if the observed result would be unusual under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q3/5

2304

QUESTION ID

S62-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Logic of the test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q3/5

A simulated result at least as extreme as the observed one occurs in 34 of 200 runs. The model is

A. proved true

B. rejected

C. consistent with the data

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q4/5

2305

QUESTION ID

S62-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. consistent with the data

GRADING NOTE 17% is not unusual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain the difference between judging a normal model for a data set and judging a probability model with a simulation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q5/5

2306

QUESTION ID

S62-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The first compares the shape of data to a bell curve; the second asks whether an observed result is unusual under the model's simulated results.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-IC.A.1 • STATISTICS, INFERENCE, AND DECISIONS

Distinguish population, sample, parameter, and statistic

MAIN	Algebra II main question	**HSS-IC.A.1**
BACKWARD 1	Samples and populations	**7.SP.A.1**
BACKWARD 2	Draw inferences from random samples	**7.SP.A.2**
BACKWARD 3	Compute sample proportions and means	**6.SP.B.5.c**
FORWARD 1	Describe sampling variability	**HSS-IC.A.1**
FORWARD 2	Quantify variability with a margin of error	**HSS-IC.B.4**

SET 63 • HSS-IC.A.1

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A city surveys 400 randomly chosen residents, and 62% support a new park. Identify the population, the sample, the parameter, and the statistic.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q1/5

2309

QUESTION ID

S63-M-Q1

TYPE

Short answer

CORRECT ANSWER

Population: all city residents; sample: the 400 surveyed; parameter: the percent of all residents who support the park; statistic: 62%

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q1/5

A factory measures 50 randomly selected bolts from a day's production; their mean length is 5.02 cm. Identify the parameter and the statistic.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q2/5

2310

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

Parameter: the mean length of all bolts made that day; statistic: 5.02 cm, the sample mean

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A news website posts a poll; 80% of the 2,000 visitors who answered want a later school start. Can this be generalized to all students? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q3/5

2311

QUESTION ID
S63-M-Q3

TYPE
Short answer

CORRECT ANSWER
No: the sample is voluntary, not random, so people with strong opinions are over-represented

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q3/5

Explain why a random sample allows conclusions about the whole population.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q4/5

2312

QUESTION ID

S63-M-Q4

TYPE

Short answer

CORRECT ANSWER

Every member has a chance to be chosen, so the sample tends to represent the population and the statistic estimates the parameter without systematic bias.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A school uses the first 30 students arriving in the morning to estimate the average hours of sleep. Is this a random sample? Explain a possible bias.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q5/5

2313

QUESTION ID

S63-M-Q5

TYPE

Short answer

CORRECT ANSWER

No: early arrivers may differ (for example they may sleep less), so the estimate may be biased

GRADING NOTE Form required: decision and a plausible bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 1

Samples and populations • Grade 7 • Statistics & Probability • 7.SP.A.1

What is a population in statistics?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

2315

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The whole group of interest

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is a sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

2316

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A part of the population that is studied

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

True or false?

A random sample gives every member of the population a chance to be chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

2317

QUESTION ID

S63-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

Surveying every student in a school is

A. a statistic

B. an experiment

C. a sample

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

2318

QUESTION ID

S63-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a census

GRADING NOTE Whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

Why are samples used instead of the whole population?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

2319

QUESTION ID

S63-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The population may be too large, costly, or slow to study fully

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 2

Draw inferences from random samples • Grade 7 • Statistics & Probability • 7.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 50 students, 20 walk to school. Estimate how many of 600 students walk.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q1/5

2321

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

240

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two random samples give 40% and 44%. Is that surprising?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q2/5

2322

QUESTION ID

S63-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No: random samples vary

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q2/5

True or false?

Different random samples from the same population usually give slightly different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q3/5

2323

QUESTION ID

S63-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

From a random sample, 30% of 200 voters prefer a candidate. The best estimate for all voters is

A. 30%

B. 200

C. cannot estimate

D. 60%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q4/5

2324

QUESTION ID

S63-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 30%

GRADING NOTE Statistic estimates parameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is a larger random sample usually better?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q5/5

2325

QUESTION ID

S63-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Its statistic varies less, so it estimates the parameter more precisely

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 3

Compute sample proportions and means • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

248 of 400 people say yes. What proportion is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q1/5

2327

QUESTION ID

S63-B3-Q1

TYPE

Short answer

CORRECT ANSWER

0.62

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 5.0, 5.1, 4.9, 5.08.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q2/5

2328

QUESTION ID

S63-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5.02

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A proportion can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q3/5

2329

QUESTION ID

S63-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A part of a whole is between 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

1,600 of 2,000 is

A. 16%

B. 1.25%

C. 80%

D. 20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q4/5

2330

QUESTION ID

S63-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 80%

GRADING NOTE 1600/2000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 62% as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q5/5

2331

QUESTION ID

S63-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.62

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 1

Describe sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Two random samples of 400 residents give 62% and 59%. Explain why they differ and what each estimates.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q1/5

2333

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Random samples vary by chance; both estimate the same population percent.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Would samples of 40 or of 400 vary more from sample to sample? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q2/5

2334

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Samples of 40: smaller samples are affected more by chance

GRADING NOTE Form required: answer and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q2/5

True or false?

A parameter changes from sample to sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q3/5

2335

QUESTION ID

S63-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The parameter is fixed; the statistic varies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q3/5

Which is a statistic?

A. the population size

B. the mean height of 30 sampled students

C. the mean height of all students

D. the population proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q4/5

2336

QUESTION ID

S63-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the mean height of 30 sampled students

GRADING NOTE From a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q4/5

Explain why a large voluntary-response sample can still be worse than a small random sample.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q5/5

2337

QUESTION ID
S63-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Size does not remove bias; a voluntary sample is systematically unrepresentative.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 2

Quantify variability with a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Simulated proportions from samples of 400 with $p = 0.62$ have SD 0.024. Give a margin of error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q1/5

2339

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

About 0.048 (4.8 percentage points)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q1/5

Use the margin of error to give an interval for the percent of residents who support the park.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q2/5

2340

QUESTION ID
S63-F2-Q2

TYPE
Short answer

CORRECT ANSWER
About 57.2% to 66.8%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q2/5

True or false?

A margin of error describes chance variation, not bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q3/5

2341

QUESTION ID
S63-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Bias is not measured.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q3/5

If the sample size is multiplied by 4, the margin of error is about

A. unchanged

B. halved

C. doubled

D. divided by 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q4/5

2342

QUESTION ID

S63-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. halved

GRADING NOTE Square-root law.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain what the interval says about the population percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q5/5

2343

QUESTION ID
S63-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The population percent is plausibly within the interval; values far outside it are unlikely.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q5/5

SET 64 • HSS-IC.A.2 • STATISTICS, INFERENCE, AND DECISIONS

Decide whether a model is consistent with observed results using a simulation

MAIN	Algebra II main question	**HSS-IC.A.2**
BACKWARD 1	Probability as long-run relative frequency	**7.SP.C.6**
BACKWARD 2	Use simulations of compound events	**7.SP.C.8.c**
BACKWARD 3	Relative frequency of a group of outcomes	**6.RP.A.3.c**
FORWARD 1	Design a simulation to test a model	**HSS-IC.A.2**
FORWARD 2	Compare two treatments with a re-randomization simulation	**HSS-IC.B.5**

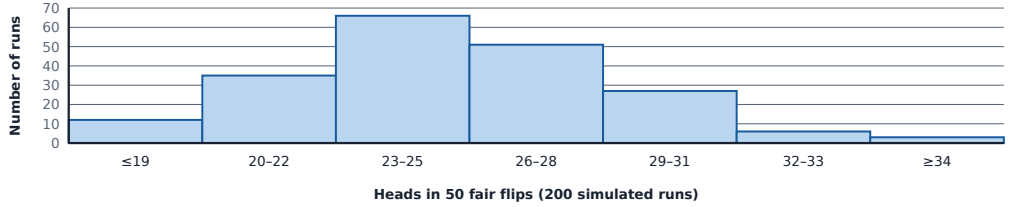
SET 64 • HSS-IC.A.2

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A coin is claimed to be fair. It landed heads 34 times in 50 flips. The histogram shows 200 simulated runs of 50 fair flips. Is the claim consistent with the result? Justify with the simulation.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q1/5

2346

QUESTION ID
S64-M-Q1

TYPE
Short answer

CORRECT ANSWER
No: only 3 of 200 runs (1.5%) had 34 or more heads, so 34 heads is very unusual for a fair coin

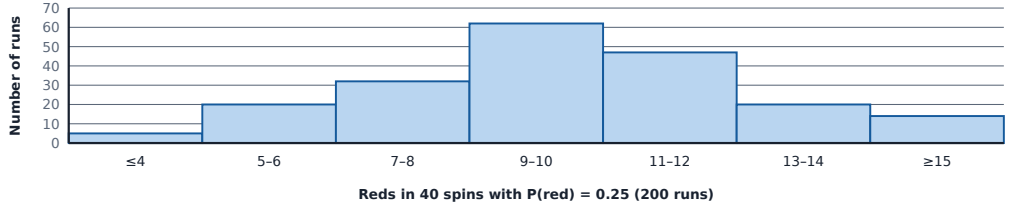
GRADING NOTE Form required: decision and the simulated proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner is claimed to land on red 25% of the time. It landed on red 13 times in 40 spins. The histogram shows 200 simulated runs. Is the claim consistent? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q2/5

2347

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 13 or more reds happened in 34 of 200 runs (17%), which is not unusual

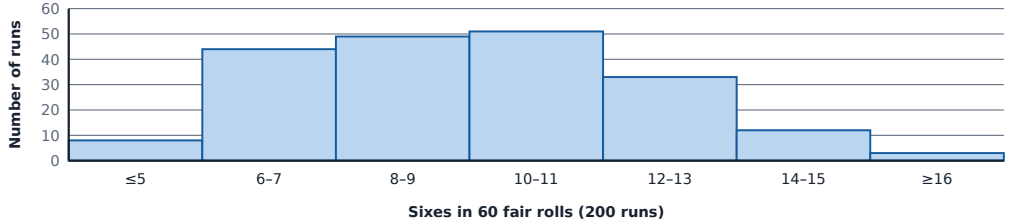
GRADING NOTE Form required: decision and the simulated proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A die is claimed to be fair. It showed 16 sixes in 60 rolls. The histogram shows 200 simulated runs. Is the claim consistent? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q3/5

2348

QUESTION ID
S64-M-Q3

TYPE
Short answer

CORRECT ANSWER
No: 16 or more sixes happened in only 3 of 200 runs (1.5%), so the result is unusual for a fair die

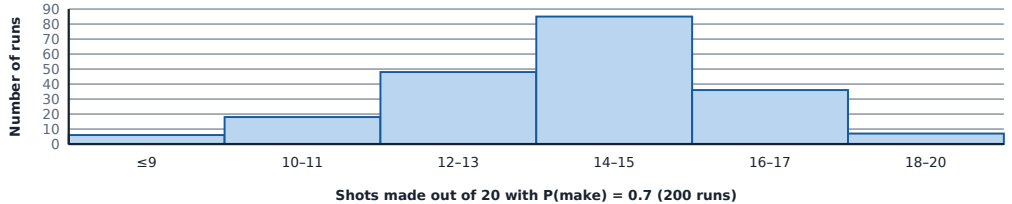
GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A player claims to make 70% of free throws. She made 11 of 20. The histogram shows 200 simulated runs of 20 shots. Is the claim consistent? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q4/5

2349

QUESTION ID

S64-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes: 11 or fewer made shots happened in 6 + 18 = 24 of 200 runs (12%), which is not unusual

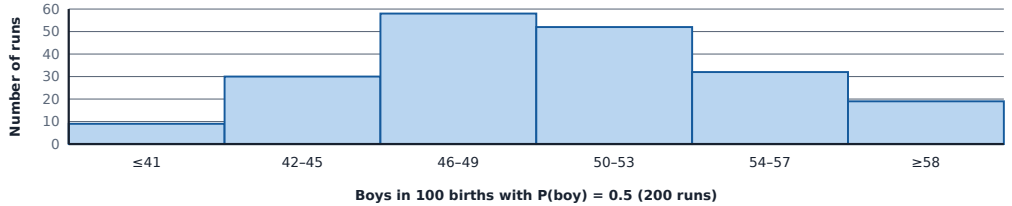
GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A model says boys and girls are equally likely. A hospital had 58 boys in 100 births. The histogram shows 200 simulated runs. Is the model consistent? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q5/5

2350

QUESTION ID
S64-M-Q5

TYPE
Short answer

CORRECT ANSWER
Yes: 58 or more boys happened in 19 of 200 runs (9.5%), which is not unusual at the 5% level

GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 1

Probability as long-run relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped 50 times. About how many heads do you expect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q1/5

2352

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner has P(red) = 0.25. About how many reds in 40 spins?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q2/5

2353

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Getting exactly the expected number every time is likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q3/5

2354

QUESTION ID

S64-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Results vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled 60 times. Expected sixes:

A. 10

B. 16

C. 6

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q4/5

2355

QUESTION ID

S64-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE 60/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is 26 heads in 50 flips not surprising?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q5/5

2356

QUESTION ID

S64-B1-Q5

TYPE

Short answer

CORRECT ANSWER

It is close to the expected 25; chance variation is normal

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 2

Use simulations of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.c

Describe how to simulate 50 fair coin flips with random digits.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q1/5

2358

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Use 50 random digits; even = heads, odd = tails

GRADING NOTE Any valid design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How could you simulate one spin of a spinner with P(red) = 0.25 using two-digit random numbers 00-99?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q2/5

2359

QUESTION ID
S64-B2-Q2

TYPE
Short answer

CORRECT ANSWER
00-24 = red;
25-99 = not red

GRADING NOTE Any valid design with 25 of the 100 numbers for red.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q2/5

True or false?

Each run of a simulation should copy the real process under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q3/5

2360

QUESTION ID

S64-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q3/5

A simulation has 200 runs. Each run represents

A. the final answer

B. one repetition of the whole experiment

C. one histogram bar

D. one flip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q4/5

2361

QUESTION ID

S64-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. one repetition of the whole experiment

GRADING NOTE Run = experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q4/5

Why are many runs used?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q5/5

2362

QUESTION ID

S64-B2-Q5

TYPE

Short answer

CORRECT ANSWER

To see the pattern of results that the model produces by chance

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 3

Relative frequency of a group of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

3 of 200 runs is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q1/5

2364

QUESTION ID

S64-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

6 + 18 runs out of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q2/5

2365

QUESTION ID

S64-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q2/5

True or false?

19 out of 200 is 19%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q3/5

2366

QUESTION ID

S64-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 19/200 = 0.095 = 9.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

34 of 200 is

A. 1.7%

B. 17%

C. 6.8%

D. 34%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q4/5

2367

QUESTION ID

S64-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17%

GRADING NOTE 34/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 2% less than 5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q5/5

2368

QUESTION ID

S64-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 1

Design a simulation to test a model • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Describe a simulation to test whether a basketball player really makes 70% of free throws, given that she made 11 of 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q1/5

2370

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Use random digits 0-9 with 0-6 = make; take 20 digits per run, count makes; repeat many runs; see how often 11 or fewer makes occur.

GRADING NOTE Any valid design with all parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A player who claims to make 70% of free throws made 11 of 20. Why should a simulation count results of 11 or fewer (as extreme or more extreme) rather than exactly 11?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q2/5

2371

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Any single exact value can be rare; the question is how unusual results this far from the claim are.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q2/5

True or false?

A result that happens in 30% of simulated runs is strong evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q3/5

2372

QUESTION ID

S64-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 30% is common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A common cutoff for "unusual" is

A. about 50% of runs

B. 100% of runs

C. about 5% of runs

D. about 25% of runs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q4/5

2373

QUESTION ID

S64-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. about 5% of runs

GRADING NOTE Convention.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q4/5

Doubling the number of runs from 200 to 400 changes what about the simulation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q5/5

2374

QUESTION ID

S64-F1-Q5

TYPE

Short answer

CORRECT ANSWER

It gives a more precise estimate of how often the result occurs under the model (the model and data are unchanged).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q5/5

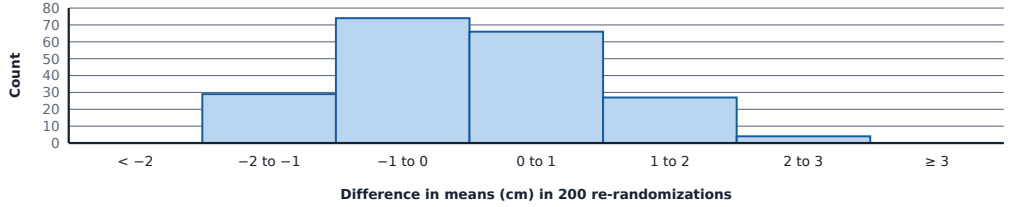
SET 64 • HSS-IC.A.2

FORWARD 2

Compare two treatments with a re-randomization simulation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, fertilized plants grew 2.9 cm more on average. In 200 re-randomizations, none of the differences was 2.9 cm or more. What do you conclude?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q1/5

2376

QUESTION ID
S64-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The difference is very unlikely by chance alone, so the fertilizer appears to increase growth

GRADING NOTE Form required: conclusion and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What model does the re-randomization assume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q2/5

2377

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

That the treatment has no effect, so group labels can be shuffled

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q2/5

True or false?

Re-randomization is a simulation of a model, like the coin-flip simulation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q3/5

2378

QUESTION ID

S64-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both test a model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q3/5

If 60 of 200 re-randomized differences were at least the observed one, the result is

A. significant

B. proof of no effect

C. not statistically significant

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q4/5

2379

QUESTION ID

S64-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. not statistically significant

GRADING NOTE 30% is common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q4/5

Explain how a re-randomization test for an experiment extends the reasoning used to test a fair-coin claim with a simulation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q5/5

2380

QUESTION ID

S64-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Both ask how often a model produces a result as extreme as the observed one; here the model is "no treatment effect".

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Identify sample surveys, experiments, and observational studies

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Statistical questions and data collection	**6.SP.A.1**
BACKWARD 2	Association is not causation	**HSS-ID.C.9**
BACKWARD 3	Random samples	**7.SP.A.1**
FORWARD 1	Identify confounding and design flaws	**HSS-IC.B.3**
FORWARD 2	Analyze the results of an experiment	**HSS-IC.B.5**

SET 65 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 100 volunteers to take a vitamin or a placebo and compare colds. What type of study is this, and what is its purpose?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q1/5

2383

QUESTION ID

S65-M-Q1

TYPE

Short answer

CORRECT ANSWER

Experiment: to test whether the vitamin causes fewer colds

GRADING NOTE Form required: type and purpose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q1/5

A pollster asks a random sample of 1,000 voters which candidate they prefer. What type of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q2/5

2384

QUESTION ID
S65-M-Q2

TYPE
Short answer

CORRECT ANSWER
Sample survey: to estimate the preference of all voters

GRADING NOTE Form required: type and purpose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q2/5

Researchers compare the hospital records of smokers and nonsmokers for lung disease. What type of study is this? Can it show causation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q3/5

2385

QUESTION ID
S65-M-Q3

TYPE
Short answer

CORRECT ANSWER
Observational study; it cannot by itself show causation because the groups were not assigned

GRADING NOTE Form required: type and the causation answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q3/5

Which type of study best answers "Does playing music while studying improve test scores?" Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q4/5

2386

QUESTION ID

S65-M-Q4

TYPE

Short answer

CORRECT ANSWER

Experiment: randomly assign students to study with or without music and compare scores

GRADING NOTE Form required: type and design idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q4/5

Which type of study best answers "What percent of students at the school use the library weekly?" Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q5/5

2387

QUESTION ID

S65-M-Q5

TYPE

Short answer

CORRECT ANSWER

Sample survey of a random sample of students

GRADING NOTE Form required: type and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 1

Statistical questions and data collection • Grade 6 • Statistics & Probability • 6.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Is "How many hours do students at our school sleep?" a statistical question? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q1/5

2389

GRADING NOTE Form required: decision and reason.

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: the answers vary

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give a data collection method for "What do voters think?"

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q2/5

2390

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Ask a sample of voters (a survey)

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q2/5

2390

True or false?

An experiment imposes a treatment on the subjects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q3/5

2391

QUESTION ID

S65-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Watching and recording without imposing treatments is

A. an experiment

B. an observational study

C. a census

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q4/5

2392

QUESTION ID

S65-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. an observational study

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is a placebo?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q5/5

2393

QUESTION ID

S65-B1-Q5

TYPE

Short answer

CORRECT ANSWER

A fake treatment with no active ingredient

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 2

Association is not causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Ice cream sales and sunburns are correlated. Does ice cream cause sunburn? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q1/5

2395

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

No: hot sunny weather increases both

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a lurking variable for "students with more books at home get higher scores".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q2/5

2396

QUESTION ID

S65-B2-Q2

TYPE

Short answer

CORRECT ANSWER

For example family income or parent education

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q2/5

True or false?

A strong correlation proves that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q3/5

2397

QUESTION ID

S65-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Correlation is not causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A variable that affects both variables in a study is

A. the placebo

B. the sample

C. a confounding (lurking) variable

D. the response

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q4/5

2398

QUESTION ID

S65-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a confounding (lurking) variable

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q4/5

What kind of study can show cause and effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q5/5

2399

QUESTION ID

S65-B2-Q5

TYPE

Short answer

CORRECT ANSWER

A randomized experiment

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 3

Random samples • Grade 7 • Statistics & Probability • 7.SP.A.1

Describe a way to choose a random sample of 50 students from a school list.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

2401

QUESTION ID

S65-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Number the students and use a random number generator to pick 50 different numbers

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

Is surveying students in the cafeteria line random?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

2402

QUESTION ID

S65-B3-Q2

TYPE

Short answer

CORRECT ANSWER

No: not every student has an equal chance

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

True or false?

A random sample tends to represent the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

2403

QUESTION ID

S65-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE No systematic bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sample is random?

A. friends of the teacher

B. students who volunteer

C. names drawn from a hat with every student's name

D. the first 50 arrivals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

2404

QUESTION ID

S65-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. names drawn from a hat with every student's name

GRADING NOTE Chance selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does a survey need a random sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

2405

QUESTION ID

S65-B3-Q5

TYPE

Short answer

CORRECT ANSWER

To generalize its results to the population

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 1

Identify confounding and design flaws • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Students who choose to take an SAT prep course score higher. Give a confounding variable and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

2407

QUESTION ID
S65-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Motivation (or income): students who choose the course may differ in ways that also raise scores.

GRADING NOTE Any valid confounder with explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Students who choose to take an SAT prep course score higher than students who do not. Redesign the study so it can show whether the course causes higher scores.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

2408

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Randomly assign volunteers to take the course or not, then compare scores.

GRADING NOTE Any valid experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

True or false?

An experiment without a control group can still show the treatment caused the result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

2409

QUESTION ID

S65-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Comparison is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, why use a placebo?

A. to make the study observational

B. to control for the effect of believing one is treated

C. to increase the sample size

D. to make the treatment stronger

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

2410

QUESTION ID

S65-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. to control for the effect of believing one is treated

GRADING NOTE Placebo effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why might an observational study be chosen instead of an experiment for smoking?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

2411

QUESTION ID

S65-F1-Q5

TYPE

Short answer

CORRECT ANSWER

It would be unethical to assign people to smoke.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 2

Analyze the results of an experiment • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

100 volunteers were randomly assigned to a vitamin or a placebo. The vitamin group had 1.2 fewer colds on average. A re-randomization found 40 of 200 differences at least that large. What do you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

2413

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Not significant: 20% of re-randomizations give a difference this large by chance

GRADING NOTE Form required: conclusion and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

In a randomized vitamin-versus-placebo experiment, the vitamin group had 1.2 fewer colds on average, and only 2 of 200 re-randomized differences were that large. What do you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

2414

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Significant: the vitamin likely reduces colds

GRADING NOTE Form required: conclusion and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

True or false?

Random assignment makes it possible to attribute a significant difference to the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

2415

QUESTION ID
S65-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Groups differ only by chance otherwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

The re-randomization model assumes

A. the groups are different sizes

B. the treatment has no effect

C. the treatment works

D. the data are normal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

2416

QUESTION ID

S65-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the treatment has no effect

GRADING NOTE Null model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

Explain why an observational study cannot be analyzed in the same causal way.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

2417

QUESTION ID

S65-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Without random assignment, a difference could come from confounding variables, not the treatment.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

SET 66 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Explain random sampling versus random assignment

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Study types and their purposes	**HSS-IC.B.3**
BACKWARD 2	Choose a random sample	**7.SP.A.1**
BACKWARD 3	Correlation and causation	**HSS-ID.C.9**
FORWARD 1	Improve an experimental design	**HSS-IC.B.3**
FORWARD 2	Use re-randomization to judge a treatment difference	**HSS-IC.B.5**

SET 66 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 500 adults is surveyed about exercise and sleep; more exercise is associated with better sleep. What can and cannot be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

2420

QUESTION ID

S66-M-Q1

TYPE

Short answer

CORRECT ANSWER

The association can be generalized to all adults (random sample), but not that exercise causes better sleep (no random assignment)

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

60 volunteers are randomly assigned to two diets; diet A loses more weight. What can and cannot be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

2421

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

Diet A caused more weight loss for these volunteers (random assignment), but it cannot be generalized to all people (not a random sample)

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

A random sample of 200 students is randomly assigned to two study methods. What can be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

2422

QUESTION ID

S66-M-Q3

TYPE

Short answer

CORRECT ANSWER

Both: cause and effect, generalized to the student population

GRADING NOTE Form required: both conclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

A teacher compares test scores of students who chose to use a review app with those who did not, in her own class. What can be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

2423

QUESTION ID

S66-M-Q4

TYPE

Short answer

CORRECT ANSWER

Neither causation nor generalization: no random assignment and no random sample; only an association in this class

GRADING NOTE Form required: the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

Describe how to randomly assign 40 subjects to two groups of 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

2424

QUESTION ID

S66-M-Q5

TYPE

Short answer

CORRECT ANSWER

Number the subjects 1-40; use a random number generator to choose 20 different numbers for group 1; the rest form group 2

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 1

Study types and their purposes • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Which type of study uses random assignment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

2426

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

An experiment

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which type of study is designed to estimate a population value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

2427

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A sample survey

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An observational study assigns treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

2428

QUESTION ID

S66-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It observes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Random assignment is used to

A. avoid measuring

B. make the sample larger

C. make the groups comparable

D. choose who is in the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

2429

QUESTION ID

S66-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. make the groups comparable

GRADING NOTE Comparability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What does random sampling help you do?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

2430

QUESTION ID

S66-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Generalize to the population

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 2

Choose a random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Describe how to choose 20 names at random from a list of 500.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

2432

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Number the names and pick 20 different random numbers from 1 to 500

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is a sample of volunteers random?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

2433

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sample of volunteers may differ from the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

2434

QUESTION ID

S66-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Self-selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

Which method gives a random sample?

A. choosing the first 20

B. drawing numbered slips from a well-mixed hat

C. asking for volunteers

D. asking friends

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

2435

QUESTION ID

S66-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. drawing numbered slips from a well-mixed hat

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why must slips in a hat be well mixed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

2436

QUESTION ID

S66-B2-Q5

TYPE

Short answer

CORRECT ANSWER

So every slip has the same chance of being drawn

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 3

Correlation and causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

An association between two variables in observational data: can it prove causation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

2438

QUESTION ID

S66-B3-Q1

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

Give a possible lurking variable for "exercise and sleep".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

2439

QUESTION ID

S66-B3-Q2

TYPE

Short answer

CORRECT ANSWER

For example age or stress

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-B3-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Random assignment controls lurking variables on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

2440

QUESTION ID

S66-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Balances groups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Causation can be concluded from

A. an observational study

B. a correlation alone

C. a randomized experiment

D. a survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q4/5

2441

QUESTION ID

S66-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q4/5

Why does a correlation not show which variable causes the other?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

2442

QUESTION ID

S66-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Either could affect the other, or a third variable could affect both

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-B3-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 1

Improve an experimental design • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers try a new diet and lose weight. Explain why a control group (for example a second, randomly assigned diet) is needed to judge the diet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

2444

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Without a comparison group we cannot tell whether weight loss was due to the diet or to other factors.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

A drug study blocks subjects by age before randomly assigning treatments. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

2445

QUESTION ID
S66-F1-Q2

TYPE
Short answer

CORRECT ANSWER
To make sure age differences do not hide or create a treatment effect; groups are compared within similar ages.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

True or false?

In a double-blind experiment, neither subjects nor evaluators know who gets the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

2446

QUESTION ID

S66-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers were randomly assigned to two diets, and diet A led to more weight loss. Which change would let the result be generalized to all adults?

A. measuring more often

B. using a placebo

C. adding more volunteers

D. randomly selecting the subjects from all adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

2447

QUESTION ID

S66-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. randomly selecting the subjects from all adults

GRADING NOTE Random sampling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

Why are many subjects better than few in an experiment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

2448

QUESTION ID

S66-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Chance differences between the groups are smaller, so real effects are easier to detect.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 2

Use re-randomization to judge a treatment difference • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers were randomly assigned to two diets; diet A lost 2.4 kg more on average. In 200 re-randomizations, 5 differences were at least 2.4 kg. Is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

2450

QUESTION ID
S66-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Yes: 2.5% of re-randomizations, which is unusual

GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers were randomly assigned to two diets, and diet A lost significantly more weight. State the conclusion in context, including to whom it applies.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

2451

QUESTION ID

S66-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Diet A caused more weight loss for these volunteers; it may not apply to all adults.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

True or false?

Re-randomization shuffles the group labels as if the treatment had no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

2452

QUESTION ID
S66-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Null model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 70 of 200 re-randomized differences were at least 2.4 kg, conclude

A. the difference could be due to chance

B. the study was observational

C. diet B works

D. diet A works

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

2453

QUESTION ID

S66-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the difference could be due to chance

GRADING NOTE 35% is common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why re-randomization is valid only because the original assignment was random.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

2454

QUESTION ID

S66-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Shuffling copies the original random assignment process; without it, the shuffle does not model how the groups were formed.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F2-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population proportion with a margin of error from a simulation

MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Statistics, parameters, and sampling variability	**HSS-IC.A.1**
BACKWARD 2	Proportions and percents	**6.RP.A.3.c**
BACKWARD 3	Standard deviation as typical distance from the mean	**HSS-ID.A.2**
FORWARD 1	Plan a sample size for a desired margin of error	**HSS-IC.B.4**
FORWARD 2	Evaluate reports that use margins of error	**HSS-IC.B.6**

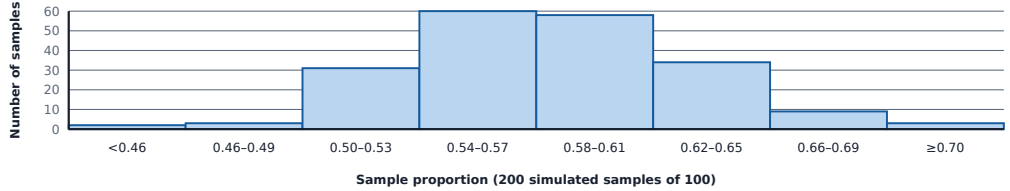
SET 67 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58 want a later start time. The histogram shows 200 simulated samples of 100 from a population with p = 0.58; their standard deviation is 0.049. Give a margin of error and an interval estimate.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

2457

QUESTION ID

S67-M-Q1

TYPE

Short answer

CORRECT ANSWER

Margin about 2(0.049) ≈ 0.10; interval about 0.48 to 0.68

GRADING NOTE Form required: margin and interval (0.098 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 300 voters, 120 support a measure. Simulated sample proportions (n = 300, p = 0.40) have standard deviation 0.0275. Give the estimate, the margin of error, and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

2458

QUESTION ID

S67-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.40 ± 0.055; about 0.345 to 0.455

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

In a random sample of 50 shoppers, 35 use a store app. Simulated proportions (n = 50, p = 0.70) have standard deviation 0.064. Give the margin of error and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

2459

QUESTION ID

S67-M-Q3

TYPE

Short answer

CORRECT ANSWER

About 0.13; about 0.57 to 0.83

GRADING NOTE Form required: both (0.128 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of the three surveys (n = 50, 100, 300), which has the smallest margin of error, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

2460

QUESTION ID

S67-M-Q4

TYPE

Short answer

CORRECT ANSWER

n = 300: larger samples produce less variable sample proportions

GRADING NOTE Form required: answer and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

A report says "52% of voters support the plan, with a margin of error of 3 percentage points." Interpret this statement.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

2461

QUESTION ID
S67-M-Q5

TYPE
Short answer

CORRECT ANSWER

The true percent of all voters who support the plan is plausibly between 49% and 55%

GRADING NOTE Any equivalent interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 1

Statistics, parameters, and sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58 want a later start time. What is the statistic?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

2463

QUESTION ID
S67-B1-Q1

TYPE
Short answer

CORRECT ANSWER
0.58 (58%)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58% want a later start time. What parameter does 58% estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

2464

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The proportion of all students who want a later start

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

True or false?

Different random samples of 100 would give different sample proportions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

2465

QUESTION ID

S67-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

A histogram of 200 simulated sample proportions (samples of 100) shows

A. the parameter exactly

B. how sample proportions vary from sample to sample

C. the population

D. one sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

2466

QUESTION ID

S67-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. how sample proportions vary from sample to sample

GRADING NOTE Sampling distribution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

Why does a random sample matter for the margin of error to be meaningful?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

2467

QUESTION ID

S67-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The margin describes chance variation in random samples; a biased sample can be off by more.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 2

Proportions and percents • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Write 120 out of 300 as a proportion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

2469

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.40

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 0.58 ± 0.10 as an interval?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

2470

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.48 to 0.68

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

35 out of 50 is 35%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

2471

QUESTION ID

S67-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 35/50 = 0.70 = 70%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

52% ± 3% gives

A. 52% to 55%

B. 49% to 55%

C. 50% to 54%

D. 49% to 52%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

2472

QUESTION ID

S67-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 49% to 55%

GRADING NOTE Subtract and add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.055 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

2473

QUESTION ID

S67-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 3

Standard deviation as typical distance from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

About what percent of mound-shaped data lie within 2 standard deviations of the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

2475

QUESTION ID
S67-B3-Q1

TYPE
Short answer

CORRECT ANSWER
About 95%

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

If the SD is 0.049, what is 2 SDs?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

2476

QUESTION ID

S67-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0.098

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A smaller standard deviation means more variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

2477

QUESTION ID

S67-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A smaller SD means the values are closer to the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which set of simulated proportions varies least?

A. SD 0.1

B. SD 0.0275

C. SD 0.049

D. SD 0.064

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

2478

QUESTION ID

S67-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. SD 0.0275

GRADING NOTE Smallest SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 0.064.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

2479

QUESTION ID

S67-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.128

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 1

Plan a sample size for a desired margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A survey of 100 students has a margin of error of about 0.10. Quadrupling the sample size roughly halves the margin of error. Estimate the margin with n = 400.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

2481

QUESTION ID

S67-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 0.05

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

About what sample size gives a margin of error of 0.05 when $p \approx 0.5$? (Use margin $\approx 2\sqrt{(p(1-p)/n)}$.)

$n \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

2482

QUESTION ID
S67-F1-Q2

TYPE
Short answer

CORRECT ANSWER
400

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Doubling the sample size halves the margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q3/5

2483

QUESTION ID

S67-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It divides it by √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q3/5

Which change most reduces the margin of error?

A. multiplying n by 9

B. doubling n

C. adding 10 people

D. repeating the same sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

2484

QUESTION ID

S67-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. multiplying n by 9

GRADING NOTE Margin divides by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two polls of the same population have margins 0.05 and 0.025. How do their sample sizes compare?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

2485

QUESTION ID

S67-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The second is about 4 times as large

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 2

Evaluate reports that use margins of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A poll shows candidate A at 51% and B at 49% with margin ±4%. A headline says "A leads." Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

2487

QUESTION ID

S67-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Not justified: the intervals overlap widely, so either could lead

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report gives 58% ± 10% from 100 students and claims "a majority of students want a later start." Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

2488

GRADING NOTE Form required: evaluation and reason.

QUESTION ID

S67-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Not clearly justified: the interval 48% to 68% includes values below 50%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

True or false?

A margin of error accounts for bias from a voluntary-response sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

2489

QUESTION ID

S67-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

Which report is best supported?

A. 45% ± 2% so a majority oppose

B. 70% ± 3% support, from a random sample

C. 51% ± 5% support, so most support

D. 60% support from an online poll

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

2490

QUESTION ID

S67-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 70% ± 3% support, from a random sample

GRADING NOTE The interval stays above 50% and the sample is random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

2490

What should a careful report include besides the percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

2491

QUESTION ID
S67-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The sample size, how the sample was chosen, and the margin of error

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population mean with a margin of error from a simulation

MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Mean of a data set	**6.SP.B.5.c**
BACKWARD 2	What resampling simulates	**HSS-IC.A.1**
BACKWARD 3	Two standard deviations and the 95% range	**HSS-ID.A.2**
FORWARD 1	Use intervals to judge claims about a mean	**HSS-IC.B.4**
FORWARD 2	Compare two group means with re-randomization	**HSS-IC.B.5**

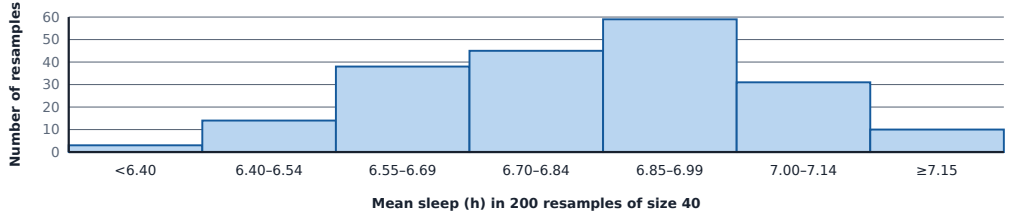
SET 68 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 40 students slept a mean of 6.8 hours. The histogram shows the means of 200 resamples of the data; their standard deviation is 0.20 hours. Give a margin of error and an interval estimate.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

2494

QUESTION ID

S68-M-Q1

TYPE

Short answer

CORRECT ANSWER

Margin about 0.4 h; interval about 6.4 to 7.2 hours

GRADING NOTE Form required: margin and interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 25 apples has mean weight 152 g. Simulated sample means have standard deviation 3.6 g. Give the margin of error and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

2495

QUESTION ID

S68-M-Q2

TYPE

Short answer

CORRECT ANSWER

About 7.2 g; about 144.8 g to 159.2 g

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 60 commuters has mean commute 24.5 min. Simulated means have standard deviation 1.4 min. Give the margin of error and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

2496

QUESTION ID

S68-M-Q3

TYPE

Short answer

CORRECT ANSWER

About 2.8 min; about 21.7 to 27.3 minutes

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

A random sample of 40 students gives the interval 6.4 to 7.2 hours for the mean hours of sleep. Interpret the interval correctly, and explain why it does not mean that 95% of students sleep between 6.4 and 7.2 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

2497

QUESTION ID

S68-M-Q4

TYPE

Short answer

CORRECT ANSWER

The mean sleep of all students is plausibly between 6.4 and 7.2 hours (not: 95% of students sleep that long)

GRADING NOTE Form required: a correct interpretation contrasted with the individual-value misreading.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 25 apples gives a margin of error of 7.2 g for the mean weight. About what margin of error do you expect from a random sample of 100 apples?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

2498

QUESTION ID

S68-M-Q5

TYPE

Short answer

CORRECT ANSWER

About 3.6 g (half)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 1

Mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 6, 7, 7.5, 6.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

2500

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6.75

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

What does a sample mean estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

2501

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The population mean

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The mean is not affected by extreme values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

2502

QUESTION ID

S68-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One extreme value can move the mean a lot; the median is resistant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The mean of 150, 152, 154 is

A. 154

B. 150

C. 152

D. 456

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

2503

QUESTION ID

S68-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 152

GRADING NOTE Sum/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

If each of 40 values increases by 0.5, how does the mean change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

2504

QUESTION ID

S68-B1-Q5

TYPE

Short answer

CORRECT ANSWER

It increases by 0.5

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 2

What resampling simulates • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why can resampling the data show how sample means vary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

2506

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Resamples act like new random samples from a population that looks like the sample

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a histogram of 200 resample means, what does one bar's height mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

2507

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

How many of the 200 resample means fell in that interval

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Sample means vary less than individual values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

2508

QUESTION ID

S68-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Averaging reduces variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 40 students slept a mean of 6.8 hours. The parameter being estimated is

A. 6.8

B. the mean sleep of all students

C. the 40 students' mean

D. the resample SD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

2509

QUESTION ID

S68-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the mean sleep of all students

GRADING NOTE Population mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 6.8 hours a parameter or a statistic?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

2510

QUESTION ID

S68-B2-Q5

TYPE

Short answer

CORRECT ANSWER

A statistic

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 3

Two standard deviations and the 95% range • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute 6.8 − 2(0.20) and 6.8 + 2(0.20).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

2512

QUESTION ID

S68-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6.4 and 7.2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 1.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

2513

QUESTION ID

S68-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

True or false?

About 95% of a mound-shaped distribution is within 2 SDs of its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

2514

QUESTION ID

S68-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

152 ± 7.2 gives

A. 144.8 to 159.2

B. 145 to 159

C. 152 to 159.2

D. 144.8 to 152

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

2515

QUESTION ID

S68-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 144.8 to 159.2

GRADING NOTE Subtract and add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 7.5 hours inside the interval 6.4 to 7.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

2516

QUESTION ID

S68-B3-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 1

Use intervals to judge claims about a mean • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A random sample gives the interval 6.4 to 7.2 hours for students' mean sleep. A health guide says students average 8 hours. Does the interval support this? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

2518

QUESTION ID

S68-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No: 8 is well outside the plausible values

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

A random sample gives the interval 144.8 g to 159.2 g for the mean apple weight. A grower claims apples average 155 g. Is the claim consistent with the interval?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

2519

QUESTION ID

S68-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 155 is inside the interval

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A value inside the interval is proved to be the true mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

2520

QUESTION ID

S68-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is only plausible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 60 commuters gives an interval for the mean commute time. Which change would most likely narrow the interval?

A. sampling 240 commuters instead of 60

B. using the median

C. rounding the data

D. sampling 30 commuters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

2521

QUESTION ID

S68-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. sampling 240 commuters instead of 60

GRADING NOTE Larger n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of students at one school gives the interval 6.4 to 7.2 hours for mean sleep. Explain why the interval should not be applied to students at a different school.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

2522

QUESTION ID

S68-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The sample came only from this school's students, so it estimates their mean only.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 2

Compare two group means with re-randomization • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, 20 students using a study app scored 6 points more on average than 20 without it. In 200 re-randomizations, 9 differences were at least 6. Is the difference significant at the 5% level?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

2524

GRADING NOTE Form required: decision and proportion.

QUESTION ID

S68-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes:

9/200 = 4.5% < 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What model do the re-randomizations represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

2525

QUESTION ID
S68-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The app has no effect on scores

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An interval for one mean and a re-randomization test for two means both use simulated variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

2526

QUESTION ID
S68-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Same idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 30 of 200 differences were at least 6, the conclusion is

A. the difference could be chance

B. the app works

C. the study failed

D. the app hurts scores

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

2527

QUESTION ID

S68-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the difference could be chance

GRADING NOTE 15%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the margin of error for one mean is related to judging a difference of two means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

2528

QUESTION ID

S68-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Both ask how much a statistic varies by chance; a difference is meaningful only if it is large compared with that variability.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

SET 69 • HSS-IC.B.5 • STATISTICS, INFERENCE, AND DECISIONS

Compare two treatments with a re-randomization simulation

MAIN	Algebra II main question	**HSS-IC.B.5**
BACKWARD 1	Compare the means of two groups	**7.SP.B.4**
BACKWARD 2	Random assignment in experiments	**HSS-IC.B.3**
BACKWARD 3	Probability as relative frequency in simulations	**7.SP.C.6**
FORWARD 1	Two-sided comparisons and design choices	**HSS-IC.B.5**
FORWARD 2	Evaluate reports of experiments	**HSS-IC.B.6**

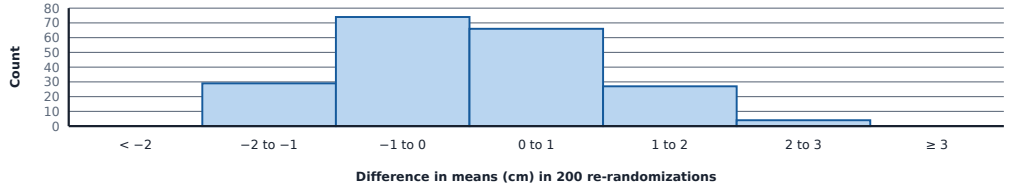
SET 69 • HSS-IC.B.5

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

10 plants got fertilizer and 10 did not (randomly assigned). Fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences; none was 2.9 or more. Is the difference significant? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

2531

QUESTION ID

S69-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: a difference of 2.9 cm almost never happens by chance alone (0 of 200), so the fertilizer appears to increase growth

GRADING NOTE Form required: decision and evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a caffeine experiment, the caffeine group's reaction time was 0.04 s faster on average. In 200 re-randomizations, 62 differences were at least 0.04 s. Is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

2532

QUESTION ID

S69-M-Q2

TYPE

Short answer

CORRECT ANSWER

No: 31% of re-randomizations give a difference that large by chance

GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Students randomly assigned to a study app scored 6 points higher on average. In 200 re-randomizations, 9 differences were at least 6. Is the difference significant at the 5% level?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

2533

QUESTION ID

S69-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes: 9/200 = 4.5%, below 5%

GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain what a re-randomization simulation assumes and how it is carried out.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

2534

QUESTION ID

S69-M-Q4

TYPE

Short answer

CORRECT ANSWER

It assumes the treatment has no effect; all responses are pooled, randomly re-split into two groups of the original sizes, and the difference in means is recorded, many times.

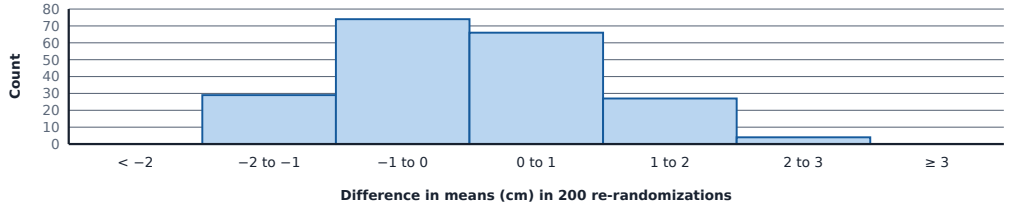
GRADING NOTE Any equivalent explanation with assumption and procedure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the histogram of 200 re-randomized differences, about how many were at least 2 cm? Would an observed difference of 2 cm be significant at the 5% level?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

2535

QUESTION ID

S69-M-Q5

TYPE

Short answer

CORRECT ANSWER

4 (the 2-to-3 bar); 4/200 = 2%, so yes

GRADING NOTE Form required: count, proportion, decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 1

Compare the means of two groups • Grade 7 • Statistics & Probability • 7.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Group A has mean 25.4 cm and group B has mean 22.5 cm. Find the difference A − B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

2537

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.9 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Group A: 4, 6, 8. Group B: 3, 5, 4. Find the difference in means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

2538

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A difference in sample means could happen even if the treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

2539

QUESTION ID

S69-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

2539

ANSWER KEY Reduced copy of the question page, with its answer

Mean A = 0.52 s and mean B = 0.48 s. A − B is

A. −0.04 s

B. 0.4 s

C. 1.00 s

D. 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

2540

QUESTION ID

S69-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.04 s

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

Why compare the difference with the variability in the groups?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

2541

QUESTION ID

S69-B1-Q5

TYPE

Short answer

CORRECT ANSWER

A difference is meaningful only if it is large relative to the natural variation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 2

Random assignment in experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

In a fertilizer experiment, 20 plants were randomly assigned to fertilizer or no fertilizer. Why were they randomly assigned?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

2543

QUESTION ID
S69-B2-Q1

TYPE
Short answer

CORRECT ANSWER
To make the groups comparable so differences can be attributed to the fertilizer

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-B2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

Describe how to randomly assign 20 plants into two groups of 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

2544

QUESTION ID

S69-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Number the plants 1-20 and use random numbers to choose 10 for fertilizer

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-B2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

True or false?

Without random assignment, a re-randomization test does not justify a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

2545

QUESTION ID

S69-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Design matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Random assignment is a feature of

A. censuses

B. experiments

C. observational studies

D. surveys

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

2546

QUESTION ID

S69-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. experiments

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What kind of conclusion does a significant result in a randomized experiment allow?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

2547

QUESTION ID

S69-B2-Q5

TYPE

Short answer

CORRECT ANSWER

That the treatment caused the difference (for these subjects)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 3

Probability as relative frequency in simulations • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

62 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

2549

QUESTION ID

S69-B3-Q1

TYPE

Short answer

CORRECT ANSWER

31%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

9 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

2550

QUESTION ID

S69-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An event that happens in 40% of simulated runs is unusual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

2551

QUESTION ID

S69-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 40% is common; unusual usually means about 5% or less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

0 of 200 runs suggests the observed result is

A. very unlikely under the model

B. certain

C. impossible

D. typical

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

2552

QUESTION ID

S69-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. very unlikely under the model

GRADING NOTE Rare, not impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 4/200 below 5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

2553

QUESTION ID

S69-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: 2%

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 1

Two-sided comparisons and design choices • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In a caffeine experiment, the observed difference in mean reaction time was 0.04 s, and the question is whether caffeine changes reaction time in either direction. How should the re-randomization results be counted?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

2555

QUESTION ID
S69-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Count differences at least 0.04 s in either direction (|difference| ≥ 0.04)

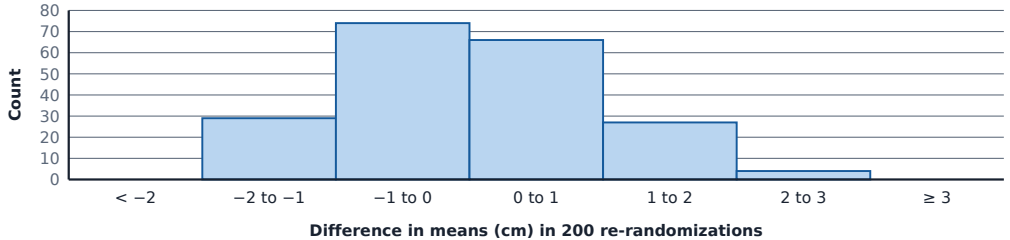
GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the fertilizer histogram, estimate how many re-randomized differences were 2 or more in absolute value.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

2556

QUESTION ID

S69-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 4 (no bar below −2, 4 in 2 to 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

Using more plants in each group usually makes the re-randomized differences less spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

2557

QUESTION ID
S69-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Less chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

A small observed difference can be significant when

A. the groups have very little natural variation

B. the data are rounded

C. the groups were not randomized

D. the sample is small

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

2558

QUESTION ID

S69-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the groups have very little natural variation

GRADING NOTE Small spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why should the decision rule (such as 5%) be chosen before seeing the data?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

2559

QUESTION ID

S69-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Choosing it afterward could bias the conclusion toward the result one wants.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 2

Evaluate reports of experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A company reports its supplement "improved memory scores by 3 points" in a study of 8 people without a control group. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

2561

QUESTION ID
S69-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Weak evidence: no control group, very small sample, no randomization, no significance test

GRADING NOTE Any two valid criticisms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says a new teaching method "significantly" raised scores in a randomized experiment, with 2 of 200 re-randomizations as extreme. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

2562

QUESTION ID
S69-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Supported: randomized, and 1% is unusual

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

True or false?

"Statistically significant" means the effect is large and important.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

2563

QUESTION ID

S69-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means unlikely by chance; it may be small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

Which report is most convincing?

A. a study with no comparison group

B. testimonials from customers

C. a randomized experiment with a control group and a small simulated p-value

D. an online poll of users

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

2564

QUESTION ID

S69-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment with a control group and a small simulated p-value

GRADING NOTE Design and evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

What information should a report of an experiment include?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

2565

QUESTION ID
S69-F2-Q5

TYPE
Short answer

CORRECT ANSWER
How subjects were assigned, the group sizes, the observed difference, and how unusual it is by chance

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSS-IC.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Evaluate claims in reports based on data

MAIN	Algebra II main question	**HSS-IC.B.6**
BACKWARD 1	Random samples and bias	**HSS-IC.A.1**
BACKWARD 2	Study design and causation	**HSS-IC.B.3**
BACKWARD 3	Read scales on graphs	**3.MD.B.3**
FORWARD 1	Rewrite claims so that they follow from the data	**HSS-IC.B.6**
FORWARD 2	Analyze decisions based on test results	**HSS-MD.B.7**

SET 70 • HSS-IC.B.6

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

An ad says "4 out of 5 dentists recommend our toothpaste." Only 5 dentists, chosen by the company, were asked. Evaluate the claim.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

2568

QUESTION ID

S70-M-Q1

TYPE

Short answer

CORRECT ANSWER

Weak: the sample is tiny and not random, so it cannot represent all dentists

GRADING NOTE Form required: evaluation with at least one valid reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "Students who eat breakfast earn higher grades, so breakfast causes higher grades." The data came from a survey. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

2569

QUESTION ID

S70-M-Q2

TYPE

Short answer

CORRECT ANSWER

Not justified: an observational survey shows association, not causation; lurking variables (such as family routines) could explain it

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

A bar graph shows sales of 50 and 52 thousand, with the vertical axis starting at 48 thousand, so one bar looks twice as tall. Evaluate the impression the graph gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

2570

QUESTION ID
S70-M-Q3

TYPE
Short answer

CORRECT ANSWER
Misleading: the truncated axis exaggerates a 4% difference to look like 100%

GRADING NOTE Form required: evaluation and the actual size of the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A TV show asks viewers to text their vote; 90% of 10,000 texts favor a policy. The show claims most citizens support it. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

2571

QUESTION ID

S70-M-Q4

TYPE

Short answer

CORRECT ANSWER

Not justified: voluntary response; texters are not representative, regardless of the large number

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A poll shows candidate A at 51% and B at 49% with a margin of error of ±4%. A headline says "A is winning." Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

2572

QUESTION ID

S70-M-Q5

TYPE

Short answer

CORRECT ANSWER

Not justified: the difference is well within the margin of error

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 1

Random samples and bias • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why is a voluntary-response sample biased?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

2574

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

People with strong opinions are more likely to respond

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

What does a random sample allow?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

2575

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Generalizing from the sample to the population

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A very large sample cannot be biased.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q3/5

2576

QUESTION ID

S70-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size does not fix bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q3/5

Which sample is most likely biased?

A. customers who chose to write reviews

B. a random sample of 1,000

C. randomly chosen customers

D. every customer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q4/5

2577

QUESTION ID

S70-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. customers who chose to write reviews

GRADING NOTE Self-selected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is "5 dentists chosen by the company" a random sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

2578

QUESTION ID

S70-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 2

Study design and causation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

What type of study is a survey of students' breakfast habits and grades?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

2580

QUESTION ID
S70-B2-Q1

TYPE
Short answer

CORRECT ANSWER
An observational study (sample survey)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

What study design could test whether breakfast causes higher grades?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

2581

QUESTION ID

S70-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A randomized experiment

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Only randomized experiments can establish cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

2582

QUESTION ID

S70-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A confounding variable is

A. the response variable

B. a variable related to both the explanatory and the response variable

C. the sample size

D. the margin of error

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

2583

QUESTION ID
S70-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. a variable related to both the explanatory and the response variable

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a possible confounder for breakfast and grades.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

2584

QUESTION ID
S70-B2-Q5

TYPE
Short answer

CORRECT ANSWER
For example sleep or family support

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 3

Read scales on graphs • Grade 3 • Measurement & Data • 3.MD.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph axis starts at 48. Bars reach 50 and 52. What are the bar values?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

2586

QUESTION ID

S70-B3-Q1

TYPE

Short answer

CORRECT ANSWER

50 and 52

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent larger is 52 than 50?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

2587

QUESTION ID

S70-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

True or false?

An axis that does not start at 0 can make differences look larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

2588

QUESTION ID

S70-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Truncation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

On a graph whose axis starts at 48, a bar of 52 looks how many times as tall as a bar of 50?

A. the same

B. 1.04 times

C. 2 times

D. 4 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

2589

QUESTION ID

S70-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2 times

GRADING NOTE Heights 4 and 2 above the axis start.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph of sales of 50 and 52 (thousand) has a vertical axis starting at 48. How should the graph be redrawn to be fair?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

2590

QUESTION ID

S70-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Start the axis at 0 (or clearly mark the break)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 1

Rewrite claims so that they follow from the data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A survey found that students who eat breakfast earn higher grades, and a report says breakfast causes higher grades. Rewrite the claim so that it is supported by the survey.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

2592

QUESTION ID
S70-F1-Q1

TYPE
Short answer

CORRECT ANSWER
"Students who eat breakfast tend to have higher grades" (an association)

GRADING NOTE Any equivalent wording without causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

Rewrite the poll headline (51% vs 49%, ±4%) so it is accurate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

2593

QUESTION ID

S70-F1-Q2

TYPE

Short answer

CORRECT ANSWER

"The race is too close to call"

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

True or false?

Saying "is associated with" instead of "causes" is appropriate for an observational study.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

2594

QUESTION ID
S70-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Careful wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which conclusion is justified by a random sample of 1,000 voters showing 62% ± 3% support?

A. the sample was biased

B. most voters support the measure

C. exactly 62% support it

D. the measure will pass by 24 points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

2595

QUESTION ID

S70-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. most voters support the measure

GRADING NOTE 59% to 65%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An ad says "4 out of 5 dentists recommend our toothpaste," based on 5 dentists chosen by the company. What would the claim need to be convincing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

2596

QUESTION ID
S70-F1-Q5

TYPE
Short answer

CORRECT ANSWER
A large random sample of dentists (and the question wording reported)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 2

Analyze decisions based on test results • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A screening test is 95% accurate for people with a disease and gives false positives 5% of the time. If 1% of 10,000 people have the disease, how many positive results are true positives?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

2598

QUESTION ID

S70-F2-Q1

TYPE

Short answer

CORRECT ANSWER

95 true positives out of 95 + 495 = 590 positives

GRADING NOTE Form required: both counts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "a positive test means you almost surely have the disease." Evaluate using your counts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

2599

QUESTION ID
S70-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Misleading: only 95/590 ≈ 16% of positives have the disease

GRADING NOTE Form required: evaluation and probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The probability of having the disease given a positive result can be low even with an accurate test.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

2600

QUESTION ID
S70-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Rare disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

2600

ANSWER KEY Reduced copy of the question page, with its answer

With 1% prevalence, P(disease | positive) is about

A. 0.99

B. 0.05

C. 0.95

D. 0.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

2601

QUESTION ID

S70-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.16

GRADING NOTE 95/590.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

What follow-up decision makes sense after a positive screening result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

2602

QUESTION ID
S70-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Confirm with a second, more accurate test before treatment

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

SET 71 • HSS-MD.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Design and judge fair decision procedures with probabilities

MAIN	Algebra II main question	**HSS-MD.B.6**
BACKWARD 1	Probabilities of equally likely outcomes	**7.SP.C.7.a**
BACKWARD 2	Sample spaces for compound events	**7.SP.C.8.b**
BACKWARD 3	Probability between 0 and 1 and redoing outcomes	**7.SP.C.5**
FORWARD 1	Design fair procedures and count expected trials	**HSS-MD.B.6**
FORWARD 2	Analyze strategies and decisions with probability	**HSS-MD.B.7**

SET 71 • HSS-MD.B.6

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Three friends choose a driver by flipping a coin twice: HH = Ana, HT = Ben, TH = Cy, TT = flip again. Is this fair? Justify with probabilities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

2605

QUESTION ID
S71-M-Q1

TYPE
Short answer

CORRECT ANSWER
Yes: HH, HT, TH are equally likely and TT is redone, so each friend has probability 1/3

GRADING NOTE Form required: decision and probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

Four people use a die: 1 = A, 2 = B, 3 = C, 4 = D, and 5 or 6 means roll again. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

2606

QUESTION ID

S71-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: each person has probability 1/4 after rerolls

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

A hat holds 3 slips with Ana's name and 2 with Ben's. One slip is drawn to choose who goes first. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

2607

QUESTION ID

S71-M-Q3

TYPE

Short answer

CORRECT ANSWER

No: P(Ana) = 3/5 and P(Ben) = 2/5

GRADING NOTE Form required: decision and probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random number generator gives a whole number 1-100. Numbers 1-33 choose A, 34-66 choose B, 67-99 choose C, and 100 means generate again. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

2608

QUESTION ID

S71-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes: each person has 33 equally likely numbers, and 100 is redone

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

Two dice are rolled; if the sum is even, player 1 wins, otherwise player 2 wins. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

2609

QUESTION ID

S71-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: 18 of the 36 outcomes have an even sum, so each player has probability 1/2

GRADING NOTE Form required: decision and probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 1

Probabilities of equally likely outcomes • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. What is P(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

2611

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 3 red and 2 blue slips. What is P(red)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

2612

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Each face of a fair die has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

2613

QUESTION ID

S71-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Uniform model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins: P(HT) is

A. 1/2

B. 1/4

C. 1/8

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

2614

QUESTION ID

S71-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/4

GRADING NOTE 4 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

2614

ANSWER KEY Reduced copy of the question page, with its answer

A spinner has 4 equal sections. What is P(one section)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

2615

QUESTION ID

S71-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 2

Sample spaces for compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

List the outcomes of flipping a coin twice.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

2617

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

HH, HT, TH, TT

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many outcomes does rolling two dice have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

2618

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

True or false?

There are 3 outcomes for two coin flips: two heads, one head, no heads.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

2619

QUESTION ID

S71-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE These are not equally likely; there are 4 equally likely outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many outcomes for two dice give an even sum?

A. 21

B. 12

C. 18

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

2620

QUESTION ID

S71-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 18

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the outcomes of rolling two dice with a sum of 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

2621

QUESTION ID

S71-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(1, 3), (2, 2), (3, 1)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 3

Probability between 0 and 1 and redoing outcomes • Grade 7 • Statistics & Probability • 7.SP.C.5

ANSWER KEY Reduced copy of the question page, with its answer

If an outcome is ignored and the experiment is redone, how does that affect the remaining outcomes' chances?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

2623

QUESTION ID

S71-B3-Q1

TYPE

Short answer

CORRECT ANSWER

They stay equally likely to each other and their probabilities are scaled up to add to 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

With rerolls on 5 and 6, what is P(1) for a die?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

2624

QUESTION ID

S71-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Probabilities of all outcomes in a sample space add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

2625

QUESTION ID

S71-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Total probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An event that can never happen has probability

A. 1

B. −1

C. 1/2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

2626

QUESTION ID

S71-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE Impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Three equally likely outcomes: what is each probability?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

2627

QUESTION ID

S71-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 1

Design fair procedures and count expected trials • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Design a fair way to choose one of 5 people using a fair coin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

2629

QUESTION ID
S71-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Flip 3 times (8 outcomes); assign 5 outcomes to the people and redo the other 3

GRADING NOTE Any valid design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three friends choose a driver by flipping a coin twice: HH, HT, and TH each pick a friend, and TT means flip again. What is the probability that a choice is made on the first try?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

2630

QUESTION ID
S71-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

True or false?

A fair procedure must give each person an equal chance, even if it may take several tries.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

2631

QUESTION ID
S71-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Fairness is about probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To choose among 6 people fairly with one die, you

A. roll twice and add

B. assign one face to each person

C. assign 1-3 to person A

D. reroll on 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

2632

QUESTION ID
S71-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. assign one face to each person

GRADING NOTE Six equally likely faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A teacher picks a student from a class of 28 with a random number generator 1-30. Design a fair rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

2633

QUESTION ID
S71-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Assign numbers 1-28 to students and regenerate on 29 or 30

GRADING NOTE Any valid design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 2

Analyze strategies and decisions with probability • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A game costs $1. You roll a die and win $4 if you roll a 6. Find the expected net gain per game.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

2635

QUESTION ID
S71-F2-Q1

TYPE
Short answer

CORRECT ANSWER
4/6 − 1 = −1/3 dollar (lose about $0.33 per game)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

A game costs $1; you roll a die and win $4 if you roll a 6. Is the game fair in the sense of expected value? What prize would make it fair?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

2636

QUESTION ID
S71-F2-Q2

TYPE
Short answer

CORRECT ANSWER
No; a $6 prize makes the expected net gain 0

GRADING NOTE Form required: decision and prize.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

True or false?

A game can be fair in who is chosen but still have a negative expected value for the player.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

2637

QUESTION ID
S71-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Two meanings of fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

With P(win) = 1/4 and prize \$8 for a \$2 ticket, the expected net gain is

A. \$2

B. −\$2

C. \$0

D. \$6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

2638

QUESTION ID
S71-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. \$0

GRADING NOTE 8/4 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain the difference between a fair choice procedure and a fair game.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

2639

QUESTION ID
S71-F2-Q5

TYPE
Short answer

CORRECT ANSWER
A fair choice gives everyone equal probability; a fair game has expected net gain 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

SET 72 • HSS-MD.B.7 • STATISTICS, INFERENCE, AND DECISIONS

Analyze decisions and strategies with probabilities

MAIN	Algebra II main question	**HSS-MD.B.7**
BACKWARD 1	Probability of compound independent events	**7.SP.C.8.a**
BACKWARD 2	Two-way tables of counts	**8.SP.A.4**
BACKWARD 3	Weighted averages of outcomes	**6.RP.A.3.c**
FORWARD 1	Investigate how changing conditions changes the decision	**HSS-MD.B.7**
FORWARD 2	Find expected values of random variables	**HSS-MD.A.2**

SET 72 • HSS-MD.B.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A disease affects 1% of 10,000 people. A test detects 95% of cases and gives false positives for 5% of healthy people. Find P(disease | positive).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

2642

QUESTION ID

S72-M-Q1

TYPE

Short answer

CORRECT ANSWER

95/590 ≈ 0.161

GRADING NOTE Form required: the probability with the counts. 95 true positives, 495 false positives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A shipment is accepted if 2 randomly tested items are both non-defective. If 10% of items are defective, what is the probability the shipment is accepted? (Assume independence.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

2643

QUESTION ID

S72-M-Q2

TYPE

Short answer

CORRECT ANSWER

$0.9 \times 0.9 = 0.81$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

A $30 warranty covers a repair that costs $200 and happens with probability 0.1. Compare the expected repair cost without the warranty to the warranty price, and decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

2644

QUESTION ID

S72-M-Q3

TYPE

Short answer

CORRECT ANSWER

Expected cost
0.1 × 200
= $20 < $30, so on average the warranty is not worth buying

GRADING NOTE Form required: expected cost, comparison, decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

Option A pays $10 with probability 0.5; option B pays $50 with probability 0.1. Compare the expected values and the risks.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

2645

QUESTION ID

S72-M-Q4

TYPE

Short answer

CORRECT ANSWER

Both have expected value $5; B is riskier (usually pays nothing)

GRADING NOTE Form required: both expected values and a risk comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A test gives +1 point for a correct answer and −1/4 point for a wrong one on 4-choice questions. Should a student guess randomly on a question? Justify with expected value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

2646

QUESTION ID

S72-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes (slightly): expected value = 1/4(1) + 3/4(−1/4) = 1/16 > 0

GRADING NOTE Form required: expected value and decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 1

Probability of compound independent events • Grade 7 • Statistics & Probability • 7.SP.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

P(non-defective) = 0.9 for each item. Find P(two non-defective).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

2648

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.81

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins: P(both heads)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

2649

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For independent events, P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

2650

QUESTION ID

S72-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiplication rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(three non-defective) with P = 0.9 each is

A. 0.9

B. 0.27

C. 0.729

D. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

2651

QUESTION ID

S72-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.729

GRADING NOTE 0.9^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find P(at least one defective) among two items when P(defective) = 0.1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

2652

QUESTION ID

S72-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0.19

GRADING NOTE Any equivalent form. 1 − 0.81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 2

Two-way tables of counts • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Of 10,000 people, 1% have a disease. How many have it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

2654

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 9,900 healthy people, 5% test positive. How many?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

2655

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

495

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

2655

True or false?

A two-way table can show test results for people with and without a disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

2656

QUESTION ID

S72-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rows and columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 95 sick and 495 healthy people test positive, the total positives are

A. 10,000

B. 590

C. 95

D. 495

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

2657

QUESTION ID
S72-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 590

GRADING NOTE Add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of positives are sick (95 of 590)? Give a decimal to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

2658

QUESTION ID

S72-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.161

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 3

Weighted averages of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 10% of $200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

2660

QUESTION ID

S72-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.5 × 10 and 0.1 × 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

2661

QUESTION ID

S72-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5 and 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1/4 \times 1 + 3/4 \times (-1/4) = 1/4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q3/5

2662

QUESTION ID

S72-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1/4 - 3/16 = 1/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

0.2 × 30 + 0.8 × 0 equals

A. 30

B. 24

C. 0.2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

2663

QUESTION ID

S72-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE Weighted sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.9 × 0 + 0.1 × 200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

2664

QUESTION ID

S72-B3-Q5

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 1

Investigate how changing conditions changes the decision • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A test detects 95% of cases and gives false positives for 5% of healthy people. In a group of 10,000 where 10% have the disease, find P(disease | positive).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

2666

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

950/1400 ≈ 0.679

GRADING NOTE Form required: the probability. 950 true and 450 false positives in 10,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A $30 warranty covers a repair that costs $200. If the repair probability is 0.2, is the warranty worth it on average?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

2667

QUESTION ID
S72-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes: expected cost $40 > $30

GRADING NOTE Form required: decision and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

True or false?

A positive result is more convincing when the disease is more common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

2668

QUESTION ID
S72-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Compare 0.161 and 0.679.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

If students could eliminate one wrong choice before guessing (3 choices left), the expected value of guessing is

A. $1/3 - 2/3 \cdot 1/4 = 1/6$

B. 0

C. $1/16$

D. $-1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

2669

QUESTION ID
S72-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $1/3 - 2/3 \cdot 1/4 = 1/6$

GRADING NOTE Higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

How many items must be tested (all non-defective) for the acceptance probability to drop below 0.5 when 10% are defective?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

2670

QUESTION ID
S72-F1-Q5

TYPE
Short answer

CORRECT ANSWER
7 items ($0.9^7 \approx 0.478$)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 2

Find expected values of random variables • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A random variable X takes values 0, 1, 2 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

2672

QUESTION ID

S72-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A raffle sells 500 tickets at $2; one prize is $400. Find the expected net gain of one ticket.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

2673

QUESTION ID
S72-F2-Q2

TYPE
Short answer

CORRECT ANSWER
400/500 − 2 = −$1.20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

True or false?

The expected value is the long-run average of many repetitions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

2674

QUESTION ID
S72-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For a fair die, E(roll) is

A. 3

B. 21

C. 3.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

2675

QUESTION ID
S72-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 3.5

GRADING NOTE 21/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q4**
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

A $30 warranty covers a $200 repair that happens with probability 0.1. Explain how deciding whether to buy it is an expected-value calculation for a random variable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

2676

QUESTION ID
S72-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The repair cost is a random variable (0 or 200); its expected value 20 is compared with the certain cost 30.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSN-CN.A.3 • ADVANCED EXTENSION (+)

Use conjugates to find moduli and quotients of complex numbers

MAIN	Algebra II main question	**HSN-CN.A.3**
BACKWARD 1	Multiply complex numbers	**HSN-CN.A.2**
BACKWARD 2	Distance from the origin	**8.G.B.8**
BACKWARD 3	Powers of i	**HSN-CN.A.1**
FORWARD 1	Prove properties of conjugates and moduli	**HSN-CN.A.3**
FORWARD 2	Write complex numbers in polar form	**HSN-CN.B.4**

SET 73 • HSN-CN.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Give the conjugate of $3 - 4i$ and find $|3 - 4i|$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

2679

QUESTION ID

S73-M-Q1

TYPE

Short answer

CORRECT ANSWER

Conjugate $3 + 4i$; $|3 - 4i| = 5$

GRADING NOTE Form required: both. $\sqrt{(9 + 16)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q1**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (2 + 3i)/(1 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

2680

QUESTION ID

S73-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2 + (5/2)i

GRADING NOTE Form required: a + bi. Multiply by (1 + i)/(1 + i).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (5 + i)/(2 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

2681

QUESTION ID

S73-M-Q3

TYPE

Short answer

CORRECT ANSWER

1 − i

GRADING NOTE Form required: a + bi. (5 + i)(2 − 3i)/13 = (13 − 13i)/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q3**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $|-5 + 12i|$ and compute $(-5 + 12i)(-5 - 12i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q4/5

2682

QUESTION ID
S73-M-Q4

TYPE
Short answer

CORRECT ANSWER
13 and 169

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q4**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1/(4 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

2683

QUESTION ID

S73-M-Q5

TYPE

Short answer

CORRECT ANSWER

4/25 − (3/25)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q5**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 1

Multiply complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Multiply (2 + 3i)(1 + i).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

2685

QUESTION ID

S73-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−1 + 5i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(3 - 4i)(3 + 4i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

2686

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a + bi)(a - bi) = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q3/5

2687

QUESTION ID

S73-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-b^2i^2 = b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q3/5

(5 + i)(2 − 3i) equals

A. 13 + 13i

B. 7 − 13i

C. 10 − 3i

D. 13 − 13i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

2688

QUESTION ID

S73-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13 − 13i

GRADING NOTE 10 − 15i + 2i + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(1 - i)(1 + i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

2689

QUESTION ID

S73-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 2

Distance from the origin • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (3, −4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

2691

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (−5, 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

2692

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

True or false?

The distance from (0, 0) to (a, b) is $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

2693

QUESTION ID

S73-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

The distance from (0, 0) to (1, 1) is

A. 0

B. 1

C. 2

D. √2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

2694

QUESTION ID

S73-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √2

GRADING NOTE √(1 + 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (4, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

2695

QUESTION ID

S73-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 3

Powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

2697

QUESTION ID

S73-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-12i \cdot i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

2698

QUESTION ID

S73-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

True or false?

$-3i^2 = -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

2699

QUESTION ID

S73-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-3i^2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is real?

A. 3i

B. $(4 + 3i)^2$

C. $(4 + 3i)(4 - 3i)$

D. $4 + 3i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

2700

QUESTION ID

S73-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(4 + 3i)(4 - 3i)$

GRADING NOTE Conjugate product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 13/13 − 13i/13 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

2701

QUESTION ID

S73-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1 − i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 1

Prove properties of conjugates and moduli • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $z \cdot \bar{z} = |z|^2$ for $z = a + bi$ (a and b real)?

A. $(a + bi)(a - bi) = a^2 + b^2i^2$, which is $|z|^2$ because $|z| = \sqrt{(a^2 + b^2i^2)}$.

B. $(a + bi)(a - bi) = a^2 - b^2$, and $|z|^2 = a^2 - b^2$.

C. $(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

D. For $z = 3 + 4i$: $(3 + 4i)(3 - 4i) = 25 = 5^2$, so the statement holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

2703

QUESTION ID

S73-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C.

$(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

GRADING NOTE Multiply, use $i^2 = -1$, and compare with $|z| = \sqrt{(a^2 + b^2)}$. The others forget $i^2 = -1$ (and misstate $|z|$), check one example, or expand with a sign error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Verify |zw| = |z||w| for z = 3 + 4i and w = 1 + i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

2704

QUESTION ID

S73-F1-Q2

TYPE

Short answer

CORRECT ANSWER

zw = −1 + 7i,
|zw| = √50 = 5√2 = 5 · √2

GRADING NOTE Form required: both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

True or false?

The conjugate of a real number is itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

2705

QUESTION ID

S73-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $z = 2 - 5i$, then $z + \bar{z}$ is

A. $4 - 10i$

B. 0

C. $-10i$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

2706

QUESTION ID

S73-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4

GRADING NOTE 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\bar{z}/|z|^2$ to write $1/z$ for $z = 1 + 2i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

2707

QUESTION ID

S73-F1-Q5

TYPE

Short answer

CORRECT ANSWER

(1 − 2i)/5 = 1/5 − (2/5)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 2

Write complex numbers in polar form • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write 1 + i in polar form r(cos θ + i sin θ).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

2709

QUESTION ID

S73-F2-Q1

TYPE

Short answer

CORRECT ANSWER

√2(cos(π/4) + i sin(π/4))

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 2(cos(π/2) + i sin(π/2)) in rectangular form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

2710

QUESTION ID

S73-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2i

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

True or false?

The modulus of a complex number is the r in its polar form.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

2711

QUESTION ID

S73-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The polar form of -3 is

A. $3(\cos \pi + i \sin \pi)$

B. $-3(\cos \pi + i \sin \pi)$

C. $3(\cos 0 + i \sin 0)$

D. $3(\cos(\pi/2) + i \sin(\pi/2))$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

2712

QUESTION ID

S73-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $3(\cos \pi + i \sin \pi)$

GRADING NOTE Angle π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain what the modulus and the angle of a complex number represent in the complex plane.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

2713

QUESTION ID
S73-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The modulus is the distance from 0; the angle is the direction measured from the positive real axis.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Graph rational functions with zeros and asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Rewrite a rational expression by inspection	**HSA-APR.D.6**
BACKWARD 2	Domain and excluded values	**HSF-IF.B.5**
BACKWARD 3	Sign of a quotient on intervals	**7.NS.A.2**
FORWARD 1	Graph rational functions with slant asymptotes	**HSF-IF.C.7.d**
FORWARD 2	Interpret a rational model's features in context	**HSF-IF.B.4**

SET 74 • HSF-IF.C.7.d

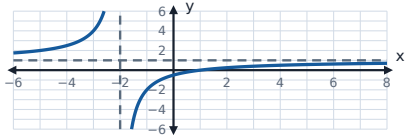
MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

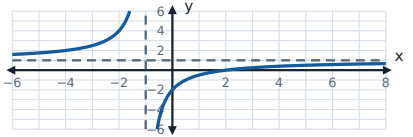
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = (x + 1)/(x − 2), with its asymptotes dashed?

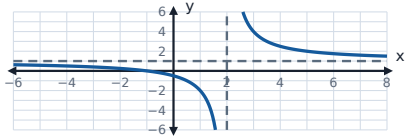
A



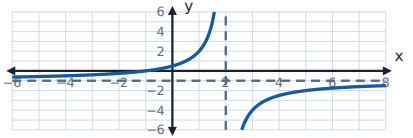
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

2716

QUESTION ID

S74-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Zero −1, y-intercept −1/2, vertical asymptote x = 2, horizontal asymptote y = 1 (ratio of leading coefficients); $y \to +\infty$ as $x \to 2^+$ and $y \to -\infty$ as $x \to 2^-$. The others show (x − 1)/(x + 2) (same y-intercept), (x − 2)/(x + 1), and the reflection.

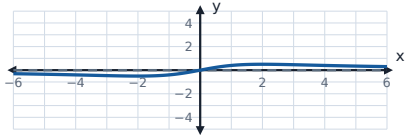
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

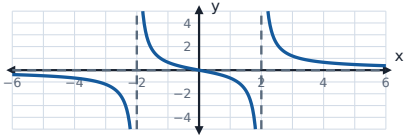
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2x/(x^2 - 4)$, with its asymptotes dashed?

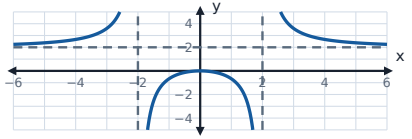
A



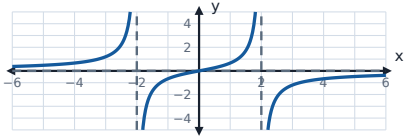
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

2717

QUESTION ID

S74-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Zero and y-intercept 0; vertical asymptotes $x = \pm 2$ (zeros of $x^2 - 4$); horizontal asymptote $y = 0$ (degree of numerator < degree of denominator); e.g. $y(3) = 6/5 > 0$ and $y(1) = -2/3 < 0$. The others show $2x/(x^2 + 4)$, $2x^2/(x^2 - 4)$, and the reflection.

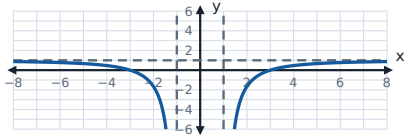
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

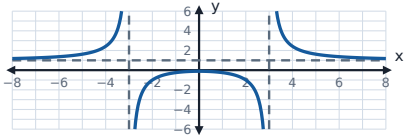
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 - 1)/(x^2 - 9)$, with its asymptotes dashed?

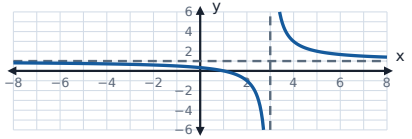
A



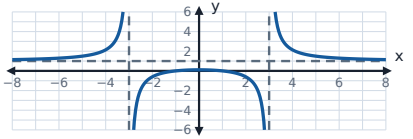
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

2718

QUESTION ID

S74-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Zeros ±1, y-intercept 1/9, vertical asymptotes $x = \pm 3$, horizontal asymptote $y = 1$; the outer branches are above $y = 1$. The others show the reciprocal $(x^2 - 9)/(x^2 - 1)$, $(x^2 + 1)/(x^2 - 9)$, and $(x - 1)/(x - 3)$.

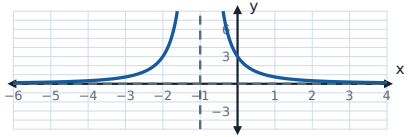
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

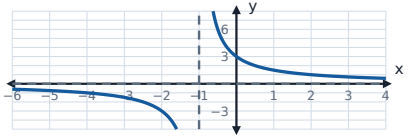
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 3/(x + 1)^2$, with its asymptotes dashed?

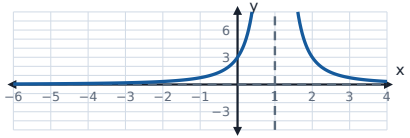
A



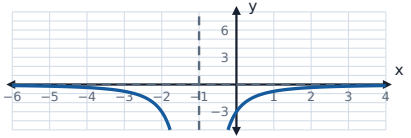
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

2719

QUESTION ID

S74-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE No zeros; y-intercept 3; vertical asymptote $x = -1$ with $y \to +\infty$ on both sides (the square is positive); horizontal asymptote $y = 0$. The others show $3/(x + 1)$, $3/(x - 1)^2$ (same y-intercept), and $-3/(x + 1)^2$.

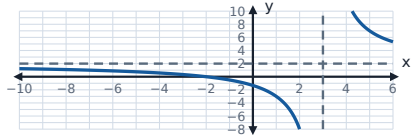
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

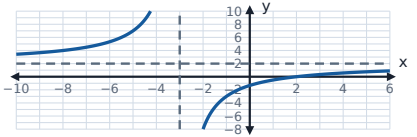
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = (2x − 4)/(x + 3), with its asymptotes dashed?

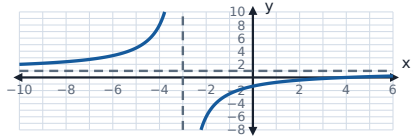
A



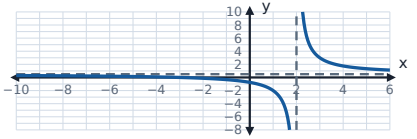
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

2720

QUESTION ID

S74-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Zero 2, y-intercept −4/3, vertical asymptote x = −3, horizontal asymptote y = 2/1 = 2. The others show (2x + 4)/(x − 3) and (x − 4)/(x + 3) (both with the same y-intercept) and the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 1

Rewrite a rational expression by inspection • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite (x + 1)/(x − 2) as 1 + r/(x − 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

2722

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 + 3/(x − 2)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite (2x − 4)/(x + 3) as 2 + r/(x + 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

2723

QUESTION ID

S74-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2 − 10/(x + 3)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

True or false?

The constant q in q + r/(x − a) is the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

2724

QUESTION ID

S74-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE r/(x − a) → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 - 1)/(x^2 - 9)$ equals

A. $1 + 8/(x^2 - 9)$

B. $1 - 8/(x^2 - 9)$

C. $1 + 1/(x^2 - 9)$

D. $1/9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

2725

QUESTION ID

S74-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $1 + 8/(x^2 - 9)$

GRADING NOTE $x^2 - 1 = (x^2 - 9) + 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor the denominator of $2x/(x^2 - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

2726

QUESTION ID

S74-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)$

GRADING NOTE Form required: factored. The vertical asymptotes are at the zeros of these factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-B1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 2

Domain and excluded values • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $(x + 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

2728

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \neq 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

2728

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $3/(x + 1)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

2729

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \neq -1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

True or false?

A rational function is undefined where its denominator is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

2730

QUESTION ID

S74-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division by 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

2730

ANSWER KEY Reduced copy of the question page, with its answer

$2x/(x^2 - 4)$ is undefined at

A. $x = \pm 2$

B. $x = 4$

C. $x = 2$ only

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

2731

QUESTION ID

S74-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x = \pm 2$

GRADING NOTE $x^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the excluded values of $(x^2 - 1)/(x^2 - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

2732

QUESTION ID

S74-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3 and −3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 3

Sign of a quotient on intervals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Is (x + 1)/(x − 2) positive or negative at x = 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

2734

QUESTION ID

S74-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Negative: 1/(−2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $2x/(x^2 - 4)$ positive or negative at $x = 3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

2735

QUESTION ID

S74-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Positive: 6/5

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

True or false?

A quotient of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

2736

QUESTION ID

S74-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For example $(-6)/(-2) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

At x = 2.1, (x + 1)/(x − 2) is

A. large and negative

B. undefined

C. zero

D. large and positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

2737

QUESTION ID

S74-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. large and positive

GRADING NOTE 3.1/0.1 = 31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $3/(x + 1)^2$ ever negative? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

2738

QUESTION ID

S74-B3-Q5

TYPE

Short answer

CORRECT ANSWER

No: a positive number over a square

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSF-IF.C.7.d

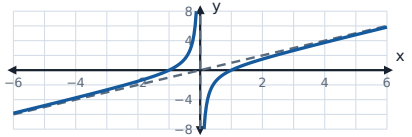
FORWARD 1

Graph rational functions with slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

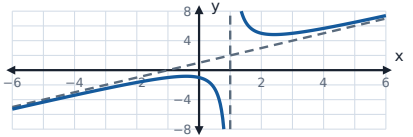
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 + 1)/x$, with its asymptotes dashed?

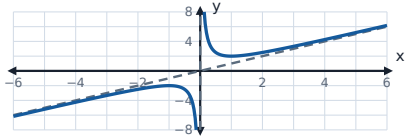
A



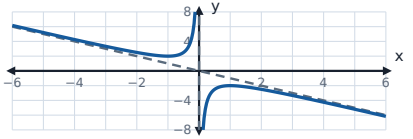
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

2740

QUESTION ID

S74-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $(x^2 + 1)/x = x + 1/x$: vertical asymptote $x = 0$, slant asymptote $y = x$, no zeros, no y-intercept, through $(1, 2)$ and $(-1, -2)$. The others show $(x^2 - 1)/x$, $(x^2 + 1)/(x - 1)$, and $-(x^2 + 1)/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slant asymptote of $y = (x^2 - x - 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

2741

QUESTION ID

S74-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = x + 1

GRADING NOTE Form required: equation. Quotient x + 1, remainder 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

True or false?

A rational function can cross its horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

2742

QUESTION ID

S74-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example $y = x/(x^2 + 1)$ crosses $y = 0$ at $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = (3x^2 + 1)/(x^2 + 4)$ has horizontal asymptote

A. none

B. $y = 3$

C. $y = 0$

D. $y = 1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

2743

QUESTION ID

S74-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 3

GRADING NOTE Ratio of leading coefficients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a rational function with vertical asymptote x = 1, horizontal asymptote y = 2, and zero at x = 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

2744

QUESTION ID
S74-F1-Q5

TYPE
Short answer

CORRECT ANSWER
For example
2(x − 3)/(x − 1)

GRADING NOTE Any valid function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

FORWARD 2

Interpret a rational model's features in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The average cost per item is A(x) = (500 + 4x)/x. Interpret the vertical and horizontal asymptotes in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

2746

QUESTION ID

S74-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 0: average cost is undefined/huge for very few items; y = 4: the average cost approaches $4 per item

GRADING NOTE Form required: both interpretations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$. Interpret its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

2747

QUESTION ID
S74-F2-Q2

TYPE
Short answer

CORRECT ANSWER
y = 0: the concentration eventually approaches 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

True or false?

For a context, only the part of a rational graph in the context domain is meaningful.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

2748

QUESTION ID

S74-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In A(x) = (500 + 4x)/x, as x grows the average cost

A. increases

B. is constant

C. approaches 0

D. decreases toward 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

2749

QUESTION ID

S74-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. decreases toward 4

GRADING NOTE 500/x → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the number of items for which the average cost is $5.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

2750

QUESTION ID

S74-F2-Q5

TYPE

Short answer

CORRECT ANSWER

500

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Distinguish holes from vertical asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 2	Simplify rational expressions and keep exclusions	**HSA-APR.D.7**
BACKWARD 3	Evaluate the simplified expression	**HSF-IF.A.2**
FORWARD 1	Construct and graph rational functions with holes	**HSF-IF.C.7.d**
FORWARD 2	Relate holes to extraneous solutions	**HSA-REI.A.2**

SET 75 • HSF-IF.C.7.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

Find every hole and vertical asymptote of $f(x) = (x^2 - 4)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

2753

QUESTION ID

S75-M-Q1

TYPE

Short answer

CORRECT ANSWER

Hole at (2, 4); no vertical asymptote

GRADING NOTE Form required: both answers. $f(x) = x + 2$ for $x \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every hole and vertical asymptote of $f(x) = (x - 3)/(x^2 - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

2754

QUESTION ID

S75-M-Q2

TYPE

Short answer

CORRECT ANSWER

Hole at (3, 1/6); vertical asymptote x = −3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

Find every hole and vertical asymptote of $f(x) = (x^2 + x - 6)/(x^2 - x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

2755

QUESTION ID
S75-M-Q3

TYPE
Short answer

CORRECT ANSWER
Hole at (2, 5/3); vertical asymptote x = −1

GRADING NOTE Form required: both. $(x + 3)(x - 2)/((x - 2)(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

Find every hole and vertical asymptote of $f(x) = (x + 1)/(x^2 - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

2756

QUESTION ID

S75-M-Q4

TYPE

Short answer

CORRECT ANSWER

Hole at (−1, −1/2); vertical asymptote x = 1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

Find every hole and vertical asymptote of $f(x) = x(x - 1)/(x(x + 2))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

2757

QUESTION ID

S75-M-Q5

TYPE

Short answer

CORRECT ANSWER

Hole at (0, −1/2); vertical asymptote x = −2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 1

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + x - 6$ and $x^2 - x - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

2759

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x − 2) and (x − 2)(x + 1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

2760

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)(x + 3)$

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 - 1 = (x - 1)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

2761

QUESTION ID

S75-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 1)^2 = x^2 - 2x + 1$; $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

2762

QUESTION ID

S75-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (x − 2)(x + 2)

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

2763

QUESTION ID

S75-B1-Q5

TYPE

Short answer

CORRECT ANSWER

x(x + 2)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 2

Simplify rational expressions and keep exclusions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (x − 3)/((x − 3)(x + 3)) and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

2765

QUESTION ID
S75-B2-Q1

TYPE
Short answer

CORRECT ANSWER
1/(x + 3); x ≠ ±3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-B2-Q1**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (x + 1)/((x − 1)(x + 1)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

2766

QUESTION ID
S75-B2-Q2

TYPE
Short answer

CORRECT ANSWER
1/(x − 1), x ≠ ±1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

After simplifying, the original excluded values are still excluded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

2767

QUESTION ID
S75-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Domain kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x(x - 1)/(x(x + 2))$ simplifies to

A. $-1/2$

B. $(x - 1)/(x + 2)$ for all x

C. $x - 1$

D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

2768

QUESTION ID
S75-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

GRADING NOTE Cancel x but keep $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x^2 - 4)/(x - 2)$ and state the excluded value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

2769

QUESTION ID
S75-B2-Q5

TYPE
Short answer

CORRECT ANSWER
x + 2, x ≠ 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 3

Evaluate the simplified expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 1/(x + 3) at x = 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

2771

QUESTION ID

S75-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate (x + 3)/(x + 1) at x = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

2772

QUESTION ID

S75-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

True or false?

The y-coordinate of a hole is the value of the simplified expression there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q3/5

2773

QUESTION ID

S75-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Limit value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 1)/(x + 2)$ at $x = 0$ is

A. 1/2

B. −1/2

C. undefined

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

2774

QUESTION ID

S75-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −1/2

GRADING NOTE −1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 1/(x − 1) at x = −1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

2775

QUESTION ID

S75-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 1

Construct and graph rational functions with holes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

Write a rational function with a hole at x = 1, a vertical asymptote x = −3, and horizontal asymptote y = 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

2777

QUESTION ID

S75-F1-Q1

TYPE

Short answer

CORRECT ANSWER

For example (x − 1)/((x − 1)(x + 3))

GRADING NOTE Any valid function.

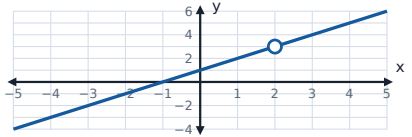
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

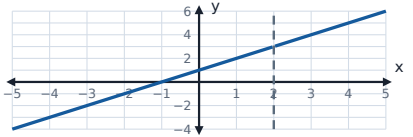
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 - x - 2)/(x - 2)$?

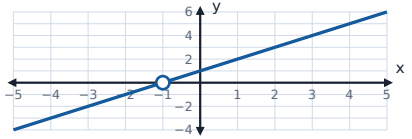
A



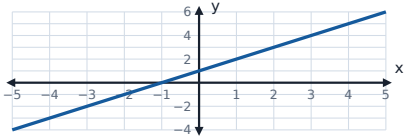
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

2778

QUESTION ID

S75-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $x^2 - x - 2 = (x - 2)(x + 1)$, so $y = x + 1$ for $x \neq 2$: the line with a hole at (2, 3). The factor cancels, so there is no asymptote at $x = 2$, and $x = -1$ is a zero, not an excluded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

True or false?

A rational function can have both a hole and a vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

2779

QUESTION ID

S75-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Different factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F1-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 4)^2/(x - 4)$ has

A. no special features

B. a hole at (4, 4)

C. a vertical asymptote $x = 4$

D. a hole at (4, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q4/5

2780

QUESTION ID

S75-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a hole at (4, 0)

GRADING NOTE Simplifies to $x - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $x = -3$ is an asymptote but $x = 3$ is a hole for $(x - 3)/(x^2 - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

2781

QUESTION ID

S75-F1-Q5

TYPE

Short answer

CORRECT ANSWER

After canceling $x - 3$, the factor $x + 3$ remains in the denominator (the values grow without bound near -3), but $x - 3$ does not (values approach 1/6 near 3).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 2

Relate holes to extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x^2 - 4)/(x - 2) = 4$. Explain the result using the hole.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

2783

QUESTION ID
S75-F2-Q1

TYPE
Short answer

CORRECT ANSWER
No solution: x + 2 = 4 gives x = 2, which is excluded (the hole at (2, 4))

GRADING NOTE Form required: answer and explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 3)/(x^2 - 9) = 1/6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

2784

QUESTION ID

S75-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No solution: $1/(x + 3) = 1/6$ gives $x = 3$, which is excluded

GRADING NOTE Form required: answer and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of f(x) = c can occur exactly at a hole whose height is c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

2785

QUESTION ID
S75-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Graph has no point there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)/(x^2 - 1) = -1/2$ has

A. no solution

B. $x = -1$

C. $x = 0$

D. $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q4/5

2786

QUESTION ID

S75-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. no solution

GRADING NOTE The only candidate is the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x^2 + x - 6)/(x^2 - x - 2) = 2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

2787

QUESTION ID

S75-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. (x + 3)/(x + 1) = 2 gives x + 3 = 2x + 2, so x = 1, which is not excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c • ADVANCED EXTENSION (+)

Compose functions and interpret compositions with domains

MAIN	Algebra II main question	**HSF-BF.A.1.c**
BACKWARD 1	Build combined models	**HSF-BF.A.1.b**
BACKWARD 2	Evaluate a function at an expression	**HSF-IF.A.2**
BACKWARD 3	Domains of radical, rational, and log functions	**HSF-IF.B.5**
FORWARD 1	Decompose functions into compositions	**HSF-BF.A.1.c**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 76 • HSF-BF.A.1.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{x}$ and $g(x) = x - 4$. Write $f(g(x))$ and $g(f(x))$ and give the domain of each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

2790

QUESTION ID

S76-M-Q1

TYPE

Short answer

CORRECT ANSWER

$f(g(x)) = \sqrt{(x - 4)}$, $x \geq 4$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 0$

GRADING NOTE Form required: both with domains.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The temperature is T(h) = 20 − 6.5h °C at height h km, and a balloon's height is h(t) = 0.3t km after t minutes. Write T(h(t)) and find the temperature at t = 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

2791

QUESTION ID

S76-M-Q2

TYPE

Short answer

CORRECT ANSWER

T(h(t)) = 20 − 1.95t; 0.5°C

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 1/(x - 1)$ and $g(x) = x^2$. Write $f(g(x))$ and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

2792

QUESTION ID

S76-M-Q3

TYPE

Short answer

CORRECT ANSWER

$1/(x^2 - 1)$; $x \neq 1, -1$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A dollar amount d converts to euros by e(d) = 0.9d, and a store's price with tax in euros is p(e) = 1.2e. Write p(e(d)) and find it for d = 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

2793

QUESTION ID
S76-M-Q4

TYPE
Short answer

CORRECT ANSWER
p(e(d)) = 1.08d; 54 euros

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$ and $g(x) = x^2 - 9$. Write $f(g(x))$ and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

2794

QUESTION ID
S76-M-Q5

TYPE
Short answer

CORRECT ANSWER
$\log_2(x^2 - 9)$; $x < -3$ or $x > 3$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 1

Build combined models • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) + g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

2796

QUESTION ID

S76-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + x + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) \cdot g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

2797

QUESTION ID

S76-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

True or false?

f(g(x)) is a different combination from f(x) · g(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

2798

QUESTION ID

S76-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Composition vs product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = 2x and g(x) = x + 3, f(x) − g(x) is

A. x − 3

B. $2x^2 + 6x$

C. x + 3

D. 3x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

2799

QUESTION ID

S76-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x − 3

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the operation in T(h(t)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

2800

QUESTION ID

S76-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Composition

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 2

Evaluate a function at an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √x. Find f(x − 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

2802

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

√(x − 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

T(h) = 20 − 6.5h. Find T(0.3t).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

2803

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20 − 1.95t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

2803

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = 1/(x - 1)$, then $f(x^2) = 1/(x^2 - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

2804

QUESTION ID
S76-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $g(x) = x^2 - 9$, then $g(3)$ is

A. -3

B. -6

C. 9

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

2805

QUESTION ID

S76-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $9 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

p(e) = 1.2e. Find p(45).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

2806

QUESTION ID

S76-B2-Q5

TYPE

Short answer

CORRECT ANSWER

54

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 3

Domains of radical, rational, and log functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{(x - 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

2808

QUESTION ID

S76-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log_2 u$ in terms of u.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

2809

QUESTION ID

S76-B3-Q2

TYPE

Short answer

CORRECT ANSWER

u > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

True or false?

$1/(x^2 - 1)$ is undefined only at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

2810

QUESTION ID

S76-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 1 = 0$ at both $x = 1$ and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

2810

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 9 > 0$ when

A. $x < -3$ or $x > 3$

B. $-3 < x < 3$

C. all x

D. $x > 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

2811

QUESTION ID

S76-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x < -3$ or $x > 3$

GRADING NOTE Outside the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{x} - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

2812

QUESTION ID

S76-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 1

Decompose functions into compositions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Write $h(x) = (3x + 1)^5$ as $f(g(x))$. Give f and g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

2814

QUESTION ID

S76-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$g(x) = 3x + 1$, $f(x) = x^5$

GRADING NOTE Any valid decomposition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $h(x) = 2^{x^2}$ as $f(g(x))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

2815

QUESTION ID

S76-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$g(x) = x^2$, $f(x) = 2^x$

GRADING NOTE Any valid decomposition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

True or false?

Every composition f(g(x)) can be decomposed in only one way.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

2816

QUESTION ID

S76-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Many decompositions exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(x) = x + 1$, $g(x) = 2x$, $k(x) = x^2$, then $f(g(k(3)))$ is

A. 37

B. 19

C. 64

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

2817

QUESTION ID

S76-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 19

GRADING NOTE $k(3) = 9$, $g = 18$, $f = 19$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a context for a composition V(r(t)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

2818

QUESTION ID

S76-F1-Q5

TYPE

Short answer

CORRECT ANSWER

For example the volume of a balloon whose radius r grows with time t

GRADING NOTE Any valid context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Verify that f(x) = 2x + 3 and g(x) = (x − 3)/2 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

2820

QUESTION ID
S76-F2-Q1

TYPE
Short answer

CORRECT ANSWER
f(g(x)) = x and g(f(x)) = x

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute f(g(x)) for $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

2821

QUESTION ID

S76-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

True or false?

If f(g(x)) = x for all x in g's domain and g(f(x)) = x for all x in f's domain, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

2822

QUESTION ID

S76-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = x + 5 and g(x) = x − 5, f(g(2)) is

A. 2

B. −3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

2823

QUESTION ID

S76-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE Inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why composition is the natural test for inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

2824

QUESTION ID

S76-F2-Q5

TYPE

Short answer

CORRECT ANSWER

An inverse undoes the function, so applying one after the other must return the input.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b • ADVANCED EXTENSION (+)

Verify inverses by composition on the stated domains

MAIN	Algebra II main question	**HSF-BF.B.4.b**
BACKWARD 1	Find inverses algebraically	**HSF-BF.B.4.a**
BACKWARD 2	Compose functions	**HSF-BF.A.1.c**
BACKWARD 3	Square roots of squares	**HSN-RN.A.2**
FORWARD 1	Show that proposed inverses fail	**HSF-BF.B.4.b**
FORWARD 2	Read inverse values from tables and graphs	**HSF-BF.B.4.c**

SET 77 • HSF-BF.B.4.b

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$ are inverses?

A. $f(x) \cdot g(x) = (2x + 3)(x - 3)/2 = 1$, so f and g are inverses.

B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

C. $f(1) = 5$ and $g(5) = 1$, so f and g are inverses.

D. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ for all real x, and one composition is enough by itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

2827

QUESTION ID
S77-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

GRADING NOTE Both compositions must simplify to x. The others check one composition only, one input only, or multiply the functions instead of composing them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = x^3 - 1$ and $g(x) = \sqrt[3]{(x + 1)}$ are inverses?

A. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$, but only for $x \geq 0$, so f and g are inverses only when $x \geq 0$.

B. $g(f(x)) = \sqrt[3]{(x^3 - 1)} + 1 = x$, so f and g are inverses.

C. $f(2) = 7$ and $g(7) = 2$, so f and g are inverses.

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

2828

QUESTION ID

S77-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

GRADING NOTE Both compositions simplify to x for every real x (cube roots are defined for negatives). The others add a false domain limit, check one input, or compose incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 2^x$ and $g(x) = \log_2 x$ are inverses, with the correct domains?

A. $f(g(x)) = 2 \cdot \log_2 x = x$ for $x > 0$.

B. $f(g(x)) = 2^{\log_2 x} = x$ and $g(f(x)) = \log_2(2^x) = x$, both for every real number x.

C. $f(3) = 8$ and $g(8) = 3$, so f and g are inverses for all x.

D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

2829

QUESTION ID
S77-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

GRADING NOTE $\log_2 x$ is defined only for $x > 0$, so $f(g(x)) = x$ holds for $x > 0$; $g(f(x)) = x$ for every real x. The others ignore the domain of $\log_2$, multiply instead of composing, or check one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 1/(x - 2)$ $(x \neq 2)$ and $g(x) = 1/x + 2$ $(x \neq 0)$ are inverses?

A. $f(3) = 1$ and $g(1) = 3$, so f and g are inverses.

B. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$, so f and g are inverses for all real x with no excluded values needed.

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

D. $g(f(x)) = 1/(x - 2) + 2 = (2x - 3)/(x - 2)$, so they are not inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

2830

QUESTION ID

S77-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

GRADING NOTE Both compositions simplify to x on the stated domains. The others drop the second composition and the domains, compute $f(x) + 2$ instead of $g(f(x))$, or check one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ with domain $x \geq 0$ and $g(x) = \sqrt{x}$. Which work verifies that they are inverses and explains why the domain restriction is needed?

A. The restriction is needed because $\sqrt{x}$ is undefined for $x > 0$.

B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

C. $f(g(x)) = (\sqrt{x})^2 = x$ and $g(f(x)) = \sqrt{(x^2)} = x$ for every real number x, so the two functions are inverses and no domain restriction is needed.

D. $g(f(x)) = \sqrt{(x^2)} = \pm x$, and the restriction chooses the + sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

2831

QUESTION ID
S77-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

GRADING NOTE $\sqrt{(x^2)} = |x|$, which equals x only when $x \geq 0$ (for example $\sqrt{((-3)^2)} = 3$). The others ignore $|x|$, state a false fact about $\sqrt{x}$, or treat $\sqrt{}$ as giving two values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 1

Find inverses algebraically • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of f(x) = 2x + 3.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

2833

QUESTION ID

S77-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = x^3 - 1$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

2834

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt[3]{(x + 1)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

True or false?

The inverse of 2^x is $\log_2 x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

2835

QUESTION ID

S77-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The inverse of f(x) = 1/(x − 2) is

A. 1/x + 2

B. 1/(x + 2)

C. 2 − 1/x

D. x − 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

2836

QUESTION ID

S77-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1/x + 2

GRADING NOTE Solve y = 1/(x − 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = x^2$ for $x \geq 0$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

2837

QUESTION ID

S77-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt{x}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 2

Compose functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 2x + 3 and g(x) = (x − 3)/2. Compute f(g(5)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

2839

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x + 3$ and $g(x) = (x - 3)/2$. Compute $g(f(5))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

2840

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-B2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

True or false?

f(g(x)) means apply g first, then f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

2841

QUESTION ID

S77-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inside out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^3$ and $g(x) = x + 1$, $f(g(x))$ is

A. $(x + 1)^3$

B. $x^3 + 1$

C. $x + 1$

D. $3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

2842

QUESTION ID

S77-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $(x + 1)^3$

GRADING NOTE Substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $\log_2(2^5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

2843

QUESTION ID
S77-B2-Q5

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 3

Square roots of squares • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt{((-3)^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

2845

QUESTION ID

S77-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

2846

QUESTION ID

S77-B3-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q3/5

2847

QUESTION ID

S77-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{(x^2)}$ equals

A. $|x|$

B. x^2

C. $-x$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

2848

QUESTION ID
S77-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $|x|$

GRADING NOTE Principal root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $\sqrt{(x^2)} = x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

2849

QUESTION ID

S77-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x ≥ 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 1

Show that proposed inverses fail • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Show by composition that f(x) = 3x − 1 and g(x) = (x − 1)/3 are not inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

2851

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = x − 2 ≠ x

GRADING NOTE Form required: a composition that is not x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ (all real x) and $g(x) = \sqrt{x}$. Which composition fails to equal x, and for which x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

2852

QUESTION ID

S77-F1-Q2

TYPE

Short answer

CORRECT ANSWER

g(f(x)) = |x| ≠ x for x < 0

GRADING NOTE Form required: the composition and the x-values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

True or false?

Checking f(g(x)) = x alone is always enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

2853

QUESTION ID

S77-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both compositions must be checked (with domains).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = e^x$ and $g(x) = \ln x$, $g(f(x)) =$

A. $|x|$

B. x only for $x > 0$

C. x for all real x

D. $e^{\ln x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

2854

QUESTION ID

S77-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x for all real x

GRADING NOTE $\ln(e^x) = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Fix the domain of $f(x) = (x - 1)^2$ so that $g(x) = 1 + \sqrt{x}$ is its inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

2855

QUESTION ID

S77-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 2

Read inverse values from tables and graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(7)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

2857

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(f^{-1}(15))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

2858

QUESTION ID
S77-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(3) = 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

True or false?

If (a, b) is on the graph of f, then (b, a) is on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

2859

QUESTION ID
S77-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(4) = 9$, then $f^{-1}(9)$ is

A. 1/4

B. 4

C. 13

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

2860

QUESTION ID

S77-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(1) = 3 and f(2) = 7. Explain how these values give $f(f^{-1}(7)) = 7$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

2861

QUESTION ID

S77-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(7) = 2$ and $f(2) = 7$, so $f(f^{-1}(7)) = 7$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c • ADVANCED EXTENSION (+)

Read values of an inverse function from graphs and tables

MAIN	Algebra II main question	**HSF-BF.B.4.c**
BACKWARD 1	Inverse functions swap inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 2	Read points from graphs and tables	**8.F.A.1**
BACKWARD 3	One-to-one functions	**HSF-BF.B.4.b**
FORWARD 1	Estimate inverse values between data points and sketch inverse...	**HSF-BF.B.4.c**
FORWARD 2	Restrict the domain to make a function invertible	**HSF-BF.B.4.d**

SET 78 • HSF-BF.B.4.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

The table shows f. Find $f^{-1}(15)$ and explain.

x	0	1	2	3	4
f(x)	1	3	7	15	31

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

2864

QUESTION ID

S78-M-Q1

TYPE

Short answer

CORRECT ANSWER

3: f(3) = 15

GRADING NOTE Form required: value and reason.

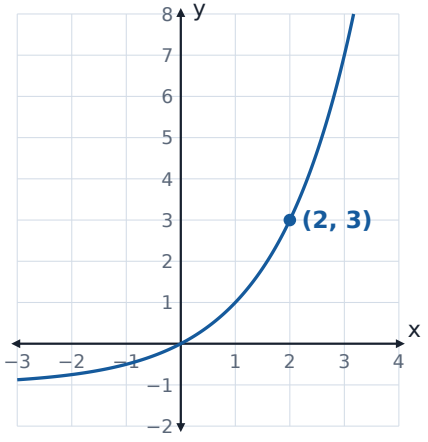
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = 2^x - 1$ is shown. Use it to find $f^{-1}(3)$.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

2865

QUESTION ID
S78-M-Q2

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. The point (2, 3) is on the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a decreasing function g. Find $g^{-1}(10)$ and $g^{-1}(g(2))$.

x	1	2	3	4	5
g(x)	40	25	16	10	4

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

2866

QUESTION ID
S78-M-Q3

TYPE
Short answer

CORRECT ANSWER
$g^{-1}(10) = 4$;
$g^{-1}(g(2)) = 2$

GRADING NOTE Form required: both.

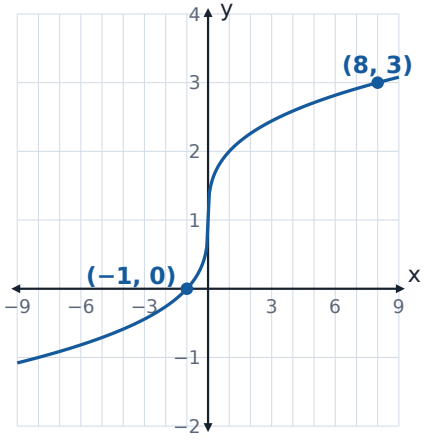
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = \sqrt[3]{x} + 1$ is shown. Use it to find $f^{-1}(0)$ and $f^{-1}(3)$.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

2867

QUESTION ID

S78-M-Q4

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(0) = -1$;

$f^{-1}(3) = 8$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The table gives a plane's altitude A (m) t minutes after takeoff. Find $A^{-1}(1200)$ and interpret it.

t (min)	0	1	2	3	4	5
A (m)	0	250	550	880	1200	1500

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

2868

QUESTION ID
S78-M-Q5

TYPE
Short answer

CORRECT ANSWER
4 minutes: the plane reaches 1200 m four minutes after takeoff

GRADING NOTE Form required: value and interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 1

Inverse functions swap inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

If $f(3) = 15$, what is $f^{-1}(15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

2870

QUESTION ID

S78-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $f(x) = 2^x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

2871

QUESTION ID

S78-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = \log_2(x + 1)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f^{-1}(y)$ is the input of f that gives the output y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

2872

QUESTION ID

S78-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

If (5, 4) is on the graph of g, which point is on g^{-1}?

A. (5, 4)

B. (1/5, 1/4)

C. (−5, −4)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

2873

QUESTION ID

S78-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 5)

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the formula to find $f^{-1}(3)$ for $f(x) = 2^x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

2874

QUESTION ID

S78-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 2

Read points from graphs and tables • Grade 8 • Functions • 8.F.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. What is f(2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

2876

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane's altitude A (m) at t = 0, 1, 2, 3, 4 minutes is 0, 250, 550, 880, 1200. What is A(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

2877

QUESTION ID

S78-B2-Q2

TYPE

Short answer

CORRECT ANSWER

880 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

True or false?

A function gives exactly one output for each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

2878

QUESTION ID

S78-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Which input of g gives 16?

A. 16

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

2879

QUESTION ID

S78-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE Read backward.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph passes through (8, 3). What is the output at input 8?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

2880

QUESTION ID

S78-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 3

One-to-one functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Why does a strictly increasing function have an inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

2882

QUESTION ID

S78-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Different inputs give different outputs, so each output comes from one input

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does a function with f(1) = 4 and f(3) = 4 have an inverse function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

2883

QUESTION ID
S78-B3-Q2

TYPE
Short answer

CORRECT ANSWER
No: 4 has two inputs

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-B3-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

True or false?

A decreasing function can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

2884

QUESTION ID

S78-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Still one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

Which table has an inverse function?

A. x: 1, 2, 3; y: 5, 7, 5

B. x: 1, 2, 3; y: 5, 7, 9

C. x: 1, 2, 3; y: 0, 1, 0

D. x: 1, 2, 3; y: 2, 2, 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

2885

QUESTION ID

S78-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x: 1, 2, 3; y: 5, 7, 9

GRADING NOTE Distinct outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(2) = 7 and f(3) = 15. Verify $f(f^{-1}(15)) = 15$ with the table.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

2886

QUESTION ID
S78-B3-Q5

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(15) = 3$ and $f(3) = 15$

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-B3-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 1

Estimate inverse values between data points and sketch inverse graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

A plane's altitude is 880 m at t = 3 min and 1200 m at t = 4 min. Estimate the time when it reaches 1000 m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

2888

QUESTION ID

S78-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 3.4 minutes

GRADING NOTE Accept 3.3 to 3.5 (linear interpolation 3 + 120/320).

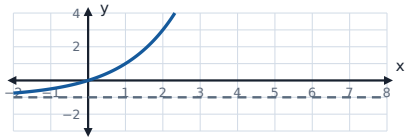
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

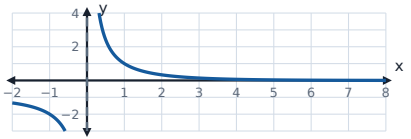
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows the inverse of $f(x) = 2^x - 1$?

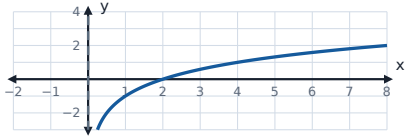
A



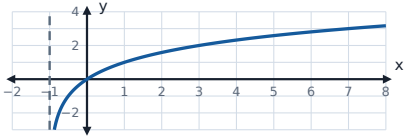
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

2889

QUESTION ID

S78-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Swap the coordinates of points of f: (0, 0), (1, 1), (2, 3) on f give (0, 0), (1, 1), (3, 2) on f^{-1}; the asymptote $y = -1$ becomes $x = -1$; $f^{-1}(x) = \log_2(x + 1)$. The others show f itself, the reciprocal $1/f(x)$, and $\log_2 x - 1$ (operations undone in the wrong order).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

True or false?

The graph of f^{-1} is the reflection of the graph of f across y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

2890

QUESTION ID
S78-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Swap coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A decreasing function has $g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Estimate $g^{-1}(20)$.

A. between 1 and 2

B. between 2 and 3

C. between 4 and 5

D. between 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

2891

QUESTION ID

S78-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. between 2 and 3

GRADING NOTE $g(2) = 25$, $g(3) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $(f^{-1})^{-1}(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

2892

QUESTION ID
S78-F1-Q5

TYPE
Short answer

CORRECT ANSWER
f(2) = 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 2

Restrict the domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

$f(x) = (x - 3)^2$ is not invertible. Give a domain restriction that makes it invertible, and write the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

2894

QUESTION ID

S78-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 3$; $f^{-1}(x) = 3 + \sqrt{x}$

GRADING NOTE Any valid restriction with its inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table of h shows $h(-2) = 4$, $h(0) = 0$, $h(2) = 4$. Explain why $h^{-1}(4)$ is not defined and how to fix it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

2895

QUESTION ID

S78-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 has two inputs; restrict h to x ≥ 0 (or x ≤ 0)

GRADING NOTE Form required: reason and fix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

True or false?

Restricting the domain can turn a non-invertible function into an invertible one.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

2896

QUESTION ID
S78-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE One-to-one on the piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ restricted to $x \le 0$ has inverse

A. $\sqrt{x}$

B. x^2

C. $-\sqrt{x}$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

2897

QUESTION ID

S78-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-\sqrt{x}$

GRADING NOTE Negative outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

Why must the restricted piece still take every output value of the original function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

2898

QUESTION ID
S78-F2-Q5

TYPE
Short answer

CORRECT ANSWER
To keep the same range, the restricted function should reach all the outputs (for example x ≥ 3 still gives all y ≥ 0).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d • ADVANCED EXTENSION (+)

Restrict a domain to make a function invertible and write the inverse

MAIN	Algebra II main question	**HSF-BF.B.4.d**
BACKWARD 1	Find the vertex of a quadratic	**HSF-IF.C.8.a**
BACKWARD 2	Write inverses of one-to-one functions	**HSF-BF.B.4.a**
BACKWARD 3	Domains and ranges	**HSF-IF.B.5**
FORWARD 1	Compare different restrictions	**HSF-BF.B.4.d**
FORWARD 2	Restrict trigonometric functions to define inverses	**HSF-TF.B.6**

SET 79 • HSF-BF.B.4.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x - 3)^2$ to $x \geq 3$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

2901

QUESTION ID

S79-M-Q1

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = \sqrt{x} + 3,\ x \geq 0$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = x^2 + 1$ to $x \le 0$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

2902

QUESTION ID

S79-M-Q2

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = -\sqrt{(x - 1)}$, $x \ge 1$

GRADING NOTE Form required: inverse (with the negative root) and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = |x + 2|$ to $x \geq -2$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

2903

QUESTION ID
S79-M-Q3

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(x) = x - 2, x \geq 0$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x + 1)^2 - 4$ to $x \geq -1$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

2904

QUESTION ID
S79-M-Q4

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(x) = -1 + \sqrt{(x + 4)}$, $x \geq -4$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = x^4$ to $x \geq 0$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

2905

QUESTION ID

S79-M-Q5

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = \sqrt[4]{x},\ x \geq 0$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 1

Find the vertex of a quadratic • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertex of $y = (x + 1)^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

2907

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, −4)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertex of $y = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

2908

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

True or false?

A parabola is increasing on one side of its vertex and decreasing on the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

2909

QUESTION ID

S79-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Turning point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The vertex of $y = (x - 3)^2$ is

A. (0, 9)

B. (0, 3)

C. (−3, 0)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

2910

QUESTION ID

S79-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (3, 0)

GRADING NOTE h = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

On which side of the vertex is $y = (x - 3)^2$ increasing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

2911

QUESTION ID

S79-B1-Q5

TYPE

Short answer

CORRECT ANSWER

x ≥ 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 2

Write inverses of one-to-one functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = (x - 3)^2$ for x when $x \geq 3$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

2913

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$3 + \sqrt{y}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = x^2 + 1$ for x when $x \leq 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

2914

QUESTION ID

S79-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-\sqrt{(y - 1)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

True or false?

When $x \leq 0$, the solution of $x^2 = 4$ is $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

2915

QUESTION ID

S79-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Choose the negative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

Solving $y = x + 2$ $(x \geq -2)$ for x gives

A. $2 - y$

B. $y + 2$

C. $y - 2$

D. $-y - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

2916

QUESTION ID

S79-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $y - 2$

GRADING NOTE Linear piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = x^4$ for x when $x \geq 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

2917

QUESTION ID

S79-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt[4]{y}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 3

Domains and ranges • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = (x - 3)^2$ for $x \geq 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

2919

QUESTION ID

S79-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = (x + 1)^2 - 4$ for $x \geq -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

2920

QUESTION ID

S79-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$y \geq -4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

True or false?

The domain of the inverse is the range of the restricted function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

2921

QUESTION ID

S79-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The range of $x^2 + 1$ for $x \leq 0$ is

A. $y \geq 0$

B. $y \geq 1$

C. all reals

D. $y \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

2922

QUESTION ID
S79-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. y ≥ 1

GRADING NOTE Minimum 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $|x + 2|$ for $x \geq -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

2923

QUESTION ID

S79-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$y \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 1

Compare different restrictions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x - 3)^2$ to $x \leq 3$. Write the inverse.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

2925

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3 - \sqrt{x}, x \geq 0$

GRADING NOTE Form required: formula and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -5(t - 2)^2 + 20$ for $0 \le t \le 4$. Which restriction gives the time as a function of height on the way down? Write it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

2926

QUESTION ID
S79-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$2 \le t \le 4$;
$t = 2 + \sqrt{((20 - h)/5)}$

GRADING NOTE Form required: restriction and inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

True or false?

A function can have more than one valid restriction that makes it invertible.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

2927

QUESTION ID

S79-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Either side of the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which restriction does not make $f(x) = x^2$ invertible?

A. $x \geq 0$

B. $x \leq 0$

C. $-1 \leq x \leq 1$

D. $1 \leq x \leq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

2928

QUESTION ID

S79-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-1 \leq x \leq 1$

GRADING NOTE Contains both ±1/2, for example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is the restriction $x \geq 0$ natural for $f(x) = x^2$ in the area formula $A = s^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

2929

QUESTION ID

S79-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Side lengths are nonnegative.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 2

Restrict trigonometric functions to define inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Why is sin x not invertible on all real numbers?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

2931

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It repeats values (for example sin 0 = sin π = 0), so it is not one-to-one.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the standard restricted domain of sin x and explain why it works.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

2932

QUESTION ID

S79-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−π/2 ≤ x ≤ π/2: sin is increasing there and takes every value from −1 to 1 once.

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

True or false?

cos x is restricted to $0 \leq x \leq \pi$ to define arccos.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

2933

QUESTION ID

S79-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Decreasing there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arcsin(1) equals

A. 0

B. 3π/2

C. π/2

D. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

2934

QUESTION ID

S79-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π/2

GRADING NOTE In the restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $0 \leq x \leq \pi$ would not work for sine.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

2935

GRADING NOTE Any equivalent explanation.

QUESTION ID

S79-F2-Q5

TYPE

Short answer

CORRECT ANSWER

sin increases then decreases on [0, π], so values repeat (sin(π/6) = sin(5π/6)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSF-BF.B.5 • ADVANCED EXTENSION (+)

Use the inverse relationship of exponents and logarithms to solve equations

MAIN	Algebra II main question	**HSF-BF.B.5**
BACKWARD 1	Write exact logarithmic solutions of exponential equations	**HSF-LE.A.4**
BACKWARD 2	Check for extraneous solutions	**HSA-REI.A.2**
BACKWARD 3	Domains of logarithmic expressions	**HSF-IF.B.5**
FORWARD 1	Use log properties and change of base in equations	**HSF-BF.B.5**
FORWARD 2	Check logarithmic solutions graphically	**HSA-REI.D.11**

SET 80 • HSF-BF.B.5

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_3 x = 4$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

2938

QUESTION ID

S80-M-Q1

TYPE

Short answer

CORRECT ANSWER

81

GRADING NOTE Any equivalent form. $x = 3^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\log_5(x - 1) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

2939

QUESTION ID

S80-M-Q2

TYPE

Short answer

CORRECT ANSWER

26

GRADING NOTE Any equivalent form. $\log_5(x - 1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ln(x + 2) = 1 exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

2940

QUESTION ID

S80-M-Q3

TYPE

Short answer

CORRECT ANSWER

e − 2

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2 x + \log_2(x - 2) = 3$ and check for extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

2941

QUESTION ID

S80-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 4; x = −2 is extraneous

GRADING NOTE Form required: solution and rejection. $x(x - 2) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve log x − log(x − 9) = 1 (base 10) and check.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

2942

QUESTION ID

S80-M-Q5

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Form required: the solution with a check. x/(x − 9) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 1

Write exact logarithmic solutions of exponential equations • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3^x = 81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

2944

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^x = 5$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

2945

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

ln 5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q2**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

log_b y = x means $b^x = y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

2946

QUESTION ID

S80-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$5^2 = 25$ written as a logarithm is

A. $\log_5 2 = 25$

B. $\log_2 25 = 5$

C. $\log_{25} 5 = 2$

D. $\log_5 25 = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

2947

QUESTION ID

S80-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $\log_5 25 = 2$

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $10^x = 1000$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

2948

QUESTION ID

S80-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 2

Check for extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = -2$ in $\log_2 x + \log_2(x - 2) = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

2950

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2(-2)$ is undefined, so -2 is extraneous

GRADING NOTE Form required: check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = 4$ in $\log_2 x + \log_2(x - 2) = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

2951

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 1 = 3; a solution

GRADING NOTE Form required: check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

True or false?

Combining logs can introduce solutions outside the original domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

2952

QUESTION ID

S80-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE log a + log b = log(ab) needs a, b > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is excluded from log(x − 9)?

A. x = 20

B. x = 100

C. x = 5

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

2953

QUESTION ID

S80-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x = 5

GRADING NOTE x − 9 > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x(x - 2) = 8$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

2954

QUESTION ID

S80-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4 or −2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 3

Domains of logarithmic expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log_5(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

2956

QUESTION ID

S80-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x > 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of ln(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

2957

QUESTION ID

S80-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x > −2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

True or false?

log 0 = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

2958

QUESTION ID

S80-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No power of 10 equals 0, so log 0 is undefined (log 1 = 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log_2 x + \log_2(x - 2)$ is

A. $x > 2$

B. $x \neq 2$

C. $x > 0$

D. all reals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

2959

QUESTION ID

S80-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x > 2

GRADING NOTE Both inputs positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

2959

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log x - \log(x - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

2960

QUESTION ID

S80-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x > 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 1

Use log properties and change of base in equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2(x + 6) - \log_2 x = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

2962

QUESTION ID

S80-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $(x + 6)/x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_4 x = \log_2 3$. (Hint: change of base.)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

2963

QUESTION ID

S80-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $\log_4 x = (\log_2 x)/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

True or false?

$\log_2(x + 6) - \log_2 x = \log_2((x + 6)/x)$ for $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q3/5

2964

QUESTION ID

S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Quotient property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\log(x + 1) + \log(x - 1) = \log 8$ has solution

A. no solution

B. $x = 8$

C. $x = 3$

D. $x = -3$ and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

2965

QUESTION ID

S80-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x = 3$

GRADING NOTE $x^2 - 1 = 8$; -3 rejected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\log_2(x - 5) = \log_2(1 - x)$ has no solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

2966

QUESTION ID
S80-F1-Q5

TYPE
Short answer

CORRECT ANSWER
It needs x > 5 and x < 1 at the same time, which is impossible.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 2

Check logarithmic solutions graphically • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the graphs of $y = \log_2 x + \log_2(x - 2)$ and $y = 3$ show the solution $x = 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

2968

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

They intersect only at x = 4; the left graph exists only for x > 2.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graph or table to approximate the solution of ln $x = 2 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

2969

QUESTION ID

S80-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.56

GRADING NOTE Accept 1.55 to 1.56 (root 1.5571).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of a log equation corresponds to no intersection point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

2970

QUESTION ID

S80-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outside the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of y = log x and y = 1 meet at

A. x = e

B. x = 10

C. x = 1

D. x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

2971

QUESTION ID

S80-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = 10

GRADING NOTE log 10 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can ln x = 2 − x not be solved exactly with logarithm rules?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

2972

QUESTION ID

S80-F2-Q5

TYPE

Short answer

CORRECT ANSWER

x appears both inside a logarithm and outside it; no algebraic step isolates x.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSF-TF.A.3 • ADVANCED EXTENSION (+)

Derive values from special triangles and relate π − x, π + x, 2π − x to x

MAIN	Algebra II main question	**HSF-TF.A.3**
BACKWARD 1	Special right triangle side ratios	**HSG-SRT.C.6**
BACKWARD 2	Unit-circle values in all quadrants	**HSF-TF.A.2**
BACKWARD 3	The Pythagorean Theorem	**8.G.B.7**
FORWARD 1	Derive more special values and the reflection identities	**HSF-TF.A.3**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 81 • HSF-TF.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A 30°−60°−90° triangle with hypotenuse 1 is half of an equilateral triangle with side 1. Which derivation of sin(π/3) and cos(π/3) is correct?

A. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the long leg, so sin(π/3) = √3/2 and cos(π/3) = 1/2.

B. The long leg is √(1 + 1/4) = √5/2, so sin(π/3) = √5/2.

C. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the short leg, so sin(π/3) = 1/2 and cos(π/3) = √3/2.

D. The legs are 1/2 and 1 − 1/2 = 1/2, so sin(π/3) = cos(π/3) = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

2975

QUESTION ID

S81-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the long leg, so sin(π/3) = √3/2 and cos(π/3) = 1/2.

GRADING NOTE Halving the base gives 1/2; the Pythagorean Theorem gives √3/2; the larger angle is opposite the longer leg. The others put π/3 opposite the short leg, subtract lengths instead of using the Pythagorean Theorem, or add instead of subtracting squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

Which derivation of tan(π/4) from a 45°−45°−90° triangle is correct?

A. The hypotenuse is twice a leg, so tan(π/4) = 1/2.

B. tan(π/4) = sin(π/4) + cos(π/4) = √2.

C. The legs are 1 and √2, so tan(π/4) = opposite/adjacent = 1/√2 = √2/2.

D. The two legs are equal, so tan(π/4) = opposite/adjacent = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

2976

QUESTION ID

S81-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. The two legs are equal, so tan(π/4) = opposite/adjacent = 1.

GRADING NOTE Equal legs make the ratio 1. The others use the hypotenuse as a leg, add instead of dividing, or use a 30°−60°−90° fact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

Use sin(π − x) = sin x to find sin(5π/6) exactly, and explain the identity with the unit circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

2977

QUESTION ID

S81-M-Q3

TYPE

Short answer

CORRECT ANSWER

sin(5π/6) = sin(π/6) = 1/2: the points at x and π − x are reflections over the y-axis, with the same y-coordinate

GRADING NOTE Form required: value and explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use cos(π + x) = −cos x to find cos(7π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

2978

QUESTION ID

S81-M-Q4

TYPE

Short answer

CORRECT ANSWER

−cos(π/6) = −√3/2

GRADING NOTE Form required: exact value and the relation used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use tan(2π − x) = −tan x to find tan(5π/3) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

2979

QUESTION ID

S81-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-\tan(\pi/3) = -\sqrt{3}$

GRADING NOTE Form required: exact value and the relation used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 1

Special right triangle side ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Give the sides of a 30°−60°−90° triangle with hypotenuse 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

2981

QUESTION ID
S81-B1-Q1

TYPE
Short answer

CORRECT ANSWER
1, √3, 2

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the hypotenuse of a 45°−45°−90° triangle with legs 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

2982

QUESTION ID
S81-B1-Q2

TYPE
Short answer

CORRECT ANSWER
√2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

True or false?

In a 30°−60°−90° triangle, the shortest side is half the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

2983

QUESTION ID

S81-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Half of an equilateral triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos 30° equals

A. √3

B. √2/2

C. 1/2

D. √3/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

2984

QUESTION ID
S81-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. √3/2

GRADING NOTE Adjacent/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give tan 60°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

2985

QUESTION ID

S81-B1-Q5

TYPE

Short answer

CORRECT ANSWER

√3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 2

Unit-circle values in all quadrants • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(5π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

2987

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−√3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B2-Q1**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

2987

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(7π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

2988

QUESTION ID

S81-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B2-Q2**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

2988

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin is positive in quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

2989

QUESTION ID

S81-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $y > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which quadrant contains 5π/3?

A. II

B. I

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

2990

QUESTION ID

S81-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. IV

GRADING NOTE Between 3π/2 and 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

2990

ANSWER KEY Reduced copy of the question page, with its answer

Give tan(π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

2991

QUESTION ID

S81-B2-Q5

TYPE

Short answer

CORRECT ANSWER

√3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B2-Q5**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 3

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 1 and leg 1/2. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

2993

QUESTION ID

S81-B3-Q1

TYPE

Short answer

CORRECT ANSWER

√3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

2993

A right isosceles triangle has hypotenuse 1. Find each leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

2994

QUESTION ID

S81-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(1/2)^2 + (\sqrt{3}/2)^2 = 2.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

2995

QUESTION ID

S81-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/4 + 3/4 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 1 and 1 give hypotenuse

A. 2

B. √3

C. √2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

2996

QUESTION ID

S81-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. √2

GRADING NOTE √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find x if $x^2 + x^2 = 1$ ($x > 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

2997

QUESTION ID

S81-B3-Q5

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 1

Derive more special values and the reflection identities • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which derivation of tan(π/6) from a special triangle is correct?

A. The opposite side is half the hypotenuse, so tan(π/6) = 1/2.

B. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg 1/2 and adjacent to the leg √3/2, so tan(π/6) = (1/2)/(√3/2) = 1/√3 = √3/3.

C. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg √3/2 and adjacent to the leg 1/2, so tan(π/6) = (√3/2)/(1/2) = √3 exactly.

D. tan(π/6) = (1/2)/(√3/2) = √3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

2999

QUESTION ID

S81-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg 1/2 and adjacent to the leg √3/2, so tan(π/6) = (1/2)/(√3/2) = 1/√3 = √3/3.

GRADING NOTE Opposite over adjacent, then simplify (1/2)/(√3/2) = 1/√3 = √3/3. The others use the angle π/3, give sin(π/6) instead, or simplify the quotient incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle argument explains why $\cos(\pi - x) = -\cos x$?

A. The points for x and $\pi - x$ are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

B. $\cos(\pi - \pi/3) = -1/2 = -\cos(\pi/3)$, so the rule always holds.

C. $\cos(\pi - x) = \cos \pi - \cos x = -1 - \cos x$.

D. The points for x and $\pi - x$ are reflections of each other over the x-axis, so their x-coordinates (the cosines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

3000

QUESTION ID

S81-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The points for x and π − x are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

GRADING NOTE Reflection over the y-axis changes the sign of x and keeps y. The others name the wrong reflection (it keeps x), distribute cosine over a difference, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(\pi + x) = -\sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

3001

QUESTION ID

S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotation by π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

cos(2π − x) equals

A. cos x

B. −cos x

C. sin x

D. −sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

3002

QUESTION ID

S81-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos x

GRADING NOTE Reflection over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(11π/6) using one of the relations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

3003

QUESTION ID

S81-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−sin(π/6) = −1/2

GRADING NOTE Form required: value and relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Explain why sin(−x) = −sin x with the unit circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

3005

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x and −x give points reflected over the x-axis; their y-coordinates are opposites.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is cos an even or odd function? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

3006

QUESTION ID
S81-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Even: cos(−x) = cos x (same x-coordinate)

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

3006

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(x + 2\pi) = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

3007

QUESTION ID

S81-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

tan(−π/4) equals

A. undefined

B. −1

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

3008

QUESTION ID

S81-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −1

GRADING NOTE Odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $2\pi - x = -x + 2\pi$ to explain why $\tan(2\pi - x) = -\tan x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

3009

QUESTION ID

S81-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$\tan(-x + 2\pi) = \tan(-x)$ by periodicity, and tan is odd, so it equals $-\tan x$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSF-TF.A.4 • ADVANCED EXTENSION (+)

Explain the symmetry and periodicity of trigonometric functions with the unit circle

MAIN	Algebra II main question	**HSF-TF.A.4**
BACKWARD 1	Explain the unit-circle definition	**HSF-TF.A.2**
BACKWARD 2	Even and odd functions	**HSF-BF.B.3**
BACKWARD 3	Reflect points over the axes	**8.G.A.3**
FORWARD 1	Use periodicity and symmetry to evaluate and simplify	**HSF-TF.A.4**
FORWARD 2	Restrict trigonometric functions so they have inverses	**HSF-TF.B.6**

SET 82 • HSF-TF.A.4

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why $\sin(-x) = -\sin x$?

A. The points for x and −x are (a, b) and (−a, b), reflections of each other over the y-axis, so their sines are opposites.

B. $\sin(-\pi/6) = -1/2 = -\sin(\pi/6)$, so the rule always holds.

C. The points for x and −x are (a, b) and (a, −b), reflections over the x-axis, so the sines b and −b are opposites.

D. A negative sign can always be moved outside a function, so $\sin(-x) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

3012

QUESTION ID

S82-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The points for x and −x are (a, b) and (a, −b), reflections over the x-axis, so the sines b and −b are opposites.

GRADING NOTE Equal arcs in opposite directions from (1, 0) give points that mirror over the x-axis. The others name the wrong reflection, use a false general rule ($\cos(-x) = \cos x$, not $-\cos x$), or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why cos(x + 2π) = cos x?

A. Adding 2π moves the point to the opposite side of the unit circle, and opposite points on the circle have the same x-coordinate, so the cosine is the same.

B. cos(x + 2π) = cos x + cos 2π = cos x + 1.

C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

D. cos(0 + 2π) = 1 = cos 0, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

3013

QUESTION ID

S82-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

GRADING NOTE The circumference is 2π, so x and x + 2π give the same point. The others distribute cosine over a sum, describe the wrong motion, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

Which unit-circle explanation shows why tan(x + π) = tan x?

A. Adding π gives the point (a, −b), a reflection over the x-axis, so tan(x + π) = −b/a = −tan x, which is not tan x.

B. Adding π gives the opposite point (−a, −b) of (a, b), and tan = (−b)/(−a) = b/a, the same ratio.

C. Adding π travels a full turn, so the point is the same.

D. tan(π/4 + π) = 1 = tan(π/4), so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

3014

QUESTION ID

S82-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Adding π gives the opposite point (−a, −b) of (a, b), and tan = (−b)/(−a) = b/a, the same ratio.

GRADING NOTE Half a turn takes (a, b) to (−a, −b); both coordinates change sign, so the ratio is unchanged. The others use the wrong point, mistake π for a full turn, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use symmetry to evaluate cos(−2π/3), and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

3015

QUESTION ID

S82-M-Q4

TYPE

Short answer

CORRECT ANSWER

−1/2: cos is even, so cos(−2π/3) = cos(2π/3) = −1/2

GRADING NOTE Form required: value and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is f(x) = sin x cos x even, odd, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

3016

QUESTION ID

S82-M-Q5

TYPE

Short answer

CORRECT ANSWER

Odd: f(−x) = (−sin x)(cos x) = −f(x)

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 1

Explain the unit-circle definition • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

What point represents angle x on the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

3018

QUESTION ID

S82-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(cos x, sin x)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which direction does a negative angle go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

3019

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Clockwise

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

3019

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

tan x = sin x/cos x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

3020

QUESTION ID

S82-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

The point for angle π is

A. (0, −1)

B. (−1, 0)

C. (0, 1)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

3021

QUESTION ID

S82-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 0)

GRADING NOTE Half turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

3022

QUESTION ID

S82-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 2

Even and odd functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Define an even function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

3024

QUESTION ID

S82-B2-Q1

TYPE

Short answer

CORRECT ANSWER

f(−x) = f(x) for every x in the domain

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Define an odd function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

3025

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

f(−x) = −f(x) for every x in the domain

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An odd function's graph is symmetric about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

3026

QUESTION ID

S82-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotation by 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

x^3 is

A. odd

B. neither

C. even

D. both

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q4/5

3027

QUESTION ID

S82-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. odd

GRADING NOTE $(-x)^3 = -x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is the product of an odd and an even function odd or even?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

3028

QUESTION ID

S82-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Odd

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 3

Reflect points over the axes • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Reflect (a, b) over the x-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

3030

QUESTION ID

S82-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(a, −b)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

3030

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (a, b) by 180° about the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

3031

QUESTION ID

S82-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(−a, −b)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

True or false?

Reflecting over the x-axis keeps the y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

3032

QUESTION ID

S82-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (a, b) → (a, −b): the x-coordinate stays and y changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

Reflecting (0.6, 0.8) over the x-axis gives

A. (0.8, 0.6)

B. (−0.6, 0.8)

C. (−0.6, −0.8)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

3033

QUESTION ID

S82-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (0.6, −0.8)

GRADING NOTE Negate y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

3033

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (√3/2, 1/2) by 180°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

3034

QUESTION ID

S82-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(−√3/2, −1/2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 1

Use periodicity and symmetry to evaluate and simplify • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(17π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

3036

QUESTION ID

S82-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE Any equivalent form. 17π/6 − 2π = 5π/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find tan(−11π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

3037

QUESTION ID

S82-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. tan(−11π/4) = tan(−11π/4 + 3π) = tan(π/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos(x + \pi) = -\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

3038

QUESTION ID

S82-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Opposite point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sin(x + \pi) + \sin x$ equals

A. 1

B. $\sin 2x$

C. $2 \sin x$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

3039

QUESTION ID

S82-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $\sin(x + \pi) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify cos(−x) + cos(x + 2π).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

3040

QUESTION ID

S82-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2cos x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 2

Restrict trigonometric functions so they have inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Explain how periodicity prevents sin x from having an inverse on all real numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

3042

QUESTION ID

S82-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each value repeats every 2π, so outputs do not determine inputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the restricted domain used for tan x and explain it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

3043

QUESTION ID

S82-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−π/2 < x < π/2: tan increases there and takes every real value once.

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

True or false?

arccos gives values between 0 and π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

3044

QUESTION ID

S82-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

arctan(−1) equals

A. 7π/4

B. 3π/4

C. −π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

3045

QUESTION ID

S82-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −π/4

GRADING NOTE In (−π/2, π/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use oddness to explain why $\arcsin(-1/2) = -\arcsin(1/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

3046

QUESTION ID

S82-F2-Q5

TYPE

Short answer

CORRECT ANSWER

sin is odd on $[-\pi/2, \pi/2]$, so $\sin(-\pi/6) = -1/2$ and $\arcsin(-1/2) = -\pi/6 = -\arcsin(1/2)$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSF-TF.B.6 • ADVANCED EXTENSION (+)

Choose and justify domain restrictions that make trigonometric functions invertible

MAIN	Algebra II main question	**HSF-TF.B.6**
BACKWARD 1	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**
BACKWARD 2	Exact values on the unit circle	**HSF-TF.A.2**
BACKWARD 3	Periodicity of trigonometric functions	**HSF-TF.A.4**
FORWARD 1	Compose trigonometric functions with their inverses	**HSF-TF.B.6**
FORWARD 2	Solve trigonometric equations with inverses	**HSF-TF.B.7**

SET 83 • HSF-TF.B.6

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Explain why sin x has no inverse on all real numbers, and give a restricted domain on which it does.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

3049

QUESTION ID
S83-M-Q1

TYPE
Short answer

CORRECT ANSWER

sin repeats values (not one-to-one); on $-\pi/2 \le x \le \pi/2$ it is increasing and takes every value in $[-1, 1]$ once

GRADING NOTE Form required: reason and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the standard restricted domain of cos x and justify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

3050

QUESTION ID

S83-M-Q2

TYPE

Short answer

CORRECT ANSWER

0 ≤ x ≤ π: cos decreases from 1 to −1 there, taking each value once

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the standard restricted domain of tan x and justify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

3051

QUESTION ID

S83-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-\pi/2 < x < \pi/2$: tan increases from $-\infty$ to ∞ there, taking each real value once

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(1/2) and explain why it is not 5π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

3052

QUESTION ID

S83-M-Q4

TYPE

Short answer

CORRECT ANSWER

π/6: arcsin outputs only angles in [−π/2, π/2], and 5π/6 is outside

GRADING NOTE Form required: value and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arccos(−1/2) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

3053

QUESTION ID

S83-M-Q5

TYPE

Short answer

CORRECT ANSWER

2π/3

GRADING NOTE Form required: exact value in [0, π].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 1

Restrict a domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Give a domain on which $f(x) = x^2$ is invertible.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

3055

QUESTION ID

S83-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$ (or $x \leq 0$)

GRADING NOTE Any valid domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

Why does the restricted function need to be increasing or decreasing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

3056

QUESTION ID

S83-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Then no output repeats, so each output has one input

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

True or false?

A function that is always increasing on an interval is one-to-one there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

3057

QUESTION ID

S83-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Monotonic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 1)^2$ is one-to-one on

A. $-1 \le x \le 3$

B. all reals

C. $x \ge 1$

D. $x \ge 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

3058

QUESTION ID

S83-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x \ge 1$

GRADING NOTE One side of the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of x^2 on $x \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

3059

QUESTION ID

S83-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt{x}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 2

Exact values on the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(π/6) and sin(5π/6).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

3061

QUESTION ID

S83-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/2 and 1/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3) and cos(4π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

3062

QUESTION ID

S83-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2 and −1/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-B2-Q2**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

3062

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\tan(\pi/4) = \tan(3\pi/4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q3/5

3063

QUESTION ID

S83-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\tan(\pi/4) = 1$ and $\tan(3\pi/4) = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q3/5

3063

ANSWER KEY Reduced copy of the question page, with its answer

Which angle in [0, π] has cos = −1/2?

A. 2π/3

B. π/3

C. 4π/3

D. 5π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

3064

QUESTION ID

S83-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2π/3

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give $\sin(-\pi/6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

3065

QUESTION ID

S83-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 3

Periodicity of trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

3067

QUESTION ID

S83-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of tan x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

3068

QUESTION ID

S83-B3-Q2

TYPE

Short answer

CORRECT ANSWER

π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

True or false?

A periodic function cannot be one-to-one on its whole domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

3069

QUESTION ID

S83-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Values repeat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin x = 0 for

A. x = π/2 + kπ

B. no x

C. x = 2kπ only

D. x = kπ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q4/5

3070

QUESTION ID

S83-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. x = kπ

GRADING NOTE Every multiple of π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q4/5

3070

ANSWER KEY Reduced copy of the question page, with its answer

Name two angles with cos = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

3071

QUESTION ID

S83-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0 and 2π (for example)

GRADING NOTE Any two valid angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 1

Compose trigonometric functions with their inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(sin(5π/6)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

3073

QUESTION ID

S83-F1-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate sin(arcsin(0.3)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

3074

QUESTION ID

S83-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

True or false?

arccos(cos x) = x for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

3075

QUESTION ID

S83-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only for 0 ≤ x ≤ π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arctan(tan(3π/4)) equals

A. −π/4

B. −3π/4

C. 3π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

3076

QUESTION ID

S83-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −π/4

GRADING NOTE Into (−π/2, π/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain and range of arccos.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

3077

QUESTION ID

S83-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Domain [−1, 1]; range [0, π]

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 2

Solve trigonometric equations with inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sin x = 0.4$ on $0 \le x < 2\pi$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

3079

QUESTION ID

S83-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.41 and 2.73

GRADING NOTE Form required: both. arcsin(0.4) and π − arcsin(0.4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\cos x = 0.4$ on $0 \le x < 2\pi$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

3080

QUESTION ID

S83-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.16 and 5.12

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

True or false?

arcsin gives only one of the solutions of sin x = c on [0, 2π).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

3081

QUESTION ID

S83-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Use symmetry for the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The second solution of $\sin x = c$ in $[0, 2\pi)$ (with $0 < c < 1$) is

A. $\pi + \arcsin c$

B. $-\arcsin c$

C. $\pi - \arcsin c$

D. $2\pi - \arcsin c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

3082

QUESTION ID

S83-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π − arcsin c

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the restricted range of arccos is used to find all solutions of cos x = c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

3083

QUESTION ID
S83-F2-Q5

TYPE
Short answer

CORRECT ANSWER
arccos gives the solution in [0, π]; the other is 2π − arccos c by symmetry.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSF-TF.B.7 • ADVANCED EXTENSION (+)

Solve trigonometric equations in models with inverse functions

MAIN	Algebra II main question	**HSF-TF.B.7**
BACKWARD 1	Evaluate inverse trigonometric functions on restricted domains	**HSF-TF.B.6**
BACKWARD 2	Parameters of sinusoidal models	**HSF-TF.B.5**
BACKWARD 3	Symmetry of unit-circle solutions	**HSF-TF.A.2**
FORWARD 1	Find all real solutions and solve equations that factor	**HSF-TF.B.7**
FORWARD 2	Solve equations with sum and double-angle identities	**HSF-TF.C.9**

SET 84 • HSF-TF.B.7

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\sin x - 1 = 0$ for $0 \le x < 2\pi$. Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

3086

QUESTION ID

S84-M-Q1

TYPE

Short answer

CORRECT ANSWER

π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3\cos x = -2$ for $0 \le x < 2\pi$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

3087

QUESTION ID

S84-M-Q2

TYPE

Short answer

CORRECT ANSWER

2.30 and 3.98

GRADING NOTE Form required: both. $\arccos(-2/3)$ and $2\pi - \arccos(-2/3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m. Find every time in $0 \le t < 30$ s when $h = 17$ m.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

3088

QUESTION ID

S84-M-Q3

TYPE

Short answer

CORRECT ANSWER

2.5 s and 12.5 s

GRADING NOTE Form required: both. $\sin(\pi t/15) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

3088

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depth is $d(t) = 5 + 4\cos(\pi t/6)$ ft. Find every time in $0 \le t < 12$ h when $d = 7$ ft.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

3089

QUESTION ID
S84-M-Q4

TYPE
Short answer

CORRECT ANSWER
2 h and 10 h

GRADING NOTE Form required: both. $\cos(\pi t/6) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\tan x = 3$ for $0 \le x < 2\pi$ to the nearest hundredth.
A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

3090

QUESTION ID
S84-M-Q5

TYPE
Short answer

CORRECT ANSWER
1.25 and 4.39

GRADING NOTE Form required: both. arctan 3 and arctan 3 + π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 1

Evaluate inverse trigonometric functions on restricted domains • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

3092

QUESTION ID

S84-B1-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arccos(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

3093

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-B1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

3093

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

arctan 3 is between 0 and π/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

3094

QUESTION ID

S84-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Positive input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arccos(−2/3) is in quadrant

A. II

B. III

C. I

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

3095

QUESTION ID

S84-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. II

GRADING NOTE Range [0, π], negative cosine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arctan 3 to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

3096

QUESTION ID

S84-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1.25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-B1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 2

Parameters of sinusoidal models • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In h(t) = 12 + 10sin(πt/15), what is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

3098

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30 s

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite h = 17 as an equation for sin(πt/15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

3099

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

sin(πt/15) = 1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$d(t) = 5 + 4\cos(\pi t/6)$ has a maximum of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q3/5

3100

QUESTION ID

S84-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The maximum is 5 + 4 = 9; 5 is the midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q3/5

3100

ANSWER KEY Reduced copy of the question page, with its answer

Isolating the cosine in 5 + 4cos(πt/6) = 7 gives

A. cos(πt/6) = 1/2

B. cos(πt/6) = 3

C. cos(πt/6) = 2

D. cos(πt/6) = 7/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q4/5

3101

QUESTION ID

S84-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos(πt/6) = 1/2

GRADING NOTE Subtract 5, divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the midline of d(t)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

3102

QUESTION ID

S84-B2-Q5

TYPE

Short answer

CORRECT ANSWER

d = 5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 3

Symmetry of unit-circle solutions • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with sin = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

3104

QUESTION ID

S84-B3-Q1

TYPE

Short answer

CORRECT ANSWER

π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with cos = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

3105

QUESTION ID

S84-B3-Q2

TYPE

Short answer

CORRECT ANSWER

π/3 and 5π/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

tan x = c has two solutions in [0, 2π) that differ by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

3106

QUESTION ID

S84-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Period π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If cos x = c has solution a in [0, π], the other in [0, 2π) is

A. 2π − a

B. π + a

C. −a

D. π − a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

3107

QUESTION ID

S84-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2π − a

GRADING NOTE Reflection over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with tan = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

3108

QUESTION ID

S84-B3-Q5

TYPE

Short answer

CORRECT ANSWER

π/4 and 5π/4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 1

Find all real solutions and solve equations that factor • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Write all real solutions of $2\sin x - 1 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

3110

QUESTION ID

S84-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \pi/6 + 2k\pi$ or $x = 5\pi/6 + 2k\pi$ (k an integer)

GRADING NOTE Form required: both families.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\sin^2 x - \sin x = 0$ for $0 \le x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

3111

QUESTION ID

S84-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0, π, π/6, 5π/6

GRADING NOTE Form required: all four. $\sin x(2\sin x - 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

True or false?

cos x = 2 has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

3112

QUESTION ID

S84-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many solutions does $\sin(2x) = 1/2$ have in $0 \le x < 2\pi$?

A. 2

B. 4

C. 8

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

3113

QUESTION ID

S84-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 2x runs over two periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\cos^2 x = 1/4$ for $0 \le x < 2\pi$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

3114

QUESTION ID

S84-F1-Q5

TYPE

Short answer

CORRECT ANSWER

π/3, 2π/3, 4π/3, 5π/3

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 2

Solve equations with sum and double-angle identities • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Use $\sin 2x = 2\sin x \cos x$ to solve $\sin 2x = \cos x$ for $0 \le x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

3116

QUESTION ID

S84-F2-Q1

TYPE

Short answer

CORRECT ANSWER

π/6, π/2, 5π/6, 3π/2

GRADING NOTE Form required: all four. $\cos x(2\sin x - 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\cos 2x = 1 - 2\sin^2 x$ to solve $\cos 2x = \sin x$ for $0 \le x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

3117

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

π/6, 5π/6, 3π/2

GRADING NOTE Form required: all three. $2\sin^2 x + \sin x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(x + \pi/2) = \cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q3/5

3118

QUESTION ID
S84-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Sum formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos 2x equals

A. $1 + 2\sin^2 x$

B. $\cos^2 x - \sin^2 x$

C. $\cos^2 x + \sin^2 x$

D. $2\cos x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

3119

QUESTION ID
S84-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $\cos^2 x - \sin^2 x$

GRADING NOTE Double angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

Explain why identities are useful for solving equations with different trigonometric functions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5

3120

QUESTION ID
S84-F2-Q5

TYPE
Short answer

CORRECT ANSWER
They rewrite the equation in one function (or factor it), so inverse functions can be applied.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q5**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5